MAT137: Calculus with Proofs
Course Notes

Compiled using Claude, ChatGPT and Google Gemini

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These notes are for the University of Toronto course **MAT137: Calculus with Proofs**. They are heavily adapted from the work of **Alfonso Gracia-Saz** via the MAT137 YouTube channel lectures and related course materials. Any remaining errors, omissions, or typos are the responsibility of the compiler(s) of these notes.

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Chapter 1

Logic, Definitions, Notation, and Proofs

1.1 Sets and Set Notation

Before we do any calculus, we need a common language. We begin with the most basic objects in mathematics—sets—and the notation we use to talk about them.

Definition 1.1 (Set). A *set* is a collection of objects. The objects that belong to a set are called its *elements*.

Remark. A logician would rightly object that the definition above is not rigorous enough for the purposes of logic and foundations. For our purposes, however, it is good enough, and we keep it simple.

A set is described by listing or characterising its elements inside curly braces. For example,

$$A = \{\, x \in \mathbb{Z} : x \text{ is even} \,\}, \qquad B = \{4,\, 5,\, 6\}.$$

The set A has infinitely many elements; the set B has exactly three.

Definition 1.2 (Element, Subset). Let S and T be sets.

(i) We write $x \in S$ to mean that x is an element of S, and $x \notin S$ to mean that it is not.

(ii) We write $S \subseteq T$ to mean that every element of S is also an element of T; in this case S is called a *subset* of T.

(iii) We write $S \subsetneq T$ to mean that $S \subseteq T$ and $S \neq T$ (a *proper subset*).

Remark (Warning). The notation for "subset" is not universal. Some books use $\subset$ to mean $\subseteq$ (subset, possibly equal), while other books use $\subset$ to mean $\subsetneq$ (proper subset). Check the conventions in whatever text you are reading. In these notes we use $\subseteq$ and $\subsetneq$ exclusively, so there is no ambiguity.

Definition 1.3 (Union, Intersection, Empty Set). Let S and T be sets.

(i) The *union* $S \cup T$ is the set of elements belonging to S, to T, or to both.

(ii) The *intersection* $S \cap T$ is the set of elements belonging to **both** S and T.

(iii) The *empty set* $\varnothing$ is the set with no elements: $\varnothing = \{\,\}$.

Example 1.1. Let $C = \{2, 4\}$ and $D = \{4, 5\}$. Then

$$C \cup D = \{2, 4, 5\}, \qquad C \cap D = \{4\}.$$

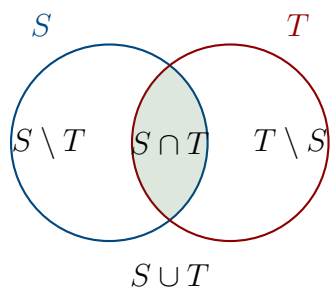

Figure 1.1: Union and intersection.

Four sets of numbers appear so frequently that they have standard names and symbols. Each is a subset of the next.

Definition 1.4 (Standard Number Sets). (i) $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ (the *natural numbers*).

(ii) $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ (the *integers*).

(iii) $\mathbb{Q} = \left\{ \dfrac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\}$ (the *rational numbers*).

(iv) $\mathbb{R}$ denotes the *real numbers*: informally, any number with a decimal expansion.

We have $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$.

Remark. Some books start $\mathbb{N}$ at 1 instead of 0. Do not argue about which convention is "right"; what matters is that you know which convention is in force whenever you do mathematics or read a book.

Remark. There is a more formal and rigorous way to define $\mathbb{R}$, but that construction is surprisingly deep and complicated; it is normally the topic of an analysis course.

1.2 Set-Builder Notation and Intervals

We now introduce a powerful notation that lets us describe sets precisely, even when an explicit list of elements is impossible.

Definition 1.5 (Set-Builder Notation). A set may be described by writing

$$\{\, x \in S : P(x) \,\} \qquad \text{or equivalently} \qquad \{\, x \in S \mid P(x) \,\},$$

which is read "the set of elements x in S **such that** $P(x)$." On the left of the colon (or bar) we state **where** we draw our elements from; on the right we state any **extra constraints**.

Remark (Warning). Use the colon or vertical bar to mean "such that" **only** inside set-builder notation. Do not use it in other contexts.

Example 1.2. Let $A = \{\, x \in \mathbb{Z} : x^2 < 6 \,\}$. Since the only integers whose squares are less than 6 are -2, -1, 0, 1, 2, we have $A = \{-2, -1, 0, 1, 2\}$.

Example 1.3. A second common usage: the left side gives the **form** of the elements, and the right side explains the notation. With A as above,

$$B = \{\, 2x : x \in A \,\} = \{-4, -2, 0, 2, 4\}.$$

Here B consists of all elements of the form $2x$ where $x \in A$.

A particularly important family of sets built with set-builder notation are the intervals.

Definition 1.6 (Intervals). Let $a, b \in \mathbb{R}$ with $a \leq b$.

$$\begin{aligned}
[a, b] &= \{\, x \in \mathbb{R} : a \leq x \leq b \,\} && \text{(closed interval)},\\
(a, b) &= \{\, x \in \mathbb{R} : a < x < b \,\} && \text{(open interval)},\\
[a, b) &= \{\, x \in \mathbb{R} : a \leq x < b \,\},\\
(a, b] &= \{\, x \in \mathbb{R} : a < x \leq b \,\}.
\end{aligned}$$

We also define unbounded intervals:

$$\begin{aligned}
[a, \infty) &= \{\, x \in \mathbb{R} : x \geq a \,\}, & (a, \infty) &= \{\, x \in \mathbb{R} : x > a \,\},\\
(-\infty, b] &= \{\, x \in \mathbb{R} : x \leq b \,\}, & (-\infty, b) &= \{\, x \in \mathbb{R} : x < b \,\}.
\end{aligned}$$

Remark. The symbol ∞ is **not** a number; it is never included as an endpoint. The notation (a, ∞) simply means "the set of all real numbers greater than a, with no upper bound."

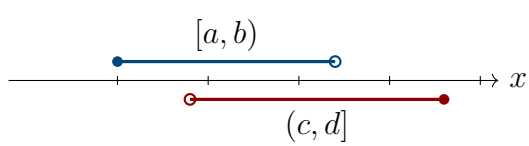

Figure 1.2: Open/closed endpoints on a number line.

1.3 Mathematical Quantifiers

*Many mathematical statements assert that something holds for **all** objects, or that **there exists** an object with a certain property. The two quantifier symbols make these assertions precise.*

Definition 1.7 (Quantifiers). (i) The symbol $\forall$ is read "*for all*" or "*for every*."

(ii) The symbol $\exists$ is read "*there exists*" or "*there is*."

Remark. In mathematics, "there exists" **always** means "there exists **at least one**." It does not mean "there exists exactly one." For instance, saying that an equation "has a solution" means it has at least one solution. If you want to say exactly one, you must say so explicitly.

Example 1.4. Consider the following statements.

(a) $\forall x \in \mathbb{R},\ x^2 \geq 0$. This is **true**; every real number has a non-negative square.

(b) $\forall x \in \mathbb{Z},\ x > \pi$. This is **false**; for example, $x = 1$ is an integer with $1 < \pi$.

Example 1.5. Consider the following statements.

(a) $\exists x \in \mathbb{R}$ such that $x^2 = 5$. This is **true**; take $x = \sqrt{5}$ (or $x = -\sqrt{5}$).

(b) $\exists x \in \mathbb{R}$ such that $x^2 = -1$. This is **false**; no real number has a negative square.

Remark. The connector "such that" adds no mathematical content; it simply makes the sentence grammatically correct. Logicians often replace it with a comma, but writing it out makes statements easier to read.

Remark (Warning). Writing "$x^2 = 5$" by itself, with no quantifier and no prior agreement about x, is **meaningless**: it is neither true nor false. Such a fragment is sometimes called a *predicate* rather than a *statement*. Every variable in a statement must be either quantified or previously fixed. If you are ever tempted to begin a definition, theorem, or proof with an equation or inequality involving an unquantified variable—do not do it.

1.4 Order of Multiple Quantifiers

When a statement involves more than one quantifier, the order in which they appear changes the meaning drastically. We illustrate with a pair of statements that differ only in the order of their quantifiers, yet one is true and the other is false.

Example 1.6. Compare:

(a) $\forall x \in \mathbb{Z},\ \exists y \in \mathbb{Z}$ such that $x < y$.

(b) $\exists y \in \mathbb{Z}$ such that $\forall x \in \mathbb{Z},\ x < y$.

Remark (Pause and Try). Before reading on, decide which statement is true and which is false, and why.

In statement (b) we claim there is a ***single*** *integer y that is greater than* ***every*** *integer x. The same y must work for all x. This is clearly false: there is no largest integer.*

In statement (a) we claim that ***for every*** *integer x, we can find* ***some*** *integer y with $x < y$. Crucially, we are allowed to use a* ***different*** *y for each x. Given any x, simply take $y = x + 1$. So statement (a) is true.*

Theorem 1.1. *For every integer x, there exists an integer y such that $x < y$.*

Remark. The moral is that the order of quantifiers tells us **what depends on what**. In (a), y is allowed to depend on x; in (b), y must be chosen independently of x. As statements grow to involve many quantifiers, keeping track of dependencies becomes essential.

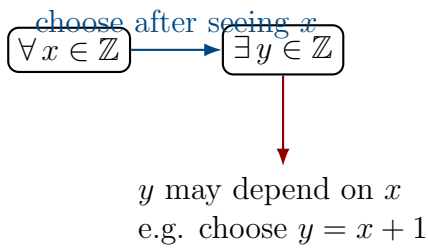

Figure 1.3: Quantifier order encodes dependence.

1.5 Simple Proofs with Quantifiers

We now write proofs of some deliberately simple statements so that we can see the building blocks clearly, before assembling them into more complex proofs later.

Theorem 1.2. *Let $A = \{2, 3, 4\}$. Then for every $x \in A$, we have $x > 0$.*

Proof. We check each element of A individually: $2 > 0$, $3 > 0$, and $4 > 0$. ■

What if the claim is false? To disprove a "for all" statement, we prove its ***negation****: a "there exists" statement.*

Exhaustive check of a finite set.

Definition 1.8 (Negation). The *negation* of a statement P is a statement that is true exactly when P is false.

The negation of "$\forall x \in S,\ P(x)$" is "$\exists x \in S$ such that $\neg P(x)$."

Theorem 1.3. *It is false that every integer is positive. Equivalently, there exists an integer x with $x \leq 0$.*

Proof. Take $x = -6$. Then $x \in \mathbb{Z}$ and $x \leq 0$. ■

Exhibit one counterexample for an existential.

*Next, a statement with **two** quantifiers, where the proof cannot proceed by exhaustive checking because there are infinitely many cases.*

Theorem 1.4. *For every integer x, there exists an integer y such that $x < y$.*

Proof. Let x be an integer (fixing an arbitrary value). Take $y = x + 1$. Then y is an integer, and $x < x + 1 = y$. ■

Fix the universally quantified variable; choose the existential one.

Remark. The word "let" in a proof means we are **fixing** an **arbitrary** value of the variable. "Arbitrary" because the proof must work for every integer; "fix" because subsequent lines may refer to x and it must carry a definite meaning. In the statement of the theorem, x is a quantified variable with no intrinsic value. In the proof, once we write "let x be an integer," it becomes fixed and we may operate with it.

Remark. The structure here—fix the "for all" variable, then choose the "there exists" variable, then verify—is a template we will reuse in much more complex proofs, including proofs involving the definition of limit.

1.6 Quantifiers and the Empty Set

A common point of confusion arises when quantifiers are applied to the empty set. There is a trap here.

Example 1.7. Consider the two statements:

(a) $\forall\, x \in \varnothing,\ x > 0$.

(b) $\exists\, x \in \varnothing$ such that $x > 0$.

One is true and the other is false.

Remark (Pause and Try). Before reading on, decide which is true and which is false.

*Statement (a) is **true**.* To prove a "for all" statement, we must check that **all** elements of the set satisfy the condition. The empty set has zero elements, so there are zero elements to check; the verification is vacuously complete. Conversely, to disprove it we would need to find one element that fails, but the empty set contains no elements at all, so we cannot.

*Statement (b) is **false**.* To prove a "there exists" statement, we must exhibit at least one element with the desired property, and no element of $\varnothing$ exists.

Theorem 1.5 (Vacuous Truth). *For any property P, the statement "$\forall\, x \in \varnothing,\ P(x)$" is true, and the statement "$\exists\, x \in \varnothing$ such that $P(x)$" is false.*

Remark. The specific condition ($x > 0$ in the example) is irrelevant. The argument depends only on the set being empty.

1.7 Conditional Statements and Implication

Conditional statements—sentences of the form "if . . . , then . . . "—appear constantly in mathematics. We now pin down their precise meaning, which is narrower than everyday English usage.

Definition 1.9 (Conditional / Implication). A *conditional statement* has the form

$$\text{If } P, \text{ then } Q, \qquad \text{equivalently written} \qquad P \implies Q.$$

The symbol $\implies$ is read "*implies*." The statement means: whenever P is true, Q must be true as well. **When P is false, we make no claim about Q at all.**

Example 1.8. "If $x > 10$, then $x > 6$" is true for all real numbers x. Consider three cases:

(i) $x = 12$: P is true, Q is true. Compatible.

(ii) $x = 8$: P is false, Q is true. Compatible (we don't care when P is false).

(iii) $x = 3$: P is false, Q is false. Also compatible.

Example 1.9. Let $A \subseteq \mathbb{R}$ and suppose we know that $x \in A \implies x > 0$.

(a) If $x \notin A$: no conclusion (the hypothesis is false).

(b) If $x > 0$: no conclusion (knowing Q is true does not tell us about P).

(c) If $x \leq 0$: we **can** conclude $x \notin A$. If x were in A, it would have to be positive, contradicting $x \leq 0$.

Theorem 1.6 (Contrapositive). *The implication $P \implies Q$ is logically equivalent to its* contrapositive $\neg Q \implies \neg P$.

Remark (Warning). $P \implies Q$ does **not** imply $\neg P \implies \neg Q$. That would be the *inverse*, which is a different statement.

Definition 1.10 (Biconditional / If and Only If). We write

$$P \iff Q$$

and read it "P *if and only if* Q" (often abbreviated *iff*). It means that **both** $P \implies Q$ and $Q \implies P$ hold. Equivalently, P and Q are always both true or both false; we say P and Q are *equivalent*.

Example 1.10. Let $n \in \mathbb{Z}$.

(a) "n is even" does **not** imply "n is a multiple of 4" (take $n = 6$). However, "n is a multiple of 4" $\implies$ "n is even."

(b) "n is even" $\iff$ "$n + 1$ is odd." Both implications hold.

Remark (Warning). In casual English, "if" sometimes means "if and only if." In mathematics, the two are **strictly distinguished**. Always check whether a statement asserts one implication or both.

Example 1.11. Is the following statement true or false?

$$\text{“If } 0 > 1, \text{ then 478,871 is prime.”}$$

The hypothesis $0 > 1$ is false. Since the "if" part is false, the entire implication is **true**, regardless of whether 478,871 is prime.

Remark. This may feel counterintuitive, but it follows directly from the definition: an implication makes a promise only when the hypothesis is true. When the hypothesis is false, the promise is vacuously kept.

1.8 Negation of Conditional Statements

A common and important fact: the negation of a conditional statement is ***never*** *another conditional statement. Understanding this is critical for writing correct proofs.*

Let $A \subseteq \mathbb{R}$. Consider the conditional:

$$x \in A \implies x > 0.$$

What does this really say? It asserts that **every** real number x falls into one of three cases:

(i) $x \notin A$ and $x > 0$,

(ii) $x \notin A$ and $x \leq 0$,

(iii) $x \in A$ and $x > 0$.

There is a **hidden quantifier**: the statement means $\forall\, x \in \mathbb{R}$, if $x \in A$ then $x > 0$.

Theorem 1.7 (Negation of a Conditional). *The negation of*

$$\forall\, x,\ (P(x) \implies Q(x))$$

is

$$\exists\, x \text{ such that } P(x) \text{ and } \neg Q(x).$$

In other words, the only way the conditional can fail is if there is some x for which the "if" part is **true** and the "then" part is **false**.

Remark (Warning). The negation of an "if... then" is **never** another "if... then." Also, many conditional statements contain a **hidden, implicit** universal quantifier. When you write the negation, you **must** write that quantifier out explicitly.

1.9 Structure of a Correct Proof

*We now look at a common **bad** proof, identify precisely what goes wrong, and then write a correct proof of the same result. This illustrates several principles that every proof must satisfy.*

Theorem 1.8 (AM–GM Inequality for Two Variables). *If x, $y \geq 0$, then*

$$\sqrt{xy} \leq \frac{x+y}{2}.$$

A common incorrect attempt proceeds as follows: start by writing the desired inequality, square both sides, rearrange, and arrive at $(x-y)^2 \geq 0$. Since a square is non-negative, conclude that the original inequality is true.

Remark (Warning). This "proof" is wrong for several reasons.

(1) It **assumes what it wants to prove**. The argument only shows: if the inequality holds, then $(x-y)^2 \geq 0$. That tells us nothing about whether the inequality itself is true.

(2) The statement of the "theorem" was just a bare inequality with no quantifiers. A theorem must say **when** or **for which** values something is true.

(3) The inequality is **not** always true: when $x = y = -1$, the left side is $\sqrt{1} = 1$ and the right side is -1, so $1 \leq -1$ is false. A correct proof must notice the need for $x, y \geq 0$.

(4) The proof contains no words, only algebra. In general, a proof must **explain** what is being done.

Proof of Theorem 1.8. Let x, $y \geq 0$. Since every square is non-negative,

$$(x-y)^2 \geq 0.$$

Expanding and rearranging:

$$\begin{aligned} x^2 - 2xy + y^2 &\geq 0, \\ x^2 + 2xy + y^2 &\geq 4xy, \\ (x+y)^2 &\geq 4xy. \end{aligned}$$

Both sides are non-negative (since $x, y \geq 0$), so we may take square roots of both sides:

$$|x+y| \geq 2\sqrt{xy}.$$

Since $x + y \geq 0$, we have $|x+y| = x+y$, and therefore

$$\sqrt{xy} \leq \frac{x+y}{2}. \qquad \blacksquare$$

*Start from what you **know** is true; end at what you want to prove.*

Remark. You may wonder: how did we *come up with* starting from $(x-y)^2 \geq 0$? The answer is **rough work**. We first take the desired inequality and manipulate it (not as a proof, but as exploration) to discover what known fact it rests on. Once we find that foundation, we write the actual proof **forward**: beginning from the known fact and ending at the desired conclusion. The rough work is not part of the proof; it is the process of *finding* the proof.

Remark. A correct proof has the following structure:

(1) The theorem states clearly **what** is being proved and **when** it holds.

(2) The proof begins with things that are **known** to be true.

(3) Each step follows logically from the previous ones.

(4) Whenever necessary, explanations accompany the algebra.

(5) The proof ends by concluding the desired result.

1.10 Writing Rigorous Definitions

We all have some geometric intuition for what it means for a function to be "increasing"—a function that "always goes up." But "always goes up" is not a mathematical definition. We now illustrate, step by step, how to turn an intuitive idea into a precise, rigorous definition.

Remark (Warning). Some students believe that the definition of "increasing" is "positive derivative." That is **wrong**. A function can be increasing even at a point where it has a corner and no derivative at all. We need a definition that captures every function we intuitively regard as increasing, including such examples.

To formalise "always goes up," compare two input values x_1 and x_2. If $x_1 < x_2$, we want $f(x_1) < f(x_2)$. But this pair of inequalities by itself is not a definition; we need to specify the function, the domain, and a quantifier.

Definition 1.11 (Increasing Function). Let I be an interval and let f be a function defined on I. We say f is *increasing on I* if

$$\forall\, x_1,\, x_2 \in I, \quad x_1 < x_2 \implies f(x_1) < f(x_2).$$

Remark. Every piece of this definition is necessary:

(1) We define "increasing **on** I," not just "increasing," because a function can be increasing on one interval and not on another.

(2) The universal quantifier $\forall$ is required; without it, we would not know whether the condition applies to all pairs or just some.

(3) The conditional $\implies$ is required. Writing "$x_1 < x_2$ and $f(x_1) < f(x_2)$" (with "and" instead of "implies") would demand that *every* pair satisfies $x_1 < x_2$, which is never true.

Without any one of these pieces, the definition would mean something different, or nothing at all.

1.11 Verifying a Definition Directly

Once we have a definition, the definition itself tells us what a proof should look like. We do not need to memorise a separate proof template: the structure of the proof is contained in the structure of the definition.

Theorem 1.9. *The function $f(x) = 3x + 7$ is increasing on $\mathbb{R}$.*

By Definition 1.11, what we want to show (WTS) is:

$$\forall\, x_1,\, x_2 \in \mathbb{R}, \quad x_1 < x_2 \implies f(x_1) < f(x_2).$$

Proof. Let $x_1,\, x_2 \in \mathbb{R}$ and assume $x_1 < x_2$. Multiplying both sides by 3 (positive, so the inequality is preserved):

$$3x_1 < 3x_2.$$

Adding 7 to both sides:

$$3x_1 + 7 < 3x_2 + 7,$$

that is, $f(x_1) < f(x_2)$. ■

Fix the universal variables; assume the hypothesis of the conditional; conclude.

Remark. We did not need to memorise a method for proving that a function is increasing. The definition of "increasing" dictated the proof structure: fix arbitrary x_1, x_2; assume $x_1 < x_2$; derive $f(x_1) < f(x_2)$. The same principle applies to **every** mathematical definition.

1.12 Disproving via Definition Negation

*We now prove that an example does **not** satisfy a definition, again using nothing more than the definition itself—specifically, its negation.*

Theorem 1.10. *The function $g(x) = \cos x$ is not increasing on $[0,\, \pi]$.*

*To say g is **not** increasing on $[0,\, \pi]$ is to negate the definition of increasing: there must exist a pair $x_1,\, x_2 \in [0,\, \pi]$ with $x_1 < x_2$ and $g(x_1) \geq g(x_2)$.*

Proof. Take $x_1 = 0$ and $x_2 = \frac{\pi}{2}$. Then $x_1 < x_2$, but

$$g(x_1) = \cos 0 = 1 \geq 0 = \cos \frac{\pi}{2} = g(x_2).$$

Hence g is not increasing on $[0,\, \pi]$. ■

Exhibit one pair violating the conditional.

Remark. The lesson applies to any concept: to show that something does **not** satisfy a definition, negate the definition and prove the negation. No special technique is needed beyond understanding the definition.

1.13 Proofs Involving Hypotheses

*Most theorems take the form "if ..., then ...": there is a **hypothesis** that we may assume, and a **conclusion** that we must prove. We illustrate with a slightly more complex proof that uses the definition of increasing together with hypotheses.*

Theorem 1.11. *Let f and g be functions defined on an interval I, and let $h = f + g$. If f and g are both increasing on I, then h is increasing on I.*

Remark. The statement before the "if" is the *set-up*: it introduces the notation. The "if" part is the *hypothesis*: something we are allowed to **assume** during the proof. The "then" part is the *conclusion*: what we must **prove**. Treat them very differently.

Proof. We want to show that h is increasing on I, i.e.

$$\forall\, x_1,\, x_2 \in I, \quad x_1 < x_2 \implies h(x_1) < h(x_2).$$

Let $x_1,\, x_2 \in I$ with $x_1 < x_2$. Since f is increasing on I,

$$f(x_1) < f(x_2).$$

Since g is increasing on I,

$$g(x_1) < g(x_2).$$

Adding these two inequalities:

$$f(x_1) + g(x_1) < f(x_2) + g(x_2),$$

that is, $h(x_1) < h(x_2)$. ■

Use both hypotheses; add the resulting inequalities.

Remark. Notice how f and g were treated differently from h. For f and g, we **assumed** they were increasing (hypothesis) and **used** that fact. For h, we **proved** it was increasing (conclusion). This is the fundamental distinction between hypothesis and conclusion in a proof.

1.14 Proof by Induction

Proof by induction is a technique especially suited to theorems that depend on a natural number n and that cannot be checked one value at a time (because there are infinitely many values), yet for which a direct proof for all n at once is not apparent.

Theorem 1.12. *For every natural number $n \geq 4$,*

$$n! > 2^n.$$

We denote the statement "$n! > 2^n$" by S_n.

Definition 1.12 (Proof by Induction). A *proof by induction* of a statement S_n for all $n \geq n_0$ consists of two steps:

(i) **Base case.** Prove S_{n_0}.

(ii) **Induction step.** Prove that for every $n \geq n_0$, $S_n \implies S_{n+1}$.

Together, these show that S_n holds for every $n \geq n_0$.

Remark. In the induction step, we are **not** proving S_n and we are **not** proving S_{n+1}. We are proving an **implication**: *if* S_n is true, *then* S_{n+1} must be true as well. By itself, the induction step does not tell us whether any S_n is true. Combined with the base case, however, the chain of implications propagates truth from n_0 onward: $S_{n_0} \implies S_{n_0+1} \implies S_{n_0+2} \implies \cdots$.

Proof of Theorem 1.12. We proceed by induction on n.

Base case ($n = 4$). We have $4! = 24$ and $2^4 = 16$, so $4! > 2^4$.

Induction step. Let $n \geq 4$ and assume $n! > 2^n$ (induction hypothesis). We want to show $(n+1)! > 2^{n+1}$.

Since $(n+1)! = (n+1) \cdot n!$ and $2^{n+1} = 2 \cdot 2^n$, we compute:

$$\begin{aligned}
(n+1) \cdot n! &> (n+1) \cdot 2^n && \text{(by the induction hypothesis)} \\
&\geq 5 \cdot 2^n && \text{(since } n \geq 4 \text{ implies } n+1 \geq 5\text{)} \\
&> 2 \cdot 2^n && \text{(since } 5 > 2\text{)} \\
&= 2^{n+1}.
\end{aligned}$$

Hence $(n+1)! > 2^{n+1}$, completing the induction step. ■

Induction hypothesis used in the first inequality.

Remark (Warning). A common misunderstanding: the induction step is **not** "prove S_{n+1}." It is "prove that $S_n \implies S_{n+1}$." We prove the statement itself only for the base case. Everything else is an implication.

1.15 Direct and Inductive Proofs

To illustrate that a theorem may have more than one proof, and that different proof strategies are sometimes available, we prove the same identity by two completely different methods.

Theorem 1.13. *For every positive integer n,*

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Let $S_n = 1 + 2 + \cdots + n$.

First proof (direct). Write the sum in two ways:

$$\begin{aligned} S_n &= 1 + 2 + 3 + \cdots + n, \\ S_n &= n + (n-1) + (n-2) + \cdots + 1. \end{aligned}$$

Adding term by term, each pair sums to $n + 1$ and there are n pairs:

$$2S_n = n(n+1).$$

Therefore $S_n = \dfrac{n(n+1)}{2}$. ■

No induction; a direct algebraic trick handles all n at once.

Second proof (by induction). We proceed by induction on n.

Base case ($n = 1$). $S_1 = 1$ and $\dfrac{1 \cdot 2}{2} = 1$. ✓

Induction step. Let n be a positive integer and assume $S_n = \dfrac{n(n+1)}{2}$ (induction hypothesis). Then

$$\begin{aligned} S_{n+1} &= S_n + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) && \text{(by the induction hypothesis)} \\ &= \frac{n(n+1) + 2(n+1)}{2} \\ &= \frac{(n+1)(n+2)}{2}, \end{aligned}$$

which is the formula with n replaced by $n + 1$. ■

Induction hypothesis used to rewrite S_n.

Remark. Both proofs are equally valid. The first is a direct proof: a single algebraic trick works for all n simultaneously, with no assumptions. The second is an inductive proof: we prove the identity for $n = 1$, then show that truth at n implies truth at $n + 1$. Which proof is "better" is a matter of taste.

Chapter 2

Limits and Continuity

2.1 Intuitive Idea of Limits

We begin not with a rigorous definition, but with examples—hand-wavy, heuristic—just to gain some intuition about what the concept of "limit" means. Once we are comfortable with the idea, we will introduce the formal, rigorous definition in a later section.

Example 2.1. Consider the function f defined by

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

The domain is all of $\mathbb{R}$ except 1; the value $f(1)$ is undefined. We may factor the numerator and cancel:

$$\frac{x^2 - 1}{x - 1} = \frac{(x-1)(x+1)}{x-1} = x + 1, \qquad x \neq 1.$$

Call $g(x) = x + 1$. Then f and g agree everywhere except at 1: $g(1) = 2$, while $f(1)$ is undefined. The graph of f is the line $y = x + 1$ with a **hole** at the point $(1, 2)$.

Looking at the graph, all the values of $f(x)$ for x close to 1 suggest that $f(1)$ "should be" 2. We say that the *limit* of $f(x)$ as x approaches 1 is 2, and write

$$\lim_{x \to 1} f(x) = 2.$$

Example 2.2. Consider the function

$$f(x) = \frac{1 - \sqrt{1+x}}{x},$$

defined for $x \geq -1$, $x \neq 0$. A table of values suggests that as x approaches 0, $f(x)$ approaches $-\frac{1}{2}$. We can verify this algebraically by multiplying numerator and denominator by the conjugate $1 + \sqrt{1+x}$:

$$\frac{1 - \sqrt{1+x}}{x} \cdot \frac{1 + \sqrt{1+x}}{1 + \sqrt{1+x}} = \frac{1 - (1+x)}{x\big(1 + \sqrt{1+x}\big)} = \frac{-1}{1 + \sqrt{1+x}}, \qquad x \neq 0.$$

The simplified function equals $-\frac{1}{2}$ at $x = 0$, confirming $\lim_{x\to 0} f(x) = -\frac{1}{2}$.

In both examples, the rough idea is the same. We summarise it informally as follows.

Remark. The statement "$\lim_{x\to a} f(x) = L$" means, roughly, that if x is very close to a but **not** equal to a, then $f(x)$ is very close to L. This is **not** a definition; the formal definition will come in Section 2.5.

2.2 Limits That Do Not Exist

Not every function has a limit at every point. In this section we examine two ways a limit can fail to exist, and introduce the notation $\lim_{x\to a} f(x) = \infty$.

Example 2.3. Let $h(x) = \sin\left(\dfrac{\pi}{2x}\right)$. The value $h(0)$ is undefined. Evaluating at $x = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$ gives

$$h(1) = 1, \quad h\left(\tfrac{1}{2}\right) = 0, \quad h\left(\tfrac{1}{3}\right) = -1, \quad h\left(\tfrac{1}{4}\right) = 0, \quad h\left(\tfrac{1}{5}\right) = 1, \quad \ldots$$

As x approaches 0, the function oscillates between -1 and 1 without settling near any single number. Therefore $\lim_{x\to 0} h(x)$ **does not exist** (DNE).

Remark. For a limit to exist, the function must approach **one single number**. Oscillation between two or more values prevents this.

Example 2.4. Let $F(x) = \dfrac{1}{(x-1)^2}$. Then $F(1)$ is undefined. When x is close to 1, $(x-1)^2$ is close to 0 but positive, so $F(x)$ grows arbitrarily large. We write

$$\lim_{x\to 1} F(x) = \infty.$$

Remark (Warning). The notation $\lim_{x\to a} f(x) = \infty$ does **not** mean the limit exists. It means the limit does not exist **in a specific way**: $f(x)$ fails to approach any single real number because it grows without bound. Infinity is **not** a real number.

2.3 One-Sided Limits

Sometimes a function approaches different values depending on the direction from which we approach a point. This leads to the notion of side limits.

Example 2.5. Let $G(x) = \dfrac{x^2 + x}{|x|}$. Then $G(0)$ is undefined. Breaking into cases:

$$G(x) = \begin{cases} x + 1 & \text{if } x > 0, \\ -x - 1 & \text{if } x < 0. \end{cases}$$

When x is close to 0 and positive, $G(x)$ is close to 1. When x is close to 0 and negative, $G(x)$ is close to -1. Since the two values disagree, $\lim_{x\to 0} G(x)$ does

not exist. However, the *one-sided limits* do exist:

$$\lim_{x\to 0^+} G(x) = 1, \qquad \lim_{x\to 0^-} G(x) = -1.$$

Remark. The notation $\lim_{x\to a^+} f(x)$ means the limit as x approaches a **from the right** (i.e. through values greater than a). Similarly, $\lim_{x\to a^-} f(x)$ means the limit **from the left**. Together, these are called the *side limits* or *one-sided limits*.

Let us summarise the different behaviours we have seen.

Remark. The limit of a function at a point may or may not exist.

(i) The limit **exists** and equals L when $f(x)$ approaches L as $x \to a$. It does not matter whether $f(a)$ is defined, undefined, or equal to L.

(ii) The limit **does not exist**, and this can happen in several ways:

- the limit is ∞ or $-\infty$ (vertical asymptote),
- the two one-sided limits exist but differ (jump),
- the function oscillates without settling (wild oscillation).

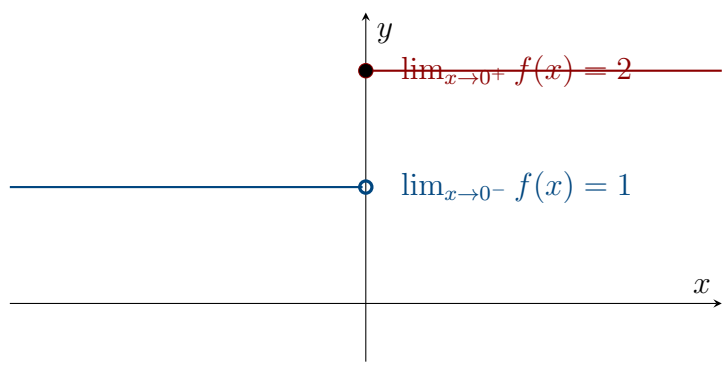

Figure 2.1: A jump discontinuity with distinct one-sided limits.

2.4 Absolute Values and Distance

This section is a short but essential preparation for the formal definition of limit. We explain how to use absolute values and inequalities to talk about distances, both geometrically and algebraically.

Definition 2.1 (Absolute Value). For any $x \in \mathbb{R}$,

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Geometrically, $|x|$ is the *distance* between x and 0 on the real line. More generally, $|x - a|$ is the distance between x and a.

Theorem 2.1 (Properties of Absolute Value). *For all $x, y \in \mathbb{R}$:*

(i) $|xy| = |x|\,|y|$ *(multiplicativity).*

(ii) $|x + y| \leq |x| + |y|$ *(triangle inequality).*

Remark (Warning). The equality $|xy| = |x|\,|y|$ holds for products, but the analogous statement for **sums** is **false**: $|x + y|$ need not equal $|x|+|y|$. Instead, we only have the triangle inequality.

The key idea we need going forward is how to express "x is close to a" using inequalities. The following equivalences should become second nature.

Proposition 2.2 (Equivalent Forms of Closeness). *Let $a \in \mathbb{R}$ and $\delta > 0$. The following statements are equivalent:*

(i) $|x - a| < \delta$.

(ii) The distance between x and a is less than δ.

(iii) $-\delta < x - a < \delta$.

(iv) $a - \delta < x < a + \delta$.

(v) $x \in (a - \delta,\, a + \delta)$.

Remark. Every time you encounter one of these inequalities, you should be able to convert it instantly to any of the other equivalent forms, and to the geometric picture of x lying in an open interval of radius δ centred at a. If you can do that, you are ready for the formal definition of limit.

2.5 Formal Definition of Limit

We now transform the rough idea "if x is close to a but not a, then $f(x)$ is close to L" into a rigorous statement. Every piece of the definition is necessary; omitting or modifying any part would change the meaning.

Definition 2.2 (Limit of a Function at a Point). Let $a, L \in \mathbb{R}$, and let f be a function defined at least on an open interval containing a, except possibly at a itself. We say

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for all $x \in \mathbb{R}$,

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Remark. The second line is an implication with a **hidden universal quantifier**: it means "for every real number x, if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$." It is a standard convention to leave the "for all x" implicit. Write it explicitly if it helps you avoid confusion.

Let us unpack the roles of ε and δ by considering functions that do and do not have limits.

Remark. Think of ε as a challenge and δ as a response. We require that for **every** $\varepsilon > 0$ (no matter how small), we can **find** a $\delta > 0$ (depending on ε) that makes the implication true. Making ε smaller forces the graph into a narrower horizontal band around L; if we can always respond with a δ that keeps the graph inside this band, the limit is L.

For a function that does **not** have limit L, there will be some small ε for which **no** δ works: no matter how close we restrict x to a, the graph escapes the band.

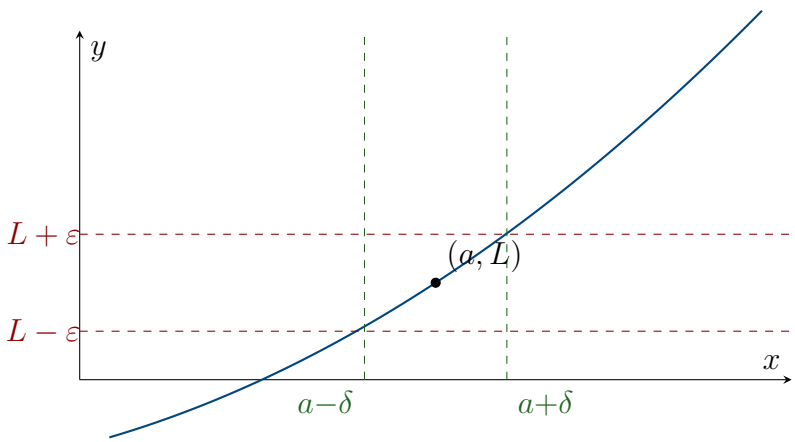

Figure 2.2: An ε-band around L and a δ-window around a.

Remark (Pause and Try). As an exercise, try to write rigorous (ε, δ)-definitions for the one-sided limits $\lim_{x\to a^+} f(x) = L$ and $\lim_{x\to a^-} f(x) = L$, and for $\lim_{x\to a} f(x) = \infty$. If you succeed, it means you truly understand the formal definition.

2.6 Limits at Infinity

We now ask: when x is very large, what happens to $f(x)$? This leads to the notion of limit at infinity.

Definition 2.3 (Limit at Infinity). Let $L \in \mathbb{R}$, and let f be a function defined at least on an interval (p, ∞) for some $p \in \mathbb{R}$. We say

$$\lim_{x\to\infty} f(x) = L$$

if for every $\varepsilon > 0$ there exists a real number M such that for all x,

$$x > M \implies |f(x) - L| < \varepsilon.$$

Example 2.6. Consider $f(x) = \dfrac{x+1}{x} = 1 + \dfrac{1}{x}$. When x is large, $\frac{1}{x}$ is close to 0, so $f(x)$ is close to 1. Therefore $\lim_{x\to\infty} f(x) = 1$.

Remark. When $\lim_{x\to\infty} f(x) = L$, the line $y = L$ is called a *horizontal asymptote* of f at ∞. A horizontal asymptote is **not** a line that the graph "never touches." The graph may cross the asymptote; it may even intersect it infinitely many times. What matters is that eventually, as $x \to \infty$, $f(x)$ stays arbitrarily close to L.

Remark. In the definition, the quantifier structure is: for every $\varepsilon > 0$, there exists M such that ... The important quantifier is ε: we need the implication to hold for **all** positive ε, but for each ε we only need to find **one** value of M. As ε shrinks, M may need to grow.

2.7 Proving Limits from the Definition

We now illustrate, with a simple example, how to prove that a function has a specific limit directly from the (ε, δ)-definition. The emphasis is on the ***structure*** *of the proof: fix ε, choose δ, assume the hypothesis, derive the conclusion.*

Theorem 2.3. $\lim_{x\to 3}(2x + 1) = 7$.

Before writing the proof, we do rough work to discover the right choice of δ. We want to show that $|(2x + 1) - 7| < \varepsilon$ whenever $0 < |x - 3| < \delta$. Observe:

$$|(2x + 1) - 7| = |2x - 6| = 2\,|x - 3|\,.$$

If $|x - 3| < \delta$, then $2\,|x - 3| < 2\delta$. Setting $2\delta = \varepsilon$, i.e. $\delta = \varepsilon/2$, will work.

Proof. Strategy: choose $\delta = \varepsilon/2$ and verify the implication.

Let $\varepsilon > 0$. Take $\delta = \varepsilon/2 > 0$. Fix any $x \in \mathbb{R}$ with $0 < |x - 3| < \delta$. Then

$$\begin{aligned} |(2x + 1) - 7| &= |2x - 6| \\ &= 2\,|x - 3| \\ &< 2\delta \\ &= \varepsilon. \end{aligned}$$

Therefore $|(2x + 1) - 7| < \varepsilon$, as required. ■

2.8 A Harder Epsilon-Delta Proof

The structure of the proof is the same as in Section 2.7, but the algebra is harder. The key new difficulty is that δ cannot simply be ε divided by something; we need to ***bound*** *a factor that depends on x.*

Theorem 2.4. $\lim_{x\to 4}(x^2 + 1) = 17$.

Rough work. We want $|(x^2 + 1) - 17| < \varepsilon$ whenever $0 < |x - 4| < \delta$. Observe:

$$\left|x^2 - 16\right| = |x - 4|\;|x + 4|\,.$$

The factor $|x+4|$ depends on x, so we cannot simply set $\delta = \varepsilon/|x+4|$—that would make δ depend on x, which is not permitted. Instead, we impose $\delta \leq 1$, which forces $x \in (3,5)$ and hence $|x+4| < 9$. Then $|x^2-16| < 9\delta$, and choosing $\delta \leq \varepsilon/9$ finishes the job.

Remark (Warning). In the structure of an (ε,δ)-proof, δ is chosen **before** x is fixed, so δ may depend on ε but **never** on x. Allowing δ to depend on x would change the meaning of the statement entirely and would not establish anything about limits.

Proof. Strategy: choose $\delta = \min\{1, \varepsilon/9\}$ and verify.

Let $\varepsilon > 0$. Take $\delta = \min\left\{1, \dfrac{\varepsilon}{9}\right\} > 0$. Fix any $x \in \mathbb{R}$ with $0 < |x-4| < \delta$.

Since $\delta \leq 1$, we have $|x-4| < 1$, so $3 < x < 5$ and therefore $7 < x+4 < 9$. In particular, $|x+4| < 9$.

Since $\delta \leq \varepsilon/9$, we have $|x-4| < \varepsilon/9$. Therefore:

$$\begin{aligned}\left|(x^2+1)-17\right| &= \left|x^2-16\right| \\ &= |x-4|\,|x+4| \\ &< \frac{\varepsilon}{9}\cdot 9 \\ &= \varepsilon.\end{aligned}$$

Thus $|(x^2+1)-17| < \varepsilon$, as required. ■

Remark. The choice $\delta \leq 1$ is arbitrary; any positive constant would work (producing a different but equally valid proof). The essential idea is to first restrict x to a bounded neighbourhood of 4 so that $|x+4|$ can be bounded by a constant.

2.9 Proving a Limit Does Not Exist

*To show that a limit does **not** exist, we negate the formal definition. This requires four quantifiers and careful handling of each.*

Definition 2.4 (Non-existence of a Limit at Infinity). We say $\lim_{x\to\infty} h(x)$ does not exist if for every $L \in \mathbb{R}$ there exists $\varepsilon > 0$ such that for every $M \in \mathbb{R}$ there exists $x \in \mathbb{R}$ satisfying

$$x > M \quad \text{and} \quad |h(x) - L| \geq \varepsilon.$$

Remark. This definition is obtained by negating the definition of $\lim_{x\to\infty} h(x) = L$ for every possible value of L. If you find the four quantifiers intimidating, revisit the negation one quantifier at a time:

- "for every ε" becomes "there exists ε,"
- "there exists M" becomes "for every M,"

- the hidden "for all x" becomes "there exists x,"
- the implication $P \Rightarrow Q$ becomes $P \wedge \neg Q$.

Theorem 2.5. $\lim_{x\to\infty} \cos(\pi x)$ *does not exist.*

Key observations: when x is an even integer, $\cos(\pi x) = 1$; when x is an odd integer, $\cos(\pi x) = -1$. So $h(x) = \cos(\pi x)$ keeps oscillating between 1 and -1 for arbitrarily large x.

Proof. Strategy: fix an arbitrary L, choose $\varepsilon = \frac{1}{2}$, and for any M exhibit an integer $x > M$ with $|\cos(\pi x) - L| \geq \frac{1}{2}$.

Let $L \in \mathbb{R}$. Take $\varepsilon = \frac{1}{2}$. Let $M \in \mathbb{R}$ be arbitrary.

Since $\varepsilon = \frac{1}{2}$, the open interval $(L - \varepsilon,\, L + \varepsilon)$ has length 1. Therefore it cannot contain **both** 1 and -1 (which are distance 2 apart). At least one of 1 and -1 lies outside $(L - \frac{1}{2},\, L + \frac{1}{2})$.

Case 1: $1 \notin (L - \frac{1}{2},\, L + \frac{1}{2})$. Choose any even integer $x > M$. Then $\cos(\pi x) = 1$, so $|\cos(\pi x) - L| = |1 - L| \geq \frac{1}{2}$.

Case 2: $-1 \notin (L - \frac{1}{2},\, L + \frac{1}{2})$. Choose any odd integer $x > M$. Then $\cos(\pi x) = -1$, so $|\cos(\pi x) - L| = |-1 - L| \geq \frac{1}{2}$.

In either case, we have found $x > M$ with $|\cos(\pi x) - L| \geq \varepsilon$. Since L was arbitrary, the limit does not exist. ■

2.10 The Limit Laws

Computing every limit from the (ε, δ)-definition would be tedious. The limit laws let us reduce complicated limits to simple building blocks. Their importance cannot be overstated: once proven, they free us from epsilon-delta arguments for a huge class of functions.

Theorem 2.6 (Basic Limits). *Let $a, c \in \mathbb{R}$.*

(i) $\lim_{x\to a} x = a$.

(ii) $\lim_{x\to a} c = c$.

Remark (Pause and Try). These are the two simplest (ε, δ)-proofs one can write. Try them as an exercise before proceeding.

Theorem 2.7 (Limit Laws). *Let $a, L, M \in \mathbb{R}$ and let f, g be functions defined on an open interval containing a, except possibly at a. Suppose $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$. Then:*

(i) ***Sum law:*** $\lim_{x\to a}\big[f(x) + g(x)\big] = L + M$.

(ii) ***Product law:*** $\lim_{x\to a}\big[f(x) \cdot g(x)\big] = L \cdot M$.

(iii) ***Quotient law:*** *If $M \neq 0$, then* $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}$.

Remark (Warning). The limit laws **only** apply when the individual limits **exist** as real numbers. If even one of the limits does not exist, we cannot draw any conclusion—positive or negative—about the sum, product, or quotient. For example, $\lim_{x\to 0} x = 0$ and $\lim_{x\to 0} \frac{3}{x}$ does not exist, yet $\lim_{x\to 0} x \cdot \frac{3}{x} = 3$. Likewise, $\frac{1}{x}$ and $\frac{2x-1}{x}$ both lack limits at 0, yet their sum $\frac{1}{x} + \frac{2x-1}{x} = \frac{2x}{x}$ has limit 2.

Example 2.7. To compute $\lim_{x\to 2}(x^4 - 3x)$, note that this polynomial is built from sums and products of x and constants. By Theorem 2.6 and iterated application of the sum and product laws:

$$\lim_{x\to 2}(x^4 - 3x) = 2^4 - 3 \cdot 2 = 10.$$

The same argument shows that for **any** polynomial p and any $a \in \mathbb{R}$, $\lim_{x\to a} p(x) = p(a)$.

2.11 Proof of the Sum Law

We prove the sum law from Theorem 2.7 as a model for how the limit laws are established. The key idea is the triangle inequality: split the error of the sum into the errors of the two summands, and make each one less than $\varepsilon/2$.

Proof. Strategy: use the hypotheses on f and g with $\varepsilon/2$ to obtain δ_1 and δ_2, then set $\delta = \min\{\delta_1, \delta_2\}$.

Let $h = f + g$. We wish to show $\lim_{x\to a} h(x) = L + M$.

Let $\varepsilon > 0$. Since $\lim_{x\to a} f(x) = L$, applying the definition with $\varepsilon/2 > 0$ yields a $\delta_1 > 0$ such that

$$0 < |x - a| < \delta_1 \implies |f(x) - L| < \frac{\varepsilon}{2}.$$

Definition of $\lim_{x\to a} f(x) = L$

Similarly, since $\lim_{x\to a} g(x) = M$, there exists $\delta_2 > 0$ such that

$$0 < |x - a| < \delta_2 \implies |g(x) - M| < \frac{\varepsilon}{2}.$$

Definition of $\lim_{x\to a} g(x) = M$

Take $\delta = \min\{\delta_1, \delta_2\} > 0$. Fix any $x \in \mathbb{R}$ with $0 < |x - a| < \delta$. Then $|x - a| < \delta_1$ and $|x - a| < \delta_2$, so both conclusions above hold. Therefore:

$$\begin{aligned} |h(x) - (L + M)| &= \big|\big(f(x) - L\big) + \big(g(x) - M\big)\big| \\ &\le |f(x) - L| + |g(x) - M| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Triangle inequality (Theorem 2.1)

Thus $|h(x) - (L + M)| < \varepsilon$, completing the proof. ■

Remark. Notice the different ways the definition of limit is used. For h, we are **proving** the limit exists, so we must fix an arbitrary ε and produce δ. For f and g, we **assume** the limits exist, so we may **choose** any ε we like (here $\varepsilon/2$) and are **guaranteed** the existence of a suitable δ.

2.12 The Squeeze Theorem

The squeeze theorem (also called the sandwich theorem) is a powerful tool for computing limits of functions that are hard to handle directly but can be trapped between two simpler functions with the same limit. We motivate it with an example that also illustrates a common pitfall.

Example 2.8. Compute $\lim_{x\to 0}\big(x^2\sin(1/x)\big)$.

*A tempting but **incorrect** argument:* write the limit as a product and apply the product law:

$$\lim_{x\to 0} x^2 \cdot \lim_{x\to 0}\sin(1/x) = 0\cdot(\cdots) = 0. \quad \textbf{WRONG!}$$

This fails because $\lim_{x\to 0}\sin(1/x)$ **does not exist** (the function oscillates wildly), so the product law does not apply.

Remark (Warning). One cannot conclude anything by writing "0 times anything is 0" in the context of limits. The product law requires **both** individual limits to exist. As a cautionary example: $\lim_{x\to 0} x = 0$ and $\lim_{x\to 0}\frac{3}{x}$ does not exist, yet $\lim_{x\to 0} x\cdot\frac{3}{x} = 3 \neq 0$.

The correct approach uses the fact that $|\sin(1/x)| \le 1$, *so* $-x^2 \le x^2\sin(1/x) \le x^2$ *for all* $x \neq 0$. *Both* $-x^2$ *and* x^2 *have limit* 0 *at* $x = 0$. *The function* $x^2\sin(1/x)$ *is "squeezed" between them and must also have limit* 0.

Theorem 2.8 (Squeeze Theorem). *Let* $a, L \in \mathbb{R}$, *and let* f, g, h *be functions defined on an open interval containing* a, *except possibly at* a. *Suppose:*

(i) $h(x) \le g(x) \le f(x)$ *for all* x *sufficiently close to* a *(but* $x \neq a$*),*

(ii) $\lim_{x\to a} h(x) = L$ *and* $\lim_{x\to a} f(x) = L$.

Then $\lim_{x\to a} g(x) = L$.

Example 2.9 *(Continuation of Example 2.8).* Since $-1 \le \sin(1/x) \le 1$ for all $x \neq 0$,

$$-x^2 \le x^2\sin(1/x) \le x^2.$$

Both $\lim_{x\to 0}(-x^2) = 0$ and $\lim_{x\to 0} x^2 = 0$. By the squeeze theorem (Theorem 2.8), $\lim_{x\to 0} x^2\sin(1/x) = 0$.

2.13 Proof of the Squeeze Theorem

We now prove Theorem 2.8. The proof follows the same pattern as the proof of the sum law: use the hypotheses on f *and* h *to find* δ_1 *and* δ_2, *then combine them.*

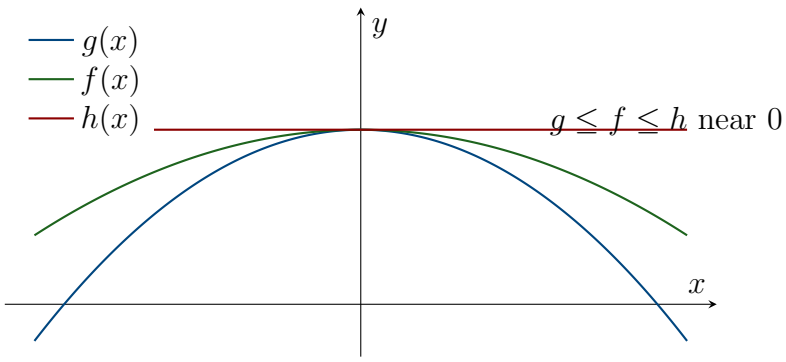

Figure 2.3: A function trapped between two functions with the same limit.

Proof. Strategy: for a given ε, use the limits of f and h to obtain δ_1 and δ_2, then set $\delta = \min\{\delta_1, \delta_2, p\}$, where p controls the region where the inequalities hold.

By hypothesis (i), there exists $p > 0$ such that for all x with $0 < |x - a| < p$,

$$h(x) \le g(x) \le f(x).$$

Let $\varepsilon > 0$. Since $\lim_{x\to a} f(x) = L$, there exists $\delta_1 > 0$ such that

$$0 < |x - a| < \delta_1 \implies |f(x) - L| < \varepsilon.$$

In particular, $f(x) < L + \varepsilon$. *Definition of* $\lim_{x\to a} f(x) = L$

Since $\lim_{x\to a} h(x) = L$, there exists $\delta_2 > 0$ such that

$$0 < |x - a| < \delta_2 \implies |h(x) - L| < \varepsilon.$$

In particular, $h(x) > L - \varepsilon$. *Definition of* $\lim_{x\to a} h(x) = L$

Take $\delta = \min\{\delta_1, \delta_2, p\} > 0$. Fix any x with $0 < |x - a| < \delta$. Then $|x - a| < \delta_2$ gives $h(x) > L - \varepsilon$; $|x - a| < p$ gives $g(x) \ge h(x)$; $|x - a| < p$ also gives $g(x) \le f(x)$; and $|x - a| < \delta_1$ gives $f(x) < L + \varepsilon$. Combining:

$$L - \varepsilon < h(x) \le g(x) \le f(x) < L + \varepsilon.$$

Therefore $|g(x) - L| < \varepsilon$, as required. ■

2.14 Definition of Continuity

When we say a function is "continuous," we mean, informally, that its graph can be sketched without lifting the pen from the paper. We now make this precise.

Definition 2.5 (Continuity at a Point). Let $a \in \mathbb{R}$, and let f be a function defined at least on an open interval containing a. We say f is *continuous at a* if

$$\lim_{x\to a} f(x) = f(a).$$

Equivalently, this means three things:

(i) $\lim_{x\to a} f(x)$ exists (and is a real number),

(ii) $f(a)$ is defined,

(iii) the limit equals the value.

If f is not continuous at a, we say it is *discontinuous* at a.

Definition 2.6 ((ε, δ)-Definition of Continuity). Under the same hypotheses, f is continuous at a if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all x,

$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

Remark. Comparing Definition 2.6 with Definition 2.2, the key difference is that we **no longer exclude** the case $x = a$. For the definition of limit, the value at a is irrelevant, so we write $0 < |x - a|$ to exclude it. For continuity, we **need** $f(a)$ to be the right value, so we allow $x = a$.

Remark. Roughly speaking:

- Limit: if x is close to a **but not** a, then $f(x)$ is close to L.
- Continuity: if x is close to a, **including** a, then $f(x)$ is close to $f(a)$.

This distinction is the source of all subtleties when composing limits versus composing continuous functions (see Section 2.16).

Definition 2.7 (Continuity on an Interval). (i) f is *continuous on the open interval* (a, b) if f is continuous at every point of (a, b).

(ii) f is *continuous on the closed interval* $[a, b]$ if f is continuous on (a, b), and in addition

$$\lim_{x\to a^+} f(x) = f(a) \qquad \text{and} \qquad \lim_{x\to b^-} f(x) = f(b).$$

(iii) If we say "f is continuous" without specifying where, we mean *continuous on its domain.*

Remark (Warning). A function may be "continuous" (meaning continuous on its domain) and yet fail to be continuous at a point not in its domain. For instance, $f(x) = 1/x$ is continuous (on $\mathbb{R} \setminus \{0\}$), but it is **not** continuous at 0 (which is not in its domain). This is not a contradiction; it is a consequence of the precise definitions.

2.15 Continuity of Elementary Functions

The following theorem is extremely useful: it tells us that almost every function we encounter in practice is continuous on its domain, so computing limits at points in the domain reduces to simple evaluation.

Theorem 2.9 (Continuity of Elementary Functions). *Any function built from polynomials, roots, trigonometric functions, exponentials, logarithms, and absolute values by means of addition, subtraction, multiplication, division, and composition is continuous on its domain.*

Remark. These functions are sometimes called *elementary functions*. The word "elementary" does not mean "simple"; it is just the name for this class.

Example 2.10. To compute $\lim_{x\to 1} \dfrac{\sqrt{x^3+\sin x}}{e^x+2}$, note that this function is elementary and $x = 1$ is in its domain. Therefore

$$\lim_{x\to 1} \frac{\sqrt{x^3+\sin x}}{e^x+2} = \frac{\sqrt{1+\sin 1}}{e+2}.$$

Remark (Warning). Theorem 2.9 only applies **on the domain**. If the point in question is **not** in the domain, one cannot simply "plug in." For example, $\sin(x)/x$ is not defined at 0, so the theorem does not tell us the limit there; a separate argument is needed (see Section 2.18).

How is Theorem 2.9 proved? The strategy proceeds in two steps.

Step 1. Prove that the "basic" building blocks—constant functions, $f(x) = x$, e^x, $\ln x$, $\sin x$, and $|x|$—are each continuous on their domains.

Step 2. Prove that sums, products, quotients, and compositions of continuous functions are continuous.

For Step 2, sums, products, and quotients follow immediately from the limit laws (Theorem 2.7). For instance:

Proof that the sum of continuous functions is continuous. Strategy: apply the sum law.

Let f and g be continuous at a. By definition, $\lim_{x\to a} f(x) = f(a)$ and $\lim_{x\to a} g(x) = g(a)$. By the sum law (Theorem 2.7):

$$\lim_{x\to a}\big[f(x)+g(x)\big] = f(a)+g(a) = (f+g)(a).$$

Hence $f+g$ is continuous at a. ■ *Sum law (2.7)*

Composition requires a separate argument, which is the subject of the next section.

2.16 Limits and Composition of Functions

One might expect that "the limit of a composition is the composition of the limits," just as for sums and products. Surprisingly, this is ***false*** *in general. Understanding why it fails—and when it can be salvaged—illuminates the crucial difference between limits and continuity.*

Remark. Recall the key distinction:

- $\lim_{x\to a} f(x) = L$: if x is close to a **but not** a, then $f(x)$ is close to L.

- f continuous at a: if x is close to a **including** a, then $f(x)$ is close to $f(a)$.

Continuity has a symmetry between input and output that the limit definition lacks. In a composition, the **output** of one function becomes the **input** of the next, and this asymmetry causes trouble.

Remark (Pause and Try). Before reading further, try to construct a function f such that $\lim_{x\to 0} f(x) = 0$ but $\lim_{x\to 0} f(f(x)) = 1$. This feels counterintuitive, but it can be done with a simple piecewise function.

Here is the false statement, stated explicitly so that you know to avoid it.

Remark (Warning — False "Theorem"). The following is **not** a theorem. Assuming $\lim_{x\to a} f(x) = L$ and $\lim_{y\to L} g(y) = M$, one **cannot** conclude that $\lim_{x\to a} g(f(x)) = M$.

*We now state the results that **are** true.*

Theorem 2.10 (Composition of Continuous Functions). *If f is continuous at a and g is continuous at $f(a)$, then $g \circ f$ is continuous at a.*

Proof. Strategy: concatenate the two continuity implications.

Since f is continuous at a: if x is close to a, then $f(x)$ is close to $f(a)$. Since g is continuous at $f(a)$: if y is close to $f(a)$, then $g(y)$ is close to $g(f(a))$. Setting $y = f(x)$, we obtain: if x is close to a, then $g(f(x))$ is close to $g(f(a))$. This says exactly that $g \circ f$ is continuous at a.

More formally: let $\varepsilon > 0$. Since g is continuous at $f(a)$, there exists $\eta > 0$ such that $|y - f(a)| < \eta \implies |g(y) - g(f(a))| < \varepsilon$. Since f is continuous at a, for this $\eta > 0$ there exists $\delta > 0$ such that $|x - a| < \delta \implies |f(x) - f(a)| < \eta$. Combining: $|x - a| < \delta \implies |g(f(x)) - g(f(a))| < \varepsilon$. ■

Theorem 2.11 (Limit of a Composition — Version 1). *Suppose $\lim_{x\to a} f(x) = L$, $\lim_{y\to L} g(y) = M$, and in addition $f(x) \neq L$ for all x in some open interval about a (except possibly $x = a$). Then $\lim_{x\to a} g(f(x)) = M$.*

Theorem 2.12 (Limit of a Composition — Version 2). *Suppose $\lim_{x\to a} f(x) = L$ and g is **continuous** at L. Then $\lim_{x\to a} g(f(x)) = g(L)$.*

Remark. Both Theorems 2.11 and 2.12 add an extra hypothesis to the false "theorem" in order to make it true. Theorem 2.11 strengthens the condition on f (requiring $f(x) \neq L$ near a), while Theorem 2.12 strengthens the condition on g (upgrading from "having a limit" to "being continuous"). Either fix suffices because each repairs the mismatch between the "then" clause of one hypothesis and the "if" clause of the other.

2.17 Types of Discontinuity

When a function is discontinuous at a point, it is traditional to classify the discontinuity as removable or non-removable. The classification tells us whether the function can be "fixed" by redefining it at a single point.

Definition 2.8 (Removable Discontinuity). A function f has a *removable discontinuity* at a if $\lim_{x\to a} f(x)$ exists but f is not continuous at a (either because $f(a)$ is undefined or because $f(a)$ differs from the limit).

Definition 2.9 (Non-Removable Discontinuity). A function f has a *non-removable discontinuity* at a if $\lim_{x\to a} f(x)$ does not exist.

Example 2.11. The function $h(x) = \dfrac{\sin x}{x}$ is undefined at 0, but $\lim_{x\to 0} h(x) = 1$ (see Section 2.18). Define a new function

$$H(x) = \begin{cases} \dfrac{\sin x}{x} & \text{if } x \neq 0, \\ 1 & \text{if } x = 0. \end{cases}$$

Then H is continuous at 0. We have "removed" the discontinuity by filling the hole in the graph.

Remark. A removable discontinuity can always be fixed: define (or redefine) the function at the point to equal the limit, and the result is continuous. A non-removable discontinuity **cannot** be fixed by redefining the function at a single point. Examples of non-removable discontinuities include jumps (the two one-sided limits exist but differ), vertical asymptotes, and wild oscillations.

Remark. In physics and applied mathematics, removable discontinuities are often silently removed. When a physicist writes $\sin(x)/x$, they usually mean the function H above, with the hole filled in. In a pure-mathematics course, we are more careful about the distinction.

2.18 The Limit of $\sin(x)/x$

This is a classic proof using nothing but high-school geometry and the squeeze theorem. It is a beautiful argument that relies on comparing the areas of two triangles and a circular sector.

Theorem 2.13. $\displaystyle\lim_{x\to 0} \frac{\sin x}{x} = 1.$

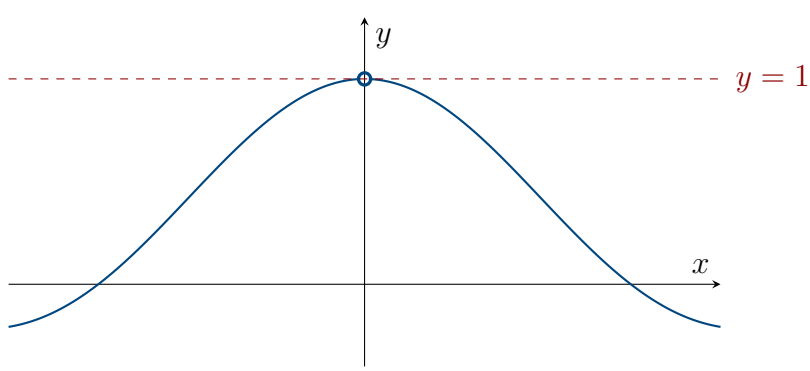

Figure 2.4: The graph of $\sin(x)/x$ near 0.

We begin by establishing a geometric inequality. Consider the unit circle with a central angle $x \in (0, \frac{\pi}{2})$, and compare three regions:

(i) The triangle with vertices at the origin, the point $(\cos x, \sin x)$, and $(\cos x, 0)$ has area $\frac{1}{2}\cos x \sin x$.

(ii) The circular sector of radius 1 and angle x has area $\frac{1}{2}x$.

(iii) The triangle with vertices at the origin, $(1, 0)$, and $(1, \tan x)$ has area $\frac{1}{2}\tan x$.

Lemma 2.14. *For $0 < x < \frac{\pi}{2}$,*

$$\cos x \leq \frac{\sin x}{x} \leq \frac{1}{\cos x}.$$

Proof. Strategy: use the area comparison, then divide through by $\sin x$ and take reciprocals.

Since the triangle (i) is contained in the sector (ii), which is contained in triangle (iii):

$$\frac{1}{2}\cos x \sin x \;\leq\; \frac{1}{2}x \;\leq\; \frac{1}{2}\tan x = \frac{1}{2}\frac{\sin x}{\cos x}.$$

Multiplying through by $\dfrac{2}{\sin x} > 0$ gives

$$\cos x \;\leq\; \frac{x}{\sin x} \;\leq\; \frac{1}{\cos x}.$$

Taking reciprocals (and reversing the inequalities) yields

$$\cos x \;\leq\; \frac{\sin x}{x} \;\leq\; \frac{1}{\cos x}.$$

The derivation assumes $0 < x < \frac{\pi}{2}$. Since all three functions $\cos x$, $\frac{\sin x}{x}$, and $\frac{1}{\cos x}$ are **even**, the same inequalities hold for $-\frac{\pi}{2} < x < 0$. ■

Proof of Theorem 2.13. Strategy: apply the squeeze theorem.

By Lemma 2.14, for $0 < |x| < \frac{\pi}{2}$,

$$\cos x \;\leq\; \frac{\sin x}{x} \;\leq\; \frac{1}{\cos x}.$$

Since cos is continuous at 0 with $\cos 0 = 1$,

$$\lim_{x\to 0} \cos x = 1 \qquad \text{and} \qquad \lim_{x\to 0} \frac{1}{\cos x} = 1.$$

Squeeze Theorem (2.8) By the squeeze theorem (Theorem 2.8), $\displaystyle\lim_{x\to 0} \frac{\sin x}{x} = 1$. ■

2.19 Techniques for Computing Limits

We review the basic toolkit for computing limits at a point. Mastering these techniques requires practice beyond what a single section can provide, but this summary identifies what to try first.

Method 1: Direct evaluation. If the function is continuous at the point, evaluate.

Example 2.12. $\lim_{x\to 2} \dfrac{x^2+1}{\sqrt{x+2}} = \dfrac{4+1}{\sqrt{4}} = \dfrac{5}{2}$.

Method 2: Algebraic manipulation. If the function is not defined at the point (often revealed by the indeterminate form 0/0), try to simplify. Common techniques include factoring, multiplying by conjugates, and cancellation. The manipulations need only be valid **near** the point, not **at** it.

Example 2.13. Compute $\lim_{x\to 2} \dfrac{x^2-4}{x-2}$.

Attempting to evaluate gives 0/0. Factor:

$$\frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2} = x+2, \qquad x \neq 2.$$

The simplified function is continuous at 2, so the limit is 4.

Example 2.14. Compute $\lim_{x\to 0} \dfrac{\sqrt{1+x}-1}{x}$.

Multiply numerator and denominator by the conjugate $\sqrt{1+x}+1$:

$$\frac{\sqrt{1+x}-1}{x} \cdot \frac{\sqrt{1+x}+1}{\sqrt{1+x}+1} = \frac{(1+x)-1}{x(\sqrt{1+x}+1)} = \frac{1}{\sqrt{1+x}+1}.$$

This is continuous at 0, so the limit is $\frac{1}{2}$.

Method 3: Reduction to a known limit. Transform the expression into one whose limit is already known, such as $\lim_{x\to 0} \frac{\sin x}{x} = 1$.

Example 2.15. Compute $\lim_{x\to 0} \dfrac{\sin(2x)}{x}$.

Write $\dfrac{\sin(2x)}{x} = 2 \cdot \dfrac{\sin(2x)}{2x}$. As $x \to 0$, $2x \to 0$, so $\dfrac{\sin(2x)}{2x} \to 1$. Therefore the limit is 2.

Example 2.16. Compute $\lim_{x\to 0} \dfrac{\sin(2x)}{\sin(3x)}$.

Multiply and divide to create the standard form:

$$\frac{\sin(2x)}{\sin(3x)} = \frac{\sin(2x)}{2x} \cdot \frac{3x}{\sin(3x)} \cdot \frac{2}{3}.$$

Each ratio has limit 1 as $x \to 0$, so the limit is $\dfrac{2}{3}$.

Example 2.17. Compute $\lim_{x\to 0} \frac{1-\cos x}{x^2}$.

Multiply and divide by $1+\cos x$:

$$\frac{1-\cos x}{x^2}\cdot\frac{1+\cos x}{1+\cos x} = \frac{1-\cos^2 x}{x^2(1+\cos x)} = \frac{\sin^2 x}{x^2}\cdot\frac{1}{1+\cos x} = \left(\frac{\sin x}{x}\right)^2\cdot\frac{1}{1+\cos x}.$$

As $x\to 0$, $\left(\frac{\sin x}{x}\right)^2 \to 1$ and $\frac{1}{1+\cos x} \to \frac{1}{2}$. Therefore the limit is $\frac{1}{2}$.

Remark. These methods can be combined. More advanced methods exist—notably L'Hôpital's rule (which requires derivatives) and Taylor series (which requires power series)—but they belong to later chapters.

2.20 Limits at Infinity of Rational Functions

For rational functions and related expressions, limits at infinity are computed by factoring out the highest power of x and using the fact that $\lim_{x\to\infty} 1/x = 0$.

Theorem 2.15 (Limit of a Polynomial at Infinity). *The limit of a polynomial as $x\to\infty$ is determined by its leading term. Specifically, if $p(x) = a_n x^n + \cdots + a_0$ with $a_n \neq 0$, then*

$$\lim_{x\to\infty} p(x) = \lim_{x\to\infty} a_n x^n.$$

Proof. Strategy: factor out x^n and use $\lim_{x\to\infty} 1/x = 0$.

Write

$$p(x) = x^n\left(a_n + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n}\right).$$

As $x\to\infty$, each term $a_k/x^{n-k} \to 0$ for $k<n$, so the factor in parentheses approaches a_n. Therefore $p(x)$ behaves like $a_n x^n$. ■

Example 2.18. $\lim_{x\to\infty}(2x^3 - x^2 + 5)$. The leading term is $2x^3 \to \infty$. So the limit is ∞.

Example 2.19. Compute $\lim_{x\to\infty} \frac{2x^2+3}{3x^2-x+1}$.

Factor out x^2 from both numerator and denominator:

$$\frac{2x^2+3}{3x^2-x+1} = \frac{x^2(2+3/x^2)}{x^2(3-1/x+1/x^2)} = \frac{2+3/x^2}{3-1/x+1/x^2} \xrightarrow{x\to\infty} \frac{2}{3}.$$

Example 2.20. Compute $\lim_{x\to\infty} \frac{x^2+\sqrt{x^4+1}}{\sqrt{x^4+x^2}+x}$.

Factor x^4 inside each square root:

$$\begin{aligned}\frac{x^2+\sqrt{x^4(1+1/x^4)}}{\sqrt{x^4(1+1/x^2)}+x} &= \frac{x^2+x^2\sqrt{1+1/x^4}}{x^2\sqrt{1+1/x^2}+x} \\ &= \frac{x^2\left(1+\sqrt{1+1/x^4}\right)}{x^2\left(\sqrt{1+1/x^2}+1/x\right)} \\ &\xrightarrow{x\to\infty} \frac{1+1}{\sqrt{1}+0} = 2.\end{aligned}$$

Alternatively, $\dfrac{1+1}{\sqrt{1}} = 2$ can also be written as $\sqrt{2}\cdot\sqrt{2} = 2$. In either case the limit is 2.

Remark. The rule of thumb is: identify the dominant term (highest power of x) in the numerator and denominator, factor it out, and simplify. This method extends naturally to expressions involving roots: a term $\sqrt{x^{2k}}$ behaves like x^k for large positive x.

2.21 The Extreme Value Theorem

We now state a deep theorem guaranteeing that continuous functions on closed intervals attain their maximum and minimum. The theorem does ***not*** *tell us how to find these extreme values—only that they exist.*

Definition 2.10 (Maximum of a Function). Let f be a function and $I \subseteq \mathbb{R}$. We say f has a *maximum on* I if there exists $c \in I$ such that $f(x) \leq f(c)$ for all $x \in I$. The value $f(c)$ is the maximum, attained *at* c. The definition of *minimum* is analogous, with the inequality reversed.

Remark. The maximum is a value (the output $f(c)$); it occurs **at** a point c in the domain. If we say "the maximum of f" without specifying the set I, we mean the maximum on the entire domain of f.

Let us build intuition for when a maximum is guaranteed to exist by examining some examples of functions that lack one.

Example 2.21. The function $f(x) = x$ on the open interval $(0, 1)$ has no maximum: f gets arbitrarily close to 1 but never attains it, because $1 \notin (0, 1)$.

Example 2.22. A function with a vertical asymptote on a half-open interval may fail to have a maximum because it is unbounded.

Remark. These examples suggest two necessary ingredients: the domain should include its endpoints (a **closed** interval), and the function should not blow up (it should be **continuous**).

Theorem 2.16 (Extreme Value Theorem). *If f is continuous on the closed and bounded interval $[a, b]$, then f has a maximum and a minimum on $[a, b]$.*

Remark. This theorem is easy to state and use but **difficult to prove**. A rigorous proof requires a careful treatment of the completeness of $\mathbb{R}$ and is typically presented in a real analysis course. For our purposes, the extensive examples above should make the statement plausible.

Remark (Warning). Both hypotheses—continuity of f and closedness/boundedness of the interval—are essential. Removing either one can produce a function without a maximum or minimum.

Remark (Pause and Try). Can you construct a function that is **continuous** on the half-open interval $(0, 1]$ and has **neither** a maximum **nor** a minimum? Hint: sketch the graph first, then find an equation.

2.22 The Intermediate Value Theorem

The intermediate value theorem formalises something you already believe: if a continuous function is negative at one point and positive at another, it must be zero somewhere in between.

Example 2.23. Consider the equation $x^5 + 9x - 6 = 0$. Let $f(x) = x^5 + 9x - 6$. Then $f(0) = -6 < 0$ and $f(1) = 4 > 0$. Since f is a polynomial and hence continuous, the function must cross zero somewhere in $(0, 1)$. By evaluating $f(0.5) < 0$, we can narrow the root to $(0.5, 1)$, and so on.

Remark. This equation cannot be solved exactly using radicals (a fact provable via Galois theory). Nevertheless, the intermediate value theorem guarantees the existence of a solution, and successive bisection can approximate it to any desired accuracy.

Theorem 2.17 (Intermediate Value Theorem). *Let f be continuous on the closed interval $[a, b]$. If M is any real number between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that $f(c) = M$.*

In the special case $M = 0$: if $f(a)$ and $f(b)$ have opposite signs and f is continuous on $[a, b]$, then f has a root in (a, b).

Remark. Like the extreme value theorem, the intermediate value theorem is easy to understand and believe but **hard to prove** rigorously. A formal proof relies on the completeness of $\mathbb{R}$ and belongs to a course in real analysis. For us, what matters is:

(i) understanding the statement,

(ii) recognising that it is a property of **continuous** functions (it can fail for discontinuous ones),

(iii) knowing when we are using it.

Remark (Warning). Continuity is essential. A step function, for instance, can jump from a negative value to a positive value without ever equalling zero.

Chapter 3

Derivatives

3.1 Definition of the Derivative

We want to define the "slope of a curve" at a point, but we only know how to define the slope of a line. We might try to define derivative as the slope of the tangent line, but what is a tangent line? Every naïve definition of one of these concepts seems to rely on another equally undefined concept. The only way to break the circle is to use limits.

Consider a curve—the graph of a function f—and two points P and Q on the curve with x-coordinates a and b, respectively. The slope of the line through P and Q is $\frac{f(b)-f(a)}{b-a}$. Now keep P fixed and slide Q toward P. As Q approaches P, the line through them becomes the tangent line at P. This motivates taking a limit.

Definition 3.1 (Derivative). Let f be a function defined at least on an open interval containing a. The *derivative of f at a* is

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a},$$

provided this limit exists. Equivalently, setting $h = x - a$,

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}.$$

These two expressions are entirely the same; it is simply a change of variable.

Definition 3.2 (Differentiable). We say that f is *differentiable at a* when the limit in Definition 3.1 exists. We say that f is *differentiable* (without qualification) when f is differentiable at every point in its domain.

Definition 3.3 (Tangent Line). Let f be differentiable at a. The *tangent line* to the graph of f at the point $(a, f(a))$ is the line through $(a, f(a))$ with slope $f'(a)$. Its equation is

$$y = f(a) + f'(a)(x - a).$$

Remark. Notice that this definition of tangent line relies on the derivative, which we defined independently using a limit. The tangent line is fully determined by a point and a slope, and the derivative supplies that slope.

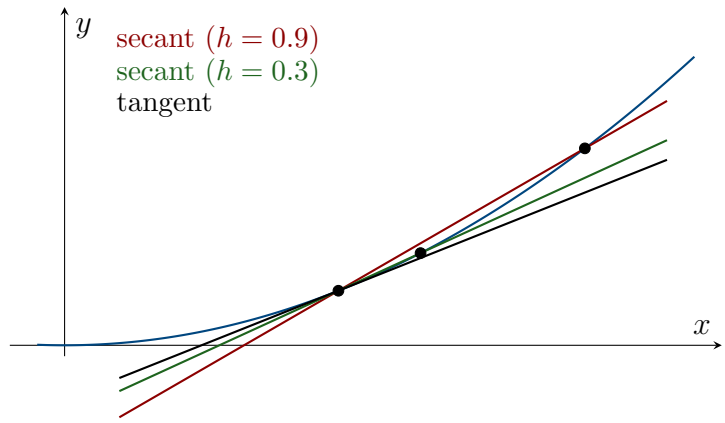

Figure 3.1: Secant lines converging to the tangent line.

Example 3.1. Consider a function f whose graph is given. To compute $f'(0)$, find the point on the graph with x-coordinate 0 and draw the tangent line there. If that line has slope 2, then $f'(0) = 2$. Similarly, if the tangent line at the point with x-coordinate -1 is horizontal, then $f'(-1) = 0$.

3.2 Derivatives from the Definition

We now illustrate how to compute a derivative directly from the limit definition. This is the foundational method; later, differentiation rules will spare us the effort of returning to the definition each time.

Example 3.2. Let $f(x) = 4x - x^2$. Compute $f'(1)$ from the definition.

Using the h-form of the definition:

$$\begin{aligned} f'(1) &= \lim_{h\to 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h\to 0} \frac{\left[4(1+h) - (1+h)^2\right] - \left[4(1) - 1^2\right]}{h} \\ &= \lim_{h\to 0} \frac{4 + 4h - 1 - 2h - h^2 - 3}{h} \\ &= \lim_{h\to 0} \frac{2h - h^2}{h} \\ &= \lim_{h\to 0} (2 - h) \\ &= 2. \end{aligned}$$

After all cancellations we are left with a continuous function, and evaluating at $h = 0$ gives $f'(1) = 2$.

Remark. Computing derivatives from the definition is tedious for any function more complicated than a simple polynomial. The **differentiation rules**, introduced in Section 3.4, provide a much faster method. We still need to prove them from the definition, but once proven, they can be reused indefinitely.

3.3 Derivative as Rate of Change

The derivative has a second fundamental interpretation: rate of change. To motivate it, consider a familiar example—velocity.

Suppose you are driving at 180 *km/h. What does that mean? It does not literally mean you will drive* 180 *km in the next hour; your velocity may change. A better interpretation: over a very small time interval* Δt*, you travel approximately* 180 *km/h*$\cdot \Delta t$*, and this approximation improves as* Δt *shrinks. There is a limit lurking here. It is impossible to define velocity—or any instantaneous rate of change—without limits.*

Definition 3.4 (Average and Instantaneous Velocity). Let x denote the position of a particle as a function of time t.

(i) The *average velocity* between times t_1 and t_2 is

$$\frac{\Delta x}{\Delta t} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}.$$

(ii) The *instantaneous velocity* at time t_1 is

$$\left.\frac{dx}{dt}\right|_{t=t_1} = \lim_{t_2 \to t_1} \frac{x(t_2) - x(t_1)}{t_2 - t_1} = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}.$$

Remark (Warning). The notation $\frac{dx}{dt}$ does **not** mean a number dx divided by a number dt. The symbols dx and dt are not numbers. Rather, $\frac{dx}{dt}$ denotes the limit of $\frac{\Delta x}{\Delta t}$ as $\Delta t \to 0$. We call it the *derivative of* x *with respect to* t.

Definition 3.5 (Instantaneous Rate of Change). More generally, if Q is a physical quantity that depends on another quantity x, the *instantaneous rate of change of* Q *with respect to* x is

$$\frac{dQ}{dx} = \lim_{\Delta x \to 0} \frac{\Delta Q}{\Delta x}.$$

Remark. The two notions of derivative—$f'(a)$ for an abstract function, and $\frac{dx}{dt}$ for a physical quantity—are the **same concept**. If position x is given by a function $f(t)$, then

$$\left.\frac{dx}{dt}\right|_{t=t_1} = \lim_{t_2 \to t_1} \frac{f(t_2) - f(t_1)}{t_2 - t_1} = f'(t_1).$$

Whether we think of derivative as slope, as instantaneous rate of change, or as instantaneous velocity, we always arrive at the same limit.

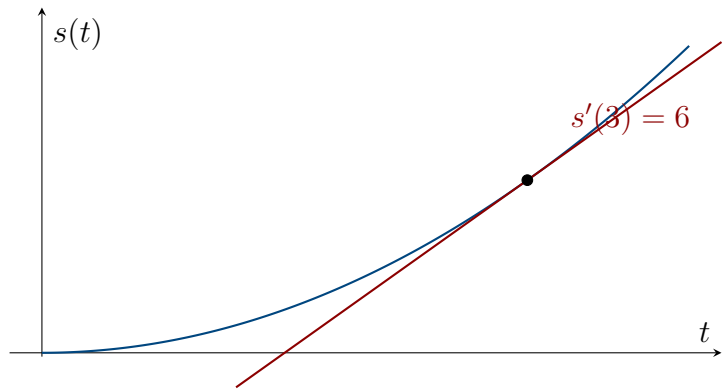

Figure 3.2: Slope of the tangent as instantaneous rate of change.

3.4 Basic Differentiation Rules

Computing every derivative from the definition is slow and often messy. The differentiation rules let us compute derivatives of common functions quickly. Each rule must be proved from the definition, but once proved it can be reused without limit.

Here are the basic rules, presented first in a "lazy" form that suppresses hypotheses. A precise statement of the product rule follows as an illustration of what the "proper" version looks like.

Theorem 3.1 (Basic Differentiation Rules — Summary). *Let f and g be differentiable functions, and let c be a real constant.*

(i) ***Constant rule.*** $\dfrac{d}{dx}[c] = 0.$

(ii) ***Power rule.*** *For every $c \in \mathbb{R}$,* $\dfrac{d}{dx}[x^c] = c\,x^{c-1}.$

(iii) ***Sum rule.*** $(f+g)' = f' + g'.$

(iv) ***Constant multiple rule.*** $(c \cdot f)' = c \cdot f'.$

(v) ***Product rule.*** $(f \cdot g)' = f'\,g + f\,g'.$

(vi) ***Quotient rule.*** $\left(\dfrac{f}{g}\right)' = \dfrac{f'\,g - f\,g'}{g^2}, \quad$ *provided* $g \neq 0.$

Theorem 3.2 (Product Rule — Precise Statement). *Let $a \in \mathbb{R}$, and let f and g be functions defined on an open interval containing a. Define $h = f \cdot g$. If f and g are both differentiable at a, then h is also differentiable at a, and*

$$h'(a) = f'(a)\,g(a) + f(a)\,g'(a).$$

Remark (Warning). The **if–then** structure matters. The product rule requires both f and g to be differentiable individually. If one or both are not differentiable, the formula does not apply, although the product may still have a derivative that must be computed by other means (e.g., the definition).

Example 3.3. Compute the derivative of $p(x) = 3x^4 - 5x^2 + 7x - 2$.

A polynomial is a sum of terms, each a constant times a power. By the sum and constant multiple rules,

$$\begin{aligned} p'(x) &= 3 \cdot 4x^3 - 5 \cdot 2x + 7 \cdot 1 - 0 \\ &= 12x^3 - 10x + 7. \end{aligned}$$

Example 3.4. Compute the derivative of $q(x) = \dfrac{x-1}{x^2+x}$.

By the quotient rule:

$$\begin{aligned} q'(x) &= \frac{(1)(x^2+x) - (x-1)(2x+1)}{(x^2+x)^2} \\ &= \frac{x^2 + x - 2x^2 - x + 2x + 1}{(x^2+x)^2} \\ &= \frac{-x^2 + 2x + 1}{(x^2+x)^2}. \end{aligned}$$

Example 3.5. The power rule works for **all** real exponents, not only positive integers.

(i) Writing $\frac{1}{x^3} = x^{-3}$, we get $\dfrac{d}{dx}[x^{-3}] = -3x^{-4} = -\dfrac{3}{x^4}$.

(ii) Writing $\sqrt{x} = x^{1/2}$, we get $\dfrac{d}{dx}\left[x^{1/2}\right] = \dfrac{1}{2}\,x^{-1/2} = \dfrac{1}{2\sqrt{x}}$.

3.5 Differentiability and Continuity

One of the first theoretical results about derivatives is that differentiability is a stronger condition than continuity: every differentiable function is continuous, but not conversely. We first give an informal argument explaining why, and then a formal proof.

Theorem 3.3 (Differentiability Implies Continuity). *Let $c \in \mathbb{R}$, and let f be a function defined on an open interval containing c. If f is differentiable at c, then f is continuous at c.*

Remark. The converse is **false**. There exist functions that are continuous at a point but not differentiable there. See Section 3.9 for examples.

Before the formal proof, here is an informal explanation. Continuity means $\lim_{x\to c} f(x) = f(c)$, or equivalently $\lim_{x\to c}[f(x) - f(c)] = 0$. When we try to compute $f'(c)$ from continuity alone, the defining limit $\frac{f(x)-f(c)}{x-c}$ gives the indeterminate form $\frac{0}{0}$, which may or may not exist. But if f is not even continuous at c, the numerator does not approach 0 while the denominator does, and the limit can never exist. So non-continuity forces non-differentiability.

Proof. We show that $\lim_{x\to c}[f(x) - f(c)] = 0$.

Multiply and divide by $x - c$:

$$\begin{aligned}\lim_{x\to c}\big[f(x) - f(c)\big] &= \lim_{x\to c}\left[\frac{f(x) - f(c)}{x - c}\cdot(x - c)\right]\\ &= \left(\lim_{x\to c}\frac{f(x) - f(c)}{x - c}\right)\cdot\left(\lim_{x\to c}(x - c)\right)\\ &= f'(c)\cdot 0\\ &= 0.\end{aligned}$$

Limit law for products; the first limit exists because f is differentiable at c.

The product law for limits applies because both individual limits exist: the first is $f'(c)$ (by the hypothesis of differentiability), and the second is 0. We are **not** claiming that 0 times "anything" is 0; we are using the fact that 0 times a **number** is 0.

Since $\lim_{x\to c}[f(x) - f(c)] = 0$, we conclude $\lim_{x\to c} f(x) = f(c)$, and f is continuous at c. ■

3.6 Proof of the Product Rule

We now prove the product rule formally. The key trick is to add and subtract a carefully chosen term in order to factor out the derivatives of f and g separately.

Proof of Theorem 3.2. We compute $h'(a)$ from the definition.

Write $h(x) = f(x)\,g(x)$, so

$$\begin{aligned}h'(a) &= \lim_{x\to a}\frac{h(x) - h(a)}{x - a}\\ &= \lim_{x\to a}\frac{f(x)\,g(x) - f(a)\,g(a)}{x - a}.\end{aligned}$$

Add and subtract $f(a)\,g(x)$ in the numerator:

$$\begin{aligned}&= \lim_{x\to a}\frac{f(x)\,g(x) - f(a)\,g(x) + f(a)\,g(x) - f(a)\,g(a)}{x - a}\\ &= \lim_{x\to a}\left[\frac{f(x) - f(a)}{x - a}\cdot g(x) + f(a)\cdot\frac{g(x) - g(a)}{x - a}\right].\end{aligned}$$

Apply the limit laws (sum and product of limits):

$$\begin{aligned}&= \left(\lim_{x\to a}\frac{f(x) - f(a)}{x - a}\right)\left(\lim_{x\to a} g(x)\right) + f(a)\left(\lim_{x\to a}\frac{g(x) - g(a)}{x - a}\right)\\ &= f'(a)\,g(a) + f(a)\,g'(a).\end{aligned}$$

Limit laws; 3.3 (differentiability implies continuity)

Three individual limits were used. The first is $f'(a)$, which exists by hypothesis. The third is $g'(a)$, which exists by hypothesis. The second is $\lim_{x\to a} g(x) = g(a)$; this holds because g is differentiable at a, and differentiability implies continuity (Theorem 3.3). ■

3.7 Proof of the Power Rule

The power rule states that $\frac{d}{dx}[x^c] = c\,x^{c-1}$ for every real number c. A proof for all real exponents requires heavy machinery (in particular, logarithmic differentiation). Here we present a proof for the case when c is a positive integer, using induction.

Theorem 3.4 (Power Rule for Positive Integers). *For every positive integer n,*

$$\frac{d}{dx}[x^n] = n\,x^{n-1}.$$

Proof. We proceed by induction on n.

Base case ($n = 1$). Define $f(x) = x$. By the definition of derivative,

$$f'(x) = \lim_{h\to 0} \frac{(x+h)-x}{h} = \lim_{h\to 0} \frac{h}{h} = \lim_{h\to 0} 1 = 1 = 1 \cdot x^0.$$

So the formula holds when $n = 1$.

Induction step. Fix a positive integer n and assume

$$\frac{d}{dx}[x^n] = n\,x^{n-1}.$$

We must show the formula holds for $n+1$. Write $x^{n+1} = x^n \cdot x$ and apply the product rule:

$$\begin{aligned}\frac{d}{dx}\left[x^{n+1}\right] &= \frac{d}{dx}[x^n] \cdot x + x^n \cdot \frac{d}{dx}[x] \\ &= n\,x^{n-1} \cdot x + x^n \cdot 1 \\ &= n\,x^n + x^n \\ &= (n+1)\,x^n.\end{aligned}$$

This is exactly the power-rule formula with exponent $n+1$. By induction, the formula holds for every positive integer n. ■

Product rule (3.2); induction hypothesis; base case

3.8 Higher-Order Derivatives

Since the derivative of a function is itself a function, we can differentiate again, and again, and so on. There are two standard systems of notation for these repeated derivatives, and it is useful to be comfortable with both.

Definition 3.6 (Higher-Order Derivatives — Lagrange Notation). Let f be a function. Its successive derivatives are denoted:

(i) f' — the first derivative (or simply the derivative),

(ii) f'' — the second derivative,

(iii) f''' — the third derivative,

(iv) $f^{(n)}$ — the nth derivative, for any positive integer n.

Each is defined whenever the preceding derivative exists and is itself differentiable.

Remark (Warning). The parentheses around n in $f^{(n)}$ are essential. Without them, f^n is ambiguous: for trigonometric functions, $\sin^2 x$ conventionally means $(\sin x)^2$; in other contexts, a superscript may denote iterated composition. With the parentheses, $f^{(n)}$ unambiguously means the nth derivative.

Definition 3.7 (Higher-Order Derivatives — Leibniz Notation). If y is a function of x, we write

$$\frac{dy}{dx}, \quad \frac{d^2y}{dx^2}, \quad \frac{d^3y}{dx^3}, \quad \ldots, \quad \frac{d^ny}{dx^n}$$

for the first, second, third, ..., nth derivatives of y with respect to x. Think of $\frac{d}{dx}$ as an *operator*; $\frac{d^ny}{dx^n}$ means "apply the operator $\frac{d}{dx}$ to y a total of n times."

Remark. The placement of the exponents in $\frac{d^ny}{dx^n}$ is deliberate. The n in the numerator records the number of differentiations applied to y; the n in the denominator records that all differentiations were with respect to x. This notation also keeps track of physical dimensions. For example, if x is position measured in metres and t is time in seconds, then velocity $\frac{dx}{dt}$ has units m/s and acceleration $\frac{d^2x}{dt^2}$ has units m/s^2.

Example 3.6. Let $f(x) = x^7$. Then

$$f'(x) = 7x^6, \quad f''(x) = 42x^5, \quad f'''(x) = 210x^4, \quad \ldots, \quad f^{(7)}(x) = 7!, \quad f^{(8)}(x) = 0.$$

After seven applications of the power rule, all powers of x are exhausted and every subsequent derivative is 0.

3.9 Non-Differentiable Functions

We know from Theorem 3.3 that every differentiable function is continuous. Any discontinuous function is therefore automatically non-differentiable. More interesting are functions that are continuous yet still fail to be differentiable. We examine two specific ways this can occur.

Recall: f is not differentiable at a when the limit $\lim_{x\to a} \frac{f(x)-f(a)}{x-a}$ does not exist. Like any limit, it may fail to exist for various reasons. We focus on two:

(i) *the two one-sided limits exist but are different (a* ***corner****), or*

(ii) *the limit is $\pm\infty$ (a* ***vertical tangent line****).*

Definition 3.8 (Corner). A function f has a *corner* at $x = a$ if f is continuous at a, but the limit

$$\lim_{x\to a} \frac{f(x) - f(a)}{x - a}$$

does not exist because the left-hand and right-hand limits are different (both finite).

Example 3.7. Let $f(x) = |x|$. We test differentiability at 0:

$$\lim_{x\to 0} \frac{|x| - 0}{x - 0} = \lim_{x\to 0} \frac{|x|}{x}.$$

$$\lim_{x\to 0^+} \frac{|x|}{x} = \lim_{x\to 0^+} \frac{x}{x} = 1,$$
$$\lim_{x\to 0^-} \frac{|x|}{x} = \lim_{x\to 0^-} \frac{-x}{x} = -1.$$

Since the one-sided limits disagree, $f'(0)$ does not exist. The graph has a corner at the origin: the slope from the right is 1 and the slope from the left is -1.

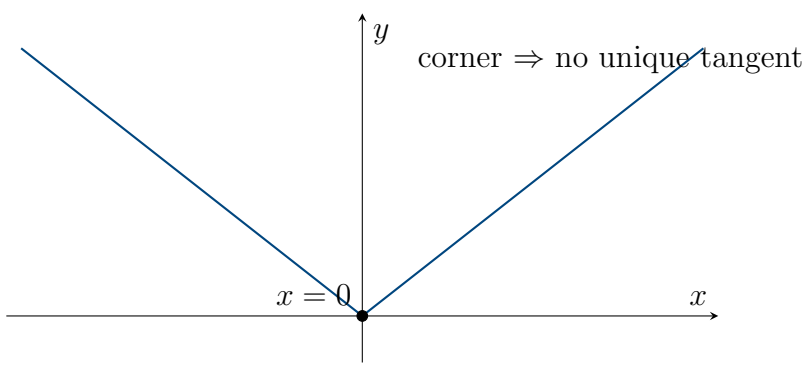

Figure 3.3: A corner: continuous but not differentiable at the corner.

Definition 3.9 (Vertical Tangent Line). A function f has a *vertical tangent line* at $x = a$ if f is continuous at a and

$$\lim_{x\to a} \frac{f(x) - f(a)}{x - a} = +\infty \quad \text{or} \quad -\infty.$$

Example 3.8. Let $g(x) = x^{1/3}$. We check differentiability at 0:

$$\lim_{x\to 0} \frac{x^{1/3} - 0}{x - 0} = \lim_{x\to 0} \frac{1}{x^{2/3}} = +\infty.$$

The derivative at 0 does not exist; the graph has a vertical tangent line at the origin.

Remark. For $x \neq 0$, the power rule gives $g'(x) = \frac{1}{3}\, x^{-2/3}$, and one may verify that $\lim_{x\to 0} g'(x) = +\infty$ as well. It is possible to prove (using L'Hôpital's rule) that if $\lim_{x\to a} f'(x) = \pm\infty$, then $\lim_{x\to a} \frac{f(x)-f(a)}{x-a} = \pm\infty$ also. In practice, computing $\lim_{x\to a} f'(x)$ is often easier.

Remark (Warning). A vertical tangent line is **not** the same as a vertical asymptote. A vertical asymptote occurs when the function itself is undefined at a point and $\lim_{x\to a} f(x) = \pm\infty$. A vertical tangent line occurs when f is continuous at a but the *derivative* tends to $\pm\infty$. Do not confuse the two.

3.10 Chain Rule Examples

The chain rule is the differentiation rule for compositions of functions. Before stating it formally, we build intuition through examples.

Example 3.9. Compute $\frac{d}{dx}\left[\sqrt{x^3+1}\right]$.

The derivative of $\sqrt{x}$ is $\frac{1}{2\sqrt{x}}$. By the chain rule, replace the argument x with x^3+1, then multiply by the derivative of the inner function:

$$\frac{d}{dx}\left[\sqrt{x^3+1}\right] = \frac{1}{2\sqrt{x^3+1}} \cdot 3x^2 = \frac{3x^2}{2\sqrt{x^3+1}}.$$

Example 3.10. Compute $\frac{d}{dx}[\sin(6x)]$.

The derivative of $\sin(x)$ is $\cos(x)$. By the chain rule:

$$\frac{d}{dx}[\sin(6x)] = \cos(6x) \cdot 6 = 6\cos(6x).$$

Example 3.11. Compute $\frac{d}{dx}[(x^2+1)^{100}]$.

Rather than expanding the polynomial, apply the chain rule. The derivative of x^{100} is $100x^{99}$, so:

$$\frac{d}{dx}\left[(x^2+1)^{100}\right] = 100(x^2+1)^{99} \cdot 2x = 200x\,(x^2+1)^{99}.$$

Remark (Pause and Try). In each example, identify the "outer" and "inner" functions, take the derivative of the outer as if the inner were the variable, and then multiply by the derivative of the inner. Practice this pattern on several compositions before moving on.

3.11 Chain Rule Statement and Proof

We now state the chain rule precisely, discuss the notation carefully, and attempt a proof.

Some students find calculations with specific functions easy but become confused when the chain rule is written for abstract functions f and g. The key is to be strict about distinguishing the ***name*** *of a function (e.g., f) from its* ***value*** *at a point (e.g., $f(x)$). The name of the composed function is $f \circ g$; its value at x is $f(g(x))$.*

Theorem 3.5 (Chain Rule). *Let $a \in \mathbb{R}$, and let f and g be functions such that g is defined on an open interval containing a and f is defined on an open interval containing $g(a)$. If g is differentiable at a and f is differentiable at $g(a)$, then $f \circ g$ is differentiable at a and*

$$(f \circ g)'(a) = f'\big(g(a)\big) \cdot g'(a).$$

Example 3.12. Revisiting $\sqrt{x^3+1}$: set $g(x) = x^3 + 1$ and $f(u) = \sqrt{u}$. Then $f(g(x)) = \sqrt{x^3+1}$, and

$$(f \circ g)'(x) = f'\big(g(x)\big) \cdot g'(x) = \frac{1}{2\sqrt{x^3+1}} \cdot 3x^2.$$

Here $f'(u) = \frac{1}{2\sqrt{u}}$, evaluated at $u = g(x) = x^3 + 1$, gives $f'(g(x)) = \frac{1}{2\sqrt{x^3+1}}$.

It is natural to try to prove the chain rule by writing the derivative from the definition and factoring in a suggestive way. The following argument is instructive, despite a gap that prevents it from being fully rigorous.

Proof attempt. By the definition of derivative,

$$(f \circ g)'(a) = \lim_{x \to a} \frac{f(g(x)) - f(g(a))}{x - a}.$$

Multiply and divide by $g(x) - g(a)$:

$$= \lim_{x \to a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a}.$$

If both limits exist and the product law applies, the second factor tends to $g'(a)$. For the first factor, setting $u = g(x)$, as $x \to a$ we have $u \to g(a)$ (since differentiability implies continuity), and the limit becomes $f'(g(a))$. *Limit laws; 3.3*

Gap. In the second step we divided by $g(x) - g(a)$. We know $x \neq a$, but $g(x)$ may equal $g(a)$ for values of x arbitrarily close to a, causing division by zero. For "most" functions encountered in practice this does not happen, but pathological examples exist. A fully rigorous proof avoids this division entirely; such proofs typically use more sophisticated tools and are usually presented in a multivariable-calculus or analysis course. ■

Remark. In Leibniz notation, with $u = g(x)$ and $y = f(u)$, the chain rule reads

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

This is suggestive—it looks as if we "cancel the du"—but remember that $\frac{dy}{du}$ is not a fraction of numbers. The notation is a convenient mnemonic, not a cancellation.

3.12 Trigonometric Derivatives

We derive the derivatives of the six trigonometric functions. The strategy is to do the hard work only once—prove the derivative of sin *from the definition—and then obtain the rest using algebraic identities, the chain rule, and the quotient rule.*

Theorem 3.6 (Derivatives of Trigonometric Functions). *(i)* $\dfrac{d}{dx}[\sin x] = \cos x$.

(ii) $\dfrac{d}{dx}[\cos x] = -\sin x.$

(iii) $\dfrac{d}{dx}[\tan x] = \sec^2 x.$

(iv) $\dfrac{d}{dx}[\cot x] = -\csc^2 x.$

(v) $\dfrac{d}{dx}[\sec x] = \sec x \tan x.$

(vi) $\dfrac{d}{dx}[\csc x] = -\csc x \cot x.$

Remark. The derivatives of cos, cot, and csc are obtained from those of sin, tan, and sec by inserting a minus sign and swapping sin $\leftrightarrow$ cos, tan $\leftrightarrow$ cot, sec $\leftrightarrow$ csc. Memorise the first three; the other three can be re-derived in seconds.

The proof strategy is as follows. ***Step 1:*** *Prove* $\frac{d}{dx}[\sin x] = \cos x$ *directly from the definition, using two previously established limits.* ***Step 2:*** *Write* $\cos x = \sin\left(\frac{\pi}{2} - x\right)$ *and apply the chain rule.* ***Step 3:*** *Express the remaining four functions as quotients of* sin *and* cos*, and use the quotient rule.*

Proof of (i). We compute $\frac{d}{dx}[\sin x]$ from the definition.

By the angle-addition formula $\sin(x+h) = \sin x \cos h + \cos x \sin h$:

$$\begin{aligned}\frac{d}{dx}[\sin x] &= \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h\to 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \lim_{h\to 0} \left[\sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}\right].\end{aligned}$$

For the purpose of this limit, x is fixed, so $\sin x$ and $\cos x$ are constants. By the limit laws:

$$\begin{aligned}&= \sin x \cdot \lim_{h\to 0} \frac{\cos h - 1}{h} + \cos x \cdot \lim_{h\to 0} \frac{\sin h}{h} \\ &= \sin x \cdot 0 + \cos x \cdot 1 \\ &= \cos x.\end{aligned}$$

$\lim_{h\to 0} \frac{\sin h}{h} = 1$ *and* $\lim_{h\to 0} \frac{\cos h - 1}{h} = 0$ *(proved by elementary geometry)*

The two key limits, $\lim_{h\to 0} \frac{\sin h}{h} = 1$ and $\lim_{h\to 0} \frac{\cos h - 1}{h} = 0$, were established earlier using only geometry and trigonometry, with no reliance on derivatives. ■

Remark (Pause and Try). Derive the formulas for $\frac{d}{dx}[\cos x]$, $\frac{d}{dx}[\tan x]$, and $\frac{d}{dx}[\sec x]$ using the chain rule and quotient rule. These are quick exercises and worth doing at least once.

Remark (Warning). It is tempting to compute $\lim_{h\to 0} \frac{\sin h}{h}$ using L'Hôpital's rule. But L'Hôpital's rule requires knowing $\frac{d}{dx}[\sin x] = \cos x$, which in turn requires knowing that $\lim_{h\to 0} \frac{\sin h}{h} = 1$. Using L'Hôpital's rule here is circular. This limit must be established by elementary means first.

3.13 Implicit Differentiation

Implicit differentiation is not a new rule; it is a clever application of the chain rule. When a relation between x and y is given implicitly—that is, not solved for y—we can still compute $\frac{dy}{dx}$ by differentiating both sides with respect to x, treating y as an unknown function of x.

To build intuition, consider the equation of a circle: $x^2+y^2 = 25$. We could solve for y explicitly as $y = \pm\sqrt{25 - x^2}$ and differentiate using the chain rule. However, this requires handling the positive and negative cases separately and dealing with messy square roots.

Instead, we can differentiate the original equation directly. The key is to remember that y is a function of x. When we differentiate a term with y, we must apply the chain rule.

Example 3.13. Find $\frac{dy}{dx}$ for the circle $x^2 + y^2 = 25$.

Differentiate both sides with respect to x:

$$\begin{aligned} \frac{d}{dx}[x^2 + y^2] &= \frac{d}{dx}[25] \\ 2x + 2y\frac{dy}{dx} &= 0 \end{aligned}$$

Notice the $2y\frac{dy}{dx}$ term: the derivative of the outside function $(\cdot)^2$ is $2y$, and we multiply by the derivative of the inside function y, which is $\frac{dy}{dx}$.

Now, isolate $\frac{dy}{dx}$:

$$\begin{aligned} 2y\frac{dy}{dx} &= -2x \\ \frac{dy}{dx} &= -\frac{2x}{2y} = -\frac{x}{y} \end{aligned}$$

This gives us a formula for the slope at any point (x, y) on the circle.

General Technique Outline

From this intuition, we can generalize the process of implicit differentiation into four concrete steps:

1. **Differentiate both sides** of the equation with respect to x. Treat y as a differentiable function of x, applying the chain rule whenever differentiating terms involving y.

2. **Collect all terms** containing $\frac{dy}{dx}$ on one side of the equation and move all other terms to the opposite side.

3. **Factor out** $\frac{dy}{dx}$ from the terms containing it.

4. **Solve for** $\frac{dy}{dx}$ by dividing both sides by the factored expression.

Let's apply this to a more complex curve where solving for y explicitly would be nearly impossible.

Example 3.14. Find the slope of the tangent line to the curve

$$2x^3 + x^2y - xy^3 = 2$$

at the point $(1, 1)$.

Step 1: Differentiate both sides with respect to x. We differentiate each term carefully, using the product rule for x^2y and xy^3. Writing y' for $\frac{dy}{dx}$:

- $\frac{d}{dx}[2x^3] = 6x^2$
- $\frac{d}{dx}[x^2y] = 2xy + x^2y'$ (Product rule: derivative of x^2 times y, plus x^2 times derivative of y)
- $\frac{d}{dx}[-xy^3] = -y^3 - 3xy^2y'$ (Product rule and chain rule)
- $\frac{d}{dx}[2] = 0$

Putting it all together:

$$6x^2 + 2xy + x^2y' - y^3 - 3xy^2y' = 0$$

Step 2: Substitute the point $(1, 1)$ to find the slope. Since we only need the slope at a specific point, we can substitute $x = 1$ and $y = 1$ immediately. This is often much easier than solving for y' algebraically first.

$$\begin{aligned} 6(1)^2 + 2(1)(1) + (1)^2y' - (1)^3 - 3(1)(1)^2y' &= 0 \\ 6 + 2 + y' - 1 - 3y' &= 0 \end{aligned}$$

Step 3 & 4: Collect terms and solve for y'.

$$\begin{aligned} 7 - 2y' &= 0 \\ 2y' &= 7 \\ y' &= \frac{7}{2} \end{aligned}$$

The slope of the tangent line at $(1, 1)$ is $\frac{7}{2}$.

Remark. Implicit differentiation is useful even when one *could* solve for y explicitly. Keeping the equation implicit typically leads to simpler calculations. The technique works because, near the point of interest, y is locally a function of x (by the implicit function theorem), and the chain rule applies to every occurrence of y.

Chapter 4

Transcendental Functions

4.1 Functions and Their Components

There are a few concepts related to the notion of function that can get confusing. Strictly speaking, they all have very precise definitions, but in some contexts we relax those definitions, abuse notation, and use the terms to mean something slightly different. Sometimes students may learn the same theorem in two apparently contradictory ways, depending on whether it appears in a calculus class or in a linear algebra class. There is no contradiction once we understand the conventions.

Definition 4.1 (Function). A *function* consists of three pieces:

(i) A set of inputs called the *domain*.

(ii) A set of potential outputs called the *codomain*.

(iii) A *rule* that matches each input in the domain to exactly one output in the codomain.

If the function is named f, the domain is A, and the codomain is B, we write $f\colon A \to B$. For each $x \in A$, the symbol $f(x)$ denotes the corresponding element of B.

Remark (Warning). The **name** of the function is f, **not** $f(x)$. The expression $f(x)$ is an element in the codomain, not the function itself.

Definition 4.2 (Range). Let $f\colon A \to B$ be a function. The *range* (or *image*) of f is the set of **actual** outputs:

$$\mathrm{Range}(f) = \{f(x) : x \in A\}.$$

The range is always a subset of the codomain, but need not equal it.

Remark. The codomain is chosen beforehand; the range is determined after the rule is defined. To specify a function, one needs only the domain, the codomain, and the rule. The range can be computed from this information.

In calculus, we adopt conventions that simplify the way we talk about functions.

Remark (Calculus Convention for Domain and Codomain). In single-variable calculus, it is common to define a function by giving only the rule (usually an equation). By default, we assume:

(i) The domain is the **largest subset** of $\mathbb{R}$ on which the rule makes sense.

(ii) The codomain is $\mathbb{R}$.

For example, the function defined by $g(x) = \frac{1}{x^2}$ implicitly has domain $\mathbb{R} \setminus \{0\}$ and codomain $\mathbb{R}$.

Remark (Warning). This convention saves time because we only work with one specific type of function, but in other areas of mathematics—such as abstract algebra—domains and codomains may not even be sets of numbers, and one must specify them carefully.

Remark. Two functions with the same domain, the same rule, and the same range, but **different codomains**, are strictly speaking different functions. In calculus, we often treat the codomain as irrelevant and identify such functions. In other areas of mathematics, this distinction can be important.

Remark. The names introduced here—domain, codomain, range—are the most commonly used names in mathematics. However, in some areas of computer science, the word "range" is used to mean codomain, and "image" is used for what we call range. When working with others, make sure you agree on notation.

4.2 Inverse Functions

The idea behind the inverse of a function is simple: swap inputs and outputs. If we reverse all the arrows in the diagram of a function, we obtain its inverse—provided the result is again a function.

Definition 4.3 (Inverse Function — Rigorous Version). Let $f\colon A \to B$ be a function. The *inverse* of f is a function $f^{-1}\colon B \to A$ defined by

$$x = f^{-1}(y) \quad \Longleftrightarrow \quad y = f(x), \qquad \text{for all } x \in A \text{ and all } y \in B.$$

Two things can go wrong when we try to construct an inverse. First, an element of the codomain might not be in the range, leaving it with no preimage. Second, two different inputs might produce the same output, forcing the inverse to be "multi-valued." We give names to functions that avoid each of these failures.

Definition 4.4 (Surjective). A function $f\colon A \to B$ is *surjective* (or *onto*) when the range equals the full codomain:

$$\text{for every } y \in B, \text{ there exists } x \in A \text{ such that } f(x) = y.$$

Definition 4.5 (Injective). A function $f\colon A \to B$ is *injective* (or *one-to-one*) when different inputs produce different outputs:

$$\text{for all } x_1, x_2 \in A, \quad x_1 \neq x_2 \implies f(x_1) \neq f(x_2).$$

Equivalently, $f(x_1) = f(x_2) \implies x_1 = x_2$.

Theorem 4.1 (Existence of Inverses). *A function $f\colon A \to B$ has an inverse if and only if f is both injective and surjective.*

This is the rigorous story. In calculus, however, we take a shortcut, entirely bypassing the surjectivity problem.

Remark (Calculus Convention for Inverses). In calculus, we almost always shrink the codomain to match the range without saying so explicitly. This is why most calculus textbooks almost never use the words "codomain" or "surjective." Under this convention, a function has an inverse if and only if it is **injective**.

Definition 4.6 (Inverse Function — Calculus Version). Let f be a one-to-one function with domain A and range C. The *inverse function* f^{-1} has domain C and range A, and is defined by

$$x = f^{-1}(y) \quad \iff \quad y = f(x), \qquad \text{for all } x \in A \text{ and all } y \in C.$$

Equivalently, f and f^{-1} satisfy the composition identities

$$f^{-1}\big(f(x)\big) = x \quad \text{for all } x \in A, \qquad f\big(f^{-1}(y)\big) = y \quad \text{for all } y \in C.$$

4.3 Inverse Function Examples

Let us study several examples of pairs of inverse functions. These illustrate the general theory, and the last example previews the subtlety that arises with inverse trigonometric functions.

Example 4.1 *(Linear Function).* Let $f(x) = 2x + 1$. The domain and range are both $\mathbb{R}$, so an inverse exists. From the definition,

$$y = 2x + 1 \quad \implies \quad x = \frac{y-1}{2},$$

so $f^{-1}(y) = \dfrac{y-1}{2}$. One can verify the composition identities directly:

$$f^{-1}\big(f(x)\big) = \frac{(2x+1)-1}{2} = x,$$
$$f\big(f^{-1}(y)\big) = 2 \cdot \frac{y-1}{2} + 1 = y.$$

Example 4.2 *(Exponentials and Logarithms).* The exponential function e^x and the natural logarithm $\ln(x)$ are inverses of each other. Their defining relation is

$$y = e^x \quad \iff \quad x = \ln(y), \qquad y > 0.$$

The domain of e^x is $\mathbb{R}$ and its range is $(0, \infty)$; the domain of ln is $(0, \infty)$ and its range is $\mathbb{R}$.

This statement can be used to **define** one function from the other. In high school, one typically takes the exponential as given and defines ln through the relation above. In analysis, it is common to start with ln (constructed as an integral) and define e^x as its inverse.

The graphs of e^x and $\ln(x)$ are reflections of each other across the line $y = x$. This is a general property: switching from a function to its inverse swaps the roles of x and y, which is geometrically a reflection across the diagonal.

The composition identities are

$$\ln(e^x) = x \quad \text{for all } x \in \mathbb{R}, \qquad e^{\ln(y)} = y \quad \text{for all } y > 0.$$

Example 4.3 (Restricting the Domain of x^2). Consider $f(x) = x^2$. This function is **not** one-to-one: $f(3) = f(-3) = 9$. Therefore f does not have an inverse.

However, we can *restrict* f to the smaller domain $[0, \infty)$. The restriction is one-to-one and its range is $[0, \infty)$, so it has an inverse. That inverse is the square root:

$$g(x) = \sqrt{x}.$$

Note that g is **not** the inverse of the original f; it is the inverse of the *restriction* of f to $[0, \infty)$.

This is reflected in the composition identities. We have $\sqrt{x^2} = x$ only when $x \geq 0$; in general, $\sqrt{x^2} = |x|$.

Remark. If we have a function that is not one-to-one, we can restrict it to a smaller domain and construct an inverse of the restriction. The definition of the inverse and the properties it satisfies must then be modified accordingly. This idea becomes essential when defining inverse trigonometric functions.

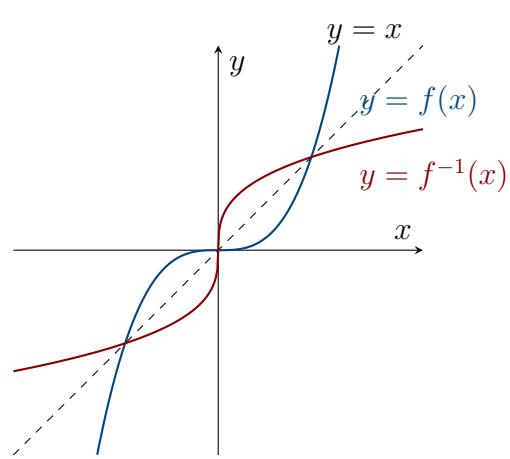

Figure 4.1: A function and its inverse reflect across $y = x$.

4.4 Derivatives of Inverse Functions

If a function is differentiable and has an inverse, must the inverse also be differentiable? The answer is: not necessarily, unless we add an extra condition.

Consider a differentiable, one-to-one function f whose graph has a horizontal tangent at some point P. Reflecting the graph across the diagonal $y = x$ to obtain the graph of f^{-1} turns that horizontal tangent into a vertical tangent at the corresponding point Q. A vertical tangent has infinite slope, so f^{-1} is not differentiable there. This is the only obstruction.

Theorem 4.2 (Differentiability of the Inverse). *Let f be a function defined on an interval I. If f has an inverse, f is differentiable, and f' is never zero on I, then f^{-1} is also differentiable.*

Remark (Warning). This result should not be confused with the **Inverse Function Theorem** from multivariable calculus, which is a much more powerful and more complicated statement. The theorem above is a simpler, single-variable result.

Once we know both f and f^{-1} are differentiable, we can find the derivative of f^{-1} by differentiating the identity $f\big(f^{-1}(y)\big) = y$.

Theorem 4.3 (Derivative of the Inverse). *Under the hypotheses of Theorem 4.2, for y in the domain of f^{-1},*

$$\big(f^{-1}\big)'(y) = \frac{1}{f'\big(f^{-1}(y)\big)}.$$

Equivalently, setting $x = f^{-1}(y)$ (so that $y = f(x)$),

$$\big(f^{-1}\big)'(y) = \frac{1}{f'(x)}.$$

Proof. We know that $f\big(f^{-1}(y)\big) = y$ for all y in the domain of f^{-1}. Differentiating both sides with respect to y and applying the chain rule on the left gives

$$f'\big(f^{-1}(y)\big) \cdot \big(f^{-1}\big)'(y) = 1.$$

Setting $x = f^{-1}(y)$, this becomes $f'(x) \cdot \big(f^{-1}\big)'(y) = 1$, and solving yields

Chain rule applied to $f \circ f^{-1}$.

$$\big(f^{-1}\big)'(y) = \frac{1}{f'(x)}.$$

Note that f and f^{-1} are evaluated at **different** points: f' is evaluated at x, while $\big(f^{-1}\big)'$ is evaluated at $y = f(x)$. ■

4.5 Derivative of the Exponential

We now try to compute the derivative of an exponential function from the limit definition, and in the process discover a defining property of the number e.

There are many exponential functions, one for each positive base a. If $a > 1$, then a^x is increasing; if $0 < a < 1$, then a^x is decreasing. In either case, the domain is $\mathbb{R}$ and the range is $(0, \infty)$.

Proposition 4.4 (Derivative of a^x — Heuristic). *Fix a positive number a and define $f(x) = a^x$. Then*

$$f'(x) = \lim_{h\to 0} \frac{a^{x+h} - a^x}{h} = a^x \cdot \lim_{h\to 0} \frac{a^h - 1}{h}.$$

*The limit $L_a = \lim_{h\to 0} \frac{a^h-1}{h}$ depends only on a, not on x. Therefore the derivative of a^x is a **constant multiple** of itself:*

$$\frac{d}{dx}\left[a^x\right] = L_a \cdot a^x.$$

Definition 4.7 (The Number e). We define e to be the unique positive real number for which $L_e = 1$, i.e.,

$$\lim_{h\to 0} \frac{e^h - 1}{h} = 1.$$

Equivalently, e is the unique base for which the exponential is its own derivative:

$$\frac{d}{dx}\left[e^x\right] = e^x.$$

Remark (Warning). This derivation, while illuminating, assumes several things that require justification: that the limit L_a exists for every positive a, that there exists a value of a for which $L_a = 1$, and indeed that a^x is well-defined for arbitrary real x. These issues are addressed in Section 4.6. Nevertheless, the derivation is helpful for understanding where the number e comes from and why exponentials have the derivative they do.

4.6 Defining Exponentials Rigorously

It may surprise you to learn that defining exponentials rigorously is much harder than what you were told in high school. For most practical purposes this is not something to worry about too much, but it is interesting to know that it is a genuinely difficult problem.

Recall the high-school definition of logarithms: $\log_a(x) = y$ means $a^y = x$. This defines the logarithm as the *inverse* of the exponential, so it is only valid if exponentials are already well-defined.

Let us review how powers are defined, starting with rational exponents.

Remark (Defining Powers with Rational Exponents). Let $a > 0$. Powers with rational exponents are built in stages:

(i) **Positive integer exponents.** a^n ($n \in \mathbb{Z}^+$) is defined as iterated multiplication.

(ii) **Reciprocal exponents.** $a^{1/n}$ is defined as $\sqrt[n]{a}$, the unique positive number whose n-th power is a.

(iii) **General positive rationals.** $a^{n/m}$ is defined as $\left(a^n\right)^{1/m}$, using steps (i) and (ii).

(iv) **Zero exponent.** $a^0 = 1$.

(v) **Negative exponents.** $a^{-c} = \dfrac{1}{a^c}$ for $c > 0$.

This defines a^c for every rational c. There are no difficulties so far.

Remark (Honest Acknowledgement of Difficulty). How do we define a^c when c is irrational? For example, what is 2^π? We cannot use iterated multiplication, roots, or quotients to define it. There is no easy way around this: one must enter "serious analysis territory" to even say what 2^π means.

*One natural idea is to approximate π by rational numbers—*3, 3.1, 3.14, 3.141, ... *—compute 2^c for each, and define 2^π as the limit of the resulting sequence. This is probably what early mathematicians were implicitly assuming. But it is surprisingly hard to prove that (a) the limit exists, and (b) it does not depend on the choice of rational approximation.*

The key identity

$$a^c = e^{c\ln(a)}$$

reduces exponentials with arbitrary base to exponentials with base e. It therefore suffices to define e^x (or equivalently $\ln(x)$) rigorously.

Remark (Three Approaches to Defining e^x). In analysis, there are at least three common approaches:

(i) **Differential equations.** Define e^x as the unique solution to the initial value problem $y' = y$, $y(0) = 1$. The Picard–Lindelöf theorem guarantees existence and uniqueness. This is natural but expensive: the theorem itself is sophisticated.

(ii) **Power series.** Define $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$. This generalises beautifully to complex numbers, but proving the algebraic properties of the exponential from this definition requires the full theory of power series.

(iii) **Integral definition of** ln. Define $\ln(x) = \int_1^x \frac{1}{t}\,dt$ for $x > 0$, and then define e^x as the inverse of ln. This requires only the Fundamental Theorem of Calculus, making it by far the most accessible approach.

Remark. A full rigorous construction proving all the details is the province of an analysis course. The goal here is simply to understand *why* defining exponentials is not trivial; every calculus student should appreciate this.

4.7 Derivative of the Logarithm

We know that the derivative of e^x is itself. How can this help us find the derivative of $\ln(x)$? Since e^x and $\ln(x)$ are inverses, the identity $e^{\ln(x)} = x$ provides a relation between the two functions. Differentiating both sides will yield a relation between their derivatives.

Theorem 4.5 (Derivative of $\ln$). *For all $x > 0$,*

$$\frac{d}{dx}\big[\ln(x)\big] = \frac{1}{x}.$$

Proof. Start from $e^{\ln(x)} = x$. Differentiate both sides with respect to x. On the left, the chain rule gives

$$e^{\ln(x)} \cdot \frac{d}{dx}\big[\ln(x)\big] = 1.$$

Chain rule applied to $e^{\ln(x)}$.

Since $e^{\ln(x)} = x$, this becomes

$$x \cdot \frac{d}{dx}\big[\ln(x)\big] = 1.$$

Solving gives $\dfrac{d}{dx}\big[\ln(x)\big] = \dfrac{1}{x}$. ■

Remark. This same trick—differentiating a composition identity—can be used to find the derivative of **any** function, provided we know the derivative of its inverse.

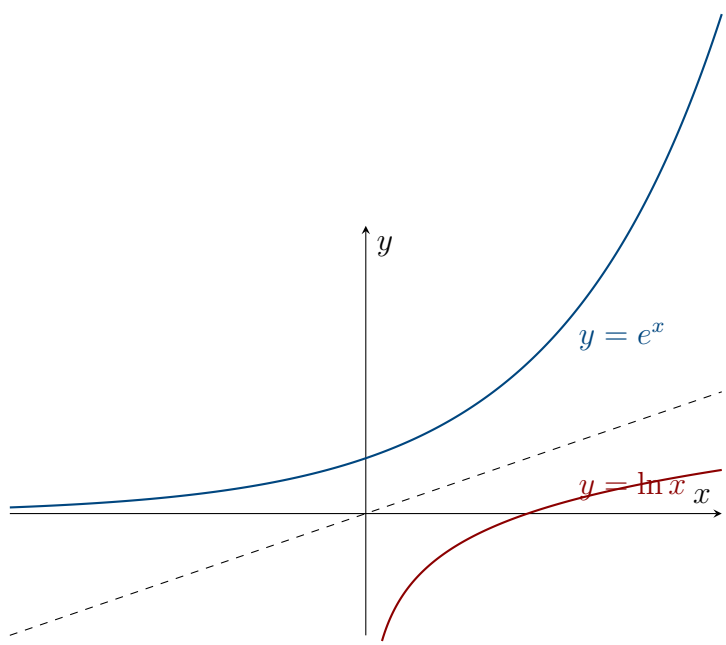

Figure 4.2: The graphs of e^x and $\ln x$.

4.8 Derivatives of General Exponentials

We know the derivative of e^x, but what about a^x for an arbitrary positive base a? A common strategy in mathematics: reduce the general problem to a special case we have already solved.

Theorem 4.6 (Derivative of a^x). *For every $a > 0$,*

$$\frac{d}{dx}\left[a^x\right] = a^x \ln(a).$$

Proof. Any positive number a can be written as $a = e^{\ln(a)}$. Therefore

$$a^x = \left(e^{\ln(a)}\right)^x = e^{x\ln(a)}.$$

The derivative of $e^{x\ln(a)}$ with respect to x is, by the chain rule,

Rewrite a^x using $a = e^{\ln(a)}$.

$$e^{x\ln(a)} \cdot \ln(a) = a^x \ln(a).$$

■

Remark. When $a = e$, we get $\ln(e) = 1$, and the formula reduces to $\frac{d}{dx}[e^x] = e^x$, recovering the known identity.

Remark (Pause and Try). You already know that $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$. What is the derivative of $\log_a(x)$ for an arbitrary positive base $a \neq 1$? As a hint, use the change-of-base identity $\log_a(x) = \frac{\ln(x)}{\ln(a)}$.

4.9 Logarithmic Differentiation

We use logarithmic differentiation to compute the derivative of a function that is raised to the power of another function—a situation in which neither the power rule nor the exponential rule directly applies.

Example 4.4 (A Function Raised to a Function). Let $f(x) = \left(\cos(x)\right)^{\sin(x)}$. This expression is **not** simply a power (like x^3), because the base depends on x. Nor is it simply an exponential (like 2^x), because the exponent also depends on x.

Neither the power rule nor the exponential derivative rule applies on its own. We need a different strategy.

Method 1: Rewrite as an exponential with base e. Using the identity $u^v = e^{v\ln u}$,

$$f(x) = e^{\sin(x)\cdot\ln(\cos(x))}.$$

Differentiating using the chain rule,

$$\begin{aligned} f'(x) &= e^{\sin(x)\cdot\ln(\cos(x))} \cdot \frac{d}{dx}\left[\sin(x)\cdot\ln(\cos(x))\right] \\ &= \left(\cos(x)\right)^{\sin(x)} \cdot \left[\cos(x)\cdot\ln(\cos(x)) + \sin(x)\cdot\frac{-\sin(x)}{\cos(x)}\right]. \end{aligned}$$

Method 2: Logarithmic differentiation. Take logarithms of both sides:

$$\ln\big(f(x)\big) = \sin(x) \cdot \ln\big(\cos(x)\big).$$

The exponent has disappeared. Differentiate both sides using implicit differentiation on the left and the product rule on the right:

$$\frac{f'(x)}{f(x)} = \cos(x) \cdot \ln(\cos(x)) + \sin(x) \cdot \frac{-\sin(x)}{\cos(x)}.$$

Solving for $f'(x)$ recovers the same answer as Method 1.

Remark. The two methods are the **same method in disguise**. Method 2 (logarithmic differentiation) is often more convenient because taking logarithms turns products into sums, quotients into differences, and powers into products, simplifying the expression before differentiation.

4.10 Power Rule for Real Exponents

In an earlier chapter we proved the power rule for positive integer exponents. We now prove it for ***any*** *real exponent, using logarithms.*

Theorem 4.7 (Power Rule — General Version). *For any $c \in \mathbb{R}$ and for all $x > 0$,*

$$\frac{d}{dx}\big[x^c\big] = c\,x^{c-1}.$$

Remark. It may feel like overkill to use logarithms to prove the power rule—are powers not more "basic" than logarithms? For *rational* exponents, alternative proofs exist that avoid logarithms entirely. But for *real* exponents, the identity $x^c = e^{c\ln(x)}$ is not merely a useful trick; it is **the definition** of a power with a real exponent.

Proof. For $x > 0$, write $x^c = e^{c\ln(x)}$. By the chain rule,

$$\begin{aligned}\frac{d}{dx}\big[x^c\big] &= \frac{d}{dx}\big[e^{c\ln(x)}\big] \\ &= e^{c\ln(x)} \cdot \frac{c}{x} \\ &= x^c \cdot \frac{c}{x} \\ &= c\,x^{c-1}.\end{aligned}$$

Uses $\frac{d}{dx}[e^u] = e^u \cdot u'$ and $\frac{d}{dx}[\ln x] = \frac{1}{x}$.

■

Remark. This proof works for **all** exponents at once—natural, rational, real, and even complex. The price is that we must first develop the theory of logarithms and establish the derivative formulas for e^x and $\ln(x)$. The proof only *looks* short because the hard work was done beforehand.

4.11 Logarithm Notation

There is a notational controversy surrounding logarithms. All mathematicians agree that the natural logarithm—logarithm in base e—is the most important, but we cannot agree on what to call it.

If one fixes a favourite base b, then exponentials and logarithms in any other base can be expressed in terms of base b. So why not choose the *best* base and use it exclusively?

Mathematicians realised early that the base e is special:

- Exponentials with base e arise naturally as limits.
- The functions e^x and $\ln(x)$ have **simpler derivatives** than exponentials or logarithms in any other base.
- e^x and $\ln(x)$ have elegant power-series expansions.

Natural logarithm is natural, and it is the only logarithm one should ever use.

Remark. Historically, the decimal system made logarithms in base 10 (the "common logarithm") the most useful for hand computation using log tables and slide rules. The notation $\log(x)$ without a base came to mean $\log_{10}(x)$. Today, nobody computes by hand, and modern mathematicians argue that log should be reserved for the natural logarithm. Engineers and many calculus textbooks continue to use ln.

In summary: whenever you see $\ln(x)$ or $\log(x)$ without a specified base, both *probably* mean logarithm in base e. When reading a book or working with others, check which convention is in use.

4.12 The Arcsine Function

Some books and most calculators call this function "sine inverse." We prefer the name "arcsine" because arcsine is ***not*** *the inverse function of sine—the situation is more subtle.*

The function $\sin\colon \mathbb{R} \to [-1,1]$ is **not** one-to-one: for example, $\sin(0) = \sin(\pi) = 0$. Therefore sine does not have an inverse function.

Definition 4.8 (Arcsine). The function *arcsine* is defined as the inverse of the **restriction** of sine to the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Equivalently,

$$x = \arcsin(y) \quad \Longleftrightarrow \quad y = \sin(x), \qquad \text{for all } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \text{ and all } y \in [-1,1].$$

The domain of arcsin is $[-1,1]$ and its range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Remark. The choice of the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is a convention. We could have restricted sine to any interval on which it is one-to-one and preserves the full range $[-1,1]$. The symmetric interval around 0 is the standard choice, but it is arbitrary.

Since arcsin and the restriction of sin are inverses, the composition identities hold on the correct domains:

$$\arcsin\big(\sin(x)\big) = x \quad \text{for all } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \qquad \sin\big(\arcsin(y)\big) = y \quad \text{for all } y \in [-1, 1].$$

Remark (Warning). These identities are **not** true outside the specified domains. For instance, $\sin(\arcsin(2))$ is undefined, and $\arcsin(\sin(2)) \neq 2$.

Example 4.5. What is $\arcsin\left(\frac{1}{2}\right)$? Call the answer t. Then $\sin(t) = \frac{1}{2}$ and $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. The only such angle is $t = \frac{\pi}{6}$.

Example 4.6. What is $\arcsin\big(\sin(2)\big)$? Since $2 \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, we cannot simply say the answer is 2.

Let $t = \arcsin(\sin(2))$. By definition, $\sin(t) = \sin(2)$ and $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. This is a trigonometry problem. From the symmetry of the sine graph, the distance from $\frac{\pi}{2}$ to t equals the distance from 2 to $\frac{\pi}{2}$:

$$\frac{\pi}{2} - t = 2 - \frac{\pi}{2} \quad \Longrightarrow \quad t = \pi - 2.$$

Therefore $\arcsin(\sin(2)) = \pi - 2$.

Remark (Pause and Try). What is $\arcsin(\sin(6))$? Hint: the answer is neither 6 nor $\pi - 6$.

4.13 Derivative of Arcsine

We use the same strategy as for ln*: differentiate a composition identity relating* sin *and* arcsin*, then solve for the unknown derivative.*

Theorem 4.8 (Derivative of arcsin). *For all $x \in (-1, 1)$,*

$$\frac{d}{dx}\big[\arcsin(x)\big] = \frac{1}{\sqrt{1 - x^2}}.$$

Proof. Start from $\sin\big(\arcsin(x)\big) = x$. Differentiating both sides with respect to x and applying the chain rule on the left gives

$$\cos\big(\arcsin(x)\big) \cdot \frac{d}{dx}\big[\arcsin(x)\big] = 1.$$

Chain rule applied to $\sin \circ \arcsin$.

Solving yields $\frac{d}{dx}[\arcsin(x)] = \frac{1}{\cos(\arcsin(x))}$.

It remains to simplify $\cos(\arcsin(x))$. Let $\theta = \arcsin(x)$, so $\sin(\theta) = x$. Using the identity $\cos^2(\theta) = 1 - \sin^2(\theta)$,

$$\cos(\theta) = \pm\sqrt{1 - x^2}.$$

Since $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ by definition of arcsin, we have $\cos(\theta) \geq 0$, so we choose the positive sign. Therefore

$$\frac{d}{dx}\big[\arcsin(x)\big] = \frac{1}{\sqrt{1 - x^2}}.$$

The sign is determined by the restriction interval $[-\pi/2, \pi/2]$.

■

Remark. The positive sign arises specifically because arcsin was defined using the restriction of sin to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, where cos is non-negative. A different restriction could yield a different sign.

4.14 Arctangent and Arccosine

Having defined arcsin *and computed its derivative, we now do the same for* arctan *and* arccos*. The ideas are identical, so we proceed more quickly.*

Definition 4.9 (Arctangent). The function tan is not one-to-one on its natural domain. We restrict it to $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (an **open** interval, since tan is undefined at the endpoints). The *arctangent* function is the inverse of this restriction:

$$x = \arctan(y) \quad \Longleftrightarrow \quad y = \tan(x), \qquad \text{for all } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \text{ and all } y \in \mathbb{R}.$$

The domain of arctan is $\mathbb{R}$ and its range is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Remark. Notice that the restriction interval for arctan is *open*, in contrast to the *closed* interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ used for arcsin. Since the range of tan on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is all of $\mathbb{R}$, the domain of arctan is $\mathbb{R}$.

Definition 4.10 (Arccosine). The function cos is not one-to-one on $\mathbb{R}$. We restrict it to $[0, \pi]$, and define the *arccosine* function as the inverse of this restriction:

$$x = \arccos(y) \quad \Longleftrightarrow \quad y = \cos(x), \qquad \text{for all } x \in [0, \pi] \text{ and all } y \in [-1, 1].$$

The domain of arccos is $[-1, 1]$ and its range is $[0, \pi]$.

Remark. For arccos, we cannot choose a domain for cos that is symmetric about 0—that would not produce a one-to-one function. The interval $[0, \pi]$ is the standard choice.

Remark (Pause and Try). Can you sketch the graphs of arctan and arccos? For each, reflect the graph of the restricted trigonometric function across the line $y = x$.

Theorem 4.9 (Derivatives of Inverse Trigonometric Functions). *(i)*

$$\frac{d}{dx}\big[\arcsin(x)\big] = \frac{1}{\sqrt{1-x^2}}, \qquad \textit{for } x \in (-1, 1).$$

(ii) $$\frac{d}{dx}\big[\arccos(x)\big] = \frac{-1}{\sqrt{1-x^2}}, \qquad \textit{for } x \in (-1, 1).$$

(iii) $$\frac{d}{dx}\big[\arctan(x)\big] = \frac{1}{1+x^2}, \qquad \textit{for all } x \in \mathbb{R}.$$

Remark. The derivative of arccos is the same as that of arcsin but with an additional **minus sign**. The particular signs arise from the specific restrictions chosen; different restrictions could produce different results.

Remark. It is worth memorising the derivatives of arcsin and arctan in particular. They appear frequently in integration, where one recognises integrands of the form $\frac{1}{\sqrt{1-x^2}}$ or $\frac{1}{1+x^2}$ as derivatives of inverse trigonometric functions.

Remark. One could similarly define arcsecant, arccosecant, and arccotangent, but in practice these three functions are rarely used. For most purposes, arcsin, arccos, and arctan suffice.

Chapter 5

The Mean Value Theorem and Applications

5.1 Motivation for the MVT

The Mean Value Theorem is perhaps the most interesting result in calculus. And yet, when you look at the statement, at first sight it appears not very interesting, technical, boring... honestly, it appears useless. So why do we care? Or should we care?

The goal of the Mean Value Theorem is to help us obtain information about a function using something about its derivative. When we know something about the derivative, what can we say about the function?

Consider the following natural question. We know that constant functions have zero derivative. Are those the only ones? Are there other functions with zero derivative? It appears not, and we would like to prove this.

Theorem 5.1 (Zero Derivative Implies Constant — Preview). *Let f be a function defined on an open interval (a, b). If $f'(x) = 0$ for every $x \in (a, b)$, then f is constant.*

This is the simplest example of a theorem in which we use something about the derivative to conclude something about the function. Since it is the simplest, let us try to prove it first... Surprisingly, we do not even know how to start! Despite looking very simple, this is not easy to prove.

Remark (Pause and Try). Pause and try to prove Theorem 5.1. There is no clear way to do it. In fact, we do not know **any** way to prove this simple theorem without using the Mean Value Theorem.

Here is a partial list of results that all rely on the Mean Value Theorem:

- If a function has zero derivative, it must be constant.
- If a function has positive derivative, it must be increasing.
- All integration methods (finding antiderivatives up to a constant).

- L'Hôpital's Rule.
- The Fundamental Theorem of Calculus.
- Taylor's Theorem and Taylor series with remainder.
- Equality of mixed partial derivatives in multivariable calculus.

Everything important in calculus, whether big or small, requires the Mean Value Theorem to prove it.

But reaching the Mean Value Theorem is quite an investment. Before we are ready to state and prove it, we need a simplified version called Rolle's Theorem, which is interesting on its own but whose main purpose is as a stepping stone. And in order to prove Rolle's Theorem, we need two other results: the Extreme Value Theorem, which we already know, and the Local Extreme Value Theorem, which we will study next.

Remark. Should you care about the Mean Value Theorem? It depends on your goals. If you just want to *use* calculus tools, you can skip it—you will never use it directly. But if you want to *prove* the theorems and understand why they are true, then you absolutely need the MVT, since there is little or nothing you can prove without it.

5.2 Local Extrema and Critical Points

Before defining local extremum, let us look at an example. Consider a function f defined on $[0, 5]$. This function has a minimum of 1 at $x = 3$ and a maximum of 5 at $x = 5$ (an endpoint—nothing wrong with that). But there is a third interesting point: the value there is 3, and the function takes values greater than 3 elsewhere, so it is not a maximum. However, if we stay close to that point, it looks like a maximum. We call it a local *maximum.*

Definition 5.1 (Maximum and Local Maximum). Let f be a function with domain I, and let $c \in I$.

(i) We say f has a *maximum* at c when $f(x) \leq f(c)$ for all $x \in I$.

(ii) We say f has a *local maximum* at c when there exists $\delta > 0$ such that

$$|x - c| < \delta \implies f(x) \leq f(c).$$

Analogous definitions hold for *minimum* and *local minimum.*

Remark. Some terminology:

- The word *extremum* means "maximum or minimum."
- Some authors say *global extremum* instead of simply *extremum* to emphasise the contrast with local extremum. The two phrases mean the same thing.

- The plurals are *extrema*, *maxima*, and *minima* (irregular Latin plurals).

Remark (Warning). According to the definition above, endpoints **never** count as local extrema, because the function needs to be defined on both sides of c. This convention is used by the majority of calculus textbooks, and we adopt it here. However, the majority of analysis textbooks use a different convention that *does* include endpoints. When reading other sources, be sure to check which convention is in use. There is one specific point later where this distinction is very important.

Now that we have our definitions, let us move towards the Local Extreme Value Theorem. What can we say about the derivative of a function at a local extremum? Looking at graphs, the tangent line at a local extremum appears to be horizontal (derivative zero)... unless the function has a corner, in which case the derivative does not exist. These seem to be the only two options.

Theorem 5.2 (Local Extreme Value Theorem). *Let f be a function with domain an interval I, and let c be an interior point of I. If f has a local extremum at c, then $f'(c) = 0$ or $f'(c)$ does not exist.*

Remark. Given the definition of local extremum that we use (which excludes endpoints), the hypothesis that c is an interior point is already implicit. We state it explicitly anyway, to emphasise it. If one uses the alternative definition that allows endpoints, then this hypothesis is **essential** and must not be omitted.

Definition 5.2 (Critical Point). Let f be a function defined on an interval I. We say that c is a *critical point* of f when c is an interior point of I and either $f'(c) = 0$ or $f'(c)$ does not exist.

With this terminology, Theorem 5.2 simply says that **local extrema are always critical points**.

Remark (Warning). The converse is **not** true. A critical point is not always a local extremum.

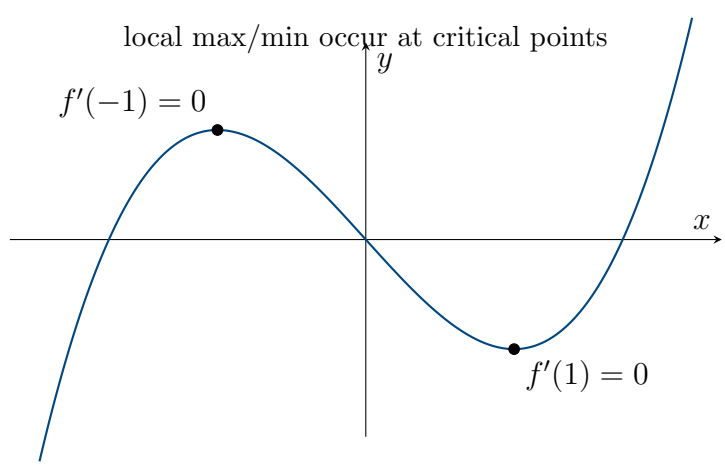

Figure 5.1: A smooth function with critical points at a local max and min.

5.3 Proof of the Local EVT

We now prove the Local Extreme Value Theorem stated in Section 5.2. The argument treats the case of a local maximum; the case of a local minimum is entirely analogous (reverse the inequalities).

Proof of Theorem 5.2 (local maximum case). Suppose f has a local maximum at an interior point c. The derivative, if it exists, is

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

We wish to show that either this limit does not exist or it equals zero. Assume the limit exists; we show it must be zero.

Examining the quotient from each side separately.

Consider the **denominator**. When $x \to c^+$, the quantity $(x-c)$ is positive. When $x \to c^-$, it is negative.

Consider the **numerator**. Since f has a local maximum at c, we have $f(x) \leq f(c)$ for all x sufficiently close to c. Therefore $f(x) - f(c) \leq 0$ for x near c, regardless of the side of approach.

Combining:

- From the right: the quotient $\dfrac{f(x) - f(c)}{x - c}$ is $\dfrac{(\leq 0)}{(> 0)} \leq 0$, so the right-hand limit is ≤ 0.
- From the left: the quotient is $\dfrac{(\leq 0)}{(< 0)} \geq 0$, so the left-hand limit is ≥ 0.

Since we assumed the full limit exists, the two one-sided limits must exist and be equal. A quantity that is both ≤ 0 and ≥ 0 must be zero. Therefore $f'(c) = 0$. ■

Remark. We needed c to be an interior point because we examined the limit from both sides. If c were an endpoint, the argument would fail.

Remark (Pause and Try). Write the proof for a local minimum. It is very similar: change the direction of the inequalities in the numerator. Everything else works the same way.

5.4 Finding Global Extrema

The Local Extreme Value Theorem is not only a stepping stone towards the Mean Value Theorem; it is also useful on its own. Here we illustrate, with an example, how to use it to find the global maximum and minimum of a function.

Example 5.1 *(Extrema of a Polynomial on a Closed Interval).* Find the extrema (and local extrema) of

$$f(x) = x^3 - 3x^2 - 9x + 35 \qquad \text{on } [-4, 4].$$

Step 1. The function f is continuous on the closed, bounded interval $[-4, 4]$. By the Extreme Value Theorem, f must have a maximum and a minimum on this interval.

Remark (Warning). Some students like to skip this first step, but it is **essential**. Without it, everything else fails: the function might not even have a maximum or a minimum.

Step 2. Since f has a maximum and a minimum, each must occur at an **endpoint** or at an **interior point**. If it occurs at an interior point, it is also a local extremum, so by Theorem 5.2 it must be a critical point.

Therefore, the plan is:

(i) Find all endpoints and critical points.

(ii) Evaluate f at each of these candidate points.

(iii) The largest value is the maximum; the smallest is the minimum.

Remark (Warning). This method only works for a **continuous function on a closed, bounded interval**. Otherwise, the function may not have a maximum or minimum, and applying this method carelessly may give the wrong answer.

Calculations. The endpoints are $x = -4$ and $x = 4$. For the critical points, compute the derivative:

$$f'(x) = 3x^2 - 6x - 9 = 3(x + 1)(x - 3).$$

Since f is a polynomial, f' exists everywhere, so the critical points are the solutions of $f'(x) = 0$: namely $x = -1$ and $x = 3$.

Evaluating f at all four candidate points:

x	-4	-1	3	4
$f(x)$	-41	40	8	15

The largest value is $f(-1) = 40$, so f has a **maximum of** 40 **at** $x = -1$. The smallest value is $f(-4) = -41$, so f has a **minimum of** -41 **at** $x = -4$.

From this information, we can also guess the shape of the graph: f has a local maximum at $x = -1$ and a local minimum at $x = 3$. Strictly speaking, this last observation is not yet fully rigorous, but geometrically it makes perfect sense. We will learn how to argue this rigorously when we study monotonicity.

5.5 Rolle's Theorem

Rolle's Theorem is an intermediate step that will allow us to prove the Mean Value Theorem. That is the main reason to study it. By itself, the theorem may appear a bit dry and uninteresting, but it will lead us to something much more important.

Theorem 5.3 (Rolle's Theorem). *Let f be defined on a closed, bounded interval $[a, b]$. Suppose:*

(i) f is continuous on $[a, b]$,

(ii) f is differentiable on (a, b),

(iii) $f(a) = f(b)$.

Then there exists at least one $c \in (a, b)$ such that $f'(c) = 0$.

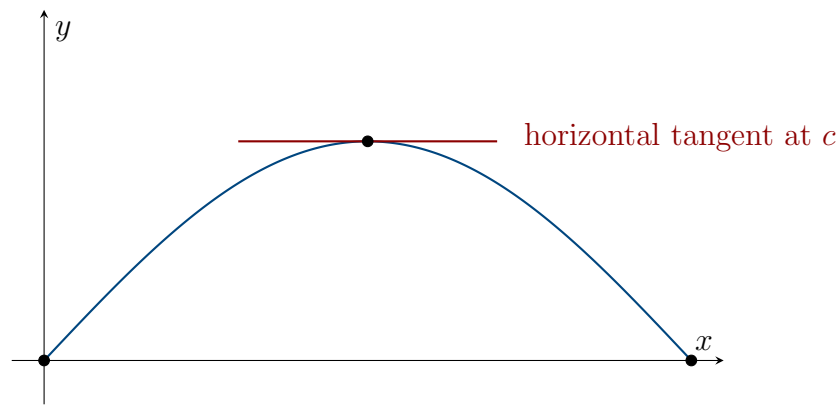

Figure 5.2: Rolle's theorem: a horizontal tangent in the interior.

*Before writing the proof, let us understand **why** each hypothesis is needed.*

Remark. **Why hypothesis (iii)?** If $f(a) = f(b)$, the graph must go up and come back down (or vice versa), forcing a local extremum with derivative zero. Without this condition, the function could be monotone and have no point with zero derivative.

Why hypothesis (ii)? Even with $f(a) = f(b)$, the function could have a local extremum at a corner, where the derivative does not exist.

Why hypothesis (i)? Even with differentiability on the interior and equal endpoint values, a jump discontinuity at an endpoint can destroy the conclusion.

Remark. One might ask: "Why not simply require differentiability on the full closed interval $[a, b]$?" That would also make the theorem true. However, it is good practice to state theorems with hypotheses as **weak** as possible, because then the theorem applies in more situations and is therefore more useful.

Proof of Rolle's Theorem. Since f is continuous on the closed, bounded interval $[a, b]$, the Extreme Value Theorem guarantees that f attains a maximum and a minimum on $[a, b]$.

Case 1: At least one extremum (maximum or minimum) occurs at an interior point $c \in (a, b)$. Then f has a local extremum at c. By the Local Extreme Value Theorem (Theorem 5.2), $f'(c)$ is either zero or does not exist. Since f is differentiable on (a, b), we conclude $f'(c) = 0$.

Using the Ex
Theorem an
Theorem 5.2

Case 2: Both the maximum and the minimum occur at endpoints. Since $f(a) = f(b)$, the maximum value and the minimum value are both equal to $f(a)$. Therefore f is constant on $[a, b]$, so $f'(c) = 0$ for every $c \in (a, b)$.

In both cases, there exists $c \in (a, b)$ with $f'(c) = 0$. ■

Remark. All three hypotheses were used in the proof: continuity on $[a, b]$ to invoke the Extreme Value Theorem, differentiability on (a, b) to conclude $f'(c) = 0$ (rather than merely "does not exist"), and $f(a) = f(b)$ to handle the case when both extrema occur at endpoints.

5.6 Counting Zeroes of Functions

The main reason we learned Rolle's Theorem is as a stepping stone to the Mean Value Theorem. Nevertheless, it has applications of its own. Here is one: it can help us count the number of zeroes of a function.

Definition 5.3 (Zero of a Function). We say that c is a *zero* of a function f when $f(c) = 0$.

Remark (Warning). We say c is a zero of the **function** f, and a solution of the **equation** $f(x) = 0$. A function by itself is not an equation.

The strategy for counting zeroes of a function uses two steps:

(i) Use the **Intermediate Value Theorem** to show that f has *at least* a certain number of zeroes.

(ii) Use **Rolle's Theorem** to show that f has *at most* a certain number of zeroes.

If the two bounds agree, we know the exact count.

The key observation linking Rolle's Theorem to zeroes is the following.

Theorem 5.4 (Zeroes and Derivatives). *Let f be continuous and differentiable on an interval I. Then the number of zeroes of f on I is at most the number of zeroes of f' on I, plus one.*

Proof. Suppose f has n zeroes on I, say $x_1 < x_2 < \cdots < x_n$. On each subinterval $[x_k, x_{k+1}]$ $(k = 1, \ldots, n-1)$, the function f satisfies the hypotheses of Rolle's Theorem: it is continuous on $[x_k, x_{k+1}]$, differentiable on (x_k, x_{k+1}), and $f(x_k) = f(x_{k+1}) = 0$.

Therefore, for each k, there exists $c_k \in (x_k, x_{k+1})$ with $f'(c_k) = 0$. Since $c_1 < c_2 < \cdots < c_{n-1}$, the derivative f' has at least $n - 1$ zeroes.

Rolle's Theorem applied to consecutive zeroes.

Equivalently, if f' has m zeroes, then f has at most $m + 1$ zeroes. ■

Example 5.2 (Exact Number of Zeroes). How many zeroes does $g(x) = x^6 + x^2 + x - 2$ have?

Lower bound (IVT). The function g is a polynomial, hence continuous on $\mathbb{R}$. We compute:

$$g(-2) = 64+4-2-2 = 64 > 0, \qquad g(0) = -2 < 0, \qquad g(1) = 1+1+1-2 = 1 > 0.$$

Since g changes sign from positive to negative on $(-2, 0)$ and from negative to positive on $(0, 1)$, the IVT guarantees at least **two** zeroes.

Upper bound (Rolle's Theorem and Theorem 5.4). Compute:

$$g'(x) = 6x^5 + 2x + 1,$$
$$g''(x) = 30x^4 + 2.$$

Since $x^4 \geq 0$ for all x, we have $g''(x) \geq 2 > 0$ for all x. So g'' has **zero** zeroes.

By Theorem 5.4 applied to g': the number of zeroes of g' is at most $0+1 = 1$.

By Theorem 5.4 applied to g: the number of zeroes of g is at most $1+1 = 2$.

Since g has at least two zeroes and at most two zeroes, g has **exactly two** zeroes.

5.7 The Mean Value Theorem

We have been working towards this theorem for a while: the Extreme Value Theorem, the Local Extreme Value Theorem, and Rolle's Theorem were all stepping stones. The Mean Value Theorem itself will appear not very interesting at first, but we care about it because it has many powerful applications.

Recall Rolle's Theorem (Theorem 5.3): if f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.

For the Mean Value Theorem, we keep the first two hypotheses (they hold for most common functions) but remove the third. The question is: with only continuity and differentiability, what conclusion can we still draw?

Consider a function satisfying the first two hypotheses, but with $f(a) \neq f(b)$. Draw the line through the points $(a, f(a))$ and $(b, f(b))$. Imagine "tilting" the picture so that this line becomes horizontal. From the tilted perspective, the endpoint values are equal, so Rolle's Theorem applies and there is a point where the tangent line is horizontal—that is, parallel to the line through the endpoints. Returning to the original picture, we conclude there exists a point where the slope of the tangent line equals the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.

Theorem 5.5 (Mean Value Theorem). *Let f be defined on a closed, bounded interval $[a, b]$. Suppose:*

(i) f is continuous on $[a, b]$,

(ii) f is differentiable on (a, b).

Then there exists at least one $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

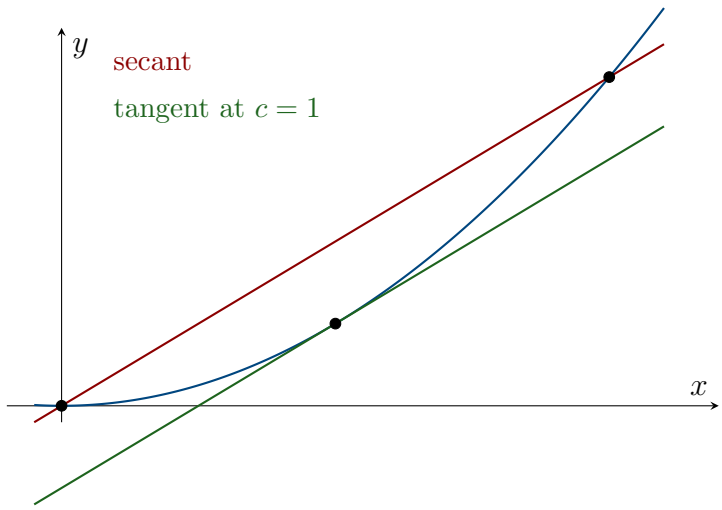

Figure 5.3: Mean Value Theorem: a tangent parallel to the secant.

Remark (Warning). The Mean Value Theorem does **not** merely assert that the equation $f'(c) = \dfrac{f(b) - f(a)}{b - a}$ holds. It says: **if** the hypotheses are satisfied, **then** there **exists** a value $c \in (a, b)$ making the equation true. The full logical structure—hypotheses, existential quantifier, and equation—is the theorem.

Remark. Like Rolle's Theorem and the Intermediate Value Theorem, the MVT is a pure **existence** result. It guarantees that at least one such c exists, but it does not tell us how to find it.

5.8 Proof of the MVT

The idea of the proof is to reduce the Mean Value Theorem to Rolle's Theorem. We construct a new function h that measures the vertical distance between the graph of f and the secant line through $(a, f(a))$ and $(b, f(b))$. Since $h(a) = h(b) = 0$, Rolle's Theorem applies to h.

Proof of the Mean Value Theorem (Theorem 5.5). Let

$$m = \frac{f(b) - f(a)}{b - a}$$

be the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.

Define a new function h on $[a, b]$ by

$$h(x) = f(x) - f(a) - m(x - a).$$

We verify the hypotheses of Rolle's Theorem for h on $[a, b]$:

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(i) h is continuous on $[a, b]$, since f is continuous and $m(x - a)$ is a polynomial.

(ii) h is differentiable on (a, b), since f is differentiable and $m(x - a)$ is a polynomial.

(iii) $h(a) = f(a) - f(a) - m \cdot 0 = 0$, and

$$h(b) = f(b) - f(a) - m(b - a) = f(b) - f(a) - \big(f(b) - f(a)\big) = 0.$$

So $h(a) = h(b) = 0$.

By Rolle's Theorem, there exists $c \in (a, b)$ such that $h'(c) = 0$.
Now observe that for any $x \in (a, b)$,

$$h'(x) = f'(x) - m.$$

Therefore $h'(c) = 0$ gives $f'(c) = m$, which is exactly

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$ ■

After all this work, we have a proof of the Mean Value Theorem. We are now ready to start profiting from it!

5.9 Zero Derivative Implies Constant

This is our first application of the Mean Value Theorem. We return to the problem raised at the very beginning: proving that constant functions are the only functions with zero derivative. This seems obvious, but it genuinely requires the Mean Value Theorem.

Theorem 5.6 (Zero Derivative on a Closed Interval). *Let f be defined on a closed interval $[a, b]$. If f is continuous on $[a, b]$ and $f'(x) = 0$ for every $x \in (a, b)$, then f is constant on $[a, b]$.*

Proof. We must show that $f(x_1) = f(x_2)$ for every $x_1, x_2 \in [a, b]$.

Fix arbitrary $x_1, x_2 \in [a, b]$. Without loss of generality, assume $x_1 < x_2$. (If $x_1 = x_2$, there is nothing to prove.)

Mean Value Theorem (5.5) applied on $[x_1, x_2]$.

We apply the Mean Value Theorem to f on the interval $[x_1, x_2]$. The hypotheses are satisfied: f is continuous on $[x_1, x_2] \subseteq [a, b]$, and f is differentiable on $(x_1, x_2) \subseteq (a, b)$.

By the MVT, there exists $c \in (x_1, x_2)$ such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

But $f'(c) = 0$ by hypothesis, so $f(x_2) - f(x_1) = 0$, that is, $f(x_1) = f(x_2)$. ■

Theorem 5.7 (Zero Derivative on an Open Interval). *Let f be defined on an open interval (a, b). If $f'(x) = 0$ for every $x \in (a, b)$, then f is constant on (a, b).*

Remark (Pause and Try). The proof is a slight modification of the proof of Theorem 5.6. Try to write it yourself.

Example 5.3 (Proving a Trigonometric Identity). Prove that there exists a constant C such that

$$\arctan\sqrt{\frac{1-x}{1+x}} = C - \frac{1}{2}\arcsin x,$$

find the value of C, and determine for which values of x the identity holds.

Solution. In principle, this is a problem in trigonometry that could be solved with identities. But calculus makes it faster.

Proving the identity is equivalent to proving that the function

$$F(x) = \arctan\sqrt{\frac{1-x}{1+x}} + \frac{1}{2}\arcsin x$$

is constant. After checking the domain (which turns out to be $(-1, 1]$), we compute the derivative. The calculation requires some work, but after simplification, one obtains $F'(x) = 0$ for all $x \in (-1, 1)$.

By Theorem 5.6 (noting that F is continuous on $(-1, 1]$ and differentiable on $(-1, 1)$), the function F is constant on $(-1, 1]$.

To find C, evaluate at $x = 0$:

$$C = F(0) = \arctan(1) + \frac{1}{2}\arcsin(0) = \frac{\pi}{4}.$$

The identity holds for all $x \in (-1, 1]$.

5.10 Antiderivatives and Integration

This is our second application of the Mean Value Theorem. It explains why integration is possible—more precisely, why adding a constant "+C" to one antiderivative produces all *antiderivatives.*

Example 5.4 (Finding All Functions with a Given Derivative). Find all functions f whose derivative is x^2. By trial and error, $\frac{1}{3}x^3$ works. Adding any constant still works: $\frac{1}{3}x^3 + C$ is also a solution. But could there be other solutions that are **not** of this form?

The answer is no, and we must prove it. The proof is short, now that we have the Mean Value Theorem.

Corollary 5.8 (Functions with the Same Derivative). *Let f and g be defined on an open interval (a, b). If $f'(x) = g'(x)$ for all $x \in (a, b)$, then there exists a constant C such that $f(x) = g(x) + C$ for all $x \in (a, b)$.*

Proof. Define $h(x) = f(x) - g(x)$. Then $h'(x) = f'(x) - g'(x) = 0$ for all $x \in (a, b)$. By Theorem 5.7, h is constant, so there exists $C \in \mathbb{R}$ with $h(x) = C$ for all x. That is, $f(x) = g(x) + C$. ■

Remark. Returning to Example 5.4: if $f'(x) = x^2$, then $f'(x) = g'(x)$ where $g(x) = \frac{1}{3}x^3$. By Corollary 5.8, there exists a constant C such that $f(x) = \frac{1}{3}x^3 + C$. These are **all** the solutions.

*This bears repeating, because it is very important. Without the Mean Value Theorem, we could only say that any function of the form $\frac{1}{3}x^3 + C$ is a solution. Thanks to the Mean Value Theorem, we can say these are **all** the solutions. This idea is at the core of every integration method: find one antiderivative, add "+C," and you have found them all. The MVT is therefore at the foundation of all integration theorems.*

5.11 Monotonicity and Derivatives

This is our third application of the Mean Value Theorem: the relationship between the sign of the derivative and whether a function is increasing or decreasing.

Remark (Warning). Saying that a function is increasing is **not** the same thing as saying it has positive derivative. These two concepts are related, but they are **not the same**. A function can be increasing everywhere yet have derivative zero at a point, or have a corner where the derivative does not exist, or even be discontinuous at infinitely many points. Positive derivative is not the *definition* of increasing.

Definition 5.4 (Increasing and Decreasing Functions). Let f be a function defined on an interval I.

(i) f is *increasing* on I when, for every $x_1, x_2 \in I$,

$$x_1 < x_2 \implies f(x_1) < f(x_2).$$

(ii) f is *non-decreasing* on I when, for every $x_1, x_2 \in I$,

$$x_1 < x_2 \implies f(x_1) \leq f(x_2).$$

Analogous definitions hold for *decreasing* and *non-increasing* (reverse the second inequality).

Remark. This terminology is not universal. Some books use "strictly increasing" for what we call "increasing," and "increasing" for what we call "non-decreasing." When reading other sources or working with others, make sure you agree on the definitions.

Theorem 5.9 (Positive Derivative Implies Increasing — Open Interval). *Let f be defined on an open interval (a, b). If $f'(x) > 0$ for every $x \in (a, b)$, then f is increasing on (a, b).*

Theorem 5.10 (Positive Derivative Implies Increasing — Closed Interval). *Let f be defined on a closed, bounded interval $[a, b]$. If f is continuous on $[a, b]$ and $f'(x) > 0$ for every $x \in (a, b)$, then f is increasing on $[a, b]$.*

Remark (Pause and Try). The proofs of Theorems 5.9 and 5.10 are direct applications of the Mean Value Theorem, following the same pattern as the proof of Theorem 5.6. Fix $x_1 < x_2$ in the interval, apply the MVT on $[x_1, x_2]$, and use $f'(c) > 0$ to conclude $f(x_2) - f(x_1) > 0$. Try writing them out.

Here is a summary of the results linking the sign of the derivative to monotonicity on an open interval (a, b):

Derivative condition	**Conclusion**
$f'(x) = 0$ for all $x \in (a, b)$	f is constant on (a, b)
$f'(x) > 0$ for all $x \in (a, b)$	f is increasing on (a, b)
$f'(x) < 0$ for all $x \in (a, b)$	f is decreasing on (a, b)

At an isolated point where the derivative is zero or does not exist, the function could be increasing, decreasing, or have a local extremum. The derivative at a single point does not determine the local behaviour; one must examine the sign of the derivative on intervals.

5.12 Intervals of Monotonicity

Let us illustrate, with an example, how to find the intervals where a function is increasing or decreasing using the theorems of the previous section.

Example 5.5 (Monotonicity of a Polynomial). Find the intervals on which $f(x) = 8x^5 + 5x^4 - 20x^3$ is increasing or decreasing.

Remark (Pause and Try). Pause and try to solve this problem before reading the solution.

Step 1: Compute and factor the derivative.

$$f'(x) = 40x^4 + 20x^3 - 60x^2 = 20x^2(2x + 3)(x - 1).$$

Step 2: Find the critical points. Setting $f'(x) = 0$ gives $x = 0$, $x = -\frac{3}{2}$, and $x = 1$.

Step 3: Determine the sign of f' on each sub-interval. Since f' is a product of three factors $20x^2$, $(2x + 3)$, and $(x - 1)$, we analyse the sign of each:

	$\left(-\infty, -\frac{3}{2}\right)$	$-\frac{3}{2}$	$\left(-\frac{3}{2}, 0\right)$	0	$(0, 1)$	1	$(1, \infty)$
$20x^2$	$+$	$+$	$+$	0	$+$	$+$	$+$
$2x + 3$	$-$	0	$+$	$+$	$+$	$+$	$+$
$x - 1$	$-$	$-$	$-$	$-$	$-$	0	$+$
$f'(x)$	$+$	0	$-$	0	$-$	0	$+$

Step 4: Conclude monotonicity.

- On $\left(-\infty, -\frac{3}{2}\right)$, $f'(x) > 0$, so f is increasing.
- On $\left(-\frac{3}{2}, 0\right)$ and $(0, 1)$, $f'(x) < 0$, so f is decreasing on each.
- On $(1, \infty)$, $f'(x) > 0$, so f is increasing.

Since f is continuous at the critical points, we may include them in the intervals of monotonicity (by Theorem 5.10 and its decreasing analogue):

- f is **increasing** on $\left(-\infty, -\frac{3}{2}\right]$.
- f is **decreasing** on $\left[-\frac{3}{2}, 1\right]$ (since $f' < 0$ on $\left(-\frac{3}{2}, 0\right) \cup (0, 1)$ and $f'(0) = 0$, the function is decreasing on the entire interval $\left[-\frac{3}{2}, 1\right]$).
- f is **increasing** on $[1, \infty)$.

Step 5: Classify the critical points.

At $x = -\frac{3}{2}$: the derivative is zero, the function is increasing to the left and decreasing to the right, so f has a **local maximum**.

At $x = 1$: the derivative is zero, the function is decreasing to the left and increasing to the right, so f has a **local minimum**.

At $x = 0$: the derivative is zero, but the function is decreasing on both sides. This is **not** a local extremum; it is an *inflection point* (with a horizontal tangent line).

Remark. We include $x = 1$ in both the decreasing interval $\left[-\frac{3}{2}, 1\right]$ and the increasing interval $[1, \infty)$. This is not a contradiction: a function is increasing or decreasing on an *interval*, not at a *point*.

Chapter 6

Further Applications of Derivatives and Limits

6.1 A Related Rates Problem

A related rates problem is an application of calculus to a real-life situation where there are multiple quantities that depend on each other, and we need to find a relation between their derivatives. Here is our first example.

Example 6.1 *(Plane and Radar Station).* A plane flying horizontally at an altitude of 10 km passes directly above a radar station. A bit later, the radar station measures that the distance between the plane and the station is 20 km and is increasing at a rate of 1000 km/h. What is the speed of the plane?

Remark (Pause and Try). Pause and try to solve the problem. Even if you cannot fully solve it, spending time understanding the question, getting started, and sketching a picture will make the following explanation much more useful.

Step 1: Modelling. We transform the word problem into a calculus question. Begin by drawing a picture. The radar station is on the ground, and the plane flies horizontally above it.

Introduce variables:

- Let $h = 10\,\text{km}$ be the altitude of the plane (a **constant**).
- Let z be the distance between the radar station and the plane (this is what the radar measures).
- Let x be the horizontal position of the plane.

Both x and z depend on time t. We **want** $\dfrac{dx}{dt}$ at the instant when $z = 20\,\text{km}$ and $\dfrac{dz}{dt} = 1000\,\text{km/h}$.

Since the radar station, the point directly below the plane, and the plane itself form a right triangle, the Pythagorean theorem gives the relation

$$z^2 = x^2 + h^2.$$

Step 2: Solving. We need a relation between $\frac{dx}{dt}$ and $\frac{dz}{dt}$. Differentiate both sides of the relation with respect to t (using the chain rule):

$$2z\,\frac{dz}{dt} = 2x\,\frac{dx}{dt}.$$

Solving for the desired quantity:

$$\frac{dx}{dt} = \frac{z}{x}\,\frac{dz}{dt}.$$

We know $z = 20$ and $\frac{dz}{dt} = 1000$, and we can solve for x from the original equation:

$$x = \sqrt{z^2 - h^2} = \sqrt{20^2 - 10^2} = \sqrt{300} = 10\sqrt{3}\,\text{km}.$$

Substituting:

$$\frac{dx}{dt} = \frac{20}{10\sqrt{3}} \cdot 1000 = \frac{2000}{\sqrt{3}}\,\text{km/h}.$$

The speed of the plane is $\frac{2000}{\sqrt{3}}$ km/h.

6.2 Related Rates: Shadow Problem

Here is a second example of a related rates problem. The modelling step is the key: we must identify what is constant, what depends on time, find an equation relating the quantities, and then differentiate implicitly.

Example 6.2 (Prisoner's Shadow). A prisoner runs in a straight line towards the wall of a prison. A spotlight on the ground, 30 m from the wall, points at the prisoner. The prisoner is 1.8 m tall. How fast is the height of the shadow on the wall changing when the prisoner is 20 m from the wall and running at 8 m/s?

Remark (Pause and Try). Pause and try to solve this problem. Even if you cannot fully solve it, drawing a picture and understanding the statement will make the explanation much more useful.

Step 1: Modelling. Draw a picture with the spotlight on the ground, the wall, and the prisoner between them. Introduce variables:

- $a = 30\,\text{m}$: the distance from the spotlight to the wall (**constant**).
- $h = 1.8\,\text{m}$: the height of the prisoner (**constant**).
- x: the distance from the prisoner to the spotlight (depends on time).
- y: the height of the shadow on the wall (depends on time).

We **want** $\dfrac{dy}{dt}$ at the instant when $x = 10\,\text{m}$ (since the prisoner is $20\,\text{m}$ from the wall and $a = 30$) and $\dfrac{dx}{dt} = 8\,\text{m/s}$.

By similar triangles (the triangle with height h and base x, and the triangle with height y and base a):

$$\frac{h}{x} = \frac{y}{a}.$$

Step 2: Solving. Differentiate both sides implicitly with respect to t. Since h and a are constants:

$$h \cdot \left(-\frac{1}{x^2}\right)\frac{dx}{dt} = \frac{1}{a}\,\frac{dy}{dt}.$$

Solving:

$$\frac{dy}{dt} = -\frac{ah}{x^2}\,\frac{dx}{dt}.$$

Substituting:

$$\frac{dy}{dt} = -\frac{(30)(1.8)}{10^2} \cdot 8 = -4.32\,\text{m/s}.$$

The rate of change is negative, confirming that the shadow height is *decreasing* as the prisoner approaches the wall.

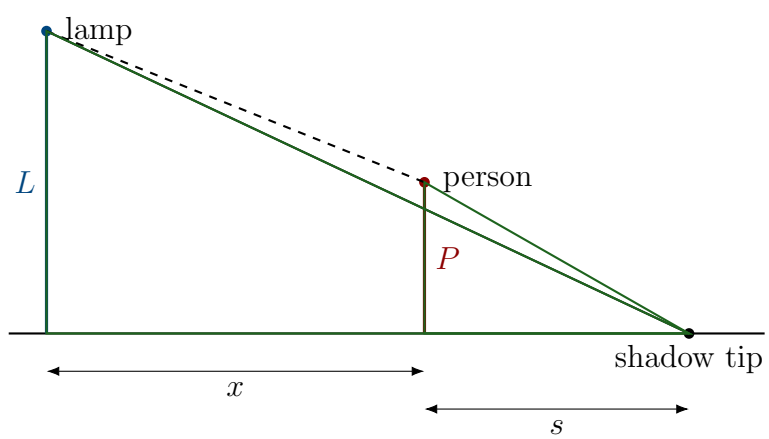


Figure 6.1: Shadow geometry for a classic related-rates setup.

6.3 Optimization: Minimizing Surface Area

An optimization problem is an application of calculus to a physical situation where we want to make a certain quantity as large or as small as possible.

Example 6.3 (Open Box with Fixed Volume). We want to build an open box (four side walls and a bottom, no top) with a square base. The volume must be $500\,\text{cm}^3$. What dimensions minimize the total surface area (and hence the material cost)?

Remark (Pause and Try). Pause and try to solve the problem before continuing. Even a partial attempt will make the explanation much more useful.

Step 1: Modelling. Let x be the side length of the square base and y the height. The total surface area (bottom plus four sides) is

$$S = x^2 + 4xy.$$

The volume constraint is $x^2y = 500$, so $y = \dfrac{500}{x^2}$. Substituting:

$$S(x) = x^2 + \frac{2000}{x}, \qquad x > 0.$$

Step 2: Solving. The function S is differentiable on $(0, \infty)$. Compute the derivative:

$$S'(x) = 2x - \frac{2000}{x^2} = \frac{2(x^3 - 1000)}{x^2}.$$

Setting $S'(x) = 0$ gives $x^3 = 1000$, i.e. $x = 10$. This is the only critical point.

Remark (Warning). Many students stop here and declare the answer, but we are looking for the **minimum**, not just a critical point. We must justify that the minimum actually occurs at $x = 10$.

When $x < 10$, we have $S'(x) < 0$ (decreasing), and when $x > 10$, we have $S'(x) > 0$ (increasing). Therefore S decreases all the way to $x = 10$ and increases afterwards, so the global minimum occurs at $x = 10$.

The optimal dimensions are $x = 10\,\text{cm}$ and $y = \dfrac{500}{100} = 5\,\text{cm}$.

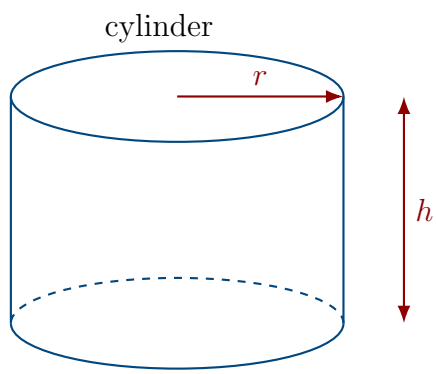


Figure 6.2: A cylinder labeled with radius r and height h.

6.4 Optimization with Endpoints

In this example, the domain is a closed and bounded interval, so we can appeal to the Extreme Value Theorem.

Example 6.4 (Rowing and Running). A woman at point A on one side of a straight river, 3 km wide, wants to reach point B, 8 km downstream on the opposite side. She can row at 6 km/h and run at 8 km/h. Where should she land to reach B as soon as possible?

Remark (Pause and Try). Pause and try to set up the problem: draw a picture, introduce a variable for the landing point, and write a formula for the total travel time.

Step 1: Modelling. Let C be the point where she lands, and let x be the distance from the point directly across from A to C. Then the rowing distance is $\sqrt{x^2+9}$ and the running distance is $8-x$. The total time is

$$T(x) = \frac{\sqrt{x^2+9}}{6} + \frac{8-x}{8}, \qquad x \in [0,8].$$

Step 2: Solving. The function T is continuous on the closed, bounded interval $[0,8]$. By the Extreme Value Theorem, T has a minimum, and this minimum occurs either at a critical point or at an endpoint.

Remark. We **must** include endpoints. If she could row much faster than she could run, the optimal path would be directly from A to B ($x=8$). If she could run much faster, going straight across ($x=0$) would be optimal. So the minimum *could* be an endpoint, and we cannot dismiss them.

Computing the derivative and setting it to zero, after some algebra we find one critical point inside the domain:

$$x = \frac{9}{\sqrt{7}}\,\text{km}.$$

Compare the three candidate values:

x	0	$\frac{9}{\sqrt{7}}$	8
$T(x)$	1.5 h	$1+\frac{\sqrt{7}}{8} \approx 1.33$ h	$\frac{\sqrt{73}}{6} \approx 1.42$ h

The smallest value is at the critical point. She should land $\frac{9}{\sqrt{7}}$ km downstream.

6.5 Indeterminate Forms

Before introducing L'Hôpital's Rule, we must understand what an indeterminate form is—and, just as importantly, what it is ***not****.*

Consider the limit $\lim_{x\to 0} \frac{f(x)}{g(x)}$ where both f and g are continuous and $f(0) = g(0) = 0$. Substituting gives $\frac{0}{0}$, which is not a number. We have an *indeterminate form.*

Definition 6.1 (Indeterminate Form). We say that a limit $\lim_{x\to a} \dfrac{f(x)}{g(x)}$ has the indeterminate form $\frac{0}{0}$ when $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$. This means that, **based on this information alone**, we cannot determine the value of the limit.

To see why, consider these three examples, all of type $\frac{0}{0}$ as $x \to 0$:

$$\lim_{x\to 0} \frac{2x}{x} = 2, \qquad \lim_{x\to 0} \frac{x^2}{x} = 0, \qquad \lim_{x\to 0} \frac{x}{x^3} = \infty.$$

The same hypotheses yield different answers each time.

Remark (Warning). Three important clarifications:

(i) We are talking about **limits**. We are never actually dividing zero by zero. For each value of x, we have something *close to* zero divided by something *close to* zero.

(ii) Indeterminate form does **not** mean the limit is undefined or does not exist. It only means we do not yet know the answer and must do more work.

(iii) Indeterminate forms arise because of a **tension** between two parts of the function. In $\frac{0}{0}$, the numerator "thinks" the quotient should be zero, while the denominator "thinks" it should be $\pm\infty$. The outcome depends on which approaches zero faster.

The most common indeterminate forms are:

$$\frac{0}{0}, \quad \frac{\pm\infty}{\pm\infty}, \quad 0 \cdot \infty, \quad \infty - \infty, \quad 0^0, \quad \infty^0, \quad 1^{\pm\infty}.$$

Remark. Why is $\dfrac{1}{0}$ not on the list? Because we *can* draw a conclusion: if $\lim f(x) = 1$ and $\lim g(x) = 0$, then $\lim \left|\frac{f(x)}{g(x)}\right| = \infty$. The exact answer ($+\infty$ or $-\infty$) depends on the **sign** of the denominator near the point, but we are not stuck—we can say something.

Example 6.5. Compute $\lim_{x\to 3^+} \dfrac{1}{(x-3)(2-x)}$.

This is of the form $\frac{1}{0}$. As $x \to 3^+$: $(x-3) > 0$ and $(2-x) < 0$, so the denominator is negative. Therefore the limit is $-\infty$.

6.6 L'Hôpital's Rule

L'Hôpital's Rule is a powerful tool for computing limits that produce indeterminate forms of type $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$. It tells us that, under certain conditions, we may replace the numerator and denominator by their derivatives.

Theorem 6.1 (L'Hôpital's Rule). *Let f and g be functions, and let a be a real number (or $\pm\infty$). Suppose:*

(i) The limit $\lim_{x\to a} \frac{f(x)}{g(x)}$ is an indeterminate form of type $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$.

(ii) f and g are differentiable for x near a (but not necessarily at a).

(iii) $g(x) \neq 0$ and $g'(x) \neq 0$ for x near a (but not necessarily at a).

(iv) $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists, or equals $+\infty$, or equals $-\infty$.

Then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}.$$

The theorem also holds for one-sided limits.

Remark. Condition (iv) is the one we often forget, but it is essential. If $\lim \frac{f'}{g'}$ does not exist (and is not $\pm\infty$), then L'Hôpital's Rule gives **no information** about $\lim \frac{f}{g}$. The original limit may still exist.

Remark. The proof relies on a variant of the Mean Value Theorem and is quite long (there are separate cases for $\frac{0}{0}$ and $\frac{\infty}{\infty}$, and for each type of limit). It appears in every standard calculus textbook.

6.7 L'Hôpital's Rule Examples

Let us work through several examples, from beginning to end, to see how L'Hôpital's Rule is used in practice.

Example 6.6. Compute $\lim_{x\to\infty} \frac{x}{\ln x}$.

As $x \to \infty$: numerator $\to \infty$, denominator $\to \infty$. This is $\frac{\infty}{\infty}$, so L'Hôpital's Rule applies. Differentiating:

$$\lim_{x\to\infty} \frac{x}{\ln x} \overset{\text{L'H}}{=} \lim_{x\to\infty} \frac{1}{1/x} = \lim_{x\to\infty} x = \infty.$$

The second limit is ∞, so the use of L'Hôpital was legal, and the original limit is ∞.

Example 6.7. Compute $\lim_{x\to 0} \frac{\cos x - \cos(2x)}{xe^x - x}$.

As $x \to 0$: numerator $\to 1 - 1 = 0$, denominator $\to 0 - 0 = 0$. This is $\frac{0}{0}$. Apply L'Hôpital:

$$\overset{\text{L'H}}{=} \lim_{x\to 0} \frac{-\sin x + 2\sin(2x)}{e^x + xe^x - 1}.$$

As $x \to 0$: numerator $\to 0$, denominator $\to 1 + 0 - 1 = 0$. Still $\frac{0}{0}$. Apply L'Hôpital again:

$$\overset{\text{L'H}}{=} \lim_{x\to 0} \frac{-\cos x + 4\cos(2x)}{2e^x + xe^x}.$$

Now as $x \to 0$: numerator $\to -1 + 4 = 3$, denominator $\to 2 + 0 = 2$. The limit is $\frac{3}{2}$, which is a number; therefore both applications of L'Hôpital were legal, and the original limit is $\frac{3}{2}$.

Example 6.8. Compute $\lim_{x\to 1} \frac{x^3 - 2x + 1}{x^2 + 3x + 2}$.

As $x \to 1$: numerator $\to 1 - 2 + 1 = 0$, denominator $\to 1 + 3 + 2 = 6$. This is $\frac{0}{6}$, which is **not** an indeterminate form. We simply evaluate:

$$\lim_{x\to 1} \frac{x^3 - 2x + 1}{x^2 + 3x + 2} = \frac{0}{6} = 0.$$

Remark (Warning). In Example 6.8 we are **not allowed** to use L'Hôpital's Rule because there is no indeterminate form. If you try to use it anyway, you will get a wrong answer.

6.8 Failure of L'Hôpital's Rule

L'Hôpital's Rule is very powerful, but it cannot always be used. Here is an example where it fails silently—if we do not notice, we get a wrong answer.

Example 6.9. Compute $\lim_{x\to\infty} \frac{x + \sin x}{2x + \cos x}$.

Checking the form. As $x \to \infty$, $\sin x$ oscillates between -1 and 1, but $x + \sin x \geq x - 1 \to \infty$, so the numerator has limit ∞. By the same reasoning, the denominator has limit ∞. We have $\frac{\infty}{\infty}$.

Attempting L'Hôpital. Differentiating:

$$\lim_{x\to\infty} \frac{1 + \cos x}{2 - \sin x}.$$

The function $\frac{1 + \cos x}{2 - \sin x}$ is *periodic* with period 2π. A periodic, non-constant function never has a limit at infinity. Therefore this limit **does not exist** (and is not $\pm\infty$).

By Condition (iv) of Theorem 6.1, L'Hôpital's Rule does not apply. It gives no information.

So how do we compute the original limit? Without L'Hôpital, we can use a direct argument.

Divide numerator and denominator by x:

$$\frac{x + \sin x}{2x + \cos x} = \frac{1 + \frac{\sin x}{x}}{2 + \frac{\cos x}{x}}.$$

By the Squeeze Theorem, $\dfrac{\sin x}{x} \to 0$ and $\dfrac{\cos x}{x} \to 0$ as $x \to \infty$ (since $\sin x$ and $\cos x$ are bounded). Therefore

$$\lim_{x\to\infty} \frac{x + \sin x}{2x + \cos x} = \frac{1+0}{2+0} = \frac{1}{2}.$$

Remark (Warning). The limit *does* exist and equals $\frac{1}{2}$, even though L'Hôpital's Rule could not detect it. This situation is uncommon, but it is important to know that it can happen.

6.9 The Zero-Times-Infinity Form

L'Hôpital's Rule applies only to quotients. When we encounter $0 \cdot \infty$, we must first rewrite the product as a quotient.

Example 6.10. Compute $\lim\limits_{x\to\infty} x\big(1 - e^{2/x}\big)$.

Remark (Pause and Try). Pause and try to compute this limit. At least, try to understand what the difficulty is.

As $x \to \infty$: the first factor has limit ∞, and $e^{2/x} \to e^0 = 1$, so the second factor has limit 0. We have $\infty \cdot 0$, an indeterminate form.

Do not be tempted to say this is 0.

Rewrite x as $\dfrac{1}{1/x}$:

$$x\big(1 - e^{2/x}\big) = \frac{1 - e^{2/x}}{1/x}.$$

Now as $x \to \infty$: numerator $\to 0$ and denominator $\to 0$. This is $\dfrac{0}{0}$, and we may apply L'Hôpital:

$$\overset{\text{L'H}}{=} \lim_{x\to\infty} \frac{-e^{2/x} \cdot (-2/x^2)}{-1/x^2} = \lim_{x\to\infty} \frac{2e^{2/x} \cdot (1/x^2)}{1/x^2} = \lim_{x\to\infty} 2e^{2/x} \cdot (-1) = -2.$$

Wait—let us be more careful. Differentiating the numerator:

$$\frac{d}{dx}\big(1 - e^{2/x}\big) = -e^{2/x} \cdot \left(-\frac{2}{x^2}\right) = \frac{2e^{2/x}}{x^2}.$$

Differentiating the denominator:

$$\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}.$$

Therefore:

$$\lim_{x\to\infty} \frac{2e^{2/x}/x^2}{-1/x^2} = \lim_{x\to\infty} (-2e^{2/x}) = -2.$$

Since -2 is a number, the use of L'Hôpital was legal, and the original limit is -2.

Remark. This trick always works for $0 \cdot \infty$: if $\lim f = 0$ and $\lim g = \infty$, rewrite $f \cdot g$ as $\dfrac{f}{1/g}$ (giving $\frac{0}{0}$) or as $\dfrac{g}{1/f}$ (giving $\frac{\infty}{\infty}$).

6.10 The Infinity-Minus-Infinity Form

There is no single trick that works for all limits of the form $\infty - \infty$. The strategy is to manipulate the expression algebraically until it becomes a product or quotient to which we can apply other tools.

Example 6.11. Compute $\lim_{x\to\infty}\left(\sqrt{x^2-x}-x\right)$.

As $x \to \infty$, the square root behaves like x (since the leading term inside is x^2), so both $\sqrt{x^2-x}$ and x go to ∞. We have $\infty - \infty$.

Do not be tempted to say this is 0.

Method 1 (factoring, then L'Hôpital). Factor x^2 inside the square root:

$$\sqrt{x^2-x}-x = x\sqrt{1-\frac{1}{x}}-x = x\left(\sqrt{1-\frac{1}{x}}-1\right).$$

Now $\lim x = \infty$ and $\lim\left(\sqrt{1-1/x}-1\right) = 0$, so we have $\infty \cdot 0$. Convert to a quotient:

$$\frac{\sqrt{1-1/x}-1}{1/x}.$$

As $x \to \infty$, both numerator and denominator approach 0. Apply L'Hôpital:

$$\overset{\text{L'H}}{=} \lim_{x\to\infty} \frac{\dfrac{1}{2\sqrt{1-1/x}}\cdot\dfrac{1}{x^2}}{-\dfrac{1}{x^2}} = \lim_{x\to\infty}\frac{-1}{2\sqrt{1-1/x}} = -\frac{1}{2}.$$

Method 2 (conjugate). Multiply and divide by the conjugate $\sqrt{x^2-x}+x$:

$$\sqrt{x^2-x}-x = \frac{(\sqrt{x^2-x})^2-x^2}{\sqrt{x^2-x}+x} = \frac{x^2-x-x^2}{\sqrt{x^2-x}+x} = \frac{-x}{\sqrt{x^2-x}+x}.$$

Factor x from the denominator:

$$\frac{-x}{x\left(\sqrt{1-1/x}+1\right)} = \frac{-1}{\sqrt{1-1/x}+1} \to \frac{-1}{1+1} = -\frac{1}{2}.$$

Both methods give $-\dfrac{1}{2}$.

6.11 The One-to-Infinity Form

Of all indeterminate forms, 1^∞ is probably the most controversial—the one students argue about the most. Let us explain why it really is indeterminate, and why our intuition may be off.

When we write 1^∞, we do **not** mean "the number 1 raised to the power infinity." We mean numbers *close to* 1 raised to very large exponents.

Proposition 6.2. *1^∞ is an indeterminate form: limits of the type $\lim f(x)^{g(x)}$ with $\lim f = 1$ and $\lim g = \infty$ can take any positive value.*

Proof. Let c be any nonzero real number, and consider

$$\lim_{x\to 0^+} (e^x)^{c/x}.$$

Here $\lim_{x\to 0^+} e^x = 1$ and $\lim_{x\to 0^+} c/x = \pm\infty$ (depending on the sign of c). So this is of type $1^{\pm\infty}$. But

$$(e^x)^{c/x} = e^{xc/x} = e^c,$$

which is constant, so the limit is e^c. By choosing different values of c, we obtain any positive number. ■

Why is it hard to accept that 1^∞ is indeterminate? If the base were the ***constant*** *1, then $1^x = 1$ for all x, and the limit would certainly be 1. But the base is not constant—it is a function approaching 1. If the base is slightly greater than 1, the exponent pushes the result towards ∞; if slightly less, towards 0. The base wants the answer to be 1; the exponent wants it to be ∞ or 0. That tension is what makes it indeterminate.*

6.12 Exponential Indeterminate Forms

The exponential indeterminate forms (0^0, ∞^0, 1^∞, $1^{-\infty}$) all require the same trick: take a logarithm to convert the power into a product or quotient, compute the limit of the logarithm, then exponentiate.

Example 6.12. Compute $\lim_{x\to 0^+} (1-x)^{1/x}$.

As $x \to 0$: the base $(1-x) \to 1$ and the exponent $1/x \to \infty$. This is 1^∞.

Do not say the answer is 1.

Step 1: Compute the limit of the logarithm. Let $F(x) = (1-x)^{1/x}$. Then

$$\ln F(x) = \frac{1}{x}\ln(1-x) = \frac{\ln(1-x)}{x}.$$

As $x \to 0$: numerator $\to \ln 1 = 0$ and denominator $\to 0$. This is $\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x\to 0^+} \frac{\ln(1-x)}{x} \overset{\text{L'H}}{=} \lim_{x\to 0^+} \frac{-1/(1-x)}{1} = \lim_{x\to 0^+} \frac{-1}{1-x} = -1.$$

Step 2: Recover the original limit. Since $\ln F(x) \to -1$ and the exponential function is continuous:

$$\lim_{x\to 0^+} F(x) = \lim_{x\to 0^+} e^{\ln F(x)} = e^{-1} = \frac{1}{e}.$$

Remark. The same strategy works for all exponential indeterminate forms:

(i) Set $F(x) = f(x)^{g(x)}$ and compute $\ln F(x) = g(x) \ln f(x)$.

(ii) Compute $\lim \ln F(x)$ (this will be a product or quotient, to which L'Hôpital may apply).

(iii) Conclude $\lim F(x) = e^{\lim \ln F(x)}$.

6.13 Concavity and Inflection Points

The concavity of a function refers to the shape of its graph. There are several equivalent ways to define it for nice (differentiable) functions. We will discuss three approaches, all of which agree when the function is differentiable.

As motivation, consider the family of graphs $y = x^a$ for various values of a. When $a > 1$ the curves look like deformed parabolas, all with the same "shape." When $a < 1$ (and $a > 0$) they have a different shape. We say these two families have different concavity.

We call curves of the first type *concave up* and curves of the second type *concave down*.

Remark (Warning). Some older textbooks (and the French mathematical tradition) use *concave* and *convex* instead of *concave up* and *concave down*, but which is which may depend on the author. We prefer *concave up* and *concave down*.

Three candidate definitions for concave up:

(1) **Secant segments above the graph.** For any two points on the graph, the segment of the secant line between them lies above the graph.

(2) **Tangent lines below the graph.** At every point, the tangent line lies below the graph.

(3) **Slopes increasing.** The derivative f' is an increasing function.

(For concave down, reverse "above" and "below," and replace "increasing" by "decreasing.")

For differentiable functions, all three are **equivalent**. The proof relies on the Mean Value Theorem. Definition (3) is the simplest and is the one used by most calculus textbooks.

Definition 6.2 (Concavity). Let f be differentiable on an interval I.

(i) f is *concave up* on I when f' is increasing on I.

(ii) f is *concave down* on I when f' is decreasing on I.

Definition 6.3 (Inflection Point). An interior point c of the domain of f is called an *inflection point* if f changes concavity at c: there exists an interval centred at c such that f is concave up on one side and concave down on the other.

Inflection points are to concavity what local extrema are to monotonicity. A local extremum is where the function changes from increasing to decreasing (or vice versa); an inflection point is where it changes from concave up to concave down (or vice versa).

At an inflection point, the tangent line *crosses* the graph—it lies above the graph on one side and below on the other.

Theorem 6.3 (Second-Derivative Test for Concavity). *Let f be twice differentiable on an open interval I.*

(i) If $f''(x) > 0$ for all $x \in I$, then f is concave up on I.

(ii) If $f''(x) < 0$ for all $x \in I$, then f is concave down on I.

Theorem 6.4 (Necessary Condition for Inflection Points). *If f is twice differentiable on an interval and has an inflection point at an interior point c, then $f''(c) = 0$.*

Remark (Warning). The converse of Theorem 6.4 is **not** true. $f''(c) = 0$ does **not** imply that c is an inflection point. These are merely **candidate** points, just as $f'(c) = 0$ gives candidates for local extrema. More generally, $f''(c) = 0$ or $f''(c)$ does not exist gives the candidates for inflection points.

6.14 Curve Sketching with Derivatives

We now illustrate, with a detailed example, how to use the first and second derivatives to study the monotonicity, concavity, and shape of a function.

Before we begin, recall the tools at our disposal. On an open interval:

- $f' > 0 \implies f$ is increasing; $\quad f' < 0 \implies f$ is decreasing.
- $f'' > 0 \implies f$ is concave up; $\quad f'' < 0 \implies f$ is concave down.

On a closed interval, we require the derivative condition only on the interior, plus continuity at the endpoints. Candidates for local extrema satisfy $f' = 0$ or f' DNE; candidates for inflection points satisfy $f'' = 0$ or f'' DNE.

Example 6.13 (Full Curve Sketch). Let $f(x) = x^6(x+3)^3$. Find the intervals of increase/decrease, local extrema, intervals of concavity, inflection points, and sketch the graph.

First derivative. After computation and factoring:

$$f'(x) = 3x^5(x+3)^2(3x+6) = 9x^5(x+3)^2(x+2).$$

The zeros of f' are $x = 0$, $x = -3$, and $x = -2$. We determine the sign of f' on each subinterval by tracking which factors change sign:

- For $x > 0$: $f'(x) > 0$.
- At $x = 0$: the factor x^5 changes sign, so f' changes sign.

- At $x = -2$: the factor $(x + 2)$ changes sign.
- At $x = -3$: the factor $(x + 3)^2$ does **not** change sign (even exponent).

Sign chart for f':

	$(-\infty, -3)$	-3	$(-3, -2)$	-2	$(-2, 0)$	0	$(0, \infty)$
f'	$+$	0	$+$	0	$-$	0	$+$

Therefore f is increasing on $(-\infty, -2]$ (the sign at -3 does not change, so we join the intervals), decreasing on $[-2, 0]$, and increasing on $[0, \infty)$.

At $x = -2$ (increasing $\to$ decreasing): **local maximum**.
At $x = 0$ (decreasing $\to$ increasing): **local minimum**.

Second derivative. After computation and factoring:

$$f''(x) = \cdots = 9x^4(x + 3)(2x + 3)(2x + 5) \cdot (\text{positive constant}).$$

The zeros of f'' are at $x = 0$, $x = -3$, $x = -\frac{3}{2}$, and $x = -\frac{5}{2}$. Tracking which factors change sign:

- x^4: even power, no sign change at 0.
- $(x + 3)$: changes sign at -3.
- $(2x + 3)$: changes sign at $-\frac{3}{2}$.
- $(2x + 5)$: changes sign at $-\frac{5}{2}$.

Concavity:

- $(-\infty, -3)$: concave down.
- $(-3, -\frac{5}{2})$: concave up.
- $(-\frac{5}{2}, -\frac{3}{2})$: concave down.
- $(-\frac{3}{2}, \infty)$: concave up.

Inflection points occur at $x = -3$, $x = -\frac{5}{2}$, and $x = -\frac{3}{2}$ (where concavity changes).

Special points:

- $x = -3$: inflection point with horizontal tangent ($f'(-3) = 0$), function increasing on both sides.
- $x = -\frac{5}{2}$: inflection point, function increasing with positive slope.
- $x = -2$: local maximum.
- $x = -\frac{3}{2}$: inflection point, function decreasing with negative slope.
- $x = 0$: local minimum.

Using the monotonicity and concavity information together on each subinterval, we sketch the graph, paying attention to the shape near each special point.

6.15 Asymptotes

An asymptote is a line that becomes arbitrarily close to a curve as we move away from the origin in some direction. This single idea unifies vertical, horizontal, and slant asymptotes.

Definition 6.4 (Asymptote — Informal). A line L is an *asymptote* of a curve C if the distance between L and C approaches 0 as we move along C in some direction away from the origin.

Remark (Warning). If you learned in high school that "an asymptote is a line the curve approaches but never touches," the "never touches" part is **incorrect**. A graph may cross its asymptote, even infinitely many times, and the line is still an asymptote as long as the distance tends to 0.

A function may have multiple asymptotes. For instance, a single graph can have one horizontal asymptote as $x \to -\infty$, two vertical asymptotes, and a slant asymptote as $x \to +\infty$.

Vertical Asymptotes

Definition 6.5 (Vertical Asymptote). The line $x = a$ is a *vertical asymptote* of f if at least one of the following holds:

$$\lim_{x\to a^+} f(x) = \pm\infty, \qquad \lim_{x\to a^-} f(x) = \pm\infty.$$

Horizontal Asymptotes

Definition 6.6 (Horizontal Asymptote). The line $y = L$ is a *horizontal asymptote* of f if

$$\lim_{x\to+\infty} f(x) = L \qquad \text{or} \qquad \lim_{x\to-\infty} f(x) = L.$$

Slant (Oblique) Asymptotes

Definition 6.7 (Slant Asymptote). The line $y = mx+b$ is a *slant asymptote* of f if

$$\lim_{x\to+\infty} \big[f(x) - (mx+b)\big] = 0 \qquad \text{or} \qquad \lim_{x\to-\infty} \big[f(x) - (mx+b)\big] = 0.$$

Remark. One may view a horizontal asymptote as a special case of a slant asymptote (with $m = 0$).

6.16 Asymptotes of Rational Functions

We illustrate, with a detailed example, how to find all asymptotes of a rational function.

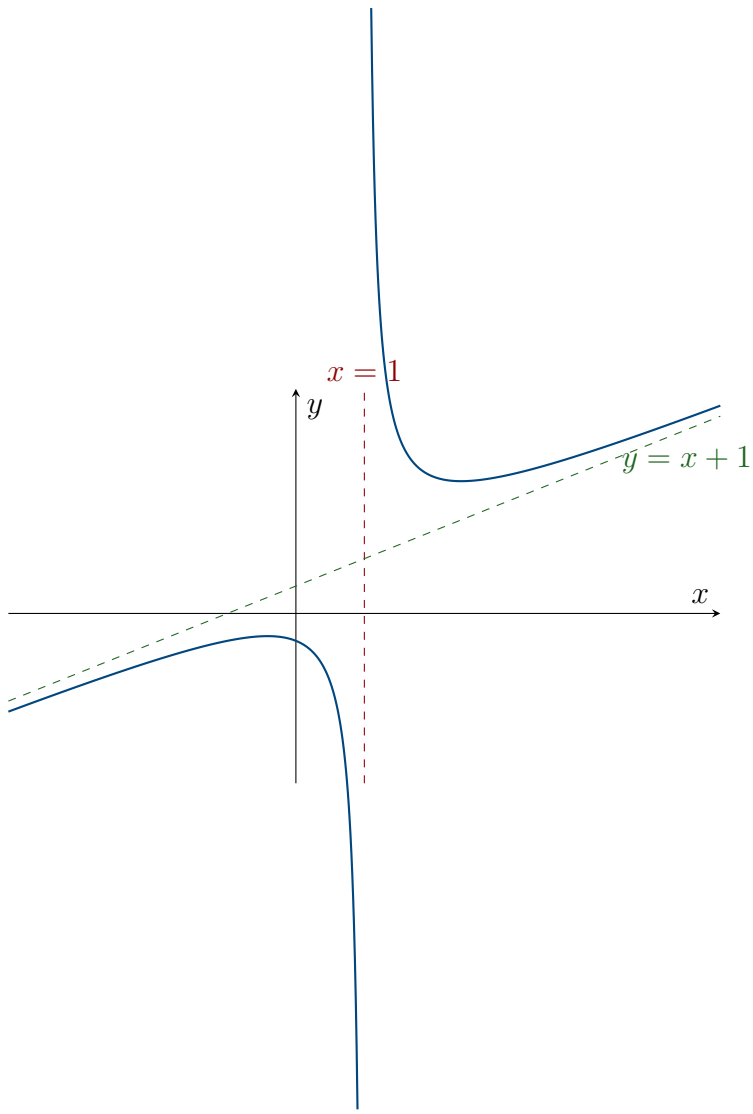


Figure 6.3: A rational function with a vertical asymptote and a slant asymptote.

Example 6.14. Find all asymptotes of

$$f(x) = \frac{x^3 + 1}{x^3 - 2x^2} = \frac{x^3 + 1}{x^2(x - 2)}.$$

Vertical asymptotes. The function is rational, continuous wherever the denominator is nonzero. The denominator vanishes at $x = 0$ and $x = 2$.

At $x = 0$: As $x \to 0$, the numerator approaches 1 (positive) and the factor $(x - 2)$ approaches -2 (negative). The factor x^2 is positive on both sides and vanishes. So the full function is negative near 0:

$$\lim_{x \to 0^+} f(x) = -\infty, \qquad \lim_{x \to 0^-} f(x) = -\infty.$$

Vertical asymptote: $x = 0$.

At $x = 2$: The numerator approaches 9 (positive), x^2 approaches 4 (positive), and $(x - 2)$ changes sign:

$$\lim_{x\to 2^+} f(x) = +\infty, \qquad \lim_{x\to 2^-} f(x) = -\infty.$$

Vertical asymptote: $x = 2$.

Horizontal asymptotes. Compute limits at $\pm\infty$ by factoring the leading term:

$$\frac{x^3+1}{x^3-2x^2} = \frac{x^3(1+1/x^3)}{x^3(1-2/x)} = \frac{1+1/x^3}{1-2/x} \to \frac{1}{1} = 1.$$

The same holds for $x \to -\infty$. Horizontal asymptote: $y = 1$ (on both sides).

This function has **no** slant asymptotes (since horizontal asymptotes exist on both sides).

Remark. The asymptotes alone are not enough to sketch the graph precisely. For a complete sketch, we should also compute the first derivative (for monotonicity) and find intercepts. But whatever the graph looks like, it must respect the asymptotic behaviour found above.

6.17 Slant Asymptote Computation

Not every function with asymptotes has horizontal ones. Here is an example with a slant asymptote that is neither vertical nor horizontal.

Example 6.15. Find all asymptotes of

$$g(x) = x + 2 + \frac{1}{e^{x^2}+1}.$$

Vertical asymptotes. The domain is all of $\mathbb{R}$: the denominator $e^{x^2}+1$ is always positive. Since g is continuous on $\mathbb{R}$, there are **no** vertical asymptotes.

Horizontal asymptotes. As $x \to \pm\infty$, $x^2 \to \infty$, so $e^{x^2} \to \infty$ and $\dfrac{1}{e^{x^2}+1} \to 0$. Therefore

$$\lim_{x\to\pm\infty} g(x) = \lim_{x\to\pm\infty} \big(x+2\big) = \pm\infty.$$

There are **no** horizontal asymptotes.

Slant asymptote. We observed that $g(x) \approx x + 2$ for large $|x|$. Check:

$$g(x) - (x+2) = \frac{1}{e^{x^2}+1} \to 0 \quad \text{as } x \to +\infty \text{ (and as } x \to -\infty).$$

Since this difference tends to 0, the line $y = x + 2$ is a **slant asymptote** on both sides.

Moreover, $\dfrac{1}{e^{x^2}+1} > 0$ always, so the graph lies *slightly above* the line $y = x + 2$.

6.18 Slant Asymptotes via L'Hôpital

This example is significantly harder than the previous ones. The function has ***two different*** *slant asymptotes, one as $x \to +\infty$ and one as $x \to -\infty$. The computation requires L'Hôpital's Rule.*

Example 6.16. Find the slant asymptote of $h(x) = x \arctan(x)$ as $x \to +\infty$.

Conjecture. Since $\lim_{x\to\infty} \arctan(x) = \dfrac{\pi}{2}$, the function $h(x) \approx \frac{\pi}{2}\, x$ for large x. Perhaps $y = \frac{\pi}{2}\, x$ is the asymptote?

Testing the conjecture. Compute:

$$h(x) - \frac{\pi}{2}\, x = x\left(\arctan(x) - \frac{\pi}{2}\right).$$

As $x \to \infty$: the first factor $\to \infty$ and the second $\to 0$. This is $\infty \cdot 0$. Convert to a quotient:

$$\frac{\arctan(x) - \frac{\pi}{2}}{1/x}.$$

Numerator $\to 0$, denominator $\to 0$: we have $\dfrac{0}{0}$. Apply L'Hôpital:

$$\overset{\text{L'H}}{=} \lim_{x\to\infty} \frac{\dfrac{1}{1+x^2}}{-\dfrac{1}{x^2}} = \lim_{x\to\infty} \frac{-x^2}{1+x^2} = -1.$$

So $\lim_{x\to\infty} \left[h(x) - \frac{\pi}{2}\, x\right] = -1 \neq 0$. The line $y = \frac{\pi}{2}\, x$ is **not** an asymptote.

Correcting the conjecture. Since the difference approaches -1, we can write:

$$h(x) - \left(\frac{\pi}{2}\, x - 1\right) \to 0 \quad \text{as } x \to \infty.$$

Therefore the slant asymptote as $x \to +\infty$ is $y = \dfrac{\pi}{2}\, x - 1$.

Remark (Pause and Try). The function $h(x) = x \arctan(x)$ has a **second**, different slant asymptote as $x \to -\infty$. The process is similar. Try to find it: consider $\lim_{x\to-\infty} \arctan(x) = -\dfrac{\pi}{2}$ and repeat the argument. You will obtain a different line.

Remark. Even if nobody has taught you a systematic algorithm for finding slant asymptotes, you can still find them as long as you understand the definition: the line $y = mx + b$ is a slant asymptote precisely when $\lim\big[f(x) - (mx + b)\big] = 0$. Be willing to try a conjecture, test it, and adjust if necessary.

Chapter 7

The Definition of the Integral

7.1 Geometric Motivation

The definition of the integral is quite long, and we will not reach it in this section. Rather, we introduce the main ideas we will need. In a later section we formalize these ideas into the actual definition.

We begin with a geometric interpretation. Consider a function f that is positive and continuous on an interval $[a, b]$. The region between the graph of f and the x-axis is shaded, and we would like to define the *area* of this region to be the integral.

Definition 7.1 (Integral — Informal). The expression

$$\int_a^b f(x)\,dx$$

is read "the integral from a to b of $f(x)$, with respect to x," where f is a function and a, b are real numbers. It is sometimes abbreviated as $\int_a^b f$. We call this a *definite integral*; it is a number.

Remark. For now, the symbol dx is merely a piece of notation indicating that the variable of integration is x. We will later give it more meaning.

Remark (Warning). This is **not** a formal definition, because we do not yet know what the area of an arbitrary region is. In general, we only know how to compute the area of very simple shapes such as rectangles and triangles. We take this geometric picture as **motivation** for what we want the integral to represent.

We now spend some time trying to come up with a method to calculate the area of an arbitrary region. Once we have that method, we will use the resulting calculation as the definition of integral.

There are three key steps in this construction.

Step 1: Estimate with rectangles. The simplest shape whose area we know is the rectangle. Given the region under f over $[a, b]$, we draw a rectangle contained in the region, whose height is the minimum value m of f on $[a, b]$,

and a rectangle containing the region, whose height is the maximum value M. Then

$$m(b-a) \;\le\; \int_a^b f(x)\,dx \;\le\; M(b-a).$$

These are very rough estimates unless f is nearly constant.

Step 2: Cut into slices. We improve the estimation by dividing the region into vertical slices. For example, we cut the interval into four subintervals and estimate each slice separately with a rectangle below (under-estimate) and a rectangle above (over-estimate). The resulting approximation is better: the error is smaller.

Step 3: Take more and thinner slices. As we increase the number of slices and make them very thin, the estimation becomes better and better.

We now have a plan: cut into slices, approximate each slice with rectangles, and then, in the limit, take many very thin slices. However, we are not ready for the formal definition yet. We need two tools first.

1. **Sigma notation.** When we use many slices and approximate each one with a rectangle, we must sum the areas of all those rectangles. The so-called sigma notation makes writing long, complicated sums much easier.

2. **Supremum and infimum.** We have casually used the maximum and minimum of f for the heights of the rectangles, but many nice functions do not attain maximum or minimum values. The notions of supremum and infimum are similar to maximum and minimum, but more versatile; they will replace the maximum and minimum in the formal definition.

7.2 Sigma Notation

Sigma notation is a convenient notation for writing complicated sums in a compact way.

Example 7.1. The sum $\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}$ can be written in sigma notation as

$$\sum_{i=3}^{7} \frac{1}{i}.$$

Each term has the form $\frac{1}{i}$, where i ranges from 3 to 7 in steps of 1. The letter i under the Σ indicates the *summation index*, $i = 3$ is the first value, and 7 is the last value.

More generally, we write the sum $a_1 + a_2 + \cdots + a_N$ as

$$\sum_{i=1}^{N} a_i,$$

where a_i is any expression depending on an integer i. This notation is especially useful when a_i is complicated.

Remark. The summation index is a *dummy index*: it does not carry any intrinsic meaning. We could equally write $\sum_{k=1}^{N} a_k$ and it would denote exactly the same sum. Any letter (or symbol) may be used, as long as it is not already in use for something else.

Example 7.2 (Dummy Index). Let k be a fixed constant. Compute each of the following:

(i) $\displaystyle\sum_{i=1}^{3} \frac{i}{k}$,

(ii) $\displaystyle\sum_{k=1}^{3} \frac{i}{k}$,

(iii) $\displaystyle\sum_{\mu=1}^{3} \frac{i}{k}$.

Solution.

(i) The summation index is i. For the purpose of the sum, k is a constant:

$$\frac{1}{k} + \frac{2}{k} + \frac{3}{k} = \frac{6}{k}.$$

(ii) The summation index is k. For the purpose of the sum, i is a constant:

$$\frac{i}{1} + \frac{i}{2} + \frac{i}{3} = \frac{11i}{6}.$$

(iii) The summation index is μ. Both i and k are constants, so the quantity $\frac{i}{k}$ does not change from term to term:

$$\frac{i}{k} + \frac{i}{k} + \frac{i}{k} = \frac{3i}{k}.$$

We conclude with two basic algebraic properties of sums.

Proposition 7.1 (Properties of Sigma Notation). *Let c be a constant and let a_i, b_i depend on the index i. Then:*

(i) ***Constant factor:*** $\displaystyle\sum_{i=1}^{N} c\, a_i = c \sum_{i=1}^{N} a_i.$

(ii) ***Sum rule:*** $\displaystyle\sum_{i=1}^{N} (a_i + b_i) = \sum_{i=1}^{N} a_i + \sum_{i=1}^{N} b_i.$

Remark. These are not new results. The first is the distributive property ("take a common factor"), and the second follows from commutativity and associativity of addition. They are simply old facts expressed in the new notation.

7.3 Supremum and Infimum of Sets

The supremum and infimum are concepts similar to the maximum and minimum, but more useful. In this section we define them for sets; in the next section we extend the concepts to functions.

The notion of maximum is often insufficient. Let us explain why with an example.

Example 7.3. The closed interval $[0, 2]$ has a maximum: the number 2, which is the largest element in the set. However, the open interval $(0, 2)$ does **not** have a maximum, because 2 is not in the set. Yet 2 still "behaves like a maximum" in some sense. We need a concept that describes what the number 2 is to both $[0, 2]$ and $(0, 2)$.

Definition 7.2 (Maximum). A number c is the *maximum* of a set A when $c \in A$ and $x \leq c$ for every $x \in A$.

Definition 7.3 (Upper Bound). Let A be a set of real numbers. A number c is an *upper bound* of A when $x \leq c$ for every $x \in A$.

Example 7.4. For both $[0, 2]$ and $(0, 2)$, the number 2 is an upper bound. But it is not the only one: 2.1, π, and 100 are all upper bounds. On the other hand, the set of integers $\mathbb{Z}$ has no upper bound at all.

What is special about the number 2*? Of all the upper bounds of* $[0, 2]$ *or* $(0, 2)$*, the number* 2 *is the* ***smallest****. This is the key idea.*

Definition 7.4 (Supremum / Least Upper Bound). A number c is the *supremum* (or *least upper bound*) of a set A when:

(i) c is an upper bound of A, and

(ii) every upper bound b of A satisfies $b \geq c$.

We write $c = \sup A$.

Example 7.5. Both $[0, 2]$ and $(0, 2)$ have $\sup = 2$. The set $\mathbb{Z}$ does not have a supremum, because it does not have an upper bound to begin with.

Remark. When the supremum of a set happens to be an element of the set, then it is the maximum. The closed interval $[0, 2]$ has maximum 2. The open interval $(0, 2)$ does not have a maximum. Neither does $\mathbb{Z}$.

Definition 7.5 (Bounded Above). A set A is *bounded above* when it has at least one upper bound.

Remark (Pause and Try). By reversing the direction of all inequalities in the definitions above, define: *lower bound*, *infimum* (greatest lower bound), *minimum*, and *bounded below*.

Definition 7.6 (Infimum, Lower Bound, and Related Notions). Let A be a set of real numbers.

(i) A number c is a *lower bound* of A when $c \leq x$ for every $x \in A$.

(ii) A number c is the *infimum* (or *greatest lower bound*) of A, written $c = \inf A$, when c is a lower bound of A and every lower bound b of A satisfies $b \leq c$.

(iii) A number c is the *minimum* of A when $c \in A$ and c is the infimum of A.

(iv) A is *bounded below* when it has at least one lower bound.

(v) A is *bounded* when it is both bounded above and bounded below.

Theorem 7.2 (Least Upper Bound Principle). *If a set $A \subseteq \mathbb{R}$ is non-empty and bounded above, then A has a supremum.*

Remark. This result is perhaps the most important property of the real numbers. It is the main difference between $\mathbb{R}$ and $\mathbb{Q}$. In an analysis course, where one constructs $\mathbb{R}$ rigorously, proving this property is a fundamental step. For now, we accept it without proof; some authors treat it as an **axiom**.

Remark (Pause and Try). State the analogous result for lower bounds: if a set A is non-empty and bounded below, then A has an infimum.

7.4 Supremum and Infimum of Functions

In the previous section we defined the supremum and infimum of a set. In this section we extend these concepts to functions. The idea is actually simple: the supremum (or infimum) of a function is just the supremum (or infimum) of its range.

Definition 7.7 (Supremum of a Function). Let f be a function defined on a domain I. The *supremum of f on I* is

$$\sup_{x \in I} f(x) \;=\; \sup\{\, f(x) : x \in I \,\},$$

i.e. the supremum of the range of f restricted to I. If the domain I is not specified, it is understood to be the full domain of f.

Remark. Every other concept — upper bound, maximum, infimum, minimum, bounded above, bounded below — extends to functions in exactly the same way: we apply the set-theoretic definition to the range of the function.

Example 7.6. Consider a function f whose graph passes through the values -1 and 1 on the interval $[-2, 2]$, with range $[-1, 1]$. Then

$$\sup_{x \in [-2,2]} f(x) = \sup[-1, 1] = 1.$$

Example 7.7. Let g be a function whose graph on $[0, 2]$ approaches but never reaches the value 1 from below, and attains the value -1 at some point. Then g is bounded on $[0, 2]$, with $\sup_{[0,2]} g = 1$ (but g has no maximum), and $\inf_{[0,2]} g = -1$ (which is also a minimum).

We mention two important theorems connecting supremum and maximum for functions.

Theorem 7.3 (Least Upper Bound Principle for Functions). *If f is bounded above on a non-empty domain I, then $\sup_{x\in I} f(x)$ exists.*

This is an immediate consequence of the Least Upper Bound Principle for sets (Theorem 7.2).

Theorem 7.4 (Extreme Value Theorem). *If f is continuous on a closed, bounded interval $[a, b]$, then f attains a maximum and a minimum on $[a, b]$.*

Remark. These two theorems are related. Both provide a sufficient condition guaranteeing that a function has a supremum (first theorem) or, more strongly, a maximum (second theorem).

7.5 The Darboux Integral

We now take the ideas from Section 7.1 and convert them into a formal, rigorous definition. Throughout this section, f is a bounded function defined on a closed interval $[a, b]$. We assume ***only*** *that f is bounded — we do not need continuity or any other hypothesis.*

We recall the plan: cut the region into slices, approximate with rectangles below and above, and then "take the limit" by making the slices very thin.

Step 1: Partitions

Definition 7.8 (Partition). A *partition* of the interval $[a, b]$ is a finite set $P \subseteq [a, b]$ that contains both endpoints a and b.

In other words, P consists of a, b, and finitely many other points from $[a, b]$.

Example 7.8. The set $\{0, 0.2, 0.5, 0.9, 1\}$ is a partition of $[0, 1]$.

Every partition determines a way to split $[a, b]$ into consecutive sub-intervals. When we write a partition $P = \{x_0, x_1, x_2, \ldots, x_n\}$, we adopt the convention that the points are listed in increasing order:

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

Step 2: Lower and Upper Sums

Given a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$, on each sub-interval $[x_{i-1}, x_i]$ we define:

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x),$$

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x),$$

$$\Delta x_i = x_i - x_{i-1}.$$

Remark. We use the *infimum* rather than the minimum for m_i because not every bounded function attains its minimum. Since f is bounded on $[a, b]$, it is bounded on every sub-interval, so the infimum and supremum exist by the Least Upper Bound Principle (Theorem 7.2).

Definition 7.9 (Lower and Upper Sums). Let f be a bounded function on $[a, b]$ and P a partition of $[a, b]$. The *P-lower sum* and *P-upper sum* of f are

$$L_P(f) = \sum_{i=1}^{n} m_i \, \Delta x_i, \qquad U_P(f) = \sum_{i=1}^{n} M_i \, \Delta x_i.$$

Geometrically, $L_P(f)$ is the total area of the rectangles ***below*** *the graph of f (the under-estimate), and $U_P(f)$ is the total area of the rectangles* ***above*** *the graph (the over-estimate).*

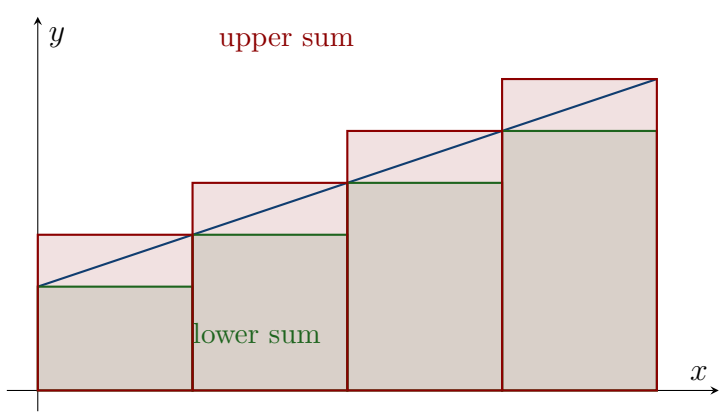


Figure 7.1: Upper and lower sums on a fixed partition.

Step 3: Lower and Upper Integrals

The integral must be a number lying between all lower sums and all upper sums. We need every lower sum to be at most every upper sum; this will be proved in Section 7.6. Assuming this fact, we proceed.

Lemma 7.5. *For any two partitions P and Q of $[a, b]$,*

$$L_P(f) \leq U_Q(f).$$

That is, every lower sum is less than or equal to every upper sum.

The proof is given in Section 7.6.

As we use finer partitions, the lower sums increase and the upper sums decrease, each approaching the area we want to compute. Rather than taking a "limit" of partitions (which are not numbers), we proceed more elegantly: we define two integrals, one from each side.

Definition 7.10 (Lower and Upper Integrals). Let f be a bounded function on $[a, b]$. The *lower integral* of f is

$$\underline{\int_a^b} f \;=\; \sup\big\{\, L_P(f) : P \text{ is a partition of } [a, b] \,\big\},$$

and the *upper integral* of f is

$$\overline{\int_a^b} f \;=\; \inf\big\{\, U_P(f) : P \text{ is a partition of } [a, b] \,\big\}.$$

Remark. The set of lower sums is non-empty (there is at least one partition) and bounded above (by any upper sum, via Lemma 7.5), so the supremum exists. Similarly, the set of upper sums is non-empty and bounded below, so the infimum exists.

Since every lower sum is at most every upper sum, it follows that

$$\underline{\int_a^b} f \;\leq\; \overline{\int_a^b} f.$$

Definition 7.11 (Integrability). A bounded function f on $[a, b]$ is *integrable* on $[a, b]$ when

$$\underline{\int_a^b} f \;=\; \overline{\int_a^b} f.$$

In this case, the common value is called the *integral of f from a to b*, denoted

$$\int_a^b f(x)\, dx.$$

If the lower and upper integrals are not equal, we say f is *non-integrable* on $[a, b]$, and the integral is undefined.

Theorem 7.6 (Continuous Functions are Integrable). *Every continuous function on a closed interval $[a, b]$ is integrable on $[a, b]$.*

Remark. The proof is quite technical and is typically covered in an analysis course. If f is not continuous, it may or may not be integrable.

7.6 Properties of Darboux Sums

We now list and justify three important properties of lower and upper sums. Throughout this section, f is a bounded function on $[a, b]$.

Proposition 7.7 (Property 1: Same Partition). *For any partition P of $[a, b]$,*

$$L_P(f) \leq U_P(f).$$

Proof. On each sub-interval $[x_{i-1}, x_i]$, the infimum m_i is at most the supremum M_i, so $m_i \, \Delta x_i \leq M_i \, \Delta x_i$. Summing over all sub-intervals gives $L_P(f) \leq U_P(f)$. ■

Can we compare lower and upper sums for ***different*** *partitions? There is a sense in which some partitions are "better" than others.*

Definition 7.12 (Refinement). Given two partitions P and Q of $[a, b]$, we say Q is *finer than* P (or Q is a *refinement* of P) when $P \subseteq Q$.

Remark. Being finer does not merely mean having more points. Q must contain **all** the points of P, plus possibly additional ones.

What happens when we add a single extra point to a partition? The extra point splits one sub-interval into two, creating an extra slice. The lower sum increases (or stays the same), because the infimum on a smaller sub-interval is at least as large as the infimum on the original sub-interval. Adding several points one at a time produces the same effect.

Proposition 7.8 (Property 2: Refinement). *Let P and Q be partitions of $[a, b]$ with Q finer than P. Then*

$$L_P(f) \leq L_Q(f) \quad \text{and} \quad U_Q(f) \leq U_P(f).$$

That is, finer partitions produce larger lower sums and smaller upper sums.

What if we have two partitions and neither is finer than the other? We cannot, in general, compare their lower sums or their upper sums individually, but we can always compare ***any*** *lower sum with* ***any*** *upper sum.*

Proposition 7.9 (Property 3: Any Lower Sum vs. Any Upper Sum). *For any two partitions P and Q of $[a, b]$,*

$$L_P(f) \leq U_Q(f).$$

Proof. Define $R = P \cup Q$. Then R is a partition that is finer than both P and Q. By Property 2 (Proposition 7.8):

$$L_P(f) \leq L_R(f) \leq U_R(f) \leq U_Q(f).$$

The middle step uses Property 1 (Proposition 7.7). ■

The middle inequality is Property 1 (7.7); the outer inequalities are Property 2 (7.8).

Remark. Property 3 is exactly Lemma 7.5, which we assumed when defining the integral in Section 7.5. The three properties together fully justify the picture: all lower sums sit to the left, all upper sums sit to the right, the lower integral lies between them on the left, and the upper integral lies between them on the right.

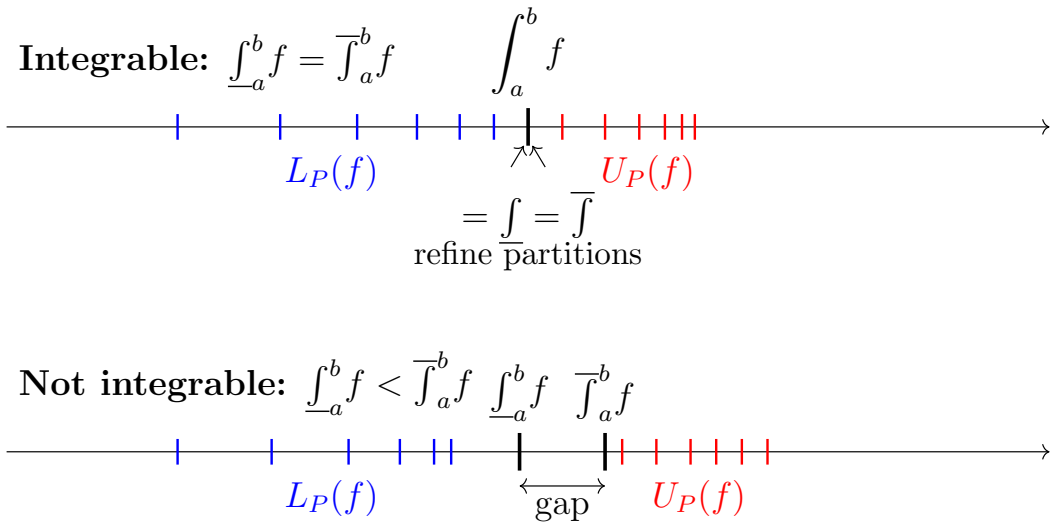


Figure 7.2: Lower sums and upper sums viewed as numbers on the real line: meeting implies integrability; a gap implies non-integrability.

7.7 An Integrable Function

We illustrate how to use the definition of integral to prove that a function is integrable.

Example 7.9. Define $f\colon [-1,1] \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x = 0, \\ 0 & \text{if } x \neq 0. \end{cases}$$

We will show that f is integrable on $[-1,1]$ and $\displaystyle\int_{-1}^{1} f(x)\,dx = 0$.

Lower sums. Let P be any partition of $[-1,1]$. On every sub-interval, the infimum of f is 0 (regardless of whether 0 is one of the partition points). Therefore $L_P(f) = 0$ for every partition P. Hence

$$\underline{\int}_{-1}^{1} f = \sup\{0\} = 0.$$

Upper sums. Let P be an arbitrary partition of $[-1,1]$. The supremum of f on a sub-interval is 1 if the sub-interval contains 0, and 0 otherwise. Exactly

one sub-interval (or two, if $0 \in P$) contributes, and the corresponding upper rectangle has height 1 and some positive width w. Therefore $U_P(f) = w$.

The width w can be made arbitrarily small (by placing partition points close to 0), but it must remain positive. With the simplest partition $P = \{-1, 1\}$, we get $w = 2$. Thus

$$\big\{ U_P(f) : P \text{ is a partition of } [-1, 1] \big\} = (0, 2].$$

Hence

$$\overline{\int_{-1}^{1}} f = \inf(0, 2] = 0.$$

Since the lower and upper integrals are both 0, the function f is integrable on $[-1, 1]$ and $\displaystyle\int_{-1}^{1} f(x)\, dx = 0$.

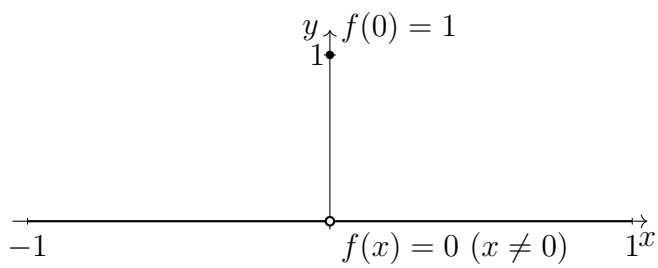


Figure 7.3: The function f equals 0 everywhere except at $x = 0$, where $f(0) = 1$.

7.8 A Non-Integrable Function

We now illustrate how to use the definition of integral to prove that a function is non-integrable.

Example 7.10 *(The Dirichlet Function).* Define $g\colon [0, 1] \to \mathbb{R}$ by

$$g(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

We will show that g is **not** integrable on $[0, 1]$.

The entire argument rests on one key observation.

Remark. For any real numbers $a < b$, the interval (a, b) contains both rational and irrational numbers.

Remark (Pause and Try). Pause and convince yourself that this observation is true, because the rest of the argument depends on it.

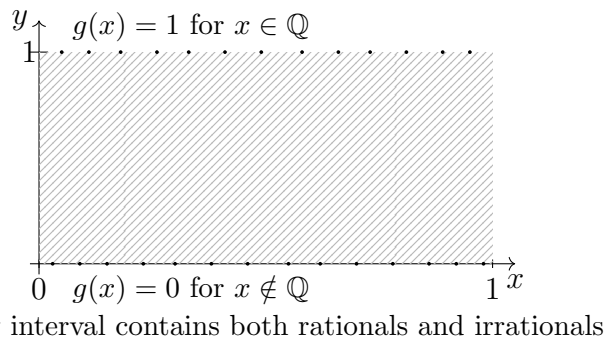


Figure 7.4: Dirichlet function intuition: on every interval, values 0 and 1 both occur (densely).

Consider any partition P of $[0,1]$, and look at any sub-interval $[x_{i-1}, x_i]$. Since this sub-interval contains both a rational number (where $g = 1$) and an irrational number (where $g = 0$), we have:

$$m_i = \inf_{[x_{i-1},\, x_i]} g = 0, \qquad M_i = \sup_{[x_{i-1},\, x_i]} g = 1.$$

Therefore:

$$L_P(g) = \sum_{i=1}^{n} 0 \cdot \Delta x_i = 0, \qquad U_P(g) = \sum_{i=1}^{n} 1 \cdot \Delta x_i = \sum_{i=1}^{n} \Delta x_i = 1.$$

This is true for **all** partitions. Hence

$$\underline{\int_0^1} g = \sup\{0\} = 0, \qquad \overline{\int_0^1} g = \inf\{1\} = 1.$$

Since $0 \neq 1$, the function g is not integrable on $[0,1]$.

Remark. There are alternative ways to define the integral. In the *Lebesgue* theory of integration, the Dirichlet function *is* integrable (with integral 0). The Lebesgue integral is more powerful but more complex; it is beyond the scope of this course.

7.9 Integrals as Limits

Can we compute the lower or upper integral of a function as a limit of lower or upper sums? It looks like we should be able to, but it is not clear how: a limit as ***what*** *approaches* ***what****?*

A limit as the partitions "become finer"? We cannot naïvely take the limit as the number of points in the partition approaches infinity, because we also need the width of every sub-interval to approach 0*.*

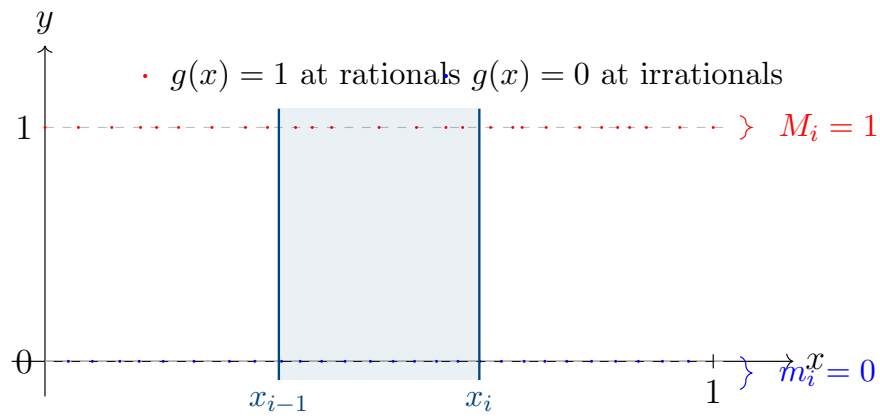


Figure 7.5: Dense oscillation intuition: upper sums stay high while lower sums stay low.

Definition 7.13 (Norm of a Partition). Given a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$, the *norm* (or *mesh*) of P is

$$\|P\| = \max_{1 \le i \le n} \Delta x_i,$$

the largest of the widths of the sub-intervals.

Remark. The norm is a measure of how "good" a partition is. If $\|P\|$ is small, then **all** sub-intervals have small width, hence all the approximating rectangles are very thin.

Theorem 7.10 (Lower Integral as a Limit). *Let f be a bounded function on $[a, b]$. Then*

$$\underline{\int_a^b} f \;=\; \lim_{\|P\| \to 0} L_P(f).$$

More precisely: for every $\varepsilon > 0$, there exists $\delta > 0$ such that for every partition P of $[a, b]$ with $\|P\| < \delta$,

$$\left| \underline{\int_a^b} f - L_P(f) \right| < \varepsilon.$$

Similarly, $\overline{\int_a^b} f = \lim_{\|P\| \to 0} U_P(f)$.

In practice, however, considering all partitions is still cumbersome. The following theorem allows us to use a single convenient sequence of partitions.

Theorem 7.11 (Limits Along a Sequence of Partitions). *Let f be a bounded function on $[a, b]$, and let $(P_n)_{n=1}^{\infty}$ be a sequence of partitions of $[a, b]$ such that $\|P_n\| \to 0$ as $n \to \infty$. Then*

$$\underline{\int_a^b} f = \lim_{n \to \infty} L_{P_n}(f), \qquad \overline{\int_a^b} f = \lim_{n \to \infty} U_{P_n}(f).$$

Example 7.11. The simplest choice is the *uniform partition* P_n that breaks $[a, b]$ into n sub-intervals of equal length $\dfrac{b-a}{n}$. Then $\|P_n\| = \dfrac{b-a}{n} \to 0$.

Remark. This expression is a limit in the familiar sense: a limit as a certain number (n) approaches infinity. For studying the theory of integration, the original definition (as a supremum/infimum) is both simpler and more useful — it makes it easier to prove further properties and theorems. But it is good to know that the integral can also be interpreted as a limit.

7.10 Riemann Sums

Riemann sums are an alternative way to calculate the integral of a function. They are not very convenient for proving theorems, but they are often useful for computations.

Throughout this section, f is a bounded, integrable function on $[a, b]$.

So far, we have estimated the area of each slice with a rectangle whose height is the infimum (lower sum) or the supremum (upper sum) of f on that sub-interval. What if, instead, we choose a ***random*** *point on the graph of f over each sub-interval and use the value of f at that point as the height of the rectangle?*

Definition 7.14 (Riemann Sum). Let f be a bounded function on $[a, b]$, let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$, and for each $i = 1, \ldots, n$, choose a *sample point* $x_i^* \in [x_{i-1}, x_i]$. The *Riemann sum* for f and P (with these sample points) is

$$S_P^*(f) = \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i.$$

Remark. We say "*a* Riemann sum," not "*the* Riemann sum." Even for a fixed function and partition, the Riemann sum depends on the choice of sample points x_i^*. The star in S_P^* indicates this dependence.

Remark. Since $m_i \le f(x_i^*) \le M_i$ on each sub-interval, every Riemann sum satisfies

$$L_P(f) \le S_P^*(f) \le U_P(f).$$

Theorem 7.12 (Integral as a Limit of Riemann Sums). *Let f be bounded and integrable on $[a, b]$. Let (P_n) be a sequence of partitions of $[a, b]$, each having n sub-intervals, with $\|P_n\| \to 0$, and for each P_n, choose sample points x_i^* on each sub-interval. Then*

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i.$$

Proof. From Theorem 7.11, we know $L_{P_n}(f) \to \displaystyle\int_a^b f$ and $U_{P_n}(f) \to \displaystyle\int_a^b f$ as $n \to \infty$ (using the fact that f is integrable, so the lower and upper integrals

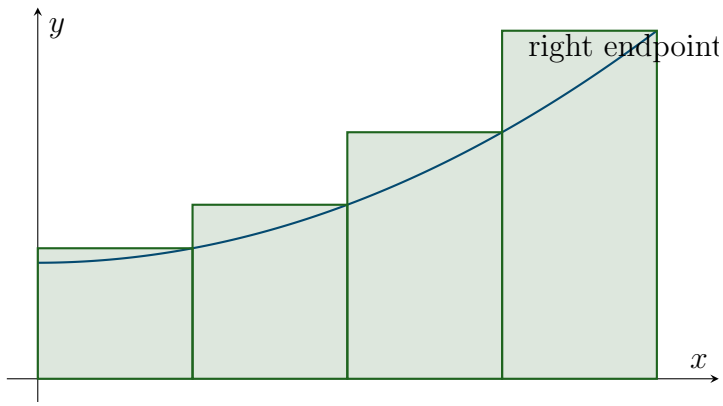


Figure 7.6: A right-endpoint Riemann sum approximation.

both equal the integral). Since $L_{P_n}(f) \leq S^*_{P_n}(f) \leq U_{P_n}(f)$, the Squeeze Theorem gives $S^*_{P_n}(f) \to \int_a^b f$. ■

Remark (Warning). **This method requires knowing in advance that f is integrable.** If f is continuous on $[a, b]$, then f is integrable by Theorem 7.6, and Riemann sums may be used. For an arbitrary bounded function whose integrability is unknown, the limit of Riemann sums may fail to exist or may depend on the choices made.

Suggestive notation. The Riemann sum formula

$$\sum_{i=1}^{n} f(x_i^*)\,\Delta x_i \xrightarrow{n\to\infty} \int_a^b f(x)\,dx$$

suggests that in the limit, the sum $\sum$ turns into the integral $\int$, and the finite width Δx turns into the "infinitesimally small" dx. This is why some authors say dx represents an infinitesimally small width.

Example 7.12 (Integral of $f(x) = x$). We compute $\int_0^1 x\,dx$ using Riemann sums. (The function $f(x) = x$ is continuous, hence integrable.)

Choose the uniform partition $P_n = \left\{0, \frac{1}{n}, \frac{2}{n}, \ldots, 1\right\}$, which has n sub-intervals of equal width $\Delta x_i = \frac{1}{n}$. On each sub-interval, choose the right endpoint as the sample point: $x_i^* = \frac{i}{n}$. The Riemann sum is

$$S^*_{P_n}(f) = \sum_{i=1}^{n} \frac{i}{n} \cdot \frac{1}{n} = \frac{1}{n^2} \sum_{i=1}^{n} i = \frac{1}{n^2} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2n}.$$

Taking the limit:

$$\int_0^1 x\,dx = \lim_{n\to\infty} \frac{n+1}{2n} = \frac{1}{2}.$$

This agrees with the fact that the region under $f(x) = x$ on $[0, 1]$ is a triangle of area $\frac{1}{2}$.

Remark. Riemann sums may look like a simpler way to compute integrals than the original definition, but they only work if we know the function is integrable. For developing the theory of integration and proving theorems, the original definition via lower and upper integrals is more useful.

7.11 Properties of Definite Integrals

We now list the five most important properties of definite integrals. They are all geometrically intuitive. To be rigorous, each should be proved from the definition, but we omit the proofs here.

Linearity

Theorem 7.13 (Linearity of the Integral). *Let f and g be bounded functions on $[a, b]$.*

(i) ***Sum rule.*** *If f and g are both integrable on $[a, b]$, then $f+g$ is integrable on $[a, b]$ and*

$$\int_a^b \big(f(x) + g(x)\big)\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx.$$

(ii) ***Constant multiple.*** *If f is integrable on $[a, b]$ and $c \in \mathbb{R}$, then cf is integrable on $[a, b]$ and*

$$\int_a^b c\, f(x)\, dx = c \int_a^b f(x)\, dx.$$

Remark (Warning). If f or g is non-integrable, the sum $f + g$ may or may not be integrable, and the equation does not apply. This is analogous to the situation with limit laws and differentiation rules.

Additivity over Intervals

Theorem 7.14 (Splitting the Interval). *Let f be a bounded function on $[a, c]$ with $a < b < c$. If f is integrable on $[a, b]$ and on $[b, c]$, then f is integrable on $[a, c]$ and*

$$\int_a^c f(x)\, dx = \int_a^b f(x)\, dx + \int_b^c f(x)\, dx.$$

So far, we have only defined $\displaystyle\int_a^b f$ when $a < b$. The following convention extends the definition.

Definition 7.15 (Backwards and Degenerate Integrals). If f is integrable on $[a, b]$ (with $a < b$), we define

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx.$$

In addition, for any a,

$$\int_a^a f(x)\,dx = 0.$$

Remark. With these conventions, the identity

$$\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$$

holds for **all** orderings of a, b, and c (as long as f is integrable where needed). For instance, taking $a = c$ gives $\int_a^b f + \int_b^a f = 0$, confirming that $\int_b^a f = -\int_a^b f$.

Geometric Interpretation: Signed Area

Our original motivation defined the integral as the area of the region between the graph of f and the x-axis, but we always assumed f was positive. The definition of the integral makes no such assumption — it works for any bounded function. So what does the integral represent when f is not positive?

Remark. For a **negative** function, the integral equals **minus** the area of the region between the graph and the x-axis. This is because the heights of the rectangles in the lower and upper sums are values of f, which are negative.

For a function that takes both positive and negative values on $[a, b]$, we split the region between the graph and the x-axis into the part above the axis and the part below. Then:

$$\int_a^b f(x)\,dx = (\text{area above the } x\text{-axis}) - (\text{area below the } x\text{-axis}).$$

Remark (Warning). Areas are always positive, but definite integrals may be positive, negative, or zero.

Comparison

Theorem 7.15 (Comparison of Integrals). *If f and g are integrable on $[a, b]$ and $f(x) \leq g(x)$ for all $x \in [a, b]$, then*

$$\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx.$$

Remark. This also makes sense geometrically: when f and g are non-negative, the region under the graph of f is contained in the region under the graph of g, so its area is no larger.

Remark. Practically everything one needs to do with definite integrals rests on these five properties. To a certain extent, if one is not concerned with rigorous definitions and proofs, one could forget the formal definition and take these properties as a starting point.

Chapter 8

The Fundamental Theorem of Calculus

8.1 Antiderivatives and Indefinite Integrals

There are two different things that we call integrals. We use very similar notation, but they mean something very different.

The first is the *definite integral*, $\int_a^b f(x)\,dx$, which we have already defined. It is a **number**, and geometrically it represents area.

The second is the *indefinite integral*, written

$$\int f(x)\,dx,$$

without limits of integration. This is something new.

Definition 8.1 (Antiderivative). Let f be a function defined on an interval I. A function F is an *antiderivative* of f on I if $F'(x) = f(x)$ for every $x \in I$.

Remark. We say "*an* antiderivative" because in general there will be more than one. Every function can only have one derivative, if it has one, but it can have many antiderivatives.

Definition 8.2 (Indefinite Integral). The *indefinite integral* $\int f(x)\,dx$ denotes the collection of **all** antiderivatives of f.

So these two objects are quite different. The definite integral is a number; the indefinite integral is a collection of functions. The definite integral means area; the indefinite integral means antiderivative. A priori, these two things have nothing to do with each other. But we use very similar notation, and that is because they ***are*** *related — by the Fundamental Theorem of Calculus, which will come in a few sections. For now, they are two separate things.*

Example 8.1. Compute $\int x^2\,dx$.

Solution. We seek all antiderivatives of x^2. We can guess: the function $\frac{1}{3}x^3$ works, since

$$\frac{d}{dx}\left(\frac{1}{3}x^3\right) = \frac{1}{3} \cdot 3x^2 = x^2.$$

Adding any constant C also works, because $\frac{d}{dx}(C) = 0$. Moreover, these are *all* of the antiderivatives: by a consequence of the Mean Value Theorem, if two functions have the same derivative on an interval then they differ by a constant. Therefore

$$\int x^2\,dx = \frac{1}{3}x^3 + C.$$

Remark. Without the Mean Value Theorem result that two functions sharing the same derivative must differ by a constant, we could only say that we found *some* antiderivatives, not *all* of them.

Guess and check

There are many so-called integration techniques — substitution, integration by parts, and so on — which are methods for computing antiderivatives. But by far the most important integration technique is ***guess and check****.*

The method is exactly what it sounds like: make a guess for what the antiderivative should look like, take its derivative to check, and modify as necessary.

Example 8.2. Compute $\int (2x+7)^{11}\,dx$.

Solution. We want a function whose derivative is $(2x+7)^{11}$. The power rule suggests trying $(2x+7)^{12}$. Differentiating by the chain rule:

$$\frac{d}{dx}\left[(2x+7)^{12}\right] = 12(2x+7)^{11} \cdot 2 = 24(2x+7)^{11}.$$

This is our target, but with an extra factor of 24. Dividing by 24 fixes this:

$$\frac{d}{dx}\left[\frac{1}{24}(2x+7)^{12}\right] = \frac{1}{24} \cdot 24(2x+7)^{11} = (2x+7)^{11}.$$

Therefore

$$\int (2x+7)^{11}\,dx = \frac{1}{24}(2x+7)^{12} + C.$$

Remark. This method, even though it seems rudimentary, is very powerful. All the integration techniques one learns later — substitution, integration by parts, and so on — are just sophisticated forms of guessing and checking that make the guessing part easier.

Remark (Pause and Try). At this point, many textbooks present a table of elementary antiderivatives. But you do not need those formulas: if you know

the differentiation rules and are comfortable with guess and check, you can solve all standard antiderivative problems without memorising anything new. Moreover, since you can check your answers by differentiating, you will know whether they are right.

8.2 Functions Defined by Integrals

We already know two notions of integral: the definite integral (a number) and the indefinite integral (a collection of antiderivatives). We now introduce a way to use integrals to define new functions.

Definition 8.3 (Function Defined by an Integral). Let I be an interval, let $a \in I$ be a fixed point, and let f be an integrable function on I. We define a new function F by

$$F(x) = \int_a^x f(t)\,dt.$$

For each value of $x \in I$, the expression $\displaystyle\int_a^x f(t)\,dt$ is a definite integral and hence a number; we call that number $F(x)$.

Strictly speaking, this is not a new concept: it is just a definite integral where the upper endpoint of the interval becomes the variable of a new function. However, the first time one sees this, it can be confusing how it relates to and differs from an ordinary definite integral, and in particular why we use the notation that we use. Let us break this down carefully.

Example 8.3. Consider a function f whose graph is drawn in the ty-plane. Fix $a = 1$ and define

$$F(x) = \int_1^x f(t)\,dt.$$

Then:

- $F(2) = \displaystyle\int_1^2 f(t)\,dt$ is the area under f from $t = 1$ to $t = 2$.
- $F(3) = \displaystyle\int_1^3 f(t)\,dt$ is the area under f from $t = 1$ to $t = 3$.
- $F(4) = \displaystyle\int_1^4 f(t)\,dt$ is the area under f from $t = 1$ to $t = 4$.

In each case the left endpoint is fixed at 1, but the right endpoint varies. For each choice of x, we obtain a number $F(x)$. This is a valid way to define a function.

Remark (Warning). Notice that we do **not** use x as the variable of integration; we use t (or any other letter). The reason is that x already has a role: it is the endpoint of the interval, and hence the variable of the new function F.

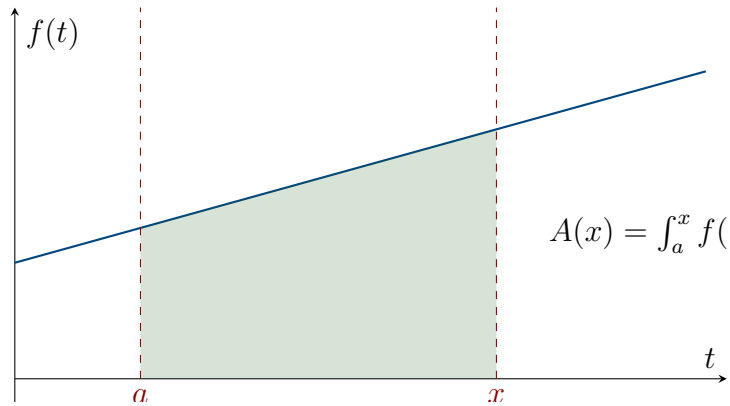


Figure 8.1: Accumulated area from a to x under f.

Recall that for a definite integral with fixed endpoints, the integration variable is a dummy variable:

$$\int_a^b f(x)\,dx = \int_a^b f(t)\,dt = \int_a^b f(s)\,ds,$$

and we may use any letter we like, **except** a or b, since they already have a meaning.

The same principle applies here. In the expression $F(x) = \displaystyle\int_a^x f(t)\,dt$, the integration variable may be t, s, u, or any letter **except** a or x. In particular, **never** write

$$F(x) = \int_a^x f(x)\,dx.$$

This is ambiguous and technically wrong, because x would have two different meanings in the same expression: the endpoint and the variable of integration.

Now that we understand how to use integrals to define new functions, what can we say about them? There is a theorem — the Fundamental Theorem of Calculus — that ties together the notions of definite integral, indefinite integral, and functions defined as integrals. It has two parts.

- **Part One** answers: what important property does a function defined as an integral always have?
- **Part Two** answers: how can we quickly compute definite integrals?

8.3 FTC Part One: Statement

The Fundamental Theorem of Calculus establishes connections between definite integrals and antiderivatives — two concepts that are defined separately and in principle are different, but are related through this theorem. We now state the first part.

Theorem 8.1 (Fundamental Theorem of Calculus, Part One). *Let I be an interval, let $a \in I$, and let f be a function with domain I. Define*

$$F(x) = \int_a^x f(t)\,dt.$$

If f is continuous on I, then F is differentiable on I and

$$F'(x) = f(x) \quad \text{for every } x \in I.$$

In other words, F is an antiderivative of f.

Remark. Note that continuity of f already guarantees integrability, so the integral defining F makes sense and F is well-defined.

We can summarise this theorem in words: a function defined as an integral is an antiderivative. This is the connection between the notion of antiderivative and the notion of integral, which were defined separately.

The theorem is often recorded as the single equation

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

Both sides are functions of x, not of t. The variable t is merely the variable of integration; it disappears.

Remark (Warning). If you attempt to change an x into a t, or vice versa, on one side of this equation, the result will not only be wrong — it will be **meaningless**.

Remark (Pause and Try). In the following examples, pause after reading each question and try to solve it before reading the solution. You do not need any extra tools; everything you need is contained in Theorem 8.1. Any answer watched (or read) without first attempting the problem is a learning opportunity wasted.

Example 8.4. Define $F(x) = \displaystyle\int_1^x e^{-t^2}\,dt$. Find $F'(2)$.

Solution. By Theorem 8.1, $F'(x) = e^{-x^2}$. Therefore $F'(2) = e^{-4}$.

Example 8.5. Construct a function g such that

$$g'(x) = \frac{1}{1+x^2+x^{10}} \quad \text{and} \quad g(2) = 5.$$

Solution. By Theorem 8.1, an antiderivative of the given function is obtained by writing an integral. We choose the lower limit $a = 2$ so that we can control the value at 2:

$$g(x) = \int_2^x \frac{1}{1+t^2+t^{10}}\,dt + 5.$$

Then $g'(x) = \dfrac{1}{1+x^2+x^{10}}$ by the theorem, and $g(2) = 0 + 5 = 5$.

Example 8.6. Define $G(x) = \displaystyle\int_{-4}^{x^2} \frac{1}{1+t^2}\,dt$. Find $G'(x)$.

Solution. Let $F(x) = \displaystyle\int_{-4}^{x} \frac{1}{1+t^2}\,dt$. By Theorem 8.1, $F'(x) = \dfrac{1}{1+x^2}$. Observe that $G(x) = F(x^2)$, so by the chain rule,

$$G'(x) = F'(x^2)\cdot 2x = \frac{2x}{1+x^4}.$$

Example 8.7. Define $H(x) = \displaystyle\int_{x^3+1}^{x^2+2x} e^{-t^2}\,dt$. Find $H'(x)$.

Solution. We split the integral at a convenient point, say 0:

$$H(x) = \int_0^{x^2+2x} e^{-t^2}\,dt - \int_0^{x^3+1} e^{-t^2}\,dt.$$

(We reversed the limits on the second integral, introducing a minus sign.) Define

$$F(u) = \int_0^u e^{-t^2}\,dt, \quad \text{so that} \quad F'(u) = e^{-u^2}.$$

Then $H(x) = F(x^2+2x) - F(x^3+1)$. By the chain rule,

$$\begin{aligned} H'(x) &= F'\big(x^2+2x\big)\cdot(2x+2) - F'\big(x^3+1\big)\cdot 3x^2 \\ &= (2x+2)\,e^{-(x^2+2x)^2} - 3x^2\,e^{-(x^3+1)^2}. \end{aligned}$$

8.4 FTC Part One: Proof

The proof of Theorem 8.1 is perhaps a bit long, but it is not too abstract.

Proof. Fix $x \in I$. We wish to show that $F'(x) = f(x)$.

Step 1: Rewrite the difference quotient. By the definition of derivative,

$$F'(x) = \lim_{h\to 0} \frac{F(x+h) - F(x)}{h}.$$

Substituting the definition of F,

$$\frac{F(x+h)-F(x)}{h} = \frac{1}{h}\left[\int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt\right] = \frac{1}{h}\int_x^{x+h} f(t)\,dt,$$

where the last step uses the additive property of integrals. Therefore it suffices to show

$$\lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\,dt = f(x). \tag{$*$}$$

We used the definition of derivative and the additive property of integrals.

Step 2: Geometric intuition. For small $h > 0$, the integral $\displaystyle\int_x^{x+h} f(t)\,dt$ represents a thin vertical strip of area under the graph of f. This strip is

approximately a rectangle of width h and height $f(x)$, so its area is roughly $f(x) \cdot h$. Dividing by h gives approximately $f(x)$, and this approximation improves as $h \to 0$.

Step 3: Rigorous bounding (for $h > 0$). Define

$$M_h = \sup_{t \in [x,\, x+h]} f(t), \qquad m_h = \inf_{t \in [x,\, x+h]} f(t).$$

Then

$$m_h \cdot h \;\le\; \int_x^{x+h} f(t)\,dt \;\le\; M_h \cdot h,$$

and dividing by $h > 0$,

$$m_h \;\le\; \frac{1}{h}\int_x^{x+h} f(t)\,dt \;\le\; M_h. \tag{$**$}$$

Step 4: Apply the Squeeze Theorem. Since f is continuous on I, the Extreme Value Theorem guarantees that f attains its maximum and minimum on $[x, x+h]$. Thus there exists $c_h \in [x, x+h]$ with $M_h = f(c_h)$, and similarly $d_h \in [x, x+h]$ with $m_h = f(d_h)$.

The integral is bounded between the infimum and supremum times the length of the interval.

As $h \to 0^+$, the point c_h is squeezed between x and $x+h$, so $c_h \to x$. Since f is continuous, $f(c_h) \to f(x)$, i.e. $M_h \to f(x)$. By the same argument, $m_h \to f(x)$.

By the Squeeze Theorem applied to $(**)$,

$$\lim_{h \to 0^+} \frac{1}{h}\int_x^{x+h} f(t)\,dt = f(x).$$

Step 5: The case $h < 0$. The argument for $h \to 0^-$ is entirely analogous (one reverses the interval and accounts for the sign). Combining both one-sided limits, we obtain $(*)$, and therefore $F'(x) = f(x)$. ■

Continuity of f is essential: without it, $f(c_h) \to f(x)$ may fail.

8.5 FTC Part Two: Statement

The first part of the Fundamental Theorem tells us what we can say about any function defined as an integral: it is an antiderivative. The second part tells us how to quickly compute a definite integral: through antiderivatives.

Theorem 8.2 (Fundamental Theorem of Calculus, Part Two). *Let $a < b$ be real numbers and let f be continuous on $[a, b]$. If G is **any** antiderivative of f on $[a, b]$, then*

$$\int_a^b f(x)\,dx = G(b) - G(a).$$

Remark. This theorem does not tell us *how* to find an antiderivative — for some functions that is easy, for others it is hard. But **if** we can find one, it lets us evaluate the definite integral very quickly, certainly much faster than computing it from the formal definition in terms of upper and lower sums.

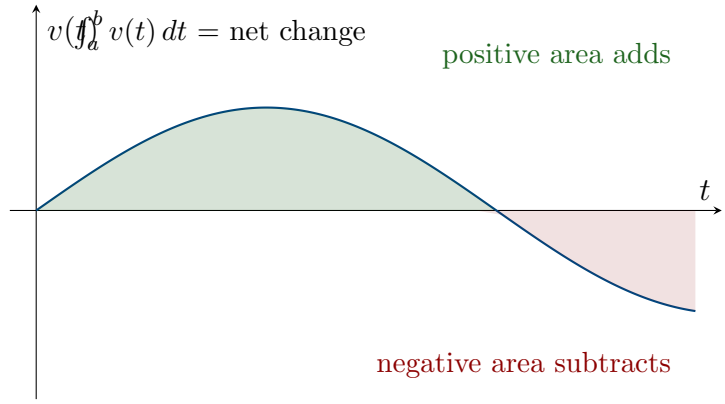


Figure 8.2: Net change as signed area under a rate function.

Definition 8.4 (Evaluation Notation). We write

$$G(x)\big|_{x=a}^{x=b} \quad \text{or simply} \quad G(x)\big|_a^b$$

to mean $G(b) - G(a)$.

Example 8.8. Compute $\displaystyle\int_1^2 x^2\,dx$.

Solution. We need an antiderivative of x^2. By inspection, $G(x) = \dfrac{x^3}{3}$ works, since $G'(x) = x^2$. By Theorem 8.2,

$$\int_1^2 x^2\,dx = \frac{x^3}{3}\bigg|_1^2 = \frac{2^3}{3} - \frac{1^3}{3} = \frac{8}{3} - \frac{1}{3} = \frac{7}{3}.$$

This is certainly much faster than computing the integral from the definition or from Riemann sums.

Remark (Pause and Try). Here are four integrals whose antiderivatives can be found by guess and check. Evaluate each using Theorem 8.2:

(i) $\displaystyle\int_0^1 3x^2\,dx$,

(ii) $\displaystyle\int_0^\pi \sin x\,dx$,

(iii) $\displaystyle\int_1^e \frac{1}{x}\,dx$,

(iv) $\displaystyle\int_0^1 e^x\,dx$.

8.6 FTC Part Two: Proof

The proof of the second part uses the first part, and is considerably shorter.

Proof. Define

$$F(x) = \int_a^x f(t)\,dt.$$

We wish to show that $F(b) = G(b) - G(a)$.

Step 1: F and G are both antiderivatives. Since f is continuous, Theorem 8.1 (FTC Part One) gives $F'(x) = f(x)$. By hypothesis, $G'(x) = f(x)$. Therefore F and G are two functions with the same derivative on $[a, b]$.

F is introduced as an auxiliary function; it is not part of the hypothesis.

Step 2: F and G differ by a constant. By the consequence of the Mean Value Theorem (if two functions have the same derivative on an interval, they differ by a constant), there exists a constant C such that

$$F(x) = G(x) + C \quad \text{for all } x \in [a, b].$$

Step 3: Determine C. Evaluate at $x = a$:

Mean Value Theorem consequence: same derivative implies constant difference.

$$F(a) = \int_a^a f(t)\,dt = 0,$$

so $0 = G(a) + C$, which gives $C = -G(a)$.

Step 4: Conclude. For every $x \in [a, b]$, $F(x) = G(x) - G(a)$. Setting $x = b$,

$$F(b) = G(b) - G(a).$$

But $F(b) = \int_a^b f(t)\,dt$, which is the definite integral we wished to evaluate. Therefore

$$\int_a^b f(x)\,dx = G(b) - G(a). \qquad \blacksquare$$

8.7 Summary and Comparison

Through this chapter and the preceding one, we have introduced various notions related to integrals. They use similar notation but mean different things, and sometimes we may confuse them. This section compares them side by side.

We have learned three concepts:

1. The **definite integral** $\int_a^b f(x)\,dx$ — a **number**.
2. The **indefinite integral** $\int f(x)\,dx$ — a **collection of functions** (the set of all antiderivatives).
3. A **function defined by an integral**, $F(x) = \int_a^x f(t)\,dt$ — a **single function**.

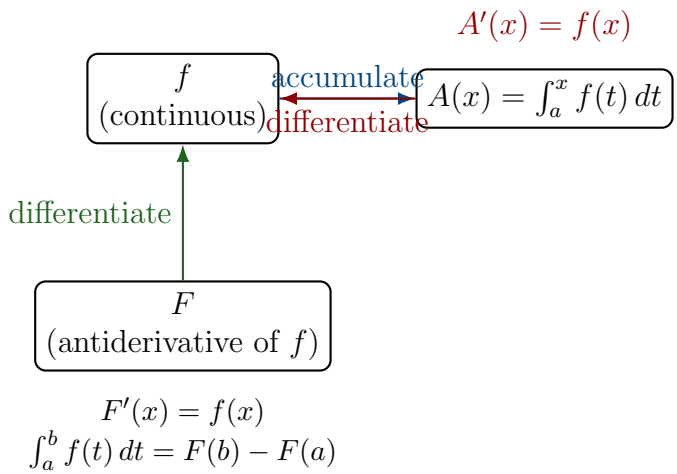


Figure 8.3: How FTC links accumulation and rates of change.

The definite integral

The definite integral $\displaystyle\int_a^b f(x)\,dx$ has both a formal definition and a geometric interpretation.

The *formal definition* is the one we have already studied: compute the upper integral (the infimum of all upper sums) and the lower integral (the supremum of all lower sums); if these two quantities agree, their common value is called the integral.

The *geometric interpretation* is simpler: if $f \geq 0$, the definite integral equals the area under the graph of f above the x-axis, from $x = a$ to $x = b$.

In practice, we want to **avoid** computing integrals from the formal definition. The Theorem 8.2 (FTC Part Two) saves us: find any antiderivative G of f, and then

$$\int_a^b f(x)\,dx = G(b) - G(a).$$

Example 8.9. $\displaystyle\int_1^2 x^2\,dx = \frac{x^3}{3}\bigg|_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3}.$

The indefinite integral

The indefinite integral $\displaystyle\int f(x)\,dx$ is the collection of all antiderivatives of f. How do we compute it? By a consequence of the Mean Value Theorem: find one antiderivative G, and then

$$\int f(x)\,dx = G(x) + C,$$

where C is an arbitrary constant (on an interval).

Example 8.10. $\displaystyle\int x^2\,dx = \frac{x^3}{3} + C.$

A function defined by an integral

Given an integrable function f on an interval I and a point $a \in I$, the expression

$$F(x) = \int_a^x f(t)\,dt$$

defines a new function. By Theorem 8.1 (FTC Part One), if f is continuous, then F is an antiderivative of f:

$$F'(x) = f(x).$$

This gives us a way to **construct** an antiderivative, even when no simpler closed-form expression exists.

Example 8.11. Define $F(x) = \displaystyle\int_3^x e^{-t^2}\,dt$. Then $F'(x) = e^{-x^2}$ and $F(3) = 0$. These two properties completely determine F, and there is no simpler way to write it.

When does the choice of variable matter?

- **Definite integral:** The result is a number. The integration variable is a dummy variable and can be called anything:

$$\int_a^b f(x)\,dx = \int_a^b f(t)\,dt.$$

- **Indefinite integral:** The result is a collection of functions *of the named variable*. If we change all x's to t's, we get a collection of functions of t, which is the same family of antiderivatives expressed in a different variable:

$$\int f(x)\,dx \quad \text{and} \quad \int f(t)\,dt \quad \text{(same antiderivatives, different variable names).}$$

- **Function defined by an integral:** The integration variable is again a dummy variable and may be called anything **except** x (or a), since those already have designated roles. Never write $\displaystyle\int_a^x f(x)\,dx$; it is ambiguous.

Chapter 9

Integration Methods

9.1 The Substitution Rule

The Substitution Rule is a method to compute antiderivatives. It is just an easy-to-use algorithm to make things a bit simpler and a bit more systematic than guess-and-check. Before presenting the general rule, let us investigate where the Substitution Rule comes from and why it works.

Motivation by example

We begin with a pair of simple antiderivatives.

Example 9.1. (a) $\displaystyle\int x^4\,dx = \frac{1}{5}x^5 + C$. This is immediate: the derivative of $\frac{1}{5}x^5$ is x^4.

(b) What about $\displaystyle\int (\ln x)^4\,dx$? One might guess $\frac{1}{5}(\ln x)^5$, but check:

$$\frac{d}{dx}\left[\frac{1}{5}(\ln x)^5\right] = \frac{1}{5}\cdot 5(\ln x)^4 \cdot \frac{1}{x} = \frac{(\ln x)^4}{x}.$$

The chain rule introduces an extra factor of $\frac{1}{x}$. So $\frac{1}{5}(\ln x)^5$ is **not** an antiderivative of $(\ln x)^4$, but it *is* an antiderivative of $\frac{(\ln x)^4}{x}$.

This observation generalises. If we know that an antiderivative of f is F, then what is an antiderivative of $f(g(x))$? We do not know in general, but an antiderivative of $f(g(x)) \cdot g'(x)$ is $F(g(x))$. The chain rule is the reason.

The theorem

Theorem 9.1 (Substitution Rule / Backwards Chain Rule). *Let F and g be differentiable functions such that $F' = f$. Then*

$$\int f(g(x)) \cdot g'(x)\,dx = F(g(x)) + C.$$

Proof. By the chain rule,

$$\frac{d}{dx}\big[F(g(x))\big] = F'(g(x)) \cdot g'(x) = f(g(x)) \cdot g'(x).$$

Chain rule applied to $F \circ g$.

Therefore $F(g(x))$ is an antiderivative of $f(g(x)) \cdot g'(x)$. ■

Remark. Despite its traditional name, the Substitution Rule should really be called the *backwards chain rule*: we take the chain rule, which is a rule about derivatives, and reverse it to get a rule about antiderivatives.

The substitution notation

In practice, we try to use Theorem 9.1 in an efficient and algorithmic way that simplifies computations. The key idea is to introduce new notation.

Suppose we want to compute $\displaystyle\int f(g(x)) \cdot g'(x)\,dx$. We set

$$u = g(x), \qquad du = g'(x)\,dx.$$

Then the integral becomes $\displaystyle\int f(u)\,du$, which may be easier to evaluate. After finding an antiderivative $F(u) + C$, we **undo the substitution** by replacing u with $g(x)$.

Remark. The expression $du = g'(x)\,dx$ is, for now, **just notation**. There is a way to give precise meaning to the differential using so-called *differential forms*, but that is beyond the scope of this course. The justification for the notation is that it produces the correct result.

Example 9.2. Compute $\displaystyle\int \frac{(\ln x)^4}{x}\,dx$.

Solution. The integrand is written in terms of $\ln x$, and the derivative $\frac{1}{x}$ already appears. Set $u = \ln x$, so $du = \frac{1}{x}\,dx$. Then

$$\int \frac{(\ln x)^4}{x}\,dx = \int u^4\,du = \frac{1}{5}u^5 + C = \frac{1}{5}(\ln x)^5 + C.$$

In general, every time the integrand is written in terms of a function $g(x)$ and the derivative $g'(x)$ already appears in the integrand, the substitution $u = g(x)$ is a good idea. After substituting, we hope to find an easier integral that we can complete.

9.2 Substitution Examples

We now work through several computational examples of the Substitution Rule, in roughly increasing order of difficulty.

Remark (Pause and Try). Before reading each solution, pause and attempt the computation yourself.

Example 9.3. Compute $\int x^2 \sin(x^3+7)\,dx$.

Solution. The argument of sin is x^3+7, and its derivative $3x^2$ is essentially present in the integrand (up to a constant). Set $u = x^3+7$, so $du = 3x^2\,dx$. Then

$$\int x^2 \sin(x^3+7)\,dx = \frac{1}{3}\int \sin(u)\,du = -\frac{1}{3}\cos(u) + C = -\frac{1}{3}\cos(x^3+7) + C.$$

Example 9.4. Compute $\int \tan(x)\,dx$.

Solution. Write $\tan x = \dfrac{\sin x}{\cos x}$. The numerator $\sin x$ is the derivative of $\cos x$ (up to a sign). Set $u = \cos x$, so $du = -\sin x\,dx$. Then

$$\int \tan(x)\,dx = \int \frac{\sin x}{\cos x}\,dx = -\int \frac{du}{u} = -\ln|u| + C = -\ln|\cos x| + C.$$

Example 9.5. Compute $\int x^3\sqrt{x^2+1}\,dx$.

Solution. The square root makes this difficult. Try $u = x^2+1$, so $du = 2x\,dx$. Separate $x^3 = x^2 \cdot x$, and note that $x^2 = u-1$. Then

$$\int x^3\sqrt{x^2+1}\,dx = \frac{1}{2}\int (u-1)\sqrt{u}\,du = \frac{1}{2}\int \left(u^{3/2} - u^{1/2}\right)du.$$

Integrating term by term:

$$\frac{1}{2}\left(\frac{u^{5/2}}{5/2} - \frac{u^{3/2}}{3/2}\right) + C = \frac{1}{5}(x^2+1)^{5/2} - \frac{1}{3}(x^2+1)^{3/2} + C.$$

Remark. Among these examples, the last one best illustrates the full power of the Substitution Rule. The first two could arguably be done by guess-and-check, but in the third example the substitution transforms a complex integrand into a simple polynomial — something that would be very difficult to guess directly.

9.3 Substitution for Definite Integrals

We have used the Substitution Rule for indefinite integrals. When computing a definite integral, there are two approaches: we can first find an antiderivative (substituting, integrating, undoing the substitution), or we can apply the substitution directly to the definite integral — including the endpoints. The second method is slightly faster because we never need to undo the substitution.

Example 9.6. Compute $\int_1^2 x\,(x^2+1)^{100}\,dx$.

Solution. Set $u = x^2+1$, so $du = 2x\,dx$. We also change the endpoints:

$$x = 1 \implies u = 2, \qquad x = 2 \implies u = 5.$$

Then

$$\int_1^2 x\,(x^2+1)^{100}\,dx = \frac{1}{2}\int_2^5 u^{100}\,du = \frac{1}{2}\cdot\frac{u^{101}}{101}\bigg|_2^5 = \frac{5^{101}-2^{101}}{202}.$$

Remark (Warning). When performing a substitution in a definite integral, you must change **everything**: the integrand, the differential, **and** the endpoints. The values 1 and 2 are values of x; after substitution, the endpoints must be the corresponding values of u.

Theorem 9.2 (Substitution Rule for Definite Integrals). *Let $a, b \in \mathbb{R}$ with $a < b$. Let g be a function that is differentiable on $[a, b]$ with g' continuous on $[a, b]$, and let f be continuous on the range of g restricted to $[a, b]$. Then*

$$\int_a^b f(g(x))\cdot g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

Proof. Let F be an antiderivative of f. By Theorem 9.1, $F(g(x))$ is an antiderivative of $f(g(x))\cdot g'(x)$. By the Fundamental Theorem of Calculus,

$$\int_a^b f(g(x))\cdot g'(x)\,dx = F(g(b)) - F(g(a)) = \int_{g(a)}^{g(b)} f(u)\,du.$$

Uses Theorem 9.1 and the FTC.

■

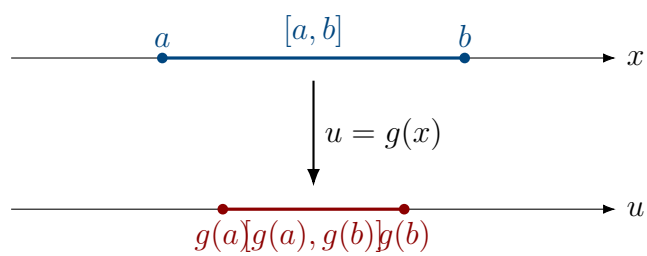


Figure 9.1: Substitution as an interval mapping.

9.4 Integration by Parts

Integration by parts is a method to help compute antiderivatives. It is an easy-to-use algorithm to make things a bit simpler and a bit more systematic than pure guess-and-check. Before presenting examples, let us explain where the formula comes from and why it works.

Motivation by example

Example 9.7. Compute $\int x\,e^x\,dx$ by guess-and-check.

Solution. We are looking for a function whose derivative is $x\,e^x$. A first guess: try $x\,e^x$ itself. By the product rule:

$$\frac{d}{dx}[x\,e^x] = 1 \cdot e^x + x\,e^x = e^x + x\,e^x.$$

The desired function $x\,e^x$ appears, but with an extra e^x. If we can find a second function whose derivative is exactly e^x, we subtract and obtain:

$$\frac{d}{dx}[x\,e^x - e^x] = (e^x + x\,e^x) - e^x = x\,e^x.$$

Therefore $\int x\,e^x\,dx = x\,e^x - e^x + C$.

The process we just used — applying the product rule, identifying an extra term, and subtracting a correction — is precisely the idea behind integration by parts. The formula below does the same thing in a more systematic way.

The formula

Theorem 9.3 (Integration by Parts / Backwards Product Rule). *Let f and g be differentiable functions. Then*

$$\int f(x)\,g'(x)\,dx = f(x)\,g(x) - \int f'(x)\,g(x)\,dx.$$

Proof. The product rule gives

$$\frac{d}{dx}\big[f(x)\,g(x)\big] = f'(x)\,g(x) + f(x)\,g'(x).$$

Taking antiderivatives of both sides and rearranging: *Product rule reversed.*

$$\int f(x)\,g'(x)\,dx = f(x)\,g(x) - \int f'(x)\,g(x)\,dx. \qquad \blacksquare$$

Remark. Despite its traditional name, integration by parts should really be called the *backwards product rule*: we take the product rule (a differentiation rule) and reverse it to obtain a rule about antiderivatives.

The $u-v$ notation

Setting $u = f(x)$, $dv = g'(x)\,dx$, so that $du = f'(x)\,dx$ and $v = g(x)$, the formula becomes

$$\int u\,dv = u\,v - \int v\,du.$$

product rule intuition: $d(uv) = u\,dv + v\,du$ (ignoring $du\,dv$)

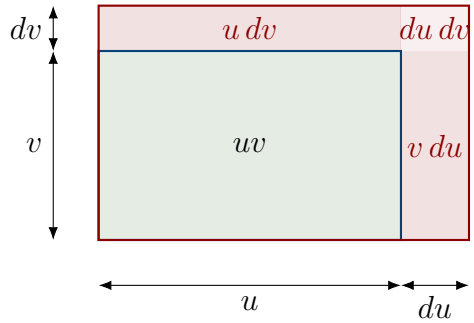


Figure 9.2: A geometric intuition behind the product rule and integration by parts.

Remark. As with the substitution notation, the expressions $du = f'(x)\,dx$ and $dv = g'(x)\,dx$ are **just notation** in this course. The formula $\int u\,dv = u\,v - \int v\,du$ is compact, easy to remember, and easy to use.

Example 9.8. Compute $\displaystyle\int x\,e^x\,dx$ using integration by parts.

Solution. We write the integral as $\int u\,dv$ and choose $u = x$, $dv = e^x\,dx$. Then $du = dx$, $v = e^x$. Applying the formula:

$$\int x\,e^x\,dx = x\,e^x - \int e^x\,dx = x\,e^x - e^x + C.$$

This is the same result as before, but obtained more quickly and systematically. When choosing u and dv, the goal is to make the transformed integral $\int v\,du$ easier than the original. The function whose derivative is simpler should be u; the function whose antiderivative is manageable should be dv.

9.5 Integration by Parts Examples

We now work through several examples of integration by parts, in roughly increasing order of difficulty.

Remark (Pause and Try). Before reading each solution, pause and attempt the computation yourself.

Example 9.9. Compute $\displaystyle\int x\cos(x)\,dx$.

Solution. For $\cos x$, both the derivative and the antiderivative are essentially the same (up to sign), so it does not matter. For x, however, the derivative (1) is simpler than the antiderivative ($\frac{1}{2}x^2$). So choose $u = x$, $dv = \cos(x)\,dx$. Then $du = dx$, $v = \sin x$.

$$\int x\cos(x)\,dx = x\sin x - \int \sin(x)\,dx = x\sin x + \cos x + C.$$

Example 9.10. Compute $\int x^2 e^{-x}\,dx$.

Solution. The derivative of x^2 is $2x$ (simpler), so take $u = x^2$, $dv = e^{-x}\,dx$. Then $du = 2x\,dx$, $v = -e^{-x}$.

$$\int x^2 e^{-x}\,dx = -x^2 e^{-x} + \int 2x\,e^{-x}\,dx.$$

The new integral is simpler, but not yet finished. Apply integration by parts a second time: $u = 2x$, $dv = e^{-x}\,dx$, $du = 2\,dx$, $v = -e^{-x}$.

$$\begin{aligned}\int x^2 e^{-x}\,dx &= -x^2 e^{-x} + \left[-2x\,e^{-x} + \int 2\,e^{-x}\,dx\right] \\ &= -x^2 e^{-x} - 2x\,e^{-x} - 2\,e^{-x} + C.\end{aligned}$$

Remark. Each application of integration by parts reduced the power of x by one while keeping the exponential. When the integrand is $x^n e^{\alpha x}$, repeated integration by parts (applied n times) will reduce the problem to an integral with no polynomial factor.

Example 9.11. Compute $\int \arctan(x)\,dx$.

Solution. This does not look like a product, but we can write it as $1 \cdot \arctan(x)$. The derivative of $\arctan(x)$ is $\dfrac{1}{1+x^2}$, which is simpler — so our priority is to *differentiate away* the arctan. Choose $u = \arctan(x)$, $dv = dx$. Then $du = \dfrac{1}{1+x^2}\,dx$, $v = x$.

$$\int \arctan(x)\,dx = x\arctan(x) - \int \frac{x}{1+x^2}\,dx.$$

The arctan is gone. For the remaining integral, note that the numerator x is (up to a constant) the derivative of the denominator $1 + x^2$. Set $t = 1 + x^2$, $dt = 2x\,dx$:

$$\int \frac{x}{1+x^2}\,dx = \frac{1}{2}\int \frac{dt}{t} = \frac{1}{2}\ln|t| + C = \frac{1}{2}\ln(1+x^2) + C.$$

Therefore

$$\int \arctan(x)\,dx = x\arctan(x) - \frac{1}{2}\ln(1+x^2) + C.$$

Remark. The absolute value is unnecessary in $\ln(1 + x^2)$ because $1 + x^2 > 0$ for all x.

9.6 Cyclic Integration by Parts

This section presents a somewhat peculiar use of integration by parts, in which the method appears to fail but actually succeeds through an algebraic trick.

Remark (Pause and Try). Before reading the solution below, try to compute $\int e^x \sin(x)\,dx$ by integration by parts. Decide whether it helps or not, and only then continue reading.

Example 9.12. Compute $\int e^x \sin(x)\,dx$.

Solution. Both e^x and $\sin x$ are essentially unchanged by differentiation — e^x stays the same, and $\sin x$ becomes $\pm\cos x$. So *whatever we choose*, the transformed integral is not any easier. Let us try anyway.

First application. Take $u = e^x$, $dv = \sin(x)\,dx$. Then $du = e^x\,dx$, $v = -\cos x$.

$$\int e^x \sin(x)\,dx = -e^x \cos x + \int e^x \cos(x)\,dx.$$

As expected, the new integral is no simpler. Nevertheless, apply integration by parts a second time, keeping the same convention (u = exponential, dv = trig): $u = e^x$, $dv = \cos(x)\,dx$, $du = e^x\,dx$, $v = \sin x$.

$$\int e^x \sin(x)\,dx = -e^x \cos x + e^x \sin x - \int e^x \sin(x)\,dx.$$

The trick. The original integral has reappeared on the right! Let $I = \int e^x \sin(x)\,dx$. Then

$$I = -e^x \cos x + e^x \sin x - I.$$

Solving for I:

$$2I = e^x(\sin x - \cos x), \qquad I = \frac{1}{2}\,e^x(\sin x - \cos x) + C.$$

Remark (Warning). When using integration by parts twice in this cyclic fashion, you must keep the same convention for which factor is u and which is dv in both applications. If you swap them the second time, you will undo the first integration by parts and obtain the useless tautology $\int e^x \sin(x)\,dx = \int e^x \sin(x)\,dx$.

Remark. The integration constant C deserves a comment. The symbol I represents a *family* of antiderivatives (a function plus an arbitrary constant). Both sides of the equation $I = (\text{something}) - I$ carry an arbitrary additive constant, but those constants need not agree. When we solve for I and write the final answer, we include $+C$ to reflect this.

9.7 Trigonometric Integrals: Odd Powers

We now consider methods for integrating products of trigonometric functions. Rather than presenting a collection of formulas to memorise, we will develop

the methods from scratch so that they can be re-derived on the spot whenever needed.

The goal is to compute integrals of the form

$$\int \sin^n x \cos^m x \, dx,$$

where n and m are non-negative integers. The strategy depends on the parity of the exponents. In this section we handle the case when at least one exponent is odd.

Example 9.13. Compute $\int \sin^5 x \cos^2 x \, dx$.

Solution. There are two substitutions one might try: $u = \sin x$ or $u = \cos x$.

Attempt 1: $u = \sin x$, $du = \cos x \, dx$. Separate one $\cos x$ for the differential. The remaining $\cos x$ must be converted to sines:

$$\cos x = \pm\sqrt{1 - \sin^2 x} = \pm\sqrt{1 - u^2}.$$

This introduces a square root — the integral becomes *harder*, not easier.

Attempt 2: $u = \cos x$, $du = -\sin x \, dx$. Separate one $\sin x$ for the differential; the remaining $\sin^4 x = (\sin^2 x)^2 = (1 - \cos^2 x)^2$ converts cleanly:

$$\int \sin^5 x \cos^2 x \, dx = -\int (1 - u^2)^2 u^2 \, du.$$

No square roots. The integrand is a polynomial in u, which we can expand and integrate term by term.

The lesson is clear. When substituting $u = \cos x$, we need one $\sin x$ for the differential; the remaining $\sin^{n-1} x$ must be an even power so that we can use $\sin^2 x = 1 - \cos^2 x$ without square roots. That requires n to be odd. Similarly, $u = \sin x$ works when m is odd.

Proposition 9.4 (Substitution Strategy for $\sin^n x \cos^m x$). *Consider $\int \sin^n x \cos^m x \, dx$.*

- *If n is odd, use $u = \cos x$. Separate one $\sin x$ for du; convert the remaining (even) powers of $\sin$ using $\sin^2 x = 1 - \cos^2 x$.*
- *If m is odd, use $u = \sin x$. Separate one $\cos x$ for du; convert the remaining (even) powers of $\cos$ using $\cos^2 x = 1 - \sin^2 x$.*

In either case the integral reduces to a polynomial in u.

Remark. If **both** n and m are even, neither substitution avoids square roots, and a different method is needed. That is the topic of Section 9.8.

These same ideas extend to products of powers of secant and tangent. Since $\sec^2 x = 1 + \tan^2 x$ and the derivatives $\frac{d}{dx}[\tan x] = \sec^2 x$ and $\frac{d}{dx}[\sec x] = \sec x \tan x$ involve the same functions, one may try either $u = \tan x$ or $u = \sec x$.

Remark (Pause and Try). Consider the three integrals $\int \sec^4 x \tan^3 x\,dx$, $\int \sec^3 x \tan^4 x\,dx$, and $\int \sec^2 x \tan^5 x\,dx$. Using the identity $\sec^2 x = 1 + \tan^2 x$ and the derivatives above, determine which substitution ($u = \sec x$ or $u = \tan x$) is helpful for each integral.

9.8 Trigonometric Integrals: Even Powers

When both exponents in $\int \sin^n x \cos^m x\,dx$ *are even, the substitutions from Section 9.7 do not help. We need a different approach.*

Remark. Is this an important problem? Historically, these integrals appeared often in science and technology. Today we have computer algebra systems, and nobody integrates by hand for practical purposes. However, there is still value: we should look at these calculations as **exercises in problem solving** — how to break a complex problem into cases and put together various algorithms. That is where their value comes from, rather than from the specific integrals themselves.

Method 1: Half-angle identities

The key identities are:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

These replace squares of sin and cos with expressions involving $\cos 2x$, which is easy to integrate.

Example 9.14. Compute $\int \sin^2 x\,dx$.

Solution. Using the half-angle identity:

$$\int \sin^2 x\,dx = \frac{1}{2}\int dx - \frac{1}{2}\int \cos(2x)\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

The same method works for higher even powers. For example, $\sin^4 x \cos^2 x = (\sin^2 x)^2 \cos^2 x$ *can be rewritten using the half-angle formulas; after expanding, each resulting term will either be easy to integrate directly, have an odd exponent (handled by Section 9.7), or have a lower even exponent (to which the half-angle formulas are applied again).*

Remark (Pause and Try). Compute $\int \sin^4 x\,dx$ using this method.

Method 2: Integration by parts

Example 9.15. Compute $\int \sin^2 x\,dx$ using integration by parts.

Solution. Write $\sin^2 x = \sin x \cdot \sin x$. Choose $u = \sin x$, $dv = \sin x\, dx$. Then $du = \cos x\, dx$, $v = -\cos x$.

$$\int \sin^2 x\, dx = -\sin x \cos x + \int \cos^2 x\, dx.$$

Now use $\sin^2 x + \cos^2 x = 1$, so $\int \cos^2 x\, dx = \int dx - \int \sin^2 x\, dx = x - \int \sin^2 x\, dx$. Substituting:

$$\int \sin^2 x\, dx = -\sin x \cos x + x - \int \sin^2 x\, dx.$$

Solving:

$$2\int \sin^2 x\, dx = x - \sin x \cos x, \qquad \int \sin^2 x\, dx = \frac{x - \sin x \cos x}{2} + C.$$

Remark. Both methods give the same answer (verify using $\sin 2x = 2 \sin x \cos x$). The integration-by-parts method can in principle be used for any even-power product, although it can easily get messy.

9.9 Integration of Secant

The integral of secant is a historically important problem. In the early years of calculus in the 17th century, computing $\int \sec x\, dx$ was needed in cartography, and mathematicians struggled for a while to find a solution.

Method 1: The textbook trick

Example 9.16. Compute $\int \sec x\, dx$ by multiplying and dividing by $\sec x + \tan x$.

Solution.

$$\int \sec x\, dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x}\, dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\, dx.$$

Set $u = \sec x + \tan x$, $du = (\sec x \tan x + \sec^2 x)\, dx$. The numerator is exactly du:

$$\int \sec x\, dx = \int \frac{du}{u} = \ln|u| + C = \ln|\sec x + \tan x| + C.$$

Remark. This method is fast and simple, but it raises an uncomfortable question: how on earth would one think to multiply and divide by $\sec x + \tan x$? This seems like too great a leap, and there is no satisfying explanation for why one would try it, other than hindsight.

Method 2: A more natural approach

The following method is longer, but each step is natural and could be discovered by any student with enough integration practice.

Write $\sec x = \dfrac{1}{\cos x}$. Thinking of this as $\cos^{-1} x$ (i.e. $\sin^0 x \cos^{-1} x$), the exponent of cosine is odd (namely -1), so by the strategy of Section 9.7, the substitution $u = \sin x$ should help.

Multiply numerator and denominator by $\cos x$:

$$\int \sec x \, dx = \int \frac{\cos x}{\cos^2 x} \, dx = \int \frac{\cos x}{1 - \sin^2 x} \, dx.$$

Set $u = \sin x$, $du = \cos x \, dx$:

$$\int \sec x \, dx = \int \frac{du}{1 - u^2}.$$

This is a rational function — a quotient of polynomials. It can be integrated using partial fraction decomposition (see Section 9.10).

Remark. This second method is longer, but every step is natural: rewrite in terms of sine and cosine, apply the standard substitution strategy, and reduce to a rational function. Once one learns partial fractions, the problem can be completed.

9.10 Partial Fractions: Distinct Factors

A rational function is a quotient of polynomials. The key idea for integrating all rational functions is that there are a few simple rational functions whose antiderivatives we already know, and we try to rewrite every other rational function in terms of those simpler pieces.

Example 9.17. Compute $\displaystyle\int \frac{x+3}{(x-1)(x+1)} \, dx$.

Solution. The denominator makes this difficult to integrate directly. However, $\dfrac{1}{x-1}$ and $\dfrac{1}{x+1}$ are each easy to integrate. We seek constants A and B such that

$$\frac{x+3}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}.$$

Reducing the right-hand side to a common denominator:

$$\frac{A(x+1) + B(x-1)}{(x-1)(x+1)} = \frac{(A+B)x + (A-B)}{(x-1)(x+1)}.$$

Matching coefficients: $A + B = 1$ and $A - B = 3$. Solving gives $A = 2$, $B = -1$.

Alternatively, from the identity $x + 3 = A(x+1) + B(x-1)$, substitute $x = 1$ to get $4 = 2A$, so $A = 2$; substitute $x = -1$ to get $2 = -2B$, so $B = -1$.

Therefore

$$\int \frac{x+3}{(x-1)(x+1)}\,dx = \int \frac{2}{x-1}\,dx - \int \frac{1}{x+1}\,dx = 2\ln|x-1| - \ln|x+1| + C.$$

This can be written as $\ln\left|\frac{(x-1)^2}{x+1}\right| + C$.

The general strategy

The same method works whenever the denominator factors into distinct linear factors. Here is the general procedure.

Given a rational function $\frac{P(x)}{Q(x)}$:

1. **Long division.** If $\deg P \geq \deg Q$, perform polynomial long division to write $\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$, where $\deg R < \deg Q$. The polynomial $S(x)$ is easy to integrate; focus on the remainder.

2. **Factor the denominator.**

3. **Partial fraction decomposition.** If $Q(x) = (x-r_1)(x-r_2)\cdots(x-r_k)$ with distinct roots, write

$$\frac{R(x)}{Q(x)} = \frac{A_1}{x-r_1} + \frac{A_2}{x-r_2} + \cdots + \frac{A_k}{x-r_k}.$$

   Determine the constants A_i by reducing to a common denominator and matching coefficients (or by evaluating at the roots).

4. **Integrate** each term: $\int \frac{A_i}{x-r_i}\,dx = A_i \ln|x-r_i| + C$.

Remark. The formal proof that this decomposition always works (with the correct number of constants) requires linear algebra — it amounts to a change of basis in a finite-dimensional vector space.

9.11 Partial Fractions: Repeated Factors

When the denominator has a repeated linear factor, the decomposition from Section 9.10 must be modified. We need additional terms to account for the multiplicity.

Example 9.18. Compute $\int \frac{x^4+1}{x^3+2x^2+x}\,dx$.

Solution.

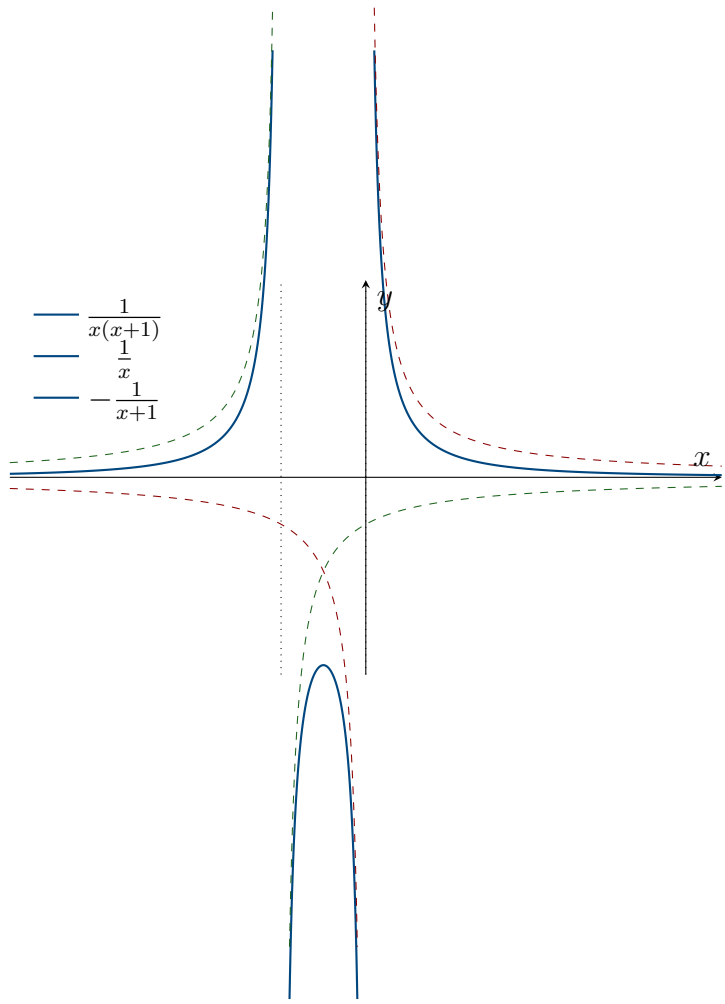


Figure 9.3: A rational function and its partial fraction pieces (qualitative shape).

Step 1: Long division. Since $\deg(x^4+1) = 4 > 3 = \deg(x^3+2x^2+x)$, we perform polynomial long division. The result is

$$\frac{x^4+1}{x^3+2x^2+x} = (x-2) + \frac{3x^2+2x+1}{x^3+2x^2+x}.$$

The polynomial $x-2$ is easy to integrate.

Step 2: Factor the denominator. $x^3+2x^2+x = x(x+1)^2$.

Step 3: Partial fractions. We seek constants A, B, C such that

$$\frac{3x^2+2x+1}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}.$$

Remark (Warning). A common mistake is to try only $\dfrac{A}{x} + \dfrac{B}{x+1}$. This has only two free constants, but the numerator on the left has three coefficients

(degree ≤ 2), so the system will generally be inconsistent. The term $\dfrac{C}{(x+1)^2}$ is needed to provide enough degrees of freedom.

Continuing the example: reducing to a common denominator and matching coefficients gives $A = 1$, $B = 2$, $C = -2$. Therefore:

$$\begin{aligned}\int \frac{x^4+1}{x^3+2x^2+x}\,dx &= \int (x-2)\,dx + \int \frac{1}{x}\,dx + \int \frac{2}{x+1}\,dx + \int \frac{-2}{(x+1)^2}\,dx \\ &= \frac{x^2}{2} - 2x + \ln|x| + 2\ln|x+1| + \frac{2}{x+1} + C.\end{aligned}$$

Remark. The antiderivative of $\dfrac{1}{(x+1)^2}$ is $\dfrac{-1}{x+1}$, because $\dfrac{d}{dx}\left[\dfrac{-1}{x+1}\right] = \dfrac{1}{(x+1)^2}$. This is just the power rule in disguise: $(x+1)^{-2}$ integrates to $\dfrac{(x+1)^{-1}}{-1}$.

9.12 Partial Fractions: Irreducible Quadratics

When the denominator contains an irreducible quadratic factor (one that does not factor over $\mathbb{R}$), the partial fraction approach requires different building blocks.

Example 9.19. Compute $\displaystyle\int \frac{3x+7}{x^2+4}\,dx$.

Solution. Since $\deg(3x+7) = 1 < 2 = \deg(x^2+4)$, no long division is needed. The denominator x^2+4 is an irreducible quadratic — it has no real roots.

Split the integrand into two pieces:

$$\int \frac{3x+7}{x^2+4}\,dx = 3\int \frac{x}{x^2+4}\,dx + 7\int \frac{1}{x^2+4}\,dx.$$

Each of these is a basic building block that we can integrate directly.

First integral. The numerator x is (up to a constant) the derivative of the denominator x^2+4. Set $u = x^2+4$, $du = 2x\,dx$:

$$\int \frac{x}{x^2+4}\,dx = \frac{1}{2}\int \frac{du}{u} = \frac{1}{2}\ln(x^2+4) + C.$$

(The absolute value is unnecessary since $x^2+4 > 0$ for all x.)

Second integral. This is related to arctan. Factor 4 from the denominator:

$$\int \frac{1}{x^2+4}\,dx = \int \frac{1}{4\left(\frac{x^2}{4}+1\right)}\,dx = \frac{1}{4}\int \frac{1}{\left(\frac{x}{2}\right)^2+1}\,dx.$$

Set $u = \frac{x}{2}$, $du = \frac{1}{2}\,dx$:

$$\frac{1}{4}\int \frac{2\,du}{u^2+1} = \frac{1}{2}\arctan(u) + C = \frac{1}{2}\arctan\left(\frac{x}{2}\right) + C.$$

Combining:

$$\int \frac{3x+7}{x^2+4}\,dx = \frac{3}{2}\ln(x^2+4) + \frac{7}{2}\arctan\left(\frac{x}{2}\right) + C.$$

Remark. When the denominator is an irreducible quadratic $x^2 + bx + c$, the two basic building blocks for integration are:

(i) $\dfrac{x + \text{const}}{x^2 + bx + c}$, where the numerator is the derivative of the denominator (handled by a substitution, yielding a logarithm);

(ii) $\dfrac{1}{(x-h)^2 + k^2}$ (after completing the square), which yields an arctangent.

Any rational function with an irreducible quadratic denominator can be split into a combination of these two types.

Chapter 10

Volumes

10.1 Volumes by Slicing

We defined integrals in order to compute areas, but there are many other applications of integrals. In this section we revisit the definition of the Riemann integral, reinterpret the notation with a different geometrical flavour, and use the same idea to set up volumes as integrals.

Reinterpreting the integral

Recall the idea behind the Riemann integral. Given a region bounded by the graph of a function f, the x-axis, and the vertical lines $x = a$ and $x = b$, we break $[a, b]$ into smaller pieces using a partition. This cuts the region into thin vertical slices. Each slice is approximated by a rectangle of width Δx and height $f(x)$, so its area is approximately $f(x) \cdot \Delta x$. Summing over all slices gives an approximation to the total area, and taking the limit as $\Delta x \to 0$ yields the definite integral.

What follows is not a fully rigorous argument, but it provides powerful geometric intuition and helps us reuse the same ideas for many applications of the integral beyond areas.

Since the integral is a limit of sums, we abuse notation and think of it as an actual sum of infinitely many, infinitesimally small pieces. Specifically, we write dx and think of it as a Δx that is **infinitesimally small**. The integral sign then represents the "sum" over infinitely many such pieces.

Consider the same region as before. Instead of finitely many slices, imagine breaking it into infinitely many slices, each of infinitesimal width dx. Focus on a single slice. Its width is dx, and since the slice is infinitesimally thin, its area is also infinitesimal; call it dA.

Remark. When we approximate such a slice by a rectangle, the error is on the order of $(dx)^2$: it is infinitesimally small *squared*. Compared with the slice itself, this error is negligible, and we may ignore it in the limit.

Therefore the area of the slice is

$$dA = f(x)\,dx,$$

and the total area is the "sum" of all such infinitesimal pieces:

$$A = \int_a^b f(x)\,dx.$$

From areas to volumes

The advantage of thinking of integrals this way is that it helps with geometric intuition, and it helps us reuse the same ideas for many other applications — for example, computing volumes.

Suppose we have a solid (picture a carrot) whose volume we wish to compute. If we approximate it crudely by a single cylinder, the result is poor. But just as we sliced a region into thin rectangles for areas, we can **slice the solid** into thin pieces. Each slice is approximately a disc, and we know how to compute the volume of a disc. Taking the limit as the thickness tends to zero, the approximation becomes exact.

Definition 10.1 (Volume by slicing — disc method). Let a solid be generated by rotating the region between the graph of a function $y = f(x)$ and the x-axis, from $x = a$ to $x = b$, around the x-axis. Cut the solid into infinitesimally thin slices perpendicular to the x-axis. Each slice is a disc of radius $R = |f(x)|$ and thickness dx. Its infinitesimal volume is

$$dV = \pi R^2\,dx = \pi\big[f(x)\big]^2\,dx,$$

and the total volume is

$$V = \int_a^b \pi\big[f(x)\big]^2\,dx.$$

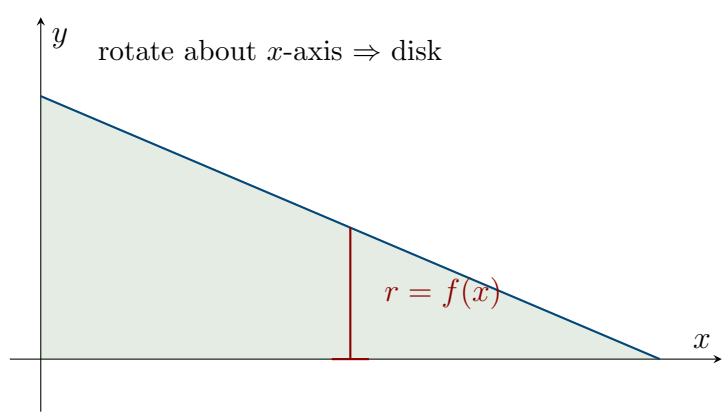


Figure 10.1: Disk method intuition: a slice rotates into a disk.

Remark. Just as with areas, the key step is identifying the infinitesimal piece (a thin disc), writing its volume in terms of the variable of integration, and then summing (integrating) over all such pieces.

A concrete example

Example 10.1. Let R be the region in the first quadrant between the graph of $y = x^2$ and the x-axis, from $x = 0$ to $x = 1$. Rotate R around the x-axis. Compute the volume of the resulting solid.

Remark (Pause and Try). Before reading the solution, draw a picture and try to understand what the solid looks like. It resembles a cone that has been bent slightly.

Solution. Slice the solid into infinitesimally thin pieces perpendicular to the x-axis. A single slice is obtained by taking a thin vertical strip of width dx from the region R and rotating it around the x-axis. This produces a disc.

The disc has thickness dx and radius equal to the height of the strip, which is $y = x^2$. Therefore the infinitesimal volume is

$$dV = \pi\left(x^2\right)^2 dx = \pi x^4\, dx.$$

Integrating from $x = 0$ to $x = 1$:

$$V = \int_0^1 \pi x^4\, dx = \pi \cdot \frac{x^5}{5}\bigg|_0^1 = \frac{\pi}{5}.$$

Remark. The strategy is always the same: slice the solid into infinitesimally thin pieces, write the volume of each piece, and integrate. The calculations may be more involved for other solids, but the underlying idea is identical — slice the carrot infinitesimally thin.

10.2 Cylindrical Shells

In the previous section, we computed volumes by slicing a solid perpendicular to the axis of revolution, producing discs (or washers). In this section, we work with a different example that requires a different approach. The key idea is still slicing the solid into infinitesimally thin pieces; we simply slice in a different direction.

When the disc method fails

Example 10.2. Let R be the region in the first quadrant between the graph of $y = 2x^4 - x^5$ and the x-axis. Rotate R around the y-axis. Compute the volume of the resulting solid.

Remark (Pause and Try). Before reading on, draw a picture and try to visualise the solid. It looks like a small volcano — a mountain with a hole in the centre. The curve $y = 2x^4 - x^5$ forms the profile, and rotating it around the y-axis produces the volcano shape.

Let us first attempt the approach from Section 10.1: slice perpendicular to the axis of revolution (the y-axis). This means slicing horizontally.

A thin horizontal strip of height dy in the region R corresponds, after rotation, to a *washer*: a disc with a hole. If the inner radius is r and the outer radius is R, the infinitesimal volume would be

$$dV = \pi\big(R^2 - r^2\big)\,dy.$$

To finish the setup, we would integrate with respect to y from 0 to the maximum value of the function, and we would need to express R and r in terms of y. That means solving

$$y = 2x^4 - x^5$$

for x in terms of y.

Remark (Warning). This equation cannot be solved for x by elementary methods. The plan was a good idea and worth trying, but ultimately we cannot complete the problem this way. We need a different approach.

The cylindrical shell method

Instead of slicing horizontally (perpendicular to the y-axis), we slice the region ***vertically*** *(parallel to the y-axis) and integrate with respect to x.*

Take a thin vertical strip of width dx in the region R. When this strip is rotated around the y-axis, it sweeps out a *cylindrical shell*: a thin hollow cylinder.

Definition 10.2 (Volume by cylindrical shells). When a thin vertical strip at position x, of height $y = f(x)$ and width dx, is rotated about the y-axis, it produces a cylindrical shell of

- **radius** $|x|$ (the distance from the strip to the axis),
- **height** $y = f(x)$,
- **thickness** dx.

Because the thickness is infinitesimally small, we may "unroll" the shell into a thin rectangular brick of dimensions $2\pi x \times f(x) \times dx$. The infinitesimal volume is therefore

$$dV = 2\pi x \cdot f(x)\,dx,$$

and the total volume is

$$V = \int_a^b 2\pi x \cdot f(x)\,dx.$$

Remark. Strictly speaking, unrolling the shell into a brick is an approximation. But because the thickness dx is infinitesimally small, the error from flattening the curvature is negligible — it is of order $(dx)^2$. If the thickness were not infinitesimal, the error could be significant.

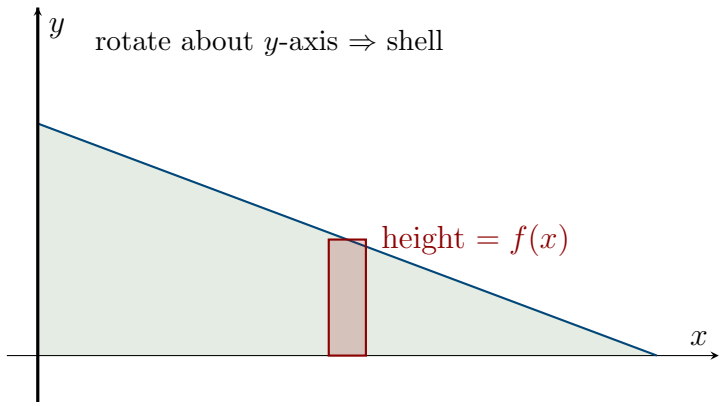


Figure 10.2: Shell method intuition: a strip rotates into a shell.

Completing the example

We return to Example 10.2. Slice the region R vertically. A thin strip at position x has width dx and height $y = 2x^4 - x^5$. Rotating this strip around the y-axis produces a cylindrical shell of radius x, height $2x^4 - x^5$, and thickness dx. Its infinitesimal volume is

$$dV = 2\pi x\big(2x^4 - x^5\big)\,dx.$$

Since $y = 2x^4 - x^5 = x^4(2 - x)$, the function is zero at $x = 0$ and $x = 2$, so we integrate from 0 to 2:

$$V = \int_0^2 2\pi x\big(2x^4 - x^5\big)\,dx = 2\pi \int_0^2 \big(2x^5 - x^6\big)\,dx.$$

This integral can now be evaluated using antiderivatives.

Comparing the two methods

Both methods are fundamentally the same idea: cut the solid into infinitesimally small pieces and integrate. We simply happen to cut in different directions.

Definition 10.3 (Washer method vs. cylindrical shell method).

- The *washer method* (or disc method): we cut the region **perpendicular** to the axis of revolution. Each piece is a disc or washer.
- The *cylindrical shell method*: we cut the region **parallel** to the axis of revolution. Each piece is a cylindrical shell.

Remark. The names are less important than the underlying principle. In any given problem, one method may lead to easier integrals than the other. When one approach produces an equation that cannot be solved, try slicing in the other direction.

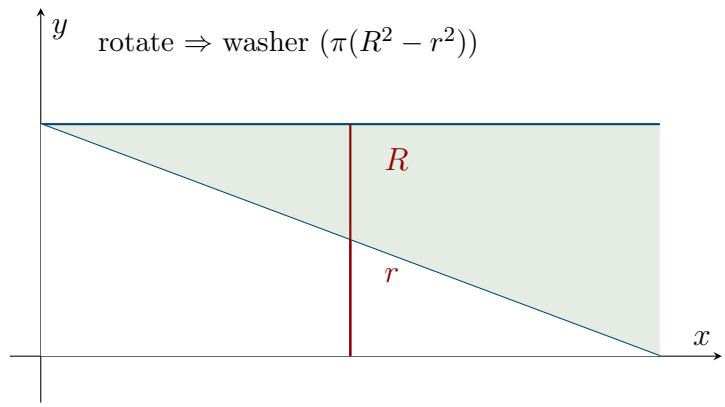


Figure 10.3: Washer method intuition: outer minus inner disk.

Chapter 11

Sequences

11.1 Sequences and Notation

Before diving into the theory of sequences, we need a precise definition and a tour of the various notational conventions that appear throughout the subject. A sequence is simply a special type of function, but by tradition we use different notation for it.

Definition 11.1 (Sequence). A *sequence* is a function whose domain is the set of all integers greater than or equal to some fixed integer n_0. That is, the domain has the form

$$\{n \in \mathbb{Z} : n \geq n_0\}$$

for some fixed $n_0 \in \mathbb{Z}$.

Remark. The most common choices are $n_0 = 0$ and $n_0 = 1$, so the domain is $\mathbb{N}$ or $\{1, 2, 3, \dots\}$. However, some sequences are naturally defined only for $n \geq 2$ or even $n \geq 7$. For instance, the sequence $\frac{1}{\ln n}$ is only defined for $n \geq 2$, because $\ln 1 = 0$.

Conventions distinguishing sequences from functions

Ultimately, sequences are functions. Nevertheless, we adopt several conventions to distinguish between a function whose domain is an interval (or a union of intervals) and a function whose domain consists only of natural numbers.

- **Variable name.** For a "function" (domain an interval) we typically use x as the variable; for a sequence we use n. There is no deep reason for this, but the convention signals that n takes only integer values.

- **Value notation.** If f is a function, we write $f(x)$. If a is a sequence, we write a_n (a subscript) rather than $a(n)$.

- **Listing terms.** Because the domain consists of consecutive integers, the values of a sequence can be listed in order:

$$a_0,\ a_1,\ a_2,\ a_3,\ \dots$$

Remark (Warning). Writing the first few terms of a sequence in curly braces, such as $\{1, 2, 4, 8, 16, \dots\}$, is common but potentially misleading: a sequence is **not** a set. The set is the *range* of the sequence; the sequence itself also specifies the *order* of its terms.

Ways to describe a sequence

There are four common ways to define a sequence.

1. **By an explicit formula.** For example, $a_n = \dfrac{2^n \cdot n!}{n+1}$. The factorial makes clear that n must be a natural number.
2. **By listing the first few terms.** For example, 1, 2, 4, 8, 16, ...
3. **By a verbal description.** For example, "let p_n be the nth prime number." This is perfectly valid when no nice closed-form formula exists.
4. **By a recurrence relation.** For example, the *Fibonacci sequence* $\{F_n\}$ is defined by

$$F_1 = 1, \quad F_2 = 1, \quad F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 3.$$

Remark (Warning). The first few terms of a sequence **never fully determine** the sequence. For instance, the terms $1, 2, 4, 8, 16, \dots$ could arise from $a_n = 2^n$, but they could equally arise from $a_n = 2^n + (n-0)(n-1)(n-2)(n-3)(n-4)$ or from many other formulas. When there is any ambiguity, always give an explicit formula or recurrence.

Notation for sequences

The most complete notation for a sequence is

$$\{a_n\}_{n=0}^{\infty}.$$

Common abbreviations include $\{a_n\}_{n\geq 0}$, $\{a_n\}_n$, $\{a_n\}$, or simply (a_n). Any of these is acceptable when unambiguous.

Remark (Warning). When there are multiple variables, be precise. For example, the sequence $\left\{\frac{n}{k}\right\}_{k=2}^{\infty}$ has variable k (not n), starting at $k = 2$. Here n is a constant. Omitting the index information would create genuine ambiguity.

11.2 Limits of Sequences

We now define the central concept of this chapter: what it means for a sequence to have a limit. If you are already familiar with the ε-δ definition of the limit of a function, the definition below will look very similar — we simply replace the domain of real numbers with the domain of natural numbers.

Motivation

Example 11.1. Consider the sequence $\left\{\dfrac{n}{n+1}\right\}_{n=0}^{\infty}$. The first few terms are

$$0,\ \frac{1}{2},\ \frac{2}{3},\ \frac{3}{4},\ \frac{4}{5},\ \ldots$$

Plotting these on the real line, we see that as n increases the terms crowd ever closer to 1. We say that the limit of the sequence is 1.

To make this precise, we think geometrically. Draw a small open interval centred at the candidate limit L. Some initial terms of the sequence may lie outside this interval, but after a certain point every remaining term lies inside. If this is true for ***every*** *open interval centred at L, no matter how small, then L is the limit.*

Definition 11.2 (Limit of a sequence). Let $\{a_n\}$ be a sequence and let $L \in \mathbb{R}$. We say that the *limit* of $\{a_n\}$ is L, and write

$$\lim_{n\to\infty} a_n = L,$$

when the following holds:

$$\forall\, \varepsilon > 0,\ \exists\, n_0 \in \mathbb{N},\ \forall\, n \in \mathbb{N}, \quad n \geq n_0 \implies |a_n - L| < \varepsilon.$$

Equivalently, $n \geq n_0$ implies $L - \varepsilon < a_n < L + \varepsilon$.

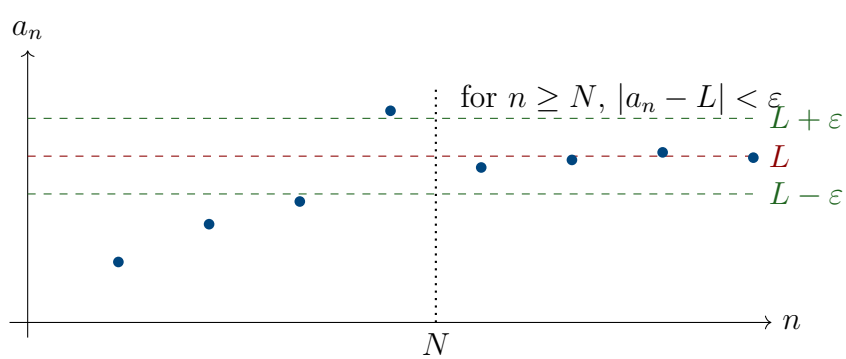


Figure 11.1: Convergence of a sequence in an ε-neighborhood.

In one sentence: a sequence $\{a_n\}$ has limit L when every open interval centred at L contains a ***tail*** *of the sequence — that is, all terms from some index onward.*

Remark. This one-sentence summary encodes all three quantifiers of the formal definition but is easier to remember and gives a clear geometric picture. It also highlights a fundamental principle: **to determine whether a sequence has a limit, only the tails of the sequence matter**. Changing the first 3, the first 12, or the first 60 terms can never affect whether the sequence converges.

Notation and terminology

When $\lim_{n\to\infty} a_n = L$, we also write $a_n \to L$ and say that the sequence *converges* to L.

Definition 11.3 (Convergent and divergent sequences). A sequence is *convergent* if it has a limit (which must be a real number). Otherwise, the sequence is *divergent*.

A divergent sequence may diverge to ∞ (e.g. $\{n^2\}$), to $-\infty$ (e.g. $\{-n\}$), or it may oscillate without tending to any value (e.g. $\{(-1)^n\}$, whose terms alternate between 1 and -1).

Remark (Pause and Try). Write a precise formal definition of "$\lim_{n\to\infty} a_n = \infty$", modelled on Definition 11.2.

11.3 Tools for Computing Limits

We do not want to appeal to the ε-definition every time we compute a limit. In this section we survey the main tools available, emphasising what carries over from limits of functions and what does not.

What works the same

Theorem 11.1 (Limit laws for sequences). *If $\{a_n\}$ and $\{b_n\}$ are convergent sequences with $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$, then*

(i) $\lim_{n\to\infty}(a_n \pm b_n) = A \pm B$,

(ii) $\lim_{n\to\infty}(a_n \cdot b_n) = A \cdot B$,

(iii) $\lim_{n\to\infty} \dfrac{a_n}{b_n} = \dfrac{A}{B}$, *provided* $B \neq 0$.

The proofs are identical to the corresponding proofs for limits of functions.

Theorem 11.2 (Squeeze Theorem for sequences). *If $a_n \leq b_n \leq c_n$ for all n (or at least for all sufficiently large n), and*

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L,$$

then $\lim_{n\to\infty} b_n = L$. Same theorem, same proof as for functions.

L'Hôpital's Rule: a caveat

*L'Hôpital's Rule does **not** directly extend to sequences, because we cannot take the derivative of a sequence. (What is the derivative of $n!$?) However, there is a useful workaround.*

Theorem 11.3 (From functions to sequences). *Let f be a function defined on $[c, \infty)$ for some integer c, and let $a_n = f(n)$ for every integer $n \geq c$.*

(i) If $\lim_{x\to\infty} f(x) = L \in \mathbb{R}$*, then* $\lim_{n\to\infty} a_n = L$*.*

(ii) If $\lim_{x\to\infty} f(x) = \infty$*, then* $\lim_{n\to\infty} a_n = \infty$*.*

Remark (Warning). If $\lim_{x\to\infty} f(x)$ does not exist (for instance because f oscillates), we **cannot** conclude anything about the sequence. The sequence may or may not converge. For example, the function $f(x) = \sin(\pi x)$ has no limit at ∞, but the sequence $a_n = \sin(\pi n) = 0$ is convergent.

Therefore, when a sequence $\{a_n\}$ "comes from" a function f (i.e. $a_n = f(n)$), we may apply L'Hôpital's Rule to f and use Theorem 11.3 to transfer the result to the sequence — provided the limit of f exists.

Composition with a continuous function

Theorem 11.4 (Continuous functions preserve sequential limits). *Let* $\{a_n\}$ *be a sequence with* $\lim_{n\to\infty} a_n = L$*, and let* f *be a function that is continuous at* L*. Then*

$$\lim_{n\to\infty} f(a_n) = f(L).$$

Remark (Pause and Try). Write a proof of Theorem 11.4. It is a great exercise because you must use **both** the definition of limit of a sequence and the definition of continuity.

Example 11.2. We compute $\lim_{n\to\infty} e^{1/n!}$. Since $\lim_{n\to\infty} \frac{1}{n!} = 0$ and the exponential function is continuous at 0, Theorem 11.4 gives

$$\lim_{n\to\infty} e^{1/n!} = e^0 = 1.$$

In summary, every tool you know for computing limits of functions as the variable approaches ∞ *carries over to sequences. The only exception is L'Hôpital's Rule, which requires extending the sequence to a function on the positive reals first.*

11.4 Monotonicity and Boundedness

In this section we introduce two structural properties of sequences — monotonicity and boundedness — and then state three theorems that describe how these properties interact with convergence.

Monotonic sequences

Definition 11.4 (Increasing and decreasing sequences). A sequence $\{a_n\}$ is

- *increasing* if $a_n < a_{n+1}$ for every n,
- *decreasing* if $a_n > a_{n+1}$ for every n,

- *non-decreasing* if $a_n \leq a_{n+1}$ for every n,
- *non-increasing* if $a_n \geq a_{n+1}$ for every n.

A sequence is *monotonic* if it satisfies any one of these four conditions.

Remark. For a sequence, "$a_n < a_{n+1}$ for every n" is equivalent to the more familiar condition "$n < m \Rightarrow a_n < a_m$." Both formulations are useful; the first is often simpler to check.

Example 11.3. The sequence $\{n^2\}$ is increasing. The sequence $\left\{\frac{1}{n+1}\right\}$ is decreasing. The sequence $1, 1, 2, 2, 3, 3, 4, 4, \ldots$ is non-decreasing but not increasing.

Definition 11.5 (Eventually monotonic). A sequence $\{a_n\}$ is *eventually decreasing* if there exists an index n_0 such that $a_n > a_{n+1}$ for every $n \geq n_0$. The terms *eventually increasing*, *eventually non-decreasing*, and *eventually monotonic* are defined similarly.

Example 11.4. Consider the sequence $\{n^3 e^{-n}\}_{n=0}^{\infty}$. Define $f(x) = x^3 e^{-x}$. Then

$$f'(x) = e^{-x}(3x^2 - x^3) = x^2 e^{-x}(3 - x).$$

Since $f'(x) < 0$ for all $x > 3$, the function f is decreasing on $[3, \infty)$. Therefore the sequence $\{n^3 e^{-n}\}$ is decreasing for $n \geq 3$; that is, it is *eventually* decreasing.

Bounded sequences

Definition 11.6 (Bounded sequence). A sequence $\{a_n\}$ is

- *bounded below* if there exists $A \in \mathbb{R}$ with $A \leq a_n$ for every n,
- *bounded above* if there exists $B \in \mathbb{R}$ with $a_n \leq B$ for every n,
- *bounded* if it is both bounded below and bounded above.

Remark. It is not worthwhile to define "eventually bounded." Can you see why? If a sequence is eventually bounded, it is already bounded: the finitely many initial terms that were excluded are themselves a finite set of real numbers, so they cannot make the sequence unbounded.

Remark (Pause and Try). Construct eight sequences, one for each combination of bounded/unbounded, monotonic/not monotonic, and convergent/divergent. Some combinations may be impossible — discover which ones.

Three fundamental theorems

The following three results describe the precise relationship between convergence, monotonicity, and boundedness. Their proofs will occupy the next two sections.

Theorem 11.5 (Convergent sequences are bounded). *If a sequence is convergent, then it is bounded.*

Theorem 11.6 (Monotone Convergence Theorem for sequences). *If a sequence is eventually monotonic and bounded, then it is convergent.*

Equivalently:

(i) If a sequence is eventually non-decreasing and bounded above, then it is convergent.

(ii) If a sequence is eventually non-increasing and bounded below, then it is convergent.

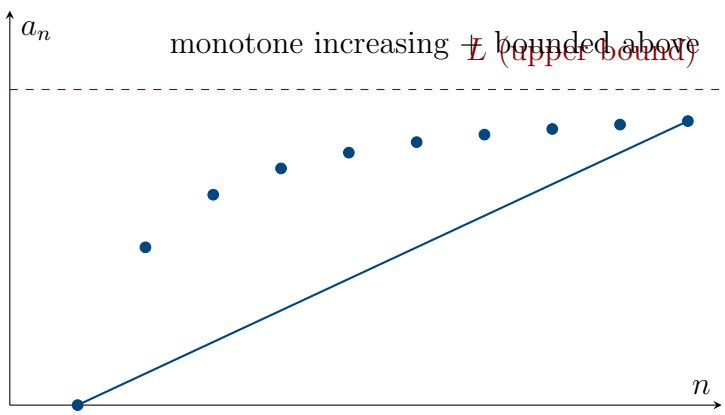


Figure 11.2: A monotone increasing sequence bounded above converges.

Remark. The word "eventually" is crucial and perfectly natural: since convergence depends only on the tail of a sequence, any hypothesis about convergence should likewise depend only on the tail. Whether the first 15 terms are out of order is irrelevant.

Theorem 11.7 (Increasing unbounded sequences diverge to ∞). *If a sequence is eventually increasing and not bounded above, then it diverges to ∞.*

Theorems 11.6 and 11.7 together tell us that the behaviour of an increasing sequence is very restricted: it is either convergent (if bounded above) or divergent to ∞ (if not bounded above). No other outcome is possible. This will be especially useful when we study series.

11.5 Boundedness of Convergent Sequences

We now prove Theorem 11.5: every convergent sequence is bounded. The key idea is simple — all but finitely many terms are close to the limit, and a finite collection of numbers is always bounded.

Proof. Let $\{a_n\}$ be a convergent sequence and let $L = \lim_{n\to\infty} a_n$.

Use the definition of limit with a specific ε.

Choose $\varepsilon = 1$ in Definition 11.2. Then there exists $n_0 \in \mathbb{N}$ such that for every $n \geq n_0$,

$$L - 1 < a_n < L + 1.$$

Construct explicit bounds A and B.

Define

$$A = \min\{L - 1,\ a_0,\ a_1,\ \ldots,\ a_{n_0-1}\},$$
$$B = \max\{L + 1,\ a_0,\ a_1,\ \ldots,\ a_{n_0-1}\}.$$

Verify that $A \leq a_n \leq B$ for every n.

Fix an arbitrary $n \in \mathbb{N}$. There are two cases.

- If $n \geq n_0$, then $A \leq L - 1 < a_n < L + 1 \leq B$.
- If $n < n_0$, then a_n appears in the lists defining A and B, so $A \leq a_n \leq B$ by construction.

In either case, $A \leq a_n \leq B$. Therefore the sequence is bounded. ■

Remark. The choice $\varepsilon = 1$ is arbitrary; any positive number would work. We choose 1 for simplicity. The essential point is that once ε is fixed, all terms from index n_0 onward are trapped in the interval $(L - \varepsilon,\ L + \varepsilon)$. Only the finitely many terms $a_0, \ldots, a_{n_0-1}$ remain, and a finite set of real numbers always has a minimum and a maximum.

11.6 The Monotone Convergence Theorem

We now prove Theorem 11.6 in the case of a non-decreasing sequence that is bounded above. The proof for a non-increasing, bounded-below sequence is entirely analogous. The key ingredient is the least-upper-bound axiom for $\mathbb{R}$.

Proof strategy

Since we want to prove convergence, the very first step must be to identify the limit. The limit is not given to us; we must construct it from the hypotheses. For an increasing, bounded-above sequence, the natural candidate is the supremum of its range.

Proof. Let $\{a_n\}$ be a non-decreasing sequence that is bounded above.

Construct the candidate limit using the least upper bound axiom.

Let $A = \{a_n : n \in \mathbb{N}\}$ be the range of the sequence. The set A is non-empty (it contains a_0) and bounded above (by hypothesis). By the least-upper-bound axiom, A has a supremum. Set

$$L = \sup A.$$

We claim that $\lim_{n\to\infty} a_n = L$.

Fix $\varepsilon > 0$ and construct n_0.

Let $\varepsilon > 0$. Since L is the supremum and $L - \varepsilon < L$, the number $L - \varepsilon$ is **not** an upper bound of A. Therefore there exists $n_0 \in \mathbb{N}$ such that

$$L - \varepsilon < a_{n_0}.$$

Fix an arbitrary $n \geq n_0$. Because the sequence is non-decreasing and $n \geq n_0$,

Show $|a_n - L| < \varepsilon$ for all $n \geq n_0$.

$$a_{n_0} \leq a_n.$$

Because every term of the sequence is at most L (by definition of supremum),

$$a_n \leq L.$$

Combining these with the inequality $L - \varepsilon < a_{n_0}$:

$$L - \varepsilon < a_{n_0} \leq a_n \leq L < L + \varepsilon.$$

Therefore $|a_n - L| < \varepsilon$, which completes the proof that $\lim_{n\to\infty} a_n = L$. ■

Remark. Observe how each hypothesis was used: "bounded above" guarantees the existence of $\sup A$; "increasing" ensures that once a single term exceeds $L - \varepsilon$, every subsequent term does as well. Without monotonicity, finding one term in $(L - \varepsilon, L + \varepsilon)$ would say nothing about the terms that follow.

Remark (Pause and Try). Write a rigorous proof of Theorem 11.7: if $\{a_n\}$ is eventually increasing and not bounded above, then $\lim_{n\to\infty} a_n = \infty$. The ideas are similar to the proof above.

11.7 The Growth Hierarchy

We close the chapter with what is sometimes called the "Big Theorem" — a hierarchy among the most common families of positive sequences. It tells us, at a glance, which terms dominate in a sum or quotient, letting us compute many limits almost instantly.

Motivation: rational sequences

Example 11.5. Compute $\displaystyle\lim_{n\to\infty} \frac{n^3 + 2n + 1}{5n^3 + 3n^2}$.

When the numerator and denominator are both polynomials, only the highest-degree terms matter. The quick answer is $\frac{n^3}{5n^3} = \frac{1}{5}$.

To justify this rigorously, factor n^3 from numerator and denominator:

$$\frac{n^3 + 2n + 1}{5n^3 + 3n^2} = \frac{n^3\left(1 + \frac{2}{n^2} + \frac{1}{n^3}\right)}{n^3\left(5 + \frac{3}{n}\right)} = \frac{1 + \frac{2}{n^2} + \frac{1}{n^3}}{5 + \frac{3}{n}}.$$

As $n \to \infty$, each correction term $\frac{2}{n^2}$, $\frac{1}{n^3}$, $\frac{3}{n}$ tends to 0, so by the limit laws the full limit is $\frac{1}{5}$.

The reason the lower-order terms vanish is that, for instance, $\lim_{n\to\infty} \frac{n}{n^3} = 0$*: the term n grows* ***much more slowly*** *than n^3. Can we extend this idea beyond polynomials?*

The "much smaller" relation

Definition 11.7. Let $\{a_n\}$ and $\{b_n\}$ be sequences with $b_n > 0$ for all n. We write

$$a_n \ll b_n$$

and say "a_n is *much smaller* than b_n" to mean

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 0.$$

Equivalently, we say "b_n is *much larger* than a_n."

Statement of the Big Theorem

Theorem 11.8 (The Big Theorem — growth hierarchy). *For any exponent $p > 0$ and any base $c > 1$,*

$$\ln n \ll n^p \ll c^n \ll n! \ll n^n.$$

In words: logarithms are much smaller than any positive power, which is much smaller than any exponential (with base > 1), which is much smaller than factorials, which are much smaller than power-exponentials.

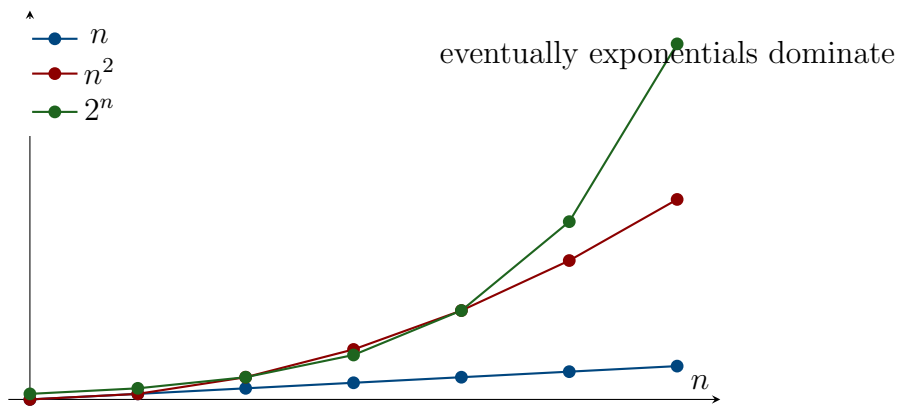


Figure 11.3: Discrete comparison of common growth rates.

Using the theorem

Example 11.6. Compute $\displaystyle\lim_{n\to\infty} \frac{e^n + 2n^{100}}{\ln n + 5e^n}$.

By Theorem 11.8, $n^{100} \ll e^n$ and $\ln n \ll e^n$. Therefore e^n is the dominant term in both numerator and denominator. Factor e^n:

$$\frac{e^n + 2n^{100}}{\ln n + 5e^n} = \frac{1 + 2\,\frac{n^{100}}{e^n}}{\frac{\ln n}{e^n} + 5}.$$

As $n \to \infty$, $\frac{n^{100}}{e^n} \to 0$ and $\frac{\ln n}{e^n} \to 0$, so the limit is $\dfrac{1}{5}$.

11.8 Proof of the Growth Hierarchy

Theorem 11.8 comprises four separate claims. The first two (logarithms vs. powers, and powers vs. exponentials) can be proved using L'Hôpital's Rule, because the sequences involved extend to functions on $[1, \infty)$*. The last two involve* $n!$*, which is defined only for natural numbers, and require different techniques.*

Remark (Pause and Try). Before reading on, try to prove that $\ln n \ll n^p$ and $n^p \ll c^n$ using L'Hôpital's Rule and Theorem 11.3.

Claim 1: $\ln n \ll n^p$ for every $p > 0$

Fix $p > 0$ and define $f(x) = \frac{\ln x}{x^p}$ for $x \geq 1$. Applying L'Hôpital's Rule (∞/∞ form):

$$\lim_{x\to\infty} \frac{\ln x}{x^p} = \lim_{x\to\infty} \frac{1/x}{p\,x^{p-1}} = \lim_{x\to\infty} \frac{1}{p\,x^p} = 0.$$

By Theorem 11.3, $\lim_{n\to\infty} \frac{\ln n}{n^p} = 0$.

Claim 2: $n^p \ll c^n$ for every $p > 0$ and $c > 1$

Fix $p > 0$ and $c > 1$. Define $f(x) = \frac{x^p}{c^x}$ for $x \geq 1$. Since both numerator and denominator tend to ∞, we apply L'Hôpital's Rule repeatedly. After $k = \lceil p \rceil$ applications, the numerator has degree at most 0 in x (it is a constant when p is an integer, or tends to 0 when p is not an integer) while the denominator still tends to ∞, so the limit is 0. By Theorem 11.3, $\lim_{n\to\infty} \frac{n^p}{c^n} = 0$.

Claim 3: $c^n \ll n!$ for every $c > 1$

Proof. Fix $c > 1$ and define $p_n = \dfrac{c^n}{n!}$ for $n \geq 0$.

Observe that

Derive a recurrence relation.

$$p_{n+1} = \frac{c^{n+1}}{(n+1)!} = \frac{c}{n+1} \cdot \frac{c^n}{n!} = \frac{c}{n+1}\, p_n.$$

For $n \geq c$ (i.e. $n + 1 > c$), we have $\frac{c}{n+1} < 1$, so $p_{n+1} < p_n$. Hence the sequence is eventually decreasing.

Show $\{p_n\}$ is eventually decreasing.

Since $p_n > 0$ for all n, the sequence is bounded below by 0. By the Monotone Convergence Theorem (Theorem 11.6), the sequence converges. Let $L = \lim_{n\to\infty} p_n$.

The sequence is bounded below and therefore convergent.

Taking limits on both sides of the recurrence $p_{n+1} = \frac{c}{n+1}\, p_n$:

Determine L from the recurrence.

$$L = \lim_{n\to\infty} \frac{c}{n+1} \cdot \lim_{n\to\infty} p_n = 0 \cdot L = 0.$$

Therefore $\lim_{n\to\infty} \frac{c^n}{n!} = 0$, as required. ■

Remark (Warning). Every step of the proof above was necessary. The final calculation (showing $L = 0$ from the recurrence) is valid **only because we already know the sequence converges**. Without first establishing convergence via the Monotone Convergence Theorem, the same algebra would merely show that the limit, *if it exists*, must be 0 — it would not rule out divergence.

Claim 4: $n! \ll n^n$

Proof. Write

$$\frac{n!}{n^n} = \frac{1 \cdot 2 \cdot 3 \cdots n}{n \cdot n \cdot n \cdots n} = \frac{1}{n} \cdot \frac{2}{n} \cdot \frac{3}{n} \cdots \frac{n}{n}.$$

Bound the product above by $1/n$.

Separate the first factor:

$$\frac{n!}{n^n} = \frac{1}{n} \cdot \prod_{k=2}^{n} \frac{k}{n}.$$

Each factor $\frac{k}{n} \leq 1$ for $k \leq n$, so the entire product satisfies

$$0 < \frac{n!}{n^n} \leq \frac{1}{n}.$$

Apply the Squeeze Theorem.

Since $\lim_{n\to\infty} 0 = 0$ and $\lim_{n\to\infty} \frac{1}{n} = 0$, the Squeeze Theorem (Theorem 11.2) gives

$$\lim_{n\to\infty} \frac{n!}{n^n} = 0.$$

■

Chapter 12

Improper Integrals

12.1 Unbounded Domains

Before we can try to compute an integral from a to ∞, we need to decide what it even means to integrate a function all the way to infinity. We motivate the definition with a concrete example and then state the general definition.

Motivating example

Example 12.1. Compute $\displaystyle\int_0^\infty \frac{1}{1+x^2}\,dx$.

Geometrically, this represents the area between the x-axis and the graph of $\frac{1}{1+x^2}$ starting at $x = 0$ and extending to infinity. We cannot compute this directly — we only know how to integrate a function between two real numbers a and b.

However, for any real number $b > 0$ we *can* compute $\int_0^b \frac{1}{1+x^2}\,dx$, and then let b grow without bound:

$$\int_0^\infty \frac{1}{1+x^2}\,dx \;=\; \lim_{b\to\infty}\int_0^b \frac{1}{1+x^2}\,dx.$$

An antiderivative of $\frac{1}{1+x^2}$ is $\arctan x$, so

$$\begin{aligned}\lim_{b\to\infty}\int_0^b \frac{1}{1+x^2}\,dx &= \lim_{b\to\infty}\Big[\arctan x\Big]_0^b \\ &= \lim_{b\to\infty}\big(\arctan b - \arctan 0\big) \\ &= \frac{\pi}{2} - 0 \;=\; \frac{\pi}{2}.\end{aligned}$$

The area under this curve, extending all the way to infinity, is $\dfrac{\pi}{2}$.

The general definition

Definition 12.1 (Improper integral — unbounded domain). Let $a \in \mathbb{R}$ and let f be continuous on $[a, \infty)$. We define

$$\int_a^\infty f(x)\,dx \;=\; \lim_{b\to\infty} \int_a^b f(x)\,dx,$$

provided the limit exists.

- Since f is continuous on $[a, b]$, the integral $\int_a^b f(x)\,dx$ exists for every $b > a$.
- If the limit exists (is a finite number), we say the improper integral is *convergent*.
- If the limit does not exist, we say the improper integral is *divergent*.

Remark. Divergence can occur in different ways: the limit may be $+\infty$, $-\infty$, or fail to exist for other reasons (e.g. oscillation).

Remark (Pause and Try). Try to compute each of the following before watching the solutions in the next few sections:

$$\int_0^\infty \frac{x}{x^2+1}\,dx, \qquad \int_0^\infty \sin x\,dx, \qquad \int_1^\infty \frac{1}{x^p}\,dx, \qquad \int_0^1 \ln x\,dx.$$

The last integral is improper in a different way — try to figure out what is different and how to define it.

12.2 A Divergent Integral

This second example looks deceptively similar to the first, yet the outcome is completely different.

Example 12.2. Determine whether $\displaystyle\int_0^\infty \frac{x}{x^2+1}\,dx$ converges or diverges.

By Definition 12.1,

$$\int_0^\infty \frac{x}{x^2+1}\,dx \;=\; \lim_{b\to\infty} \int_0^b \frac{x}{x^2+1}\,dx.$$

To find an antiderivative, note that $\frac{x}{x^2+1} = \frac{1}{2} \cdot \frac{2x}{x^2+1}$, and $2x$ is the derivative of x^2+1. Hence

$$\int_0^b \frac{x}{x^2+1}\,dx \;=\; \frac{1}{2}\Big[\ln(x^2+1)\Big]_0^b \;=\; \frac{1}{2}\ln(b^2+1) - \frac{1}{2}\ln 1 \;=\; \frac{1}{2}\ln(b^2+1).$$

As $b \to \infty$, $b^2+1 \to \infty$, and therefore $\ln(b^2+1) \to \infty$. The limit does not exist, so

$$\int_0^\infty \frac{x}{x^2+1}\,dx \quad \text{is } \textbf{divergent}.$$

Remark. The integrands $\frac{1}{x^2+1}$ and $\frac{x}{x^2+1}$ look very similar, yet the first gives a convergent integral equal to $\frac{\pi}{2}$ while the second diverges. Whether the area under a curve extending to infinity is finite or infinite depends delicately on how fast the function decays.

12.3 Oscillating Divergence

Not every divergent integral diverges to $\pm\infty$. Divergence can also result from oscillation.

Example 12.3. Determine whether $\displaystyle\int_0^\infty \sin x\,dx$ converges or diverges.

By Definition 12.1,

$$\int_0^\infty \sin x\,dx \;=\; \lim_{b\to\infty}\int_0^b \sin x\,dx \;=\; \lim_{b\to\infty}\Big[-\cos x\Big]_0^b \;=\; \lim_{b\to\infty}\big(1-\cos b\big).$$

As $b\to\infty$, $\cos b$ oscillates between -1 and 1 without settling down, so $1-\cos b$ oscillates between 0 and 2. The limit does not exist, and the integral is **divergent**.

Remark. It is tempting to argue by symmetry that the positive and negative regions of $\sin x$ cancel, giving 0. This reasoning is flawed. Each positive arch from $[2k\pi,(2k+1)\pi]$ has area 2, and each negative arch contributes -2 to the integral. As b grows, the partial integrals oscillate between 0 and 2 rather than settling to any single value. The integral is not 0 — it is simply **not a number**.

12.4 The p-Integral

The family of integrals $\int_1^\infty \frac{1}{x^p}\,dx$ is arguably the most important family of improper integrals. Its convergence behaviour is simple to state and appears repeatedly when we study comparison tests and, later, series.

Theorem 12.1 (The p-integral). *For every $p\in\mathbb{R}$,*

$$\int_1^\infty \frac{1}{x^p}\,dx \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } p>1,\\ \text{divergent } (+\infty) & \text{if } p\le 1. \end{cases}$$

When $p>1$, the value of the integral is $\dfrac{1}{p-1}$.

Proof. By Definition 12.1,

$$\int_1^\infty \frac{1}{x^p}\,dx \;=\; \lim_{b\to\infty}\int_1^b x^{-p}\,dx.$$

Case 1: $p \neq 1$**.** An antiderivative of x^{-p} is $\dfrac{x^{-p+1}}{-p+1}$, so

$$\int_1^b x^{-p}\,dx \;=\; \frac{b^{1-p}}{1-p} - \frac{1}{1-p}.$$

- If $p < 1$, then $1 - p > 0$, so $b^{1-p} \to \infty$ as $b \to \infty$, and the limit is $+\infty$.
- If $p > 1$, then $1 - p < 0$, so $b^{1-p} \to 0$ as $b \to \infty$, giving

$$\lim_{b\to\infty}\left(\frac{b^{1-p}}{1-p} - \frac{1}{1-p}\right) \;=\; 0 - \frac{1}{1-p} \;=\; \frac{1}{p-1}.$$

Case 2: $p = 1$**.** An antiderivative of $\frac{1}{x}$ is $\ln x$, so

$$\int_1^b \frac{1}{x}\,dx \;=\; \ln b - \ln 1 \;=\; \ln b \;\to\; +\infty \quad \text{as } b \to \infty.$$

Combining the cases: the integral converges if and only if $p > 1$. ■

Remark. This result is worth memorising. Once we introduce comparison tests, the p-integral becomes the standard family against which we compare other integrals. It plays exactly the same role for series later.

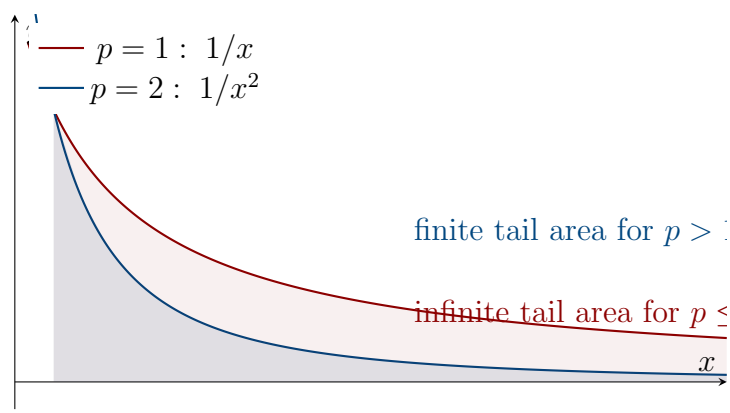


Figure 12.1: The p-integral threshold at $p = 1$.

12.5 Unbounded Integrands

So far, every improper integral has been improper because the domain extended to $\pm\infty$. There is a second, equally important type: the integrand may be unbounded on the interval of integration, typically due to a vertical asymptote at one of the endpoints.

Motivating example

Example 12.4. Compute $\displaystyle\int_0^1 \ln x\,dx$.

The function $\ln x$ is unbounded near $x = 0$ (it has a vertical asymptote there), so the integral is improper. We cannot integrate a function that is unbounded on its domain in the usual Riemann sense.

For any c with $0 < c < 1$, the function $\ln x$ is continuous on $[c, 1]$, so $\int_c^1 \ln x\,dx$ exists. We define

$$\int_0^1 \ln x\,dx \;=\; \lim_{c\to 0^+} \int_c^1 \ln x\,dx.$$

By integration by parts, an antiderivative of $\ln x$ is $x\ln x - x$. Therefore

$$\begin{aligned}\int_c^1 \ln x\,dx &= \Big[x\ln x - x\Big]_c^1 \\ &= (1\cdot\ln 1 - 1) - (c\ln c - c) \\ &= -1 - c\ln c + c.\end{aligned}$$

We need to evaluate $\lim_{c\to 0^+} c\ln c$. This is an indeterminate form of type $0\cdot(-\infty)$. Rewriting as a quotient and applying L'Hôpital's rule:

$$\lim_{c\to 0^+} c\ln c \;=\; \lim_{c\to 0^+} \frac{\ln c}{1/c} \;=\; \lim_{c\to 0^+} \frac{1/c}{-1/c^2} \;=\; \lim_{c\to 0^+}(-c) \;=\; 0.$$

Therefore

$$\int_0^1 \ln x\,dx \;=\; \lim_{c\to 0^+}(-1 - c\ln c + c) \;=\; -1 - 0 + 0 \;=\; -1.$$

The negative sign is expected: the graph of $\ln x$ lies below the x-axis on $(0, 1)$.

The general definition

Definition 12.2 (Improper integral — unbounded integrand at the left endpoint). Let $a < b$ be real numbers and let f be continuous on $(a, b]$. We define

$$\int_a^b f(x)\,dx \;=\; \lim_{c\to a^+} \int_c^b f(x)\,dx,$$

provided the limit exists. Since f is continuous on $[c, b]$ for every $c \in (a, b)$, the integral $\int_c^b f(x)\,dx$ always makes sense. As before, we say the improper integral is *convergent* when the limit exists and *divergent* otherwise.

Remark. There is, of course, a symmetric definition when the integrand is unbounded at the *right* endpoint: if f is continuous on $[a, b)$, then

$$\int_a^b f(x)\,dx \;=\; \lim_{c\to b^-} \int_a^c f(x)\,dx.$$

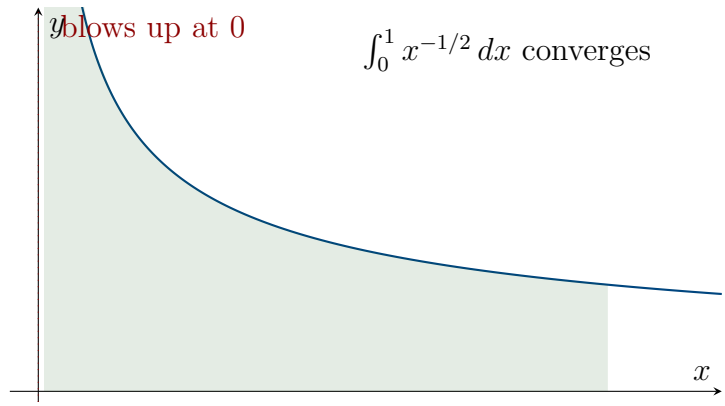


Figure 12.2: An improper integral converging despite an endpoint blow-up.

12.6 Doubly Improper Integrals

Some integrals are improper in more than one way — for instance, both endpoints may be problematic, or there may be a singularity in the interior. Such integrals require extra care to avoid fallacious cancellations.

A cautionary example

Example 12.5. Consider $\displaystyle\int_{-\infty}^{\infty} x\,dx$.

One might argue that $f(x) = x$ is odd, so the positive and negative regions cancel by symmetry, giving 0. This reasoning is **wrong**.

To see why, suppose we try to evaluate the integral by centring the domain at 0*:*

$$\lim_{R\to\infty}\int_{-R}^{R} x\,dx \;=\; \lim_{R\to\infty}\left[\frac{x^2}{2}\right]_{-R}^{R} \;=\; \lim_{R\to\infty}\left(\frac{R^2}{2}-\frac{R^2}{2}\right) \;=\; 0.$$

But if we instead centre at 1, integrating from $1-R$ to $1+R$, a similar computation gives ∞. The two choices of "symmetric" limits yield different answers.

Moreover, the substitution $u = x - 1$ transforms $\int_{-\infty}^{\infty} x\,dx$ into $\int_{-\infty}^{\infty}(u+1)\,du = \int_{-\infty}^{\infty} u\,du + \int_{-\infty}^{\infty} 1\,du$. If $I = \int_{-\infty}^{\infty} x\,dx$ were a number, we would conclude $I = I + \infty$, a contradiction.

Remark (Warning). If we ever try to cancel $+\infty$ with $-\infty$, fundamental properties such as linearity and substitution break down. The only safe course is to declare such integrals divergent.

The correct strategy

Definition 12.3 (Doubly improper integral). Suppose an integral $\int_a^b f(x)\,dx$ is improper at more than one point (e.g. at both endpoints, or at an interior point). The procedure is:

1. **Split** the domain into finitely many subintervals so that each piece is improper at **exactly one** endpoint.
2. **Evaluate** each piece independently as a separate limit.
3. The original integral is *convergent* if and only if **every** piece is convergent. If even one piece diverges, the whole integral is *divergent*.

The final value, when convergent, does not depend on how the domain is split.

Example 12.6. For $\int_{-\infty}^{\infty} x\,dx$, split at 0:

$$\int_{-\infty}^{\infty} x\,dx \;=\; \int_{-\infty}^{0} x\,dx + \int_{0}^{\infty} x\,dx \;=\; \lim_{a\to-\infty}\int_{a}^{0} x\,dx + \lim_{b\to\infty}\int_{0}^{b} x\,dx.$$

The first limit equals $-\infty$ and the second equals $+\infty$. Since both pieces must converge independently and neither does, the integral is **divergent**.

Example 12.7. The integral $\int_{0}^{\infty} \frac{1}{\sqrt{x}\,(1+x^2)}\,dx$ is improper at $x=0$ (vertical asymptote) and at $x=\infty$. Split at $x=1$:

$$\int_{0}^{\infty} \frac{1}{\sqrt{x}\,(1+x^2)}\,dx \;=\; \underbrace{\int_{0}^{1} \frac{1}{\sqrt{x}\,(1+x^2)}\,dx}_{\text{improper at } 0} \;+\; \underbrace{\int_{1}^{\infty} \frac{1}{\sqrt{x}\,(1+x^2)}\,dx}_{\text{improper at } \infty}.$$

Each piece is improper at one endpoint only and can be studied with the definitions in Sections 12.1 and 12.5. The original integral converges if and only if both pieces converge.

Example 12.8. The integral $\int_{-1}^{1} \frac{1}{x^2}\,dx$ is improper at $x=0$ (vertical asymptote inside the domain). Split at 0:

$$\int_{-1}^{1} \frac{1}{x^2}\,dx \;=\; \int_{-1}^{0} \frac{1}{x^2}\,dx + \int_{0}^{1} \frac{1}{x^2}\,dx.$$

Each piece diverges (both equal $+\infty$), so the original integral is **divergent**.

Remark (Warning). A common error is to fail to notice that the integrand is undefined or unbounded at an interior point and to blindly apply the Fundamental Theorem of Calculus across the singularity. Always check for vertical asymptotes inside the interval of integration before computing.

12.7 The Basic Comparison Test

In many cases, computing the exact value of an improper integral is difficult or impossible. Often, all we need to know is whether the integral converges or diverges. For non-negative functions, comparison provides a powerful tool.

Positive functions: only two outcomes

Proposition 12.2. *Let f be continuous and non-negative on $[a, \infty)$. Then $\int_a^\infty f(x)\,dx$ is either convergent (a finite number) or divergent to $+\infty$. No other behaviour is possible.*

Proof. Define $F(b) = \int_a^b f(x)\,dx$. By the Fundamental Theorem of Calculus, $F'(b) = f(b) \geq 0$, so F is non-decreasing. A non-decreasing function has a limit at infinity that is either finite (by the Monotone Convergence Theorem, applied to the sequence $F(n)$ and extended to all b via monotonicity) or $+\infty$ (when F is unbounded above). These are the only two possibilities. ■

Remark. Because there are only two outcomes for non-negative functions, we adopt a convenient shorthand:

$$\int_a^\infty f(x)\,dx < \infty \quad \text{means "convergent"}, \qquad \int_a^\infty f(x)\,dx = \infty \quad \text{means "divergent"}.$$

This notation is **only valid for non-negative integrands**. For functions that change sign, "not ∞" does not guarantee convergence (the integral could oscillate).

The theorem

Theorem 12.3 (Basic Comparison Test). *Let $a \in \mathbb{R}$ and let f, g be continuous on $[a, \infty)$ with $0 \leq f(x) \leq g(x)$ for all $x \geq a$. Then:*

1. *If $\displaystyle\int_a^\infty g(x)\,dx$ converges, then $\displaystyle\int_a^\infty f(x)\,dx$ converges.*
2. *If $\displaystyle\int_a^\infty f(x)\,dx$ diverges, then $\displaystyle\int_a^\infty g(x)\,dx$ diverges.*

Remark. Equivalently, using the shorthand for non-negative integrands:

- If the integral of the **bigger** function is $< \infty$, then the integral of the **smaller** function is also $< \infty$.
- If the integral of the **smaller** function is $= \infty$, then the integral of the **bigger** function is also $= \infty$.

The remaining two combinations (bigger diverges, or smaller converges) give **no information**.

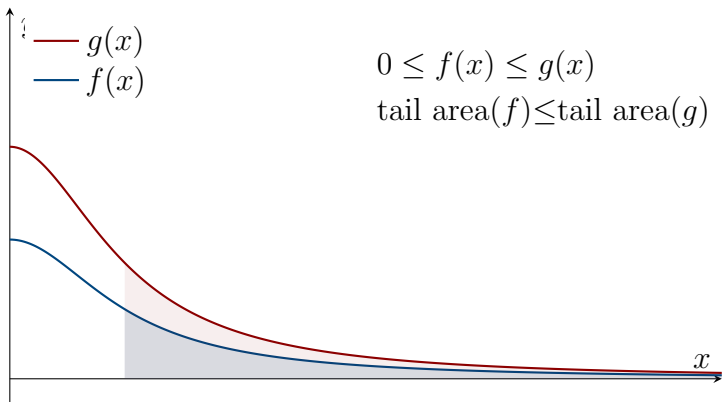


Figure 12.3: Area comparison intuition for improper integrals.

12.8 Basic Comparison Test Examples

We now illustrate the Basic Comparison Test by comparing new integrals against the p-integral family from Theorem 12.1.

Example 12.9. Determine whether $\displaystyle\int_1^\infty \frac{\sin^2 x}{x^2}\,dx$ converges or diverges.

For all $x \ge 1$,

$$0 \;\le\; \frac{\sin^2 x}{x^2} \;\le\; \frac{1}{x^2}.$$

We know $\int_1^\infty \frac{1}{x^2}\,dx$ converges because $p = 2 > 1$ (Theorem 12.1). By the Basic Comparison Test (Theorem 12.3), $\displaystyle\int_1^\infty \frac{\sin^2 x}{x^2}\,dx$ is **convergent**.

Example 12.10. Determine whether $\displaystyle\int_1^\infty \frac{\sin^2 x}{x}\,dx$ converges or diverges.

We can bound $0 \le \frac{\sin^2 x}{x} \le \frac{1}{x}$, and $\int_1^\infty \frac{1}{x}\,dx$ diverges ($p = 1$). But the Basic Comparison Test says: if the *bigger* function's integral diverges, we learn **nothing** about the smaller.

As it turns out, this integral *is* divergent, but the Basic Comparison Test alone does not establish this with the bound above.

Remark. Example 12.10 illustrates an important lesson: the Basic Comparison Test is inconclusive when the *bigger* function's integral diverges. In such cases, one must try a different bound or a different method entirely.

Example 12.11. Determine whether $\displaystyle\int_1^\infty \frac{\ln x}{x^2}\,dx$ converges or diverges.

We would like to bound $\ln x$ by a power of x. From the "Big Theorem" on the growth of logarithms,

$$\lim_{x\to\infty} \frac{\ln x}{x^{1/2}} = 0,$$

which means that for all sufficiently large x, $\ln x < x^{1/2}$.

A first attempt to compare $\ln x$ with x gives:

$$\frac{\ln x}{x^2} < \frac{x}{x^2} = \frac{1}{x},$$

but $\int_1^\infty \frac{1}{x}\,dx = \infty$, so this comparison is **inconclusive**.

Instead, use the tighter bound $\ln x < x^{1/2}$:

$$0 \;\le\; \frac{\ln x}{x^2} \;<\; \frac{x^{1/2}}{x^2} \;=\; \frac{1}{x^{3/2}}$$

for all sufficiently large x. Since $\int_1^\infty \frac{1}{x^{3/2}}\,dx$ converges ($p = \frac{3}{2} > 1$), the Basic Comparison Test gives that $\displaystyle\int_1^\infty \frac{\ln x}{x^2}\,dx$ is **convergent**.

Remark. The comparison need only hold for all sufficiently large x (say $x \ge M$ for some M). The integral from 1 to M is a finite number and does not affect convergence or divergence. This principle applies to both the Basic and Limit-Comparison Tests.

12.9 The Limit-Comparison Test

The Limit-Comparison Test provides a more flexible method of comparison. Rather than requiring a pointwise inequality $f(x) \le g(x)$, we only need the two functions to be asymptotically proportional*: their ratio must tend to a positive constant.*

Theorem 12.4 (Limit-Comparison Test). *Let $a \in \mathbb{R}$ and let f, g be positive and continuous on $[a, \infty)$. Suppose*

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L,$$

where $0 < L < \infty$ (a positive, finite number). Then

$$\int_a^\infty f(x)\,dx \;\; \textit{converges} \quad\Longleftrightarrow\quad \int_a^\infty g(x)\,dx \;\; \textit{converges.}$$

That is, the two integrals are either both convergent or both divergent.

How to use the test

The strategy is: given a complicated integrand f, identify the "dominant" behaviour of $f(x)$ for large x and let g be that simpler function. Then verify that $\lim_{x\to\infty} f(x)/g(x)$ is a positive finite number.

Example 12.12. Determine whether $\displaystyle\int_1^\infty \frac{x^2+3x}{\sqrt{x^5+1}}\,dx$ converges or diverges.

Let $f(x) = \dfrac{x^2+3x}{\sqrt{x^5+1}}$. For large x, the dominant terms are x^2 in the numerator and $\sqrt{x^5} = x^{5/2}$ in the denominator. So we expect $f(x) \approx \frac{x^2}{x^{5/2}} = \frac{1}{x^{1/2}}$. Set $g(x) = \dfrac{1}{x^{1/2}}$.

Compute the limit:

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{x^2+3x}{\sqrt{x^5+1}} \cdot x^{1/2} = \lim_{x\to\infty} \frac{x^{5/2}+3x^{3/2}}{\sqrt{x^5+1}} = 1.$$

Since $L = 1 > 0$, the Limit-Comparison Test applies. We know $\int_1^\infty \frac{1}{x^{1/2}}\,dx$ diverges ($p = \frac{1}{2} \le 1$). Therefore $\displaystyle\int_1^\infty \frac{x^2+3x}{\sqrt{x^5+1}}\,dx$ is **divergent**.

Example 12.13. Determine whether $\displaystyle\int_1^\infty \sin\left(\frac{1}{x^2}\right)\,dx$ converges or diverges.

For $x \ge 1$, the argument $\frac{1}{x^2} \in (0,1]$, so $\sin\left(\frac{1}{x^2}\right) > 0$. As $x \to \infty$, $\frac{1}{x^2} \to 0$, and we recall $\lim_{t\to 0} \frac{\sin t}{t} = 1$. Set $f(x) = \sin\left(\frac{1}{x^2}\right)$ and $g(x) = \frac{1}{x^2}$. Then

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{\sin(1/x^2)}{1/x^2} = 1.$$

Since $L = 1 > 0$ and $\int_1^\infty \frac{1}{x^2}\,dx$ converges ($p = 2 > 1$), the Limit-Comparison Test gives that $\displaystyle\int_1^\infty \sin\left(\frac{1}{x^2}\right)\,dx$ is **convergent**.

Remark (Pause and Try). What can we conclude if $L = 0$ in the Limit-Comparison Test? What if $L = \infty$? In each case, only *half* of the conclusion survives. Try to determine which half by examining the proof in the next section.

12.10 Proof of the Limit-Comparison Test

The proof reduces the Limit-Comparison Test to the Basic Comparison Test by extracting a two-sided inequality from the definition of a limit.

Rough work

The hypothesis $\lim_{x\to\infty} \frac{f(x)}{g(x)} = L$ *means that for large* x, $f(x) \approx L \cdot g(x)$. *If* f *is approximately a positive constant times* g, *it is plausible that their integrals converge or diverge together. To make this rigorous, we use the formal definition of a limit to sandwich* f *between two constant multiples of* g.

Proof. Since f and g are positive and continuous on $[a,\infty)$, Proposition 12.2 implies each of $\int_a^\infty f(x)\,dx$ and $\int_a^\infty g(x)\,dx$ is either convergent or $+\infty$. We show these two integrals have the same status.

Take $\varepsilon = \dfrac{L}{2} > 0$. By the definition of the limit, there exists $M \geq a$ such that for all $x \geq M$,

$$\left| \frac{f(x)}{g(x)} - L \right| < \frac{L}{2}.$$

Definition of limit with $\varepsilon = L/2$.

Equivalently, for all $x \geq M$,

$$\frac{L}{2}\, g(x) \;\leq\; f(x) \;\leq\; \frac{3L}{2}\, g(x).$$

Both $\frac{L}{2}$ and $\frac{3L}{2}$ are positive constants.

The inequality only holds for $x \geq M$, but this is sufficient: for any continuous function h,

$$\int_a^\infty h(x)\,dx = \underbrace{\int_a^M h(x)\,dx}_{\text{finite number}} + \int_M^\infty h(x)\,dx,$$

so the integral from a to ∞ converges if and only if the integral from M to ∞ converges.

Step 1. Assume $\int_M^\infty g(x)\,dx$ converges. Then $\int_M^\infty \frac{3L}{2}\, g(x)\,dx$ converges (a constant multiple of a convergent integral). Since $f(x) \leq \frac{3L}{2}\, g(x)$ for $x \geq M$, the Basic Comparison Test (Theorem 12.3) gives $\int_M^\infty f(x)\,dx < \infty$.

Basic Comparison Test (12.3) with $f \leq \frac{3L}{2} g$.

Step 2. Assume $\int_M^\infty f(x)\,dx$ converges. Since $\frac{L}{2}\, g(x) \leq f(x)$ for $x \geq M$, the Basic Comparison Test gives $\int_M^\infty \frac{L}{2}\, g(x)\,dx < \infty$, and hence $\int_M^\infty g(x)\,dx < \infty$.

Basic Comparison Test (12.3) with $\frac{L}{2} g \leq f$.

Since convergence from M to ∞ is equivalent to convergence from a to ∞, we conclude:

$$\int_a^\infty f(x)\,dx \text{ converges} \quad \Longleftrightarrow \quad \int_a^\infty g(x)\,dx \text{ converges.} \qquad \blacksquare$$

Remark. Examining the proof reveals two extended versions of the theorem:

- If $L = 0$: only the inequality $f(x) \leq \frac{3L}{2}\, g(x)$ survives in a useful form (with L replaced by a small ε). We can conclude: if $\int_a^\infty g(x)\,dx$ converges, then $\int_a^\infty f(x)\,dx$ converges. But the converse need not hold.

- If $L = \infty$: by symmetry (swap f and g, the reciprocal limit is 0), we can conclude: if $\int_a^\infty g(x)\,dx$ diverges, then $\int_a^\infty f(x)\,dx$ diverges. But the converse need not hold.

Chapter 13

Series

13.1 Naive Infinite Sums

A series is an infinite sum. In this section we are intentionally careless: we assume infinite sums are well-defined, behave well, and can be manipulated exactly like finite sums. The contradictions that arise will motivate the need for a rigorous definition.

A careless computation

Example 13.1. Fix a real number x and set

$$S \;=\; \sum_{n=0}^{\infty} x^n \;=\; 1 + x + x^2 + x^3 + \cdots$$

Multiplying both sides by x gives

$$xS \;=\; x + x^2 + x^3 + \cdots$$

Subtracting the second line from the first, all terms cancel except the leading 1:

$$S - xS = 1 \quad\Longrightarrow\quad S = \frac{1}{1-x}, \qquad x \neq 1.$$

This appears to give a value for every $x \neq 1$. But substituting $x = 2$ yields

$$1 + 2 + 4 + 8 + 16 + \cdots \;=\; \frac{1}{1-2} \;=\; -1,$$

which is absurd: a sum of positive terms cannot be negative.

Example 13.2. Set $T = \sum_{n=0}^{\infty}(-1)^n = 1 - 1 + 1 - 1 + \cdots$. Grouping in pairs starting from the left gives

$$(1-1) + (1-1) + \cdots \;=\; 0 + 0 + \cdots \;=\; 0.$$

But grouping differently gives

$$1 + (-1+1) + (-1+1) + \cdots \;=\; 1 + 0 + 0 + \cdots \;=\; 1.$$

We have "proved" $0 = 1$, which is obviously false.

Remark (Warning). Both contradictions arise from the same source: assuming every infinite sum represents a number, and that it can be manipulated like a finite sum. Properties such as associativity and commutativity are valid for finite sums but **may fail** for infinite sums.

The road ahead

To work with infinite sums without contradictions we need three things:

(i) A **definition** of what an infinite sum means.

(ii) A way to distinguish *convergent* series (those that represent a number) from *divergent* ones (those that do not).

(iii) **Theorems** that identify which properties of finite sums carry over to infinite sums, and which do not.

13.2 Definition of Series

In everyday English, "series" and "sequence" are nearly synonymous; in mathematics they are fundamentally different. A sequence is an infinite list; a series is an infinite sum. We now make the latter idea precise.

Motivating example

Example 13.3. Consider the sum

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

Imagine a unit square. Shade half of it (area $\frac{1}{2}$), then a quarter ($\frac{1}{4}$), then an eighth, and so on. No matter how many pieces we add, we remain inside the square, yet the un-shaded region shrinks to nothing. It seems the sum should equal 1.

Algebraically, define partial sums:

$$S_1 = \frac{1}{2}, \qquad S_2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}, \qquad S_3 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}.$$

In general, $S_k = \sum_{n=1}^{k} \frac{1}{2^n} = 1 - \frac{1}{2^k}$, so

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \lim_{k\to\infty} S_k = \lim_{k\to\infty}\left(1 - \frac{1}{2^k}\right) = 1.$$

The key idea: we never truly add infinitely many terms at once. Instead, we add the first k terms and take $k \to \infty$.

The general definition

Definition 13.1 (Series and partial sums). Let (a_n) be a sequence. The *k-th partial sum* of the series $\sum_{n=1}^{\infty} a_n$ is

$$S_k = \sum_{n=1}^{k} a_n = a_1 + a_2 + \cdots + a_k.$$

We define

$$\sum_{n=1}^{\infty} a_n = \lim_{k\to\infty} S_k,$$

provided the limit exists.

- If the limit exists (is a finite number), the series is *convergent*.
- If the limit does not exist, the series is *divergent*.

Divergence may occur because the limit is $+\infty$, $-\infty$, or because the sequence of partial sums oscillates.

Remark. The definition parallels that of improper integrals. An improper integral $\int_a^{\infty} f(x)\,dx$ is defined as $\lim_{b\to\infty} \int_a^b f(x)\,dx$: a limit of proper integrals. Similarly, a series (infinite sum) is a limit of finite sums. In both cases, the limit may or may not exist.

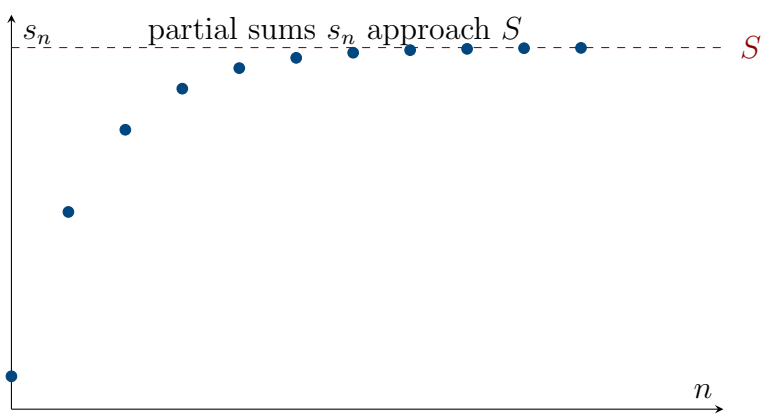


Figure 13.1: A convergent series' partial sums approach a limit.

13.3 A Telescoping Series

We compute a series directly from the definition, illustrating both the guess-and-induct approach and the more elegant partial-fraction technique.

Remark (Pause and Try). Compute $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+n}$ directly from the definition before reading on.

Example 13.4. Compute $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+n}$.

Step 1: Compute partial sums.

$$S_1 = \frac{1}{1^2+1} = \frac{1}{2}, \qquad S_2 = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}, \qquad S_3 = \frac{2}{3} + \frac{1}{12} = \frac{3}{4}.$$

The pattern $S_k = \dfrac{k}{k+1}$ can be verified by induction.

Alternative — partial fractions (no guessing). Factor $n^2 + n = n(n+1)$ and decompose:

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Therefore

$$\begin{aligned} S_k &= \sum_{n=1}^{k} \Big(\frac{1}{n} - \frac{1}{n+1}\Big) \\ &= \Big(1 - \frac{1}{2}\Big) + \Big(\frac{1}{2} - \frac{1}{3}\Big) + \cdots + \Big(\frac{1}{k} - \frac{1}{k+1}\Big) \\ &= 1 - \frac{1}{k+1} = \frac{k}{k+1}. \end{aligned}$$

Telescoping cancellation: all intermediate terms cancel, leaving only the first and last.

This is a *telescoping* sum. No induction is needed; the derivation itself constitutes a proof.

Step 2: Take the limit.

$$\sum_{n=1}^{\infty} \frac{1}{n^2+n} = \lim_{k\to\infty} \frac{k}{k+1} = 1.$$

The series is convergent and its sum is 1.

13.4 Two Divergent Series

We verify that our definition correctly identifies two series that ought to be divergent. Each diverges for a different reason, providing a useful sanity check on the definition.

Example 13.5. Show that $\displaystyle\sum_{n=1}^{\infty} 1$ is divergent.

The k-th partial sum is $S_k = k$. Therefore

$$\sum_{n=1}^{\infty} 1 = \lim_{k\to\infty} k = \infty.$$

The series is divergent (to ∞), exactly as expected.

Example 13.6. Show that $\sum_{n=0}^{\infty}(-1)^n$ is divergent.

The first few partial sums are

$$S_0 = 1, \quad S_1 = 0, \quad S_2 = 1, \quad S_3 = 0, \quad \dots$$

In general, $S_k = 1$ if k is even and $S_k = 0$ if k is odd. The sequence $1, 0, 1, 0, \dots$ has no limit — it oscillates. Therefore the series is divergent.

Remark. The series $\sum(-1)^n$ is not divergent because the partial sums tend to $\pm\infty$; it is divergent because they fail to settle down to any single value. Divergence can take many forms.

13.5 Geometric Series

The geometric series is one of the most important families of series. We compute it carefully from the definition, obtaining both the domain of convergence and the formula for the sum.

Remark (Pause and Try). Fix $x \in \mathbb{R}$. Use the definition of series to determine for which values of x the series $\sum_{n=0}^{\infty} x^n$ converges, and find its sum when it does.

Theorem 13.1 (Geometric series). *For any $x \in \mathbb{R}$,*

$$\sum_{n=0}^{\infty} x^n = \begin{cases} \dfrac{1}{1-x} & \text{if } |x| < 1, \\ \text{divergent} & \text{if } |x| \geq 1. \end{cases}$$

Proof. **Case $x = 1$.** The series becomes $\sum 1 = \infty$, which is divergent.

Case $x \neq 1$. The k-th partial sum is $S_k = 1 + x + x^2 + \cdots + x^k$. Multiplying by x:

$$xS_k = x + x^2 + \cdots + x^{k+1}.$$

Subtracting gives $S_k - xS_k = 1 - x^{k+1}$, so

$$S_k = \frac{1 - x^{k+1}}{1 - x}.$$

Valid because $x \neq 1$.

Taking the limit,

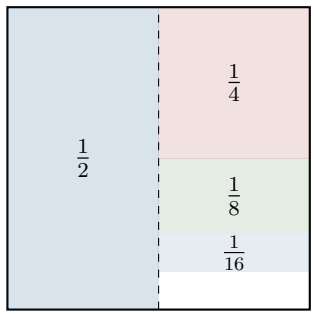
$$\sum_{n=0}^{\infty} x^n = \lim_{k\to\infty} \frac{1 - x^{k+1}}{1 - x} = \frac{1 - \lim_{k\to\infty} x^{k+1}}{1 - x}.$$

- If $|x| < 1$, then $x^{k+1} \to 0$, so the sum equals $\dfrac{1}{1-x}$.
- If $x > 1$, then $x^{k+1} \to \infty$, so the limit does not exist.

- If $x \leq -1$, then x^{k+1} oscillates (e.g. $-1, 1, -1, \ldots$ when $x = -1$), so the limit does not exist.

In every case with $|x| \geq 1$ the series is divergent. ■

Remark (Warning). The formula $\sum x^n = \frac{1}{1-x}$ is only valid when $|x| < 1$. Using it outside this range — as we did carelessly in Section 13.1 — leads to contradictions.



sum fills the square

Figure 13.2: Geometric series visualized by filling a unit square.

13.6 Linearity of Series

Not every property of finite sums extends to infinite sums. For instance, finite sums are commutative, but infinite sums are not (see Section 13.17). We single out two properties that ***do*** *carry over: addition and scalar multiplication.*

A cautionary observation

Consider the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2$. *If we rearrange the same terms (two positives, then one negative, repeating), the new sum is* $\frac{3}{2}\ln 2$, *not* $\ln 2$. *Rearranging the terms of a convergent series can change its value! This shows that we must carefully* prove *any property we wish to use for infinite sums.*

The linearity theorems

Theorem 13.2 (Sum of two series). *If* $\sum a_n$ *and* $\sum b_n$ *are both convergent, then* $\sum(a_n + b_n)$ *is also convergent and*

$$\sum_{n=1}^{\infty}(a_n + b_n) \;=\; \sum_{n=1}^{\infty} a_n \;+\; \sum_{n=1}^{\infty} b_n.$$

Proof. Write $S_k = \sum_{n=1}^{k} a_n$, $T_k = \sum_{n=1}^{k} b_n$, and $R_k = \sum_{n=1}^{k}(a_n + b_n)$. For finite sums we have $R_k = S_k + T_k$. By hypothesis, $\lim_{k\to\infty} S_k$ and $\lim_{k\to\infty} T_k$ both exist. By the limit laws,

$$\lim_{k\to\infty} R_k = \lim_{k\to\infty} S_k + \lim_{k\to\infty} T_k,$$

so $\sum(a_n + b_n)$ converges and equals the sum of the two series. ■

The key step is that $R_k = S_k + T_k$ holds for finite sums, then we pass to the limit.

Theorem 13.3 (Scalar multiple of a series). *If $\sum a_n$ is convergent and $c \in \mathbb{R}$, then $\sum c\, a_n$ is convergent and*

$$\sum_{n=1}^{\infty} c\, a_n \;=\; c \sum_{n=1}^{\infty} a_n.$$

Proof. The proof follows the same template: write S_k for the partial sums of $\sum a_n$, observe that the k-th partial sum of $\sum c\, a_n$ equals $c\, S_k$ (a property of finite sums), and apply the limit laws. ■

Remark. Both theorems require the hypothesis that the original series are convergent. Without this, the limit laws cannot be applied. This *template* recurs in almost every proof about series: (1) express the infinite sum as a limit of finite sums, (2) use a known property of finite sums, (3) pass to the limit.

13.7 Tails of Series

When studying convergence, the beginning of a series is irrelevant — only its tail matters. This simple but powerful observation makes many theorems more widely applicable.

Theorem 13.4 (Changing the starting index). *Let (a_n) be a sequence. The series $\sum_{n=0}^{\infty} a_n$ is convergent if and only if $\sum_{n=1}^{\infty} a_n$ is convergent. Moreover, if both converge,*

$$\sum_{n=0}^{\infty} a_n \;=\; a_0 + \sum_{n=1}^{\infty} a_n.$$

More generally, for any integers m_1 and m_2, the series $\sum_{n=m_1}^{\infty} a_n$ converges if and only if $\sum_{n=m_2}^{\infty} a_n$ converges.

Proof. Express each series as a limit of partial sums, use the corresponding property for finite sums, and pass to the limit. ■

Remark. Because of Theorem 13.4, we may write "$\sum a_n$ is convergent" without specifying the starting index. The starting index affects the *value* of the sum but not whether the series converges.

This observation has a useful consequence for applying convergence theorems. A typical theorem has the form:

$$\text{If for all } n \in \mathbb{N}, \text{ [some property]}, \quad \text{then } \sum a_n \text{ converges.}$$

By the tail principle, the same conclusion holds if [some property] is true only for all sufficiently large n — that is, for all $n \geq n_0$ for some $n_0 \in \mathbb{N}$. In other words, a convergence theorem that requires a condition for all n automatically yields a stronger version requiring the condition only ***eventually****. Whenever we prove a theorem of the first form, we get the second form for free.*

Definition 13.2 (Tail of a series). For the purpose of convergence, only what happens for large values of n matters. Given a series $\sum_{n=1}^{\infty} a_n$ and a positive integer n_0, the *tail* of the series starting at n_0 is the series $\sum_{n=n_0}^{\infty} a_n$. The original series and any of its tails either both converge or both diverge.

13.8 The Divergence Test

Before deploying heavy machinery, there is one quick check worth performing on every new series: does the general term tend to zero? If not, the series cannot converge.

Theorem 13.5 (Divergence Test (Necessary Condition for Convergence)). *If $\displaystyle\sum_{n=1}^{\infty} a_n$ is convergent, then $\displaystyle\lim_{n\to\infty} a_n = 0$.*

Proof. Suppose $\sum a_n$ converges; denote the k-th partial sum by S_k and its limit by S. Write

$$a_n = S_n - S_{n-1}.$$

Since $\lim_{n\to\infty} S_n = S$ and $\lim_{n\to\infty} S_{n-1} = S$, the limit laws give

$$\lim_{n\to\infty} a_n = S - S = 0. \qquad \blacksquare$$

Both S_n and S_{n-1} converge to the same limit S; their difference must converge to 0.

Remark (Warning). The **converse is false**. Having $\lim a_n = 0$ does **not** guarantee that $\sum a_n$ converges. For example, $\lim_{n\to\infty} \frac{1}{n} = 0$ but $\sum \frac{1}{n}$ diverges (the harmonic series), while $\lim_{n\to\infty} \frac{1}{n^2} = 0$ and $\sum \frac{1}{n^2}$ converges. We will establish these facts in later sections.

In practice we use the Divergence Test in its contrapositive form:

Corollary 13.6. *If $\lim_{n\to\infty} a_n \neq 0$ (including the case where the limit does not exist), then $\sum a_n$ is* ***divergent****.*

Example 13.7. The series $\displaystyle\sum_{n=0}^{\infty} \frac{\arctan n}{1+e^{-n}}$ is divergent, because

$$\lim_{n\to\infty} \frac{\arctan n}{1+e^{-n}} = \frac{\pi/2}{1+0} = \frac{\pi}{2} \neq 0.$$

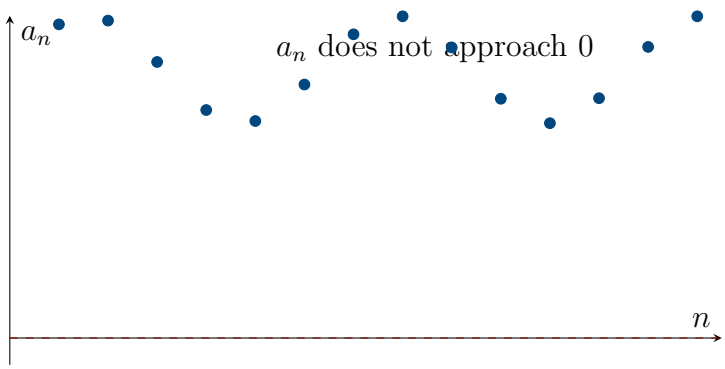


Figure 13.3: A necessary condition for convergence: $a_n \to 0$.

Remark. The Divergence Test is a quick first step. It can never prove convergence, but it can sometimes prove divergence immediately, allowing us to move on without further analysis.

13.9 Positive Series

Proving convergence for a general series can be hard. Positive series are substantially simpler: their only two behaviours are convergence and divergence to ∞. This dichotomy is what makes comparison arguments possible.

Definition 13.3 (Positive, negative, and non-negative series). A series $\sum a_n$ is called *positive* if $a_n > 0$ for all n, *negative* if $a_n < 0$ for all n, and *non-negative* if $a_n \geq 0$ for all n.

Theorem 13.7 (Dichotomy for positive series). *A positive series is either convergent or divergent to ∞. In particular, it cannot diverge by oscillation.*

Proof. If $\sum a_n$ is positive, then each $a_n > 0$, so

$$S_{k+1} = S_k + a_{k+1} > S_k.$$

The sequence (S_k) is therefore increasing. By the Monotone Convergence Theorem, an increasing sequence either converges (if it is bounded above) or diverges to ∞ (if it is unbounded). ■

Monotone Convergence Theorem applied to the increasing sequence of partial sums.

Remark. The same conclusion holds for non-negative series (replace "increasing" with "non-decreasing") and for *eventually* non-negative series, by the tail principle (Section 13.7).

Notation for positive series

Because a positive series can only converge or equal ∞, we adopt a convenient shorthand.

We write $\sum a_n < \infty$ to mean the positive series is convergent, and $\sum a_n = \infty$ to mean it is divergent.

Remark (Warning). This notation is reserved for positive (or non-negative) series **only**. An arbitrary series that is not ∞ may still be divergent by oscillation.

To prove that a positive series converges, it suffices to show it is ***not*** ∞ *— equivalently, that the partial sums are bounded above. This is less work than proving a limit exists directly. The Integral Test, the Basic Comparison Test, and the Limit Comparison Test all exploit this fact.*

13.10 The Integral Test

Series and improper integrals are defined in the same way — as limits of their "finite" counterparts. It is natural to ask whether there is a direct relationship between them. The Integral Test provides one, letting us leverage antiderivatives to study convergence of series.

Deriving the test

Let f be continuous, positive, and decreasing on $[1, \infty)$. We compare the integral $\int_1^N f(x)\,dx$ with finite sums by bounding the area under the curve using rectangles.

- **Upper bound.** On each interval $[n, n+1]$, the maximum of f (since f is decreasing) occurs at the left endpoint. Rectangles of height $f(n)$ and width 1 give
$$\int_1^N f(x)\,dx \;\le\; \sum_{n=1}^{N-1} f(n).$$
- **Lower bound.** Rectangles of height $f(n+1)$ and width 1 give
$$\int_1^N f(x)\,dx \;\ge\; \sum_{n=2}^{N} f(n).$$

Combining and letting $N \to \infty$:
$$\sum_{n=2}^{\infty} f(n) \;\le\; \int_1^{\infty} f(x)\,dx \;\le\; \sum_{n=1}^{\infty} f(n).$$
Since f is positive, each quantity is either a finite number or ∞. The two series differ only by the single term $f(1)$, so they converge or diverge together. Sandwiching the integral between them forces the integral to share their behaviour.

Theorem 13.8 (Integral Test). *Let $a \in \mathbb{R}$ and let f be continuous, positive, and decreasing on $[a, \infty)$. Then*
$$\int_a^{\infty} f(x)\,dx \quad \textit{converges} \qquad \Longleftrightarrow \qquad \sum_{n=\lceil a \rceil}^{\infty} f(n) \quad \textit{converges.}$$

Remark. We write $\int_a^\infty f(x)\,dx \sim \sum f(n)$ to indicate the integral and series are either both convergent or both divergent. This does **not** mean they are equal; when both converge, their values are generally different.

13.11 Integral Test Applications

We apply the Integral Test to two important examples. The first identifies an entire family of convergent (and divergent) series; the second handles a series that is not covered by the p-series family.

Example 13.8 *(p-series).* For $p > 0$, determine when $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges.

The function $f(x) = \frac{1}{x^p}$ is continuous, positive, and decreasing on $[1, \infty)$. By the Integral Test,

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \sim \int_1^\infty \frac{1}{x^p}\,dx.$$

From the theory of improper integrals, $\int_1^\infty \frac{1}{x^p}\,dx$ converges if and only if $p > 1$.

Theorem 13.9 (p-series test). *The series* $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ *converges if and only if* $p > 1$.

Remark. For $p \le 0$ the terms $\frac{1}{n^p}$ do not tend to zero, so the series diverges by the Divergence Test (Theorem 13.5). For $p > 0$ the convergence behaviour follows from the Integral Test and the p-integral (Theorem 12.1).

Remark. In particular, the *harmonic series* $\sum \frac{1}{n}$ (the case $p = 1$) is divergent, while $\sum \frac{1}{n^2}$ is convergent. These are the two benchmark series that the Divergence Test could not distinguish (Section 13.8).

Example 13.9. Show that $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ diverges.

The starting index is $n = 2$ because $\ln 1 = 0$. Set $f(x) = \frac{1}{x \ln x}$, which is continuous, positive, and decreasing on $[2, \infty)$. By the Integral Test,

$$\sum_{n=2}^{\infty} \frac{1}{n \ln n} \sim \int_2^\infty \frac{1}{x \ln x}\,dx.$$

An antiderivative of $\frac{1}{x \ln x}$ is $\ln(\ln x)$ (check: $\frac{d}{dx} \ln(\ln x) = \frac{1}{\ln x} \cdot \frac{1}{x}$). Therefore

$$\int_2^b \frac{1}{x \ln x}\,dx = \ln(\ln b) - \ln(\ln 2) \xrightarrow{b \to \infty} \infty.$$

The integral diverges, so by the Integral Test the series diverges as well.

13.12 Comparison Tests

The comparison tests for series are identical in spirit and in proof to the comparison tests for improper integrals. For brevity we state the results without reproving them, but the proofs transfer verbatim from the integral setting.

Both tests rest on the dichotomy for positive series (Theorem 13.7): a positive series is either convergent or ∞. To prove convergence, it suffices to show the partial sums are bounded above.

Theorem 13.10 (Basic Comparison Test). *Suppose $0 \le a_n \le b_n$ for all n (or for all sufficiently large n). Then:*

(i) If $\sum b_n$ converges, then $\sum a_n$ converges.

(ii) If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Theorem 13.11 (Limit Comparison Test). *Let $\sum a_n$ and $\sum b_n$ be positive series. If*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L, \qquad 0 < L < \infty,$$

then $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

Remark. The proof of each theorem follows the same pattern as for improper integrals: replace every integral with a series, and every proper integral with a partial sum. The details are otherwise unchanged.

13.13 Alternating Series Test

We now move beyond positive series. A large and well-behaved family consists of alternating series — those whose terms switch sign at every step. Under mild hypotheses, they must converge, and there is a clean geometric picture explaining why.

Definition 13.4 (Alternating series). A series $\sum a_n$ is *alternating* if $a_n \cdot a_{n+1} < 0$ for every n — that is, consecutive terms have opposite signs.

A typical alternating series has the form $\sum(-1)^n b_n$ or $\sum(-1)^{n+1} b_n$ with $b_n > 0$.

Theorem 13.12 (Alternating Series Test (Leibniz Test)). *Suppose the series $\sum_{n=1}^{\infty}(-1)^{n+1} b_n$ satisfies:*

(i) $b_n > 0$ for all n (the series alternates in sign),

(ii) (b_n) is decreasing,

(iii) $\lim_{n\to\infty} b_n = 0$.

Then the series is convergent.

Proof. Write $S_k = \sum_{n=1}^{k}(-1)^{n+1}b_n$. The terms are $b_1, -b_2, b_3, -b_4, \ldots$

Step 1: Ordering of partial sums. Since b_n is decreasing and positive,

$$S_{2k+2} = S_{2k} + (b_{2k+1} - b_{2k+2}) \geq S_{2k},$$
$$S_{2k+1} = S_{2k-1} - (b_{2k} - b_{2k+1}) \leq S_{2k-1}.$$

Also $S_{2k+1} = S_{2k} + b_{2k+1} > S_{2k}$. Therefore

$$S_2 \leq S_4 \leq S_6 \leq \cdots \leq S_5 \leq S_3 \leq S_1.$$

Even partial sums increase; odd partial sums decrease; every even partial sum is less than every odd one.

Step 2: Even and odd subsequences converge. The sequence (S_{2k}) is increasing and bounded above (by S_1), so by the Monotone Convergence Theorem it converges to some limit A. Similarly, (S_{2k+1}) is decreasing and bounded below (by S_2), so it converges to some limit B.

Step 3: The two limits agree. Since $S_{2k+1} = S_{2k} + b_{2k+1}$, taking limits gives $B = A + 0 = A$ (using hypothesis (iii)).

Step 4: Full sequence converges. The even and odd subsequences both converge to the same limit A. By a standard lemma (if the even-indexed and odd-indexed subsequences of a sequence both converge to the same limit, then the full sequence converges to that limit), $\lim_{k\to\infty} S_k = A$. ■

Example 13.10. The alternating harmonic series $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ converges by the Alternating Series Test, since $b_n = \frac{1}{n}$ is positive, decreasing, and has limit 0.

Example 13.11. The series $\displaystyle\sum_{n=2}^{\infty}\frac{(-1)^n}{\ln n}$ converges by the Alternating Series Test, since $b_n = \frac{1}{\ln n}$ is positive, decreasing, and has limit 0.

13.14 Alternating Series Estimation

Alternating series that satisfy the hypotheses of the Alternating Series Test are not only easy to test for convergence — they are also easy to estimate. The error made by truncating the series is bounded by the first omitted term.

Theorem 13.13 (Alternating Series Estimation Theorem). *Under the same three hypotheses as Theorem 13.12, let $S = \sum_{n=1}^{\infty}(-1)^{n+1}b_n$ and $S_k = \sum_{n=1}^{k}(-1)^{n+1}b_n$. Then*

$$|S - S_k| \;\leq\; b_{k+1}.$$

In words: the error made by approximating the infinite sum with its k-th partial sum is at most the absolute value of the first omitted term.

Proof. From the proof of Theorem 13.12, the value S lies between every consecutive pair of partial sums. In particular, S is between S_k and S_{k+1}, so $|S - S_k| \leq |S_{k+1} - S_k| = b_{k+1}$. ■

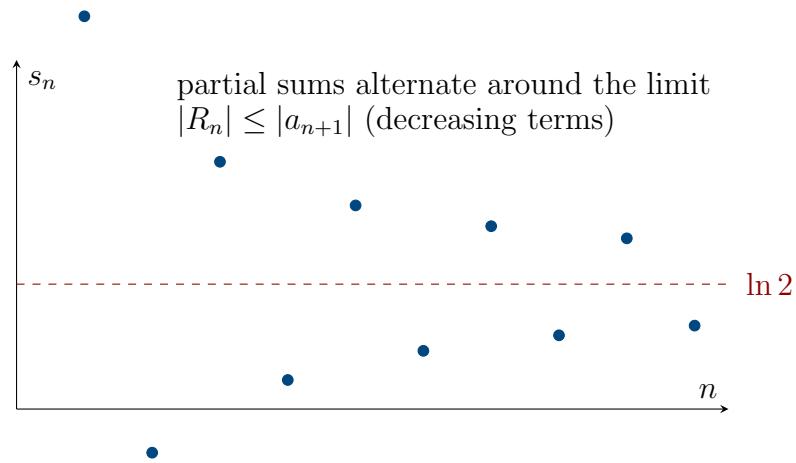


Figure 13.4: Alternating series estimation: the remainder is bounded by the next term.

Example 13.12. Estimate $S = \sum_{n=1}^{\infty} \frac{(-1)^n}{n^4}$ with error less than 0.001.

(Note: this series uses $(-1)^n$ rather than $(-1)^{n+1}$. Since $(-1)^n = -(-1)^{n+1}$, the series equals $-\sum \frac{(-1)^{n+1}}{n^4}$, which is simply the negative of an alternating series in standard form. The Alternating Series Test and its estimation theorem still apply.)

The series satisfies the hypotheses of the Alternating Series Test ($b_n = \frac{1}{n^4}$ is positive, decreasing, and tends to 0), so the estimation theorem applies. Truncating at the k-th term gives error at most $\frac{1}{(k+1)^4}$. We need

$$\frac{1}{(k+1)^4} < 0.001 \quad \Longleftrightarrow \quad (k+1)^4 > 1000.$$

The smallest integer k satisfying this is $k = 5$ (since $6^4 = 1296 > 1000$). Therefore

$$S \approx S_5 = \sum_{n=1}^{5} \frac{(-1)^n}{n^4} = -1 + \frac{1}{16} - \frac{1}{81} + \frac{1}{256} - \frac{1}{625} \approx -0.94754.$$

The exact value (computable by advanced methods) begins $-0.94703\ldots$, confirming that the error is indeed less than 0.001.

Remark. For other types of convergent series, bounding the truncation error is generally much harder. The simplicity of the alternating-series error bound is one of the reasons alternating series are so pleasant to work with.

13.15 Absolute and Conditional Convergence

When a series is neither positive nor alternating, we have few direct tools. A natural strategy is to take absolute values, reducing the problem to a pos-

itive series. This motivates the distinction between absolute and conditional convergence.

Motivation

Example 13.13. Is $\displaystyle\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$ convergent?

This series is not positive (since $\sin n$ changes sign irregularly) and not alternating. However, taking absolute values gives a positive series:

$$\sum_{n=1}^{\infty} \frac{|\sin n|}{n^2} \leq \sum_{n=1}^{\infty} \frac{1}{n^2},$$

and $\sum \frac{1}{n^2}$ converges (p-series with $p = 2$). By the Basic Comparison Test, $\sum \frac{|\sin n|}{n^2}$ converges. Can we then conclude the original series converges?

The Absolute Convergence Test

Theorem 13.14 (Absolute Convergence Test). *If* $\displaystyle\sum_{n=1}^{\infty} |a_n|$ *converges, then* $\displaystyle\sum_{n=1}^{\infty} a_n$ *converges.*

Returning to Example 13.13: since $\sum \frac{|\sin n|}{n^2}$ *converges, the Absolute Convergence Test tells us that* $\sum \frac{\sin n}{n^2}$ *converges as well.*

Remark (Warning). The converse is false. A series may converge while the series of absolute values diverges (see the definitions below).

Absolute versus conditional convergence

Definition 13.5 (Absolute and conditional convergence). A convergent series $\sum a_n$ is called:

- *absolutely convergent* if $\sum |a_n|$ also converges,
- *conditionally convergent* if $\sum |a_n|$ diverges.

Example 13.14. The alternating harmonic series $\sum \frac{(-1)^{n+1}}{n}$ is **conditionally convergent**: the series itself converges (Alternating Series Test), but $\sum \frac{1}{n}$ diverges (p-series with $p = 1$).

Example 13.15. The series $\sum \frac{(-1)^{n+1}}{n^2}$ is **absolutely convergent**: the series itself converges (Alternating Series Test), and $\sum \frac{1}{n^2}$ also converges (p-series with $p = 2$).

Remark. Absolutely convergent series are the "nice" type. They behave much like finite sums and enjoy all the properties we would expect (including commutativity of terms). Conditionally convergent series are **treacherous**: they converge, but rearranging their terms can change the sum, or even make the series diverge. See Section 13.17.

13.16 Proof of Absolute Convergence

We prove the Absolute Convergence Test (Theorem 13.14) by separating a series into its positive and negative parts and using comparison.

Positive and negative parts

Given a series $\sum a_n$*, define its* positive part $a_n^+ = \max(a_n, 0)$ *and its* negative part $a_n^- = \min(a_n, 0)$*. Write* P.T. $= \sum a_n^+$ *for the sum of the positive parts and* N.T. $= \sum a_n^-$ *for the sum of the negative parts. For finite sums,* $\sum a_n =$ P.T. + N.T. *is obvious. For infinite sums, this splitting is valid* ***only when both*** P.T. ***and*** N.T. ***converge****.*

Lemma 13.15 (Splitting a series into positive and negative parts). *If* $\sum$ P.T. *and* $\sum$ N.T. *both converge, then* $\sum a_n$ *converges and* $\sum a_n = \sum$ P.T. $+$ $\sum$ N.T.

Proof. Write the k-th partial sum as $S_k = P_k + N_k$, where P_k and N_k are the partial sums of the positive and negative parts respectively. This identity holds for finite sums. Since both $\lim P_k$ and $\lim N_k$ exist by hypothesis, the limit law gives $\lim S_k = \lim P_k + \lim N_k$. ■

Splitting $\lim(P_k + N_k) = \lim P_k + \lim N_k$ *requires both limits to exist.*

Remark. When $\sum$ P.T. $= \infty$ and $\sum$ N.T. $= -\infty$, the splitting tells us nothing: the original series may still converge or diverge, depending on the order of the terms. This is exactly what happens for conditionally convergent series — see Section 13.17.

Proof of the Absolute Convergence Test

Proof of Theorem 13.14. Assume $\sum |a_n|$ converges.

The positive terms satisfy $|a_n^+| \leq |a_n|$, where $a_n^+ = a_n$ if $a_n > 0$ and 0 otherwise. By the Basic Comparison Test, $\sum a_n^+$ converges.

Similarly, $|a_n^-| \leq |a_n|$, where $a_n^- = a_n$ if $a_n < 0$ and 0 otherwise. The series $\sum |a_n^-|$ converges by comparison, so $\sum a_n^-$ converges as well (it differs only by a sign).

Both the positive and negative parts converge. By Lemma 13.15, $\sum a_n = \sum a_n^+ + \sum a_n^-$ converges. ■

Comparison with $\sum |a_n|$ *bounds both the positive and negative parts; Lemma 13.15 recombines them.*

13.17 Rearrangements of Series

This is perhaps one of the most surprising results about infinite sums: rearranging the terms of a convergent series can change its value. We explore when this happens and why.

Example 13.16. The alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ converges to $\ln 2$. Rearranging the same terms — two positives, then one negative, repeating — produces a different convergent series with sum $\frac{3}{2} \ln 2$. Same terms, different order, different answer.

Theorem 13.16 (Rearrangements of absolutely convergent series). *If $\sum a_n$ is **absolutely convergent**, then every rearrangement of its terms converges to the same sum.*

Theorem 13.17 (Riemann Rearrangement Theorem). *If $\sum a_n$ is **conditionally convergent**, then for every $L \in \mathbb{R}$ there exists a rearrangement of the series that converges to L. There also exist rearrangements that diverge to $+\infty$, to $-\infty$, or by oscillation.*

Three properties of a conditionally convergent series $\sum a_n$ explain why this is possible:

(i) $\lim_{n\to\infty} a_n = 0$ (true for any convergent series, by the necessary condition for convergence, Theorem 13.5).

(ii) The positive terms alone sum to $+\infty$.

(iii) The negative terms alone sum to $-\infty$.

Proof sketch of Theorem 13.17. Suppose we wish to rearrange the terms so the sum equals a target value L.

1. Add positive terms (in their original order) until a partial sum first exceeds L. This is possible because the positive terms sum to ∞.

2. Add negative terms until a partial sum first drops below L. This is possible because the negative terms sum to $-\infty$.

3. Repeat, alternating between positive and negative terms.

Each time the partial sum overshoots or undershoots L, it does so by at most the term just added. Since $\lim a_n = 0$, these overshoots shrink to zero, so the partial sums converge to L. ■

Remark. This theorem is shocking: the "value" of a conditionally convergent series depends on the *order* we add its terms. In summary:

- **Absolutely convergent** series are the "good" type — they behave like finite sums.

- **Conditionally convergent** series are treacherous — they converge, but one must not rearrange their terms.

13.18 The Ratio Test

The Ratio Test determines convergence by examining how fast the terms of a series shrink. It is especially effective for series involving factorials and exponentials.

Theorem 13.18 (Ratio Test). *Let $\sum a_n$ be a series with $a_n \neq 0$ for all n. Suppose*

$$L = \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|}$$

exists (or equals ∞). Then:

*(i) If $L < 1$, the series is **absolutely convergent**.*

*(ii) If $L > 1$ (including $L = \infty$), the series is **divergent**.*

*(iii) If $L = 1$, the test is **inconclusive**.*

Idea of the proof

***Case** $L > 1$. For large n, $|a_{n+1}| > |a_n|$, so the terms grow in absolute value and cannot tend to 0. By the Divergence Test (Theorem 13.5), the series diverges.*

***Case** $L < 1$. For large n, $|a_{n+1}| \approx L \cdot |a_n|$, meaning each term is roughly a fixed fraction L of the previous one. This mirrors the behaviour of a geometric series $\sum L^n$, which converges because $|L| < 1$. A (rigorous) comparison with such a geometric series shows $\sum |a_n|$ converges, so the series is absolutely convergent.*

Remark (Pause and Try). Try to fill in the formal ε-details of the proof for the case $L < 1$: choose r with $L < r < 1$, show $|a_{n+1}| \leq r\,|a_n|$ for all large n, and compare $\sum |a_n|$ with a convergent geometric series.

13.19 Ratio Test Examples

We illustrate the Ratio Test with two examples that highlight its strengths and its limitations.

Example 13.17. Determine whether $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n\, e^{2n}}{n!\,(n+3)}$ converges.

Set $a_n = \frac{(-1)^n\, e^{2n}}{n!\,(n+3)}$. Compute

$$\begin{aligned}\frac{|a_{n+1}|}{|a_n|} &= \frac{e^{2(n+1)}}{(n+1)!\,(n+4)} \cdot \frac{n!\,(n+3)}{e^{2n}} \\ &= \frac{e^2}{n+1} \cdot \frac{n+3}{n+4}.\end{aligned}$$

Therefore

$$L = \lim_{n\to\infty} \frac{e^2}{n+1} \cdot \frac{n+3}{n+4} = 0.$$

Since $L = 0 < 1$, the Ratio Test tells us the series is **absolutely convergent**.

Remark. The Ratio Test is particularly effective when a_n involves factorials and exponentials, because the ratio $|a_{n+1}/a_n|$ simplifies these terms cleanly.

Example 13.18. Attempt to use the Ratio Test on $\displaystyle\sum_{n=1}^{\infty} \frac{n+1}{n^2+2}$.

Set $a_n = \frac{n+1}{n^2+2}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{(n+2)(n^2+2)}{((n+1)^2+2)(n+1)}.$$

Both numerator and denominator are degree-3 polynomials with leading coefficient 1, so

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = 1.$$

The Ratio Test is **inconclusive**.

For this series the Limit Comparison Test (Theorem 13.11) is the right tool. Comparing with $\sum \frac{1}{n}$*:*

$$\lim_{n\to\infty} \frac{(n+1)/(n^2+2)}{1/n} = \lim_{n\to\infty} \frac{n(n+1)}{n^2+2} = 1,$$

so the series diverges (since the harmonic series diverges).

Remark. In general, whenever a_n is a rational function of n (a quotient of polynomials), the Ratio Test gives $L = 1$ and is inconclusive. The Limit Comparison Test with an appropriate p-series is the better choice. These two tests complement each other: the Ratio Test handles factorials and exponentials; the Limit Comparison Test handles rational functions.

Chapter 14

Power Series and Taylor Series

14.1 A Motivating Example

We begin with a concrete example of a power series before defining the general notion. This example illustrates the phenomena that arise every time: a radius of convergence, an interval of convergence, and different behaviour at the endpoints.

Example 14.1. Define a function g by

$$g(x) \;=\; \sum_{n=1}^{\infty} \frac{x^n}{n \cdot 3^n}.$$

The domain of g is the set of real numbers x for which this series converges, since convergent means the series adds up to a number.

Without any calculation, we can already anticipate the shape of the answer. When x is large the terms x^n are large, and we expect divergence. When x is small, close to zero, the terms are small, and we expect convergence.

Remark (Pause and Try). Find the exact domain of g. This is an exercise in convergence tests; begin with the ratio test.

Applying the ratio test

Set $a_n = \dfrac{x^n}{n \cdot 3^n}$. Then

$$\frac{|a_{n+1}|}{|a_n|} \;=\; \frac{|x|^{n+1}}{(n+1) \cdot 3^{n+1}} \cdot \frac{n \cdot 3^n}{|x|^n} \;=\; \frac{|x|}{3} \cdot \frac{n}{n+1}.$$

Taking the limit as $n \to \infty$, the factor $\frac{n}{n+1} \to 1$, so

$$\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} \;=\; \frac{|x|}{3}.$$

- If $|x| < 3$, this limit is less than 1, and the ratio test gives **absolute convergence**.

- If $|x| > 3$, this limit is greater than 1, and the ratio test gives **divergence**.
- If $|x| = 3$, the ratio test is inconclusive; we check the endpoints individually.

Checking the endpoints

At $x = 3$:

$$\sum_{n=1}^{\infty} \frac{3^n}{n \cdot 3^n} = \sum_{n=1}^{\infty} \frac{1}{n},$$

the harmonic series, which is **divergent**.

At $x = -3$:

$$\sum_{n=1}^{\infty} \frac{(-3)^n}{n \cdot 3^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n},$$

which is **conditionally convergent** by the alternating series test.

Conclusion

The domain of g is the interval $[-3, 3)$. We call this interval the *interval of convergence*, and the number 3 (half the length of the interval) the *radius of convergence*. In the next section we will see that this picture—an interval centred at some point, absolute convergence in the interior, divergence in the exterior, and case-by-case behaviour at the endpoints—appears for every power series.

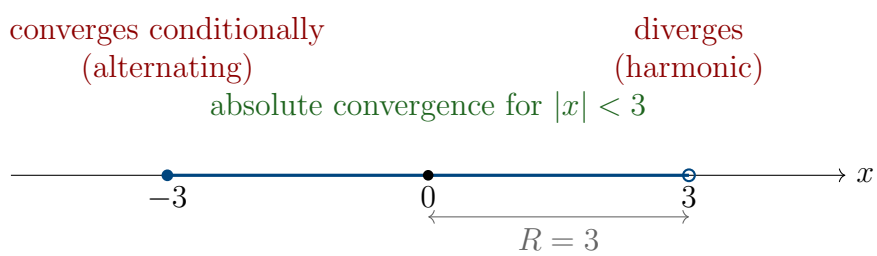


Figure 14.1: A typical power-series convergence picture: absolute inside, case-by-case at endpoints.

14.2 Definition of Power Series

Polynomials are nice. Taking derivatives, computing integrals, evaluating limits—everything is easy with polynomials. The idea behind power series is to enlarge the family of "nice, easy functions" from finite polynomials to infinite polynomials. An infinite polynomial is a series of powers, and that is why it is called a power series.

Definition 14.1 (Power series). Let $a \in \mathbb{R}$ and let $c_0, c_1, c_2, \ldots$ be real numbers. A *power series centred at a* is an expression of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n \;=\; c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

The numbers c_n are called the *coefficients*. When the power series is convergent, it defines a function of x.

Remark. If the notation is intimidating, think first of the case $a = 0$. Then the power series is simply $c_0 + c_1x + c_2x^2 + \cdots$, which looks exactly like a polynomial of infinite degree.

The domain of a function defined by a power series is the set of real numbers x for which the series converges. One value is always in the domain: $x = a$, because every term except c_0 vanishes. In the best case the domain is all of $\mathbb{R}$; in the worst case it is just the single point $\{a\}$. In general, some values of x will be in the domain and some will not.

The main theorem

The following theorem, stated in three parts, describes the structure of convergence for every power series and explains why power series are so powerful.

Theorem 14.1 (Structure of power series). *Let $\sum_{n=0}^{\infty} c_n(x-a)^n$ be a power series centred at a.*

(i) ***Interval of convergence.*** *The domain (the set of values of x for which the series converges) is always an interval centred at a. It could be a single point, a bounded interval (open, closed, or half-open), or all of $\mathbb{R}$.*

(ii) ***Nature of convergence.*** *In the* interior *of this interval, the series is* ***absolutely convergent****. In the* exterior*, it is* ***divergent****. At the endpoints, if any, anything may happen: the series may converge absolutely, converge conditionally, or diverge, independently at each endpoint.*

(iii) ***Polynomial-like behaviour.*** *In the interior of the interval of convergence, a power series can be* ***treated like a polynomial****. That is, power series may be added, multiplied, composed, differentiated, and integrated* ***term by term****, in the same way as finite polynomials. In particular, differentiation and integration do not change the radius of convergence. For sums and products of two power series, the resulting series converges at least on the intersection of the two intervals of convergence. Composition $f \circ g$ is valid when $g(x)$ lies inside the interval of convergence of f.*

Definition 14.2 (Interval and radius of convergence). The domain described in part (i) of Theorem 14.1 is called the *interval of convergence*. Its

radius (half its length) is called the *radius of convergence*, denoted R. We allow $R = 0$ (when the domain is $\{a\}$) and $R = \infty$ (when the domain is $\mathbb{R}$).

That the domain is always an interval centred at a is intuitively natural: the farther x is from a, the larger $|x - a|$ is, and the more likely the series is to diverge. The closer x is to a, the more likely it is to converge.

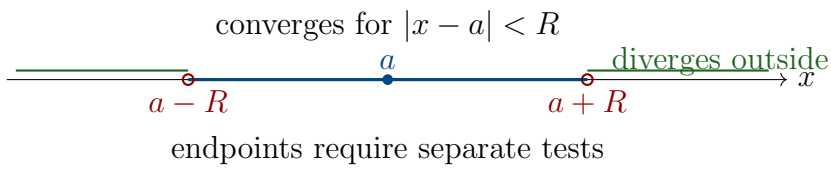


Figure 14.2: Radius of convergence for a power series centered at a.

Derivatives and antiderivatives

Part (iii) of Theorem 14.1 has an especially useful consequence for derivatives and antiderivatives. If

$$f(x) \;=\; \sum_{n=0}^{\infty} c_n\, x^n,$$

then inside the interval of convergence,

$$f'(x) \;=\; \sum_{n=1}^{\infty} n\, c_n\, x^{n-1} \qquad \text{and} \qquad \int_0^x f(t)\, dt \;=\; \sum_{n=0}^{\infty} \frac{c_n}{n+1}\, x^{n+1}.$$

These are obtained by differentiating or integrating term by term, exactly as one would do for a finite polynomial.

Remark. The proof of Theorem 14.1 is technical. The key ingredient is that in the interior of the interval of convergence, the series enjoys *absolute uniform convergence* on every compact subset. We will not develop that theory here; instead we focus on stating the theorem and learning to use it.

The road ahead

Our ultimate goal is twofold. First, try to rewrite as many functions as possible as power series, using Taylor series*. Second, once we have done that, exploit the polynomial-like behaviour of power series to make calculations—limits, integrals, estimates, differential equations—**much** easier. This programme will occupy us for the remainder of this chapter, and it is well worth the effort.*

14.3 Taylor Polynomials

Our goal is to approximate functions with polynomials. Polynomials are simple, and they make everything easy. Given a function f and a point a in its

domain, we want to find a polynomial P such that $P(x)$ is close to $f(x)$ when x is near a.

The tangent line as a first approximation

We already know one example of such an approximation: the tangent line. Of all lines through the point $(a, f(a))$, the tangent line is the one whose graph stays closest to the graph of the function near a. But what, precisely, does "closest" mean? And how can we construct even better polynomial approximations?

The remainder

Define the *remainder* (or *error*) as the difference between the function and the polynomial:

$$R(x) \;=\; f(x) - P(x).$$

We want this error to be small near a. Saying $R(a) = 0$ is too weak: many lines pass through $(a, f(a))$, and most are poor approximations. Saying $\lim_{x\to a} R(x) = 0$ is still not enough, because every line through $(a, f(a))$ satisfies this.

What makes the tangent line special is that its remainder approaches zero ***fast****. The remainder for every other line also tends to zero, but the tangent line's remainder does so faster than any other line's. This is the key idea: we need the error to approach zero* fast.

A scale for "fast"

To measure how fast a function approaches zero, we need a scale. A natural one is provided by powers of $(x - a)$.

For simplicity, consider first the case $a = 0$. As $x \to 0$, every positive power x^n approaches 0, but the larger the exponent n, the faster the approach. This gives us our yardstick: we compare the remainder with powers of x.

Definition 14.3 (Order of approximation). Let f and g be continuous at a, and let $n \in \mathbb{N}$. We say that g is an *approximation for f near a of order n* when

$$\lim_{x\to a} \frac{f(x) - g(x)}{(x-a)^n} \;=\; 0.$$

Note that for $n \geq 1$, this condition forces $f(a) = g(a)$ (otherwise the limit would not be finite).

The limit is an indeterminate form $\frac{0}{0}$: the numerator $f(x) - g(x)$ tends to 0 (because both are continuous at a and approximate each other there), and the denominator $(x-a)^n$ tends to 0. For this indeterminate form to resolve to 0, the numerator must approach 0 faster than the denominator. That is precisely the content of the definition.

Definition 14.4 (Taylor polynomial — first definition). Let $a \in \mathbb{R}$, let f be continuous at and near a, and let $n \in \mathbb{N}$. The *n-th Taylor polynomial for f at a* is a polynomial P_n of degree at most n that is an approximation for f near a of order n. That is,

$$\lim_{x \to a} \frac{f(x) - P_n(x)}{(x-a)^n} = 0.$$

Remark. The requirement that $\deg P_n \leq n$ deserves a comment. One can always find polynomials of higher degree with the same approximation property, but polynomials of lower degree are simpler, so we prefer them. Later we will show that, provided f has enough derivatives, such a polynomial always exists and is unique.

Remark (Pause and Try). Check that the first Taylor polynomial P_1 is the tangent line to f at a.

This definition explains where Taylor polynomials come from and why they are useful. However, it is not directly helpful for computations. Based on it alone, we have no recipe for actually constructing P_n. *The next section remedies this.*

14.4 Taylor Polynomials via Derivatives

We now translate the definition of Taylor polynomials into a second, equivalent form that is more concrete and better suited for computations.

Smoothness notation

Definition 14.5 (C^n functions). A function is called C^0 when it is continuous. It is C^1 when its first derivative exists and is continuous, C^2 when the first two derivatives exist and are continuous, and in general C^n when all derivatives up to and including order n exist and are continuous. A function is C^∞ when **all** of its derivatives exist and are continuous.

Remark. We will encounter many results that require different numbers of derivatives. The C^n notation makes statements shorter.

From the limit condition to matching derivatives

Recall the original definition: g is an approximation for f near a of order n when

$$L = \lim_{x \to a} \frac{f(x) - g(x)}{(x-a)^n} = 0.$$

We now investigate, under the assumption that f and g are C^n, what conditions on the derivatives of f and g are equivalent to $L = 0$.

Suppose first that $f(a) \neq g(a)$. Then the numerator has a nonzero limit while the denominator tends to 0, so L does not exist as a finite number. Hence we need $f(a) = g(a)$.

With $f(a) = g(a)$ we have the indeterminate form $\frac{0}{0}$. Apply L'Hôpital's Rule:

$$L = \lim_{x \to a} \frac{f'(x) - g'(x)}{n(x-a)^{n-1}}.$$

The same argument now forces $f'(a) = g'(a)$, and we apply L'Hôpital's Rule again. Repeating this process n times, we arrive at

$$L = \lim_{x \to a} \frac{f^{(n)}(x) - g^{(n)}(x)}{n!} = \frac{f^{(n)}(a) - g^{(n)}(a)}{n!},$$

where the last step uses the continuity of the n-th derivatives.

Theorem 14.2 (Approximation via matching derivatives). *Let f and g be C^n functions at a point a. Then the following are equivalent:*

(i) $\lim_{x \to a} \frac{f(x) - g(x)}{(x-a)^n} = 0.$ *(The function g approximates f near a of order n.)*

(ii) $f^{(k)}(a) = g^{(k)}(a)$ *for every* $k = 0, 1, 2, \ldots, n$.

In words: g is a good approximation for f near a if and only if f and g have the same first few derivatives at a. How good the approximation is depends on how many derivatives agree.

Second definition of Taylor polynomial

We can now rewrite the definition of Taylor polynomial using condition (ii) of Theorem 14.2.

Definition 14.6 (Taylor polynomial — second definition). Let $a \in \mathbb{R}$, $n \in \mathbb{N}$, and let f be a C^n function at a. The *n-th Taylor polynomial for f at a* is the polynomial P_n of degree at most n satisfying

$$P_n^{(k)}(a) = f^{(k)}(a) \qquad \text{for each } k = 0, 1, \ldots, n.$$

Remark. This is not a completely new idea. The first Taylor polynomial is a polynomial of degree at most 1 whose value at a is $f(a)$ and whose derivative at a is $f'(a)$. That is exactly the tangent line.

This definition is more useful for computations than the first: we are simply specifying the value of the polynomial and its first n derivatives at a. But we should always keep the first definition (Definition 14.4) in mind—it tells us why *Taylor polynomials are good approximations. In the next section, we take this one step further, providing an explicit formula for the coefficients.*

14.5 The Explicit Formula and Taylor Series

We have defined Taylor polynomials in two equivalent ways: first via the approximation property (Definition 14.4), then via matching derivatives (Definition 14.6). Both are useful for understanding, but neither gives us a quick recipe for computing the coefficients. Our goal now is to simplify the definition even further and obtain an explicit formula.

The case $a = 0$

A polynomial of degree at most n can be written as

$$P_n(x) \;=\; c_0 + c_1\,x + c_2\,x^2 + \cdots + c_n\,x^n.$$

We impose the matching conditions from Definition 14.6: $P_n^{(k)}(0) = f^{(k)}(0)$ for $k = 0, 1, \ldots, n$.

Remark (Pause and Try). Differentiate P_n repeatedly and evaluate at 0. Each condition determines exactly one coefficient. Try to find the formula before reading on.

Differentiating k times and setting $x = 0$ kills every term except the one of degree k:

$$P_n^{(k)}(0) \;=\; k!\,c_k.$$

The matching condition $P_n^{(k)}(0) = f^{(k)}(0)$ therefore forces

$$c_k \;=\; \frac{f^{(k)}(0)}{k!}.$$

The general case

For an arbitrary centre a, write the polynomial in powers of $(x - a)$ instead of powers of x:

$$P_n(x) \;=\; c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n.$$

This is still a generic polynomial of degree at most n; we have simply changed basis from powers of x to powers of $(x - a)$. The advantage is that the k-th derivative evaluated at a depends only on the coefficient c_k.

Remark (Pause and Try). Impose the conditions $P_n^{(k)}(a) = f^{(k)}(a)$ for $k = 0, 1, \ldots, n$ and find the coefficients. Take as much time as you need.

The same calculation gives $P_n^{(k)}(a) = k!\,c_k$, and the matching condition yields

$$c_k \;=\; \frac{f^{(k)}(a)}{k!}.$$

Third definition of Taylor polynomial

Definition 14.7 (Taylor polynomial — explicit formula). Let f be a C^n function at a. The n-th Taylor polynomial for f at a is

$$P_n(x) \;=\; \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}\,(x-a)^k.$$

Some textbooks take this formula as the starting point. We chose to begin with the more abstract approximation definition (Definition 14.4) and slowly derive the formula from it, because the abstract version explains why *Taylor polynomials are good approximations. It is best to understand all three equivalent viewpoints.*

Remark. By construction the polynomial has degree at most n: the terms up to degree n suffice to match the first n derivatives. Higher-degree terms could be added without affecting those derivatives, but they are unnecessary. Since the formula pins down every coefficient, the n-th Taylor polynomial of a given function at a given point is **unique**. This was not obvious from the earlier definitions, where it seemed possible that two different polynomials might satisfy the same conditions.

Graphical intuition

Consider a function and its Taylor polynomials at $a = 0$. The first Taylor polynomial is the tangent line—a reasonable approximation very close to 0. The second Taylor polynomial is better, and the third better still. The sixth Taylor polynomial produces a fantastic approximation, not just near 0, but on a surprisingly large interval around it. The higher the order, the better the approximation.

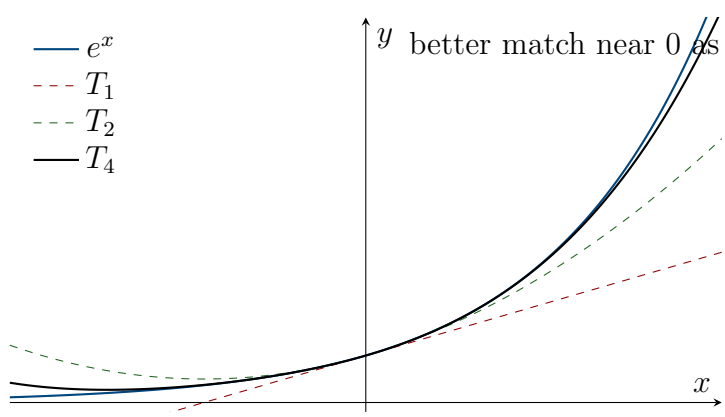


Figure 14.3: Taylor polynomial approximations to e^x near 0.

The Taylor series

This raises a natural question: what happens if we use a "polynomial of infinite degree"? Such an object is not a polynomial—it is a power series—but if every finite truncation is a good approximation, perhaps the infinite version is an ***exact*** *representation of the function.*

Definition 14.8 (Taylor series). Let f be a C^∞ function at a. The *Taylor series for f at a* is the power series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}\,(x-a)^k.$$

This is the same formula as in Definition 14.7, but with the sum extended to all degrees. All the derivatives of the power series at a agree with the corresponding derivatives of f.

Like any power series, the Taylor series makes sense only where it converges. In the ideal case it converges to $f(x)$ on some interval around a.

Remark (Warning). A function that equals its Taylor series on an interval around a is called *analytic*. Not every C^∞ function is analytic: there exist C^∞ functions whose Taylor series converges but does **not** equal the function. However, most of the common functions we encounter are analytic. Whether a given function equals its Taylor series is a deeper question, addressed in Section 14.7.

14.6 Computing Maclaurin Series

In the previous sections we derived the explicit formula for Taylor polynomials and the Taylor series. In this section we illustrate them with examples.

Definition 14.9 (Maclaurin series). When the Taylor series is centred at $a = 0$, it is called a *Maclaurin series*. That is, the Maclaurin series for f is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}\,x^k.$$

We say that a function is analytic *when it is equal to its Taylor series (see Section 14.7 for the precise definition). For analytic functions, the Taylor series is not merely an approximation—it* ***is*** *the function. Spoiler: most common functions, including exponentials and trigonometric functions, are analytic. It will take some work to prove this. For now, we focus on* finding *the Maclaurin series, without yet proving equality.*

Example 14.2 *(Maclaurin series for e^x).* Define $f(x) = e^x$. Since the derivative of e^x is itself, every derivative equals e^x, and so

$$f^{(k)}(0) = e^0 = 1 \qquad \text{for all } k \geq 0.$$

The Maclaurin series is therefore

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

This equality holds for **all** $x \in \mathbb{R}$. (The proof of analyticity is given in Section 14.8.)

Remark. If we want the Taylor series for e^x centred at a point $c \neq 0$, one convenient method is to substitute $u = x - c$. Then $e^x = e^{c+u} = e^c \cdot e^u$, and we can use the Maclaurin series for e^u:

$$e^x = e^c \sum_{k=0}^{\infty} \frac{(x-c)^k}{k!}.$$

This substitution trick works for many functions, and it is the reason we most often work with Maclaurin series.

Example 14.3 *(Maclaurin series for* $\sin x$*).* The derivatives of $\sin x$ repeat in a cycle of length four:

$$\sin x,\ \cos x,\ -\sin x,\ -\cos x,\ \sin x,\ \ldots$$

Evaluating at $x = 0$ gives the pattern $0, 1, 0, -1, 0, 1, 0, -1, \ldots$. All even-order derivatives vanish, and the odd-order derivatives alternate between 1 and -1. Therefore

$$\sin x = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} x^{2m+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

This holds for all $x \in \mathbb{R}$.

Remark (Pause and Try). Repeat the calculation for $\cos x$. You should obtain

$$\cos x = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} x^{2m} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

valid for all $x \in \mathbb{R}$.

The four fundamental Maclaurin series

To summarise, we now have four Maclaurin series that will serve as building blocks:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \qquad |x| < \infty,$$

$$\sin x = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} x^{2m+1}, \qquad |x| < \infty,$$

$$\cos x = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} x^{2m}, \qquad |x| < \infty,$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \qquad |x| < 1.$$

We will use all kinds of manipulations to derive Maclaurin series for new functions from these four.

14.7 Analytic Functions and Remainders

Fair warning: this section is packed with theorems, some quite technical. They will make the rest of this chapter considerably easier, so they are worth the effort.

Precise definition of analytic

So far we have been casually saying that a function is analytic when it equals its Taylor series. But we need to specify which *Taylor series, and on which domain.*

Definition 14.10 (Analytic function). Let f be a C^∞ function on an open interval I, and let $a \in I$. Denote by S_a the Taylor series of f centred at a. We say f is *analytic at a* when $f(x) = S_a(x)$ for all x in some open interval centred at a. We say f is *analytic* (without specifying a point) when it is analytic at every point of its domain.

Properties of analytic functions

Theorem 14.3 (Algebra of analytic functions). *The following hold:*

(i) Polynomials are analytic.

(ii) Sums, products, quotients (where the denominator is nonzero), and compositions of analytic functions are analytic.

(iii) Derivatives and antiderivatives of analytic functions are analytic.

Moreover, all operations listed in (ii) *and* (iii) *can be performed on the corresponding Taylor series term by term, exactly as one would do with polynomials. The Taylor series of a function at a point is unique.*

Remark. Because of Theorem 14.3, our strategy is very efficient. We only need to prove that the exponential and the sine function are analytic. All other elementary functions can then be obtained from these two, from polynomials, and from the operations listed above.

Remainders and analyticity

Let us make a plan for proving a function is analytic.

Let f be C^∞ on an open interval, and fix a in that interval. We can always write

$$f(x) \;=\; P_n(x) + R_n(x),$$

where P_n is the n-th Taylor polynomial and $R_n = f - P_n$ is the n-th remainder. The Taylor series is

$$S_a(x) \;=\; \lim_{n\to\infty} P_n(x),$$

since the Taylor polynomials are the partial sums of the Taylor series. Therefore

$$f(x) = S_a(x) \quad \Longleftrightarrow \quad \lim_{n\to\infty} R_n(x) = 0.$$

Remark (Warning). There are **two different limits** involving R_n that must not be confused.

- From the *definition* of Taylor polynomials, we know that for a fixed n, the remainder $R_n(x) \to 0$ as $x \to a$ (and it does so fast).
- To prove a function is *analytic*, we need the remainder $R_n(x) \to 0$ as $n \to \infty$, for a fixed value of x.

These are different limits, and the first does not imply the second.

Lagrange's Remainder Theorem

To show $R_n \to 0$ as $n \to \infty$, we need a concrete formula for R_n. This is provided by the remainder theorems—a family of results that give explicit expressions for the remainder. There are several versions (Lagrange, Cauchy, integral form); all are proved using the Mean Value Theorem and are interchangeable for our purposes.

Theorem 14.4 (Lagrange's Remainder Theorem). *Let I be an open interval, $a \in I$, $n \in \mathbb{N}$, and let f be C^{n+1} on I. Write*

$$f(x) = P_n(x) + R_n(x),$$

where P_n is the n-th Taylor polynomial for f at a. Then for every $x \in I$, there exists a number ξ between a and x such that

$$R_n(x) \;=\; \frac{f^{(n+1)}(\xi)}{(n+1)!}\,(x-a)^{n+1}.$$

Remark. This is reminiscent of the Mean Value Theorem: under certain hypotheses, there exists some intermediate point ξ satisfying an equation. The exact value of ξ is unknown, but the formula is powerful nonetheless, because it lets us **bound** R_n.

Lagrange's Remainder Theorem has two main applications. First, we can use it to prove that a function is analytic, by showing $R_n \to 0$ as $n \to \infty$ (see Section 14.8). Second, we can use it to ***estimate*** *the value of a complicated function with a Taylor polynomial and bound the error (see Section 14.11).*

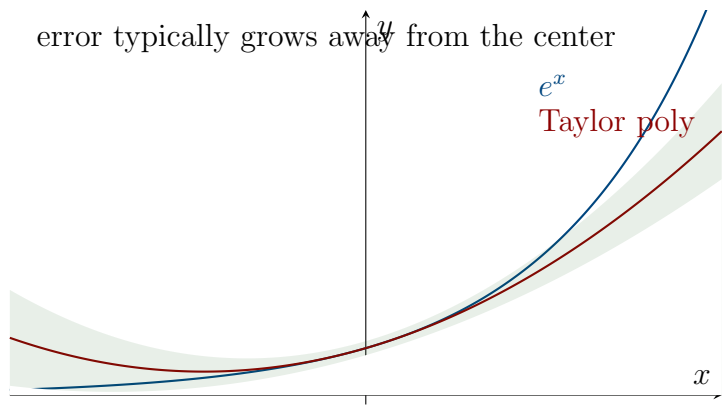


Figure 14.4: A qualitative "error band" around a Taylor approximation.

14.8 Analyticity of the Exponential

We now prove that the exponential function equals its Maclaurin series. This is a model example of how to use Lagrange's Remainder Theorem to establish analyticity.

Theorem 14.5 (e^x is analytic). *For every $x \in \mathbb{R}$,*

$$e^x \;=\; \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

Proof. Fix $x \in \mathbb{R}$. Write $f(t) = e^t$. Since every derivative of f equals e^t, the n-th Taylor polynomial at $a = 0$ is

$$P_n(x) \;=\; \sum_{k=0}^{n} \frac{x^k}{k!},$$

and $f(x) = P_n(x) + R_n(x)$. We must show $R_n(x) \to 0$ as $n \to \infty$.

By Theorem 14.4, there exists ξ between 0 and x with

$$R_n(x) \;=\; \frac{e^{\xi}}{(n+1)!}\, x^{n+1}.$$

Case $x > 0$**.** Since $0 < \xi < x$, the exponential is increasing, so $e^{\xi} \le e^{x}$. Therefore

$$|R_n(x)| \;\le\; e^{x} \cdot \frac{|x|^{n+1}}{(n+1)!}.$$

Since $\dfrac{|x|^{n+1}}{(n+1)!} \to 0$ as $n \to \infty$ (factorial growth dominates any fixed power), the Squeeze Theorem gives $R_n(x) \to 0$.

The constant e^x does not depend on n.

Case $x < 0$**.** Since $x < \xi < 0$, we have $e^{\xi} \le e^{0} = 1$. The same argument applies with the bound $|R_n(x)| \le \dfrac{|x|^{n+1}}{(n+1)!} \to 0$.

Case $x = 0$**.** Trivially $R_n(0) = 0$ for all n. ■

Remark (Pause and Try). Use a very similar argument to prove that $\sin x$ is analytic:

$$\sin x \;=\; \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!}\, x^{2m+1} \qquad \text{for all } x \in \mathbb{R}.$$

The key observation is that every derivative of sin is bounded in absolute value by 1, so $\left|f^{(n+1)}(\xi)\right| \le 1$ and $|R_n(x)| \le \dfrac{|x|^{n+1}}{(n+1)!} \to 0$.

14.9 Constructing New Power Series

We always have the slow method: compute all the derivatives, plug into the Taylor series formula, then prove analyticity with the Remainder Theorem. But there is a better way. We already have four functions written as power series (exponential, sine, cosine, geometric). Using the fact that power series can be manipulated like polynomials (Theorem 14.1), we can derive power series for new functions quickly.

Remark (Pause and Try). Before reading the solutions below, try to write each of the following functions as a power series centred at 0: e^{-x}, $x^3 \sin(x^2)$, $\dfrac{1}{x}$ centred at 3.

Example 14.4 (Maclaurin series for e^{-x}). Start from the known expansion $e^{x} = \sum_{k=0}^{\infty} \frac{x^k}{k!}$, valid for all x. Replace x by $-x$:

$$e^{-x} \;=\; \sum_{k=0}^{\infty} \frac{(-x)^k}{k!} \;=\; \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}\, x^k \;=\; 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$$

Valid for all $x \in \mathbb{R}$.

Example 14.5 (Maclaurin series for $x^3\sin(x^2)$). Begin with $\sin x = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} x^{2m+1}$, valid for all x. Replace x by x^2:

$$\sin(x^2) \;=\; \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!}\, x^{4m+2}.$$

Multiply term by term by x^3:

$$x^3\sin(x^2) \;=\; \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!}\, x^{4m+5} \;=\; x^5 - \frac{x^9}{3!} + \frac{x^{13}}{5!} - \cdots$$

Valid for all $x \in \mathbb{R}$.

Example 14.6 (Power series for $1/x$ centred at 3). Since the centre is $a = 3$, set $u = x - 3$ so that powers of u appear naturally. Then $x = u + 3$, and

$$\frac{1}{x} = \frac{1}{u+3} = \frac{1}{3}\cdot\frac{1}{1+u/3} = \frac{1}{3}\cdot\frac{1}{1-(-u/3)}.$$

Apply the geometric series with variable $-u/3$:

$$\frac{1}{x} = \frac{1}{3}\sum_{n=0}^{\infty}\Big(-\frac{u}{3}\Big)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{3^{n+1}}\,(x-3)^n.$$

This is valid when $|-u/3| < 1$, i.e. $|x-3| < 3$, i.e. $x \in (0, 6)$.

14.10 The Logarithm Series

The domain of $\ln x$ is only the positive reals. In particular, $\ln$ is not even defined at 0, so we cannot write a Maclaurin series for $\ln x$ directly. Instead, we shift the function and consider $\ln(1+x)$, whose domain includes 0.

Strategy

The function $\ln(1+x)$ does not closely resemble any of our four fundamental series, but its derivative does. Since

$$\frac{d}{dx}\ln(1+x) = \frac{1}{1+x},$$

and the right-hand side is closely related to the geometric series, we will write the derivative as a power series and then integrate to recover $\ln(1+x)$.

Example 14.7 (Maclaurin series for $\ln(1+x)$). We have

$$\frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty}(-x)^n = \sum_{n=0}^{\infty}(-1)^n\, x^n, \qquad |x| < 1.$$

Integrate term by term (valid inside the interval of convergence):

$$\ln(1+x) = \int_0^x \frac{1}{1+t}\,dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1}\,x^{n+1} = \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m}\,x^m,$$

where the second form uses the substitution $m = n+1$. Writing out the first few terms:

$$\ln(1+x) \;=\; x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots, \qquad |x| < 1.$$

Remark. When we differentiate or integrate a power series term by term, the **radius** of convergence does not change. The geometric series has radius 1, so the series for $\ln(1+x)$ also has radius 1. The behaviour at the endpoints, however, may change; in fact the series for $\ln(1+x)$ converges at $x = 1$ (it gives $\ln 2$ as the alternating harmonic series) but not at $x = -1$.

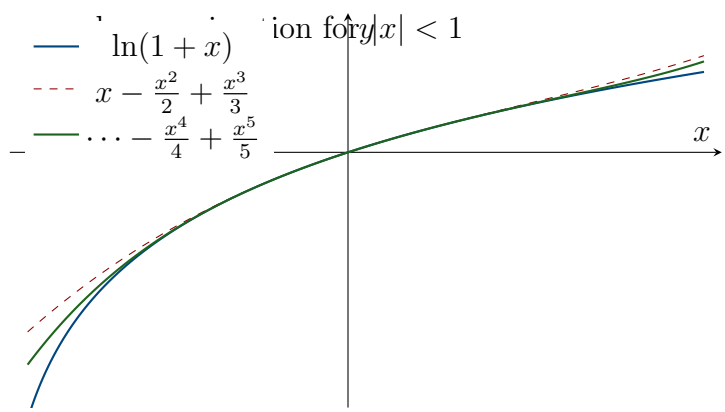


Figure 14.5: Approximating $\ln(1+x)$ by truncating its power series.

Remark (Pause and Try). Use a very similar derivation to write $\arctan x$ as a Maclaurin series. Start with the fact that $\dfrac{d}{dx}\arctan x = \dfrac{1}{1+x^2}$.

14.11 Estimation via Taylor Polynomials

One of the primary uses of Taylor series is to estimate complicated functions. Taylor polynomials were designed precisely for this purpose: they are good approximations. In this section we illustrate the method with a concrete example.

Example 14.8 (Estimating $\sqrt{e}$). We wish to estimate $\sqrt{e} = e^{1/2}$ with an error smaller than 0.001.

Since e^x equals its Maclaurin series, we have

$$e^{1/2} = \sum_{k=0}^{\infty} \frac{(1/2)^k}{k!} = P_n(\tfrac{1}{2}) + R_n(\tfrac{1}{2}),$$

where P_n is the n-th Taylor polynomial and R_n is the remainder. We need to choose n large enough that $|R_n(1/2)| < 0.001$.

Bounding the remainder. By Theorem 14.4, there exists $\xi \in (0, \frac{1}{2})$ with

$$R_n(\tfrac{1}{2}) = \frac{e^{\xi}}{(n+1)!}\left(\frac{1}{2}\right)^{n+1}.$$

Since $e^{\xi} \le e^{1/2} < 2$ (because $e < 3$, so $\sqrt{e} < \sqrt{3} < 2$), we obtain

$$\left|R_n(\tfrac{1}{2})\right| < \frac{2}{2^{n+1}\,(n+1)!} = \frac{1}{2^n\,(n+1)!}.$$

We need $2^n(n+1)! > 1000$. Checking: $2^4 \cdot 5! = 16 \cdot 120 = 1920 > 1000$. So $n = 4$ suffices.

Computing the estimate.

$$P_4(\tfrac{1}{2}) = 1 + \frac{1}{2} + \frac{(1/2)^2}{2!} + \frac{(1/2)^3}{3!} + \frac{(1/2)^4}{4!} = \frac{211}{128} \approx 1.64844.$$

The exact value begins $\sqrt{e} \approx 1.64872\ldots$, confirming that our estimate is correct to three decimal places.

Remark. This is precisely the process a calculator or computer uses to evaluate e^x: replace the function with its Taylor polynomial, compute the polynomial (a finite sum), and bound the error using the Remainder Theorem.

14.12 Integration via Power Series

Power series are, in a sense, polynomials, and nothing is easier to integrate than a polynomial. In this section we illustrate how Taylor series can compute integrals that are impossible or impractical by other means.

Example 14.9 *(An integral with no elementary antiderivative).* Compute $\displaystyle\int_0^3 e^{-x^2}\,dx$.

It can be shown (using differential Galois theory) that e^{-x^2} has no antiderivative expressible in terms of elementary functions. Power series offer a way around this obstacle.

Start with $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ and replace x by $-x^2$:

$$e^{-x^2} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}\,x^{2k}.$$

Integrate term by term:

$$\int_0^3 e^{-x^2}\,dx = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \cdot \frac{3^{2k+1}}{2k+1} = \sum_{k=0}^{\infty} \frac{(-1)^k \cdot 3^{2k+1}}{k!\,(2k+1)}.$$

This is a convergent series (the factorial in the denominator ensures that the terms eventually decrease rapidly), and truncating it to a partial sum yields an estimate to any desired accuracy.

Example 14.10 *(A simpler series than partial fractions).* Find $\displaystyle\int \frac{1}{1-x^8}\,dx$.

One could factor $1-x^8$ and decompose into partial fractions, but the result is extremely complicated. Instead, use the geometric series with variable x^8:

$$\frac{1}{1-x^8} = \sum_{n=0}^{\infty} x^{8n}, \qquad |x| < 1.$$

Integrate term by term:

$$\int \frac{1}{1-x^8}\,dx = \sum_{n=0}^{\infty} \frac{x^{8n+1}}{8n+1} + C, \qquad |x| < 1.$$

This expression is far shorter, far cleaner, and far more useful than the partial-fractions answer.

Remark. The moral: using power series to compute integrals is often preferable—sometimes because no elementary antiderivative exists, and sometimes because the elementary antiderivative, while it exists in principle, is too complicated to be useful.

14.13 Limits via Power Series

Power series provide an extremely efficient method for computing limits of indeterminate forms. The strategy is simple: expand numerator and denominator as power series, and keep only the first nonzero term.

The key observation

Consider $\displaystyle\lim_{x\to 0} \frac{2x^3+7x^5}{5x^3-x^4}$. Factor x^3 from both to get $\dfrac{2+7x^2}{5-x} \to \dfrac{2}{5}$. The only terms that matter are those with the **smallest exponent**. The same idea applies when the numerator and denominator are power series instead of polynomials.

Example 14.11 *(A straightforward limit).* Compute $\displaystyle\lim_{x\to 0} \frac{\sin x - x}{x^3}$.

Replace $\sin x$ by its Maclaurin series:

$$\sin x - x = \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right) - x = -\frac{x^3}{6} + \frac{x^5}{120} - \cdots$$

The first nonzero term is $-\frac{x^3}{6}$. Therefore

$$\lim_{x\to 0} \frac{\sin x - x}{x^3} = \lim_{x\to 0} \frac{-\frac{x^3}{6} + \text{h.o.t.}}{x^3} = -\frac{1}{6}.$$

Example 14.12 (A complicated limit). Compute $\lim_{x\to 0} \dfrac{3x^2 - e^{x^2} + \cos 2x}{x\sin x - \ln(1+x^2)}$.

This is an indeterminate form of type $\frac{0}{0}$. Using L'Hôpital's Rule would require four differentiations of a messy expression. Power series are much faster.

Numerator. Expand each piece:

$$\begin{aligned} 3x^2 &= 3x^2, \\ e^{x^2} &= 1 + x^2 + \frac{x^4}{2!} + \cdots, \\ \cos 2x &= 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \cdots = 1 - 2x^2 + \frac{2}{3}x^4 - \cdots \end{aligned}$$

Combine:

$$3x^2 - e^{x^2} + \cos 2x = (-1+1) + (3-1-2)x^2 + \Big(-\frac{1}{2} + \frac{2}{3}\Big)x^4 + \cdots = \frac{1}{6}x^4 + \text{h.o.t.}$$

The constant and x^2 terms cancel; the first nonzero term is $\frac{1}{6}x^4$.

Denominator. Expand each piece:

$$\begin{aligned} x\sin x &= x\Big(x - \frac{x^3}{3!} + \cdots\Big) = x^2 - \frac{x^4}{6} + \cdots, \\ \ln(1+x^2) &= x^2 - \frac{x^4}{2} + \cdots \end{aligned}$$

Combine:

$$x\sin x - \ln(1+x^2) = (1-1)x^2 + \Big(-\frac{1}{6} + \frac{1}{2}\Big)x^4 + \cdots = \frac{1}{3}x^4 + \text{h.o.t.}$$

Conclusion.

$$\lim_{x\to 0} \frac{3x^2 - e^{x^2} + \cos 2x}{x\sin x - \ln(1+x^2)} = \frac{1/6}{1/3} = \frac{1}{2}.$$

Remark. Here is a secret: experienced analysts and calculus instructors **rarely use L'Hôpital's Rule**. They almost always use this power-series method, because it is faster and less error-prone.

14.14 Exact Summation of Series

There are very few series whose exact value can be computed directly from the definition—mostly just geometric and a few telescoping series. Taylor series change this dramatically: with a little creativity, we can compute the exact value of many more.

Example 14.13 *(Sum of $n/2^n$).* Compute $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2^n}$.

Step 1: Interpret as a power series.

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = \sum_{n=1}^{\infty} n\, x^n \Big|_{x=1/2}.$$

Step 2: Relate to a known series. We know $\sum_{n=0}^{\infty} x^n = \dfrac{1}{1-x}$ for $|x| < 1$. Differentiating term by term:

$$\sum_{n=1}^{\infty} n\, x^{n-1} = \frac{1}{(1-x)^2}.$$

Multiply both sides by x:

$$\sum_{n=1}^{\infty} n\, x^n = \frac{x}{(1-x)^2}.$$

Step 3: Evaluate. Setting $x = \frac{1}{2}$:

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = \frac{1/2}{(1/2)^2} = \frac{1/2}{1/4} = 2.$$

Example 14.14 *(Sum of $2^n/((n+2)\cdot n!)$).* Compute $\displaystyle\sum_{n=0}^{\infty} \frac{2^n}{(n+2)\cdot n!}$.

Step 1: Interpret as a power series.

$$\sum_{n=0}^{\infty} \frac{2^n}{(n+2)\cdot n!} = \sum_{n=0}^{\infty} \frac{x^n}{(n+2)\cdot n!}\Big|_{x=2}.$$

Step 2: Relate to a known series. We know $\sum_{n=0}^{\infty} \dfrac{x^n}{n!} = e^x$, so $\sum_{n=0}^{\infty} \dfrac{x^{n+1}}{n!} = x\, e^x$. To introduce the factor $(n+2)$ in the denominator, integrate: since $\displaystyle\int x^{n+1}\, dx = \frac{x^{n+2}}{n+2}$,

$$\sum_{n=0}^{\infty} \frac{x^{n+2}}{(n+2)\cdot n!} = \int_0^x t\, e^t\, dt.$$

Integration by parts gives

$$\int_0^x t\, e^t\, dt = (x-1)e^x + 1.$$

Step 3: Extract the desired sum. Dividing by x^2:

$$\sum_{n=0}^{\infty} \frac{x^n}{(n+2)\cdot n!} = \frac{(x-1)e^x + 1}{x^2}.$$

Step 4: Evaluate at $x = 2$.

$$\sum_{n=0}^{\infty} \frac{2^n}{(n+2)\cdot n!} = \frac{(2-1)e^2 + 1}{4} = \frac{e^2+1}{4}.$$

14.15 Application to Kinetic Energy

One of the major uses of Taylor series in the sciences is to approximate complicated physical equations with simpler polynomials. In this section we illustrate this with a fundamental example from physics: kinetic energy.

Two formulas for kinetic energy

The kinetic energy of a particle—the energy it possesses because of its motion—has a different formula in classical and relativistic physics.

Classical formula.

$$T_{\text{classical}} \;=\; \frac{1}{2}\, m_0\, v^2,$$

where m_0 is the rest mass and v is the velocity.

Relativistic formula. Einstein's famous equation $E = mc^2$ gives the total energy. The kinetic energy is the total energy minus the rest energy m_0c^2:

$$T_{\text{rel}} \;=\; m_0c^2\left(\frac{1}{\sqrt{1-(v/c)^2}} - 1\right),$$

where c is the speed of light and the relativistic mass is $m = m_0/\sqrt{1-(v/c)^2}$.

These two expressions look very different, yet they describe the same physical quantity. The classical formula should be obtainable from the relativistic one as an approximation when velocities are small compared to c. This is precisely what Taylor polynomials are designed for.

Taylor expansion

Example 14.15. Set $u = (v/c)^2$ and consider the function $f(u) = (1-u)^{-1/2}$. When $v \ll c$, the variable u is small and close to 0, so we expand around $u = 0$.

First-order approximation.

$$f(0) = 1, \qquad f'(u) = \tfrac{1}{2}(1-u)^{-3/2}, \qquad f'(0) = \tfrac{1}{2}.$$

So $f(u) \approx 1 + \frac{1}{2}u$. Substituting back:

$$T_{\text{rel}} \;\approx\; m_0 c^2 \left(1 + \frac{1}{2} \cdot \frac{v^2}{c^2} - 1\right) = \frac{1}{2}\, m_0\, v^2 = T_{\text{classical}}.$$

The classical formula is exactly the first nonzero Taylor polynomial of the relativistic formula.

Second-order approximation.

$$f''(u) = \tfrac{3}{4}(1-u)^{-5/2}, \qquad f''(0) = \tfrac{3}{4}.$$

The second Taylor polynomial is $f(u) \approx 1 + \frac{1}{2}u + \frac{3}{8}u^2$. This gives the first *relativistic correction* to the classical kinetic energy:

$$T_{\text{rel}} \;\approx\; \frac{1}{2}\, m_0\, v^2 + \frac{3}{8}\, m_0\, \frac{v^4}{c^2}.$$

Higher-order Taylor polynomials yield further correction terms, each one improving the approximation for moderate velocities, until eventually one may as well use the full relativistic expression.

Remark. This is a recurring pattern throughout physics. One derives an exact equation using fundamental principles, but that equation is too complicated for practical use. Replacing certain functions by their Taylor polynomials yields simpler, approximate equations valid in appropriate regimes (here, small v/c). The classical formula for kinetic energy is not an accident—it is the inevitable low-velocity consequence of the deeper relativistic formula.

Appendix A

Precalculus Reference

This appendix collects the precalculus facts and identities used throughout the course. Results are stated without proof. Familiarity with this material is assumed from the first chapter onward.

A.1 Trigonometric Identities

Trigonometric identities pervade every part of calculus. When you differentiate $\sin x$ or $\cos x$, when you integrate powers of trigonometric functions, and when you simplify expressions after a trigonometric substitution, you rely on the identities collected here. Knowing them cold—not just recognising them, but being able to reproduce them quickly—will save you enormous time throughout the course.

Many of the identities below are consequences of just one fact: the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$. We organise them into groups so that you can see the logical relationships.

Proposition A.1 (Pythagorean Identities).

$$\begin{aligned} \sin^2\theta + \cos^2\theta &= 1 \\ 1 + \tan^2\theta &= \sec^2\theta \\ 1 + \cot^2\theta &= \csc^2\theta \end{aligned}$$

Proposition A.2 (Reciprocal and Quotient Identities).

$$\begin{aligned} \csc\theta &= \frac{1}{\sin\theta} \\ \sec\theta &= \frac{1}{\cos\theta} \\ \cot\theta &= \frac{1}{\tan\theta} \\ \tan\theta &= \frac{\sin\theta}{\cos\theta} \\ \cot\theta &= \frac{\cos\theta}{\sin\theta} \end{aligned}$$

Proposition A.3 (Angle Addition and Subtraction Formulas).

$$\begin{aligned}
\sin(\alpha \pm \beta) &= \sin\alpha\cos\beta \pm \cos\alpha\sin\beta \\
\cos(\alpha \pm \beta) &= \cos\alpha\cos\beta \mp \sin\alpha\sin\beta \\
\tan(\alpha \pm \beta) &= \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}
\end{aligned}$$

Proposition A.4 (Double Angle Formulas).

$$\begin{aligned}
\sin(2\theta) &= 2\sin\theta\cos\theta \\
\cos(2\theta) &= \cos^2\theta - \sin^2\theta \\
&= 2\cos^2\theta - 1 \\
&= 1 - 2\sin^2\theta \\
\tan(2\theta) &= \frac{2\tan\theta}{1 - \tan^2\theta}
\end{aligned}$$

Proposition A.5 (Half Angle Formulas).

$$\begin{aligned}
\sin^2\theta &= \frac{1 - \cos(2\theta)}{2} \\
\cos^2\theta &= \frac{1 + \cos(2\theta)}{2} \\
\tan^2\theta &= \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}
\end{aligned}$$

Proposition A.6 (Product-to-Sum Formulas).

$$\begin{aligned}
\sin\alpha\cos\beta &= \tfrac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)] \\
\cos\alpha\sin\beta &= \tfrac{1}{2}[\sin(\alpha+\beta) - \sin(\alpha-\beta)] \\
\cos\alpha\cos\beta &= \tfrac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)] \\
\sin\alpha\sin\beta &= \tfrac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]
\end{aligned}$$

Proposition A.7 (Sum-to-Product Formulas).

$$\begin{aligned}
\sin\alpha + \sin\beta &= 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) \\
\sin\alpha - \sin\beta &= 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right) \\
\cos\alpha + \cos\beta &= 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) \\
\cos\alpha - \cos\beta &= -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)
\end{aligned}$$

Remark. All angles in the table above are measured in radians. The notation undef indicates that the function is undefined at that angle because its denominator is zero; geometrically, the function has a vertical asymptote there.

Table A.1: Values of Trigonometric Functions at Standard Angles

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
0	0	1	0	undef	1	undef
$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3}$	2	$2/\sqrt{3}$	$\sqrt{3}$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1	$\sqrt{2}$	$\sqrt{2}$	1
$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$	$2/\sqrt{3}$	2	$1/\sqrt{3}$
$\pi/2$	1	0	undef	1	undef	0
π	0	-1	0	undef	-1	undef
$3\pi/2$	-1	0	undef	-1	undef	0
2π	0	1	0	undef	1	undef

A.2 Exponential and Logarithmic Laws

The exponential function e^x and the natural logarithm $\ln x$ are everywhere in calculus: they appear in differentiation rules, integration formulas, growth and decay models, and the definitions of many other functions. The algebraic laws listed here are the foundation on which all of that rests.

In particular, the change-of-base formula lets us reduce any exponential a^x to $e^{x\ln a}$, which is essential when differentiating or integrating exponentials with arbitrary bases in Chapters 3 and 4.

Proposition A.8 (Laws of Exponents). *For $a > 0$ and all $x, y \in \mathbb{R}$:*

$$\begin{aligned} a^x \cdot a^y &= a^{x+y} \\ \frac{a^x}{a^y} &= a^{x-y} \\ (a^x)^y &= a^{xy} \\ (ab)^x &= a^x b^x \\ a^0 &= 1 \\ a^{-x} &= \frac{1}{a^x} \end{aligned}$$

Remark. The law $(a^x)^y = a^{xy}$ is used constantly when simplifying expressions before differentiating—for instance, rewriting $\sqrt[3]{x^2}$ as $x^{2/3}$ so that the power rule applies directly.

Proposition A.9 (Laws of Logarithms). *For $a > 0$, $a \neq 1$, and $x, y > 0$:*

$$\begin{aligned}
\log_a(xy) &= \log_a x + \log_a y \\
\log_a(x/y) &= \log_a x - \log_a y \\
\log_a(x^r) &= r \log_a x \\
\log_a x &= \frac{\ln x}{\ln a} \quad \textit{(change of base)} \\
\ln(e^x) &= x \quad \textit{for all } x \in \mathbb{R} \\
e^{\ln x} &= x \quad \textit{for all } x > 0
\end{aligned}$$

Remark. The change-of-base formula $\log_a x = \dfrac{\ln x}{\ln a}$ is the key to differentiating a^x for $a \neq e$: we write $a^x = e^{x \ln a}$ and apply the chain rule. It also converts any logarithmic integral into one involving ln.

A.3 Algebraic Techniques

The algebraic manipulations below appear throughout the course: completing the square is essential for certain integrals and limit calculations, factoring identities simplify rational expressions, and the binomial theorem underpins Taylor expansions and series analysis.

Proposition A.10 (Completing the Square).

$$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \frac{b^2}{4}$$

Example A.1. Write $x^2 + 6x + 5$ in completed square form.

We isolate the quadratic and linear terms and apply the formula:

$$\begin{aligned}
x^2 + 6x + 5 &= (x^2 + 6x) + 5 \\
&= \left(x^2 + 6x + 9\right) - 9 + 5 \\
&= (x + 3)^2 - 4
\end{aligned}$$

Hence $x^2 + 6x + 5 = (x + 3)^2 - 4$, so the minimum value of the expression is -4, attained at $x = -3$.

Proposition A.11 (Factoring Identities).

$$\begin{aligned}
a^2 - b^2 &= (a - b)(a + b) \\
a^3 - b^3 &= (a - b)(a^2 + ab + b^2) \\
a^3 + b^3 &= (a + b)(a^2 - ab + b^2)
\end{aligned}$$

Proposition A.12 (Binomial Theorem). *For $n \in \mathbb{N}$ and $a, b \in \mathbb{R}$:*

$$(a + b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

The first three cases explicitly:

$$(a+b)^2 = a^2 + 2ab + b^2$$
$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Remark (Polynomial Long Division). When integrating a rational function $p(x)/q(x)$ where $\deg p \geq \deg q$, perform polynomial long division first to write the integrand as a polynomial plus a proper rational function.

For example, divide $x^3 + 2x$ by $x^2 + 1$:

$$\begin{aligned} x^3 + 2x &= (x^2+1) \cdot x + (2x - x) \\ &= (x^2+1) \cdot x + x \end{aligned}$$

We verify this step by step. We seek $q(x)$ and $r(x)$ with $\deg r < \deg(x^2 + 1) = 2$ such that $x^3 + 2x = (x^2+1)\, q(x) + r(x)$.

First, divide the leading term: $x^3 \div x^2 = x$.

Multiply and subtract:

$$\begin{aligned} x^3 + 2x - x \cdot (x^2+1) &= x^3 + 2x - x^3 - x \\ &= x \end{aligned}$$

Since $\deg(x) = 1 < 2 = \deg(x^2+1)$, we stop. Therefore:

$$\frac{x^3 + 2x}{x^2+1} = x + \frac{x}{x^2+1}$$

A.4 Inverse Functions

Inverse functions are central to Chapter 4, where we study inverse trigonometric functions (arcsin, arccos, arctan) and where we treat the natural logarithm as the inverse of the exponential. The derivative rule for inverse functions, stated below, appears in Chapter 3 and is used repeatedly whenever we differentiate an inverse function.

Understanding when a function *has* an inverse—and the role of restricting the domain to guarantee invertibility—is essential before any of those later results make sense.

Definition A.1 (Inverse Function). Let $f\colon A \to B$ be a function. A function $g\colon B \to A$ is called the *inverse* of f if

$$g(f(x)) = x \quad \text{for all } x \in A \qquad \text{and} \qquad f(g(y)) = y \quad \text{for all } y \in B.$$

When such a g exists, f is called *invertible*, and we write $g = f^{-1}$.

Remark. A function f is invertible if and only if it is *bijective*: injective (one-to-one) and surjective (onto). Geometrically, f passes the horizontal line test.

Proposition A.13 (Inverse Function Derivative Rule). *Let f be differentiable and invertible on an interval, with $f'(x) \neq 0$. Then f^{-1} is differentiable and*

$$\left(f^{-1}\right)'(y) = \frac{1}{f'(f^{-1}(y))}.$$

Remark. The proof of Proposition A.13 uses the chain rule and appears in Section 4.4.